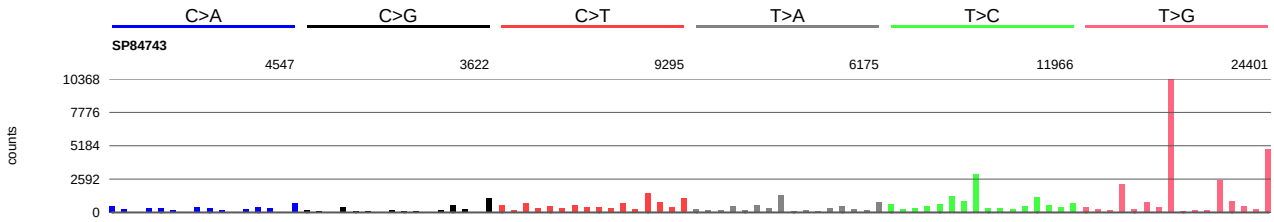
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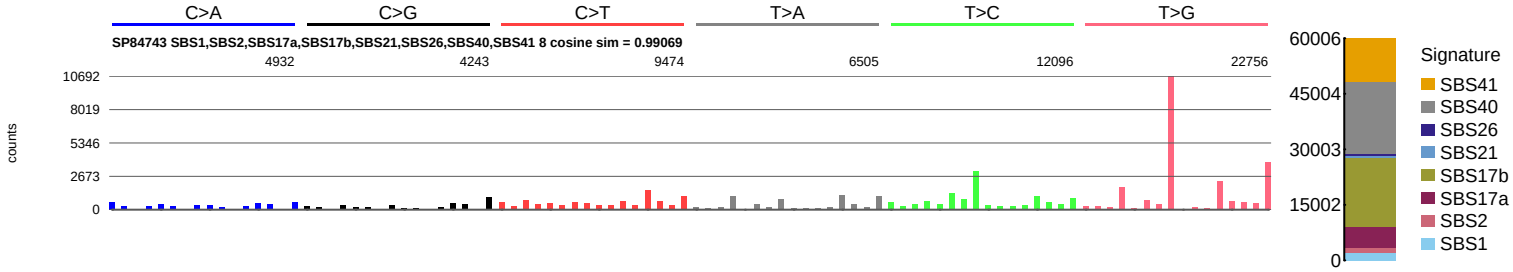
counts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SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99069
4932
4243
9474
6505
12096
22756
10692
8019
5346
2673
0
counts
60006
45004
30003
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Signature
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SBS40
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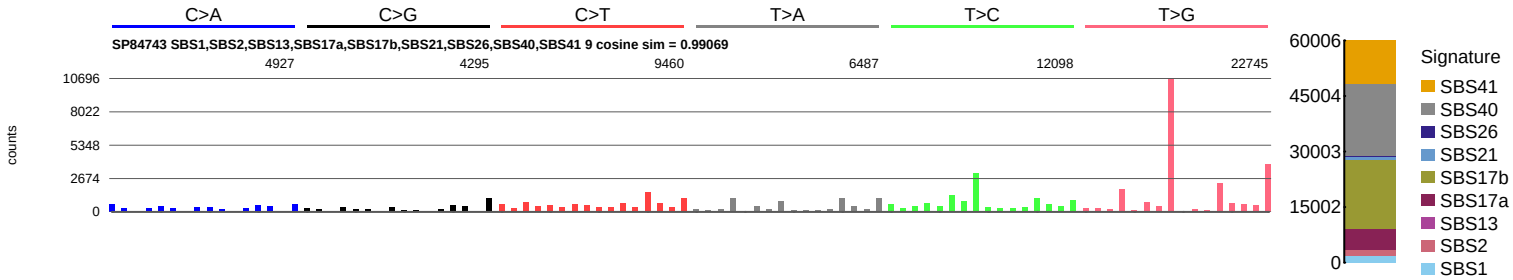
SBS21
SBS17b
SBS17a
SBS2
SBS1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4927
4295
9460
6487
12098
22745
10696
8022
5348
2674
0
counts
60006
45004
30003
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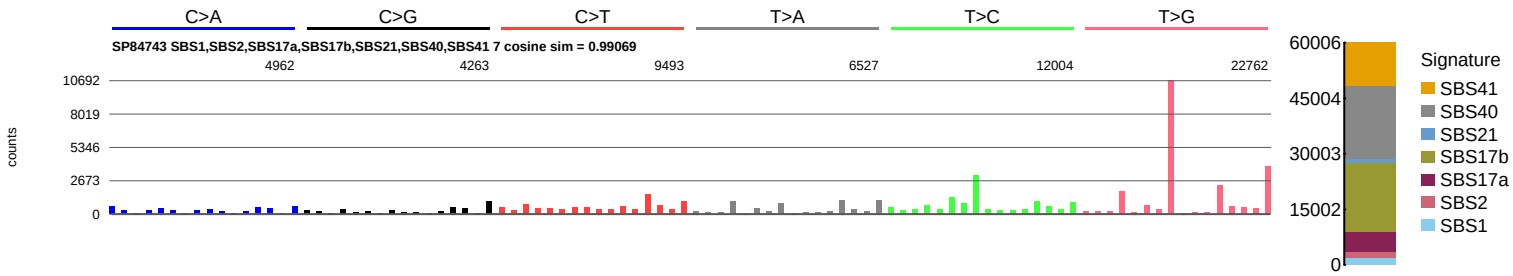
SBS2
SBS1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4263
9493
6527
12004
22762
10692
8019
5346
2673
0
counts
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SBS17b
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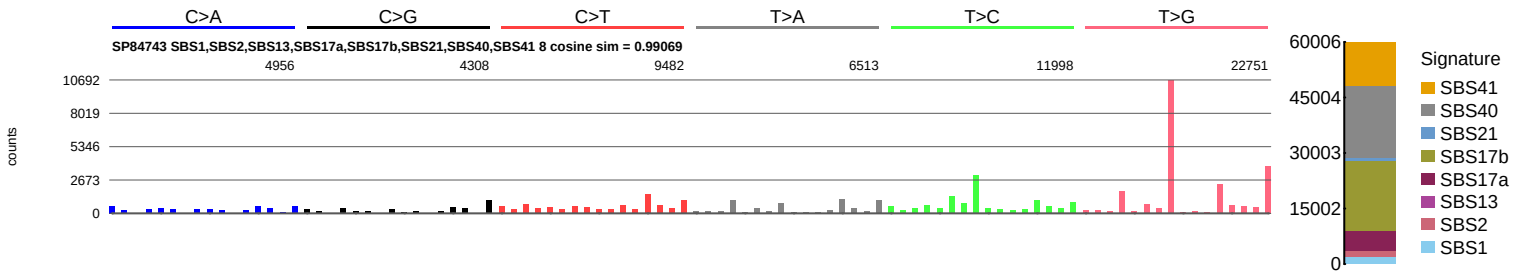
SBS2
SBS1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SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99069
4956
4308
9482
6513
11998
22751
10692
8019
5346
2673
0
counts
60006
45004
30003
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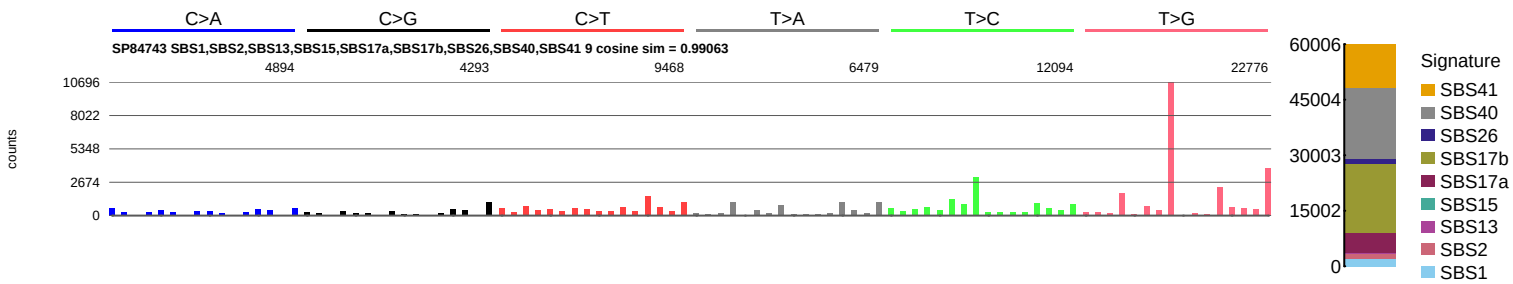
SBS17a
SBS13
SBS2
SBS1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SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.99063
4894
4293
9468
6479
12094
22776
10696
8022
5348
2674
0
counts
60006
45004
30003
15002
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SBS17b
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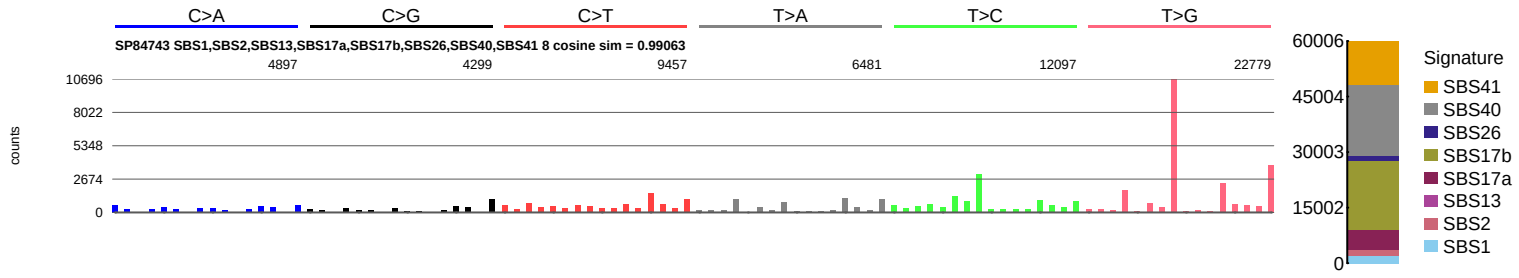
SBS15
SBS13
SBS2
SBS1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SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99063
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4299
9457
6481
12097
22779
10696
8022
5348
2674
0
counts
60006
45004
30003
15002
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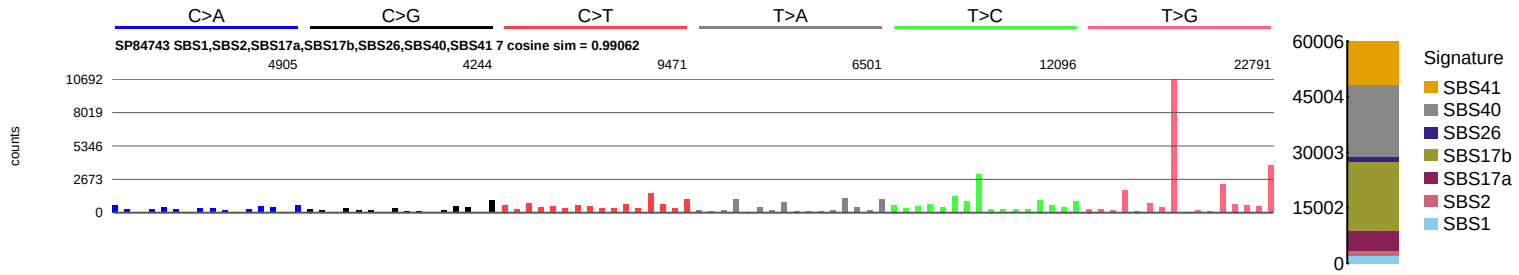
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99062
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4244
9471
6501
12096
22791
10692
8019
5346
2673
0
counts
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30003
15002
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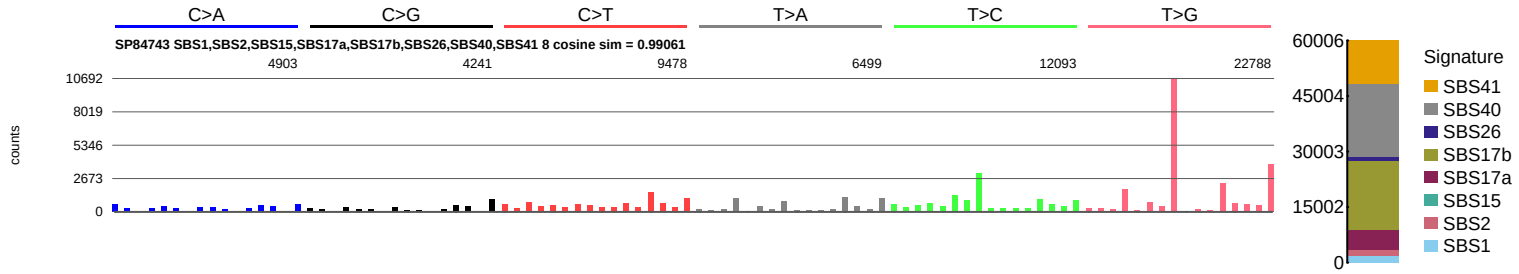
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1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SP84743 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99061
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4241
9478
6499
12093
22788
10692
8019
5346
2673
0
counts
60006
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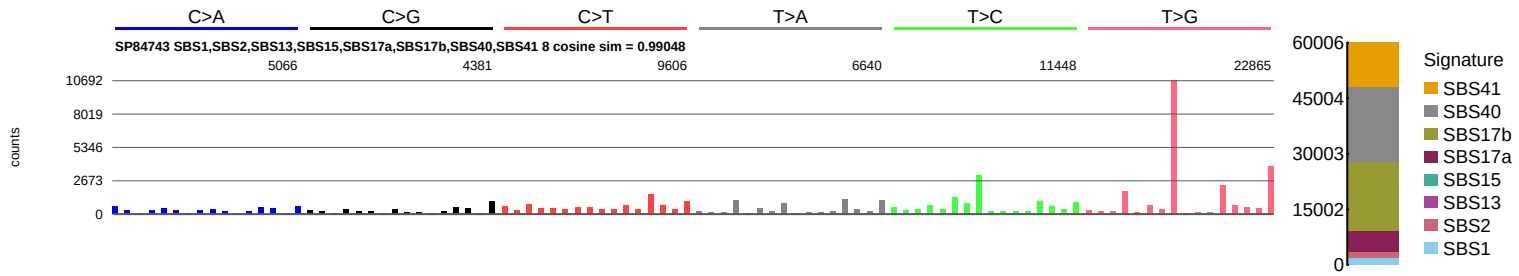
SBS2
SBS1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SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.99048
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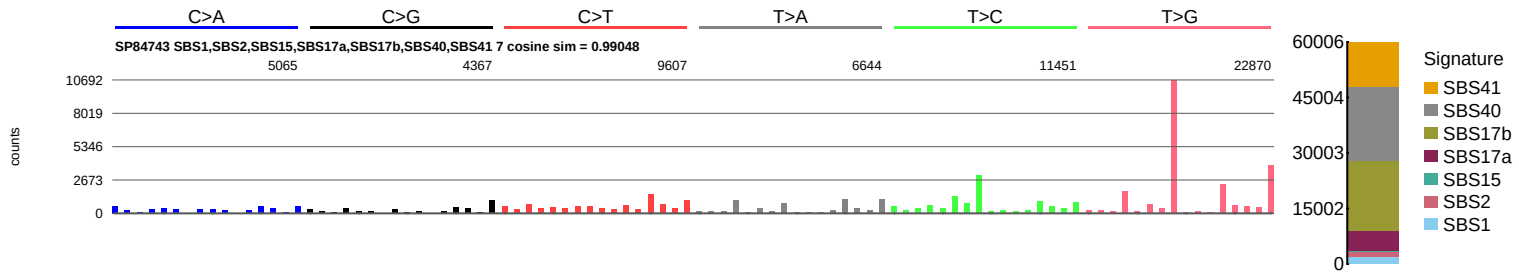
4381
9606
6640
11448
22865
10692
8019
5346
2673
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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T>C
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SP84743 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99048
5065
4367
9607
6644
11451
22870
10692
8019
5346
2673
0
counts
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45004
30003
15002
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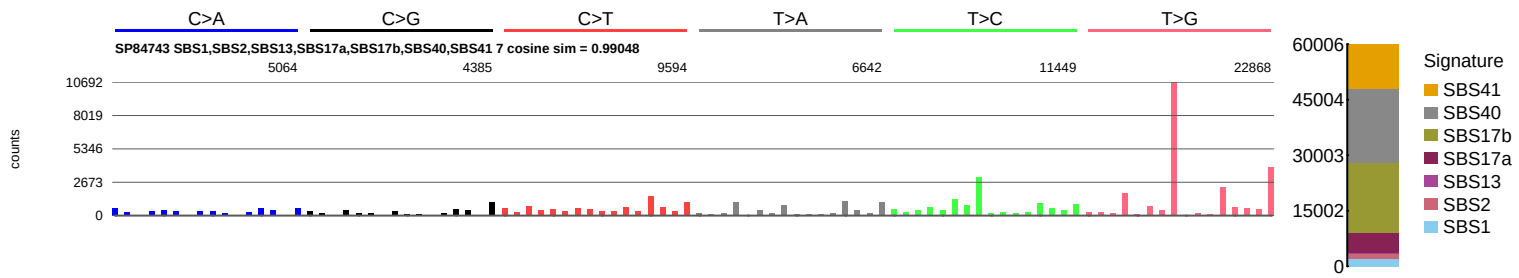
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1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SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99048
5064
4385
9594
6642
11449
22868
10692
8019
5346
2673
0
counts
60006
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30003
15002
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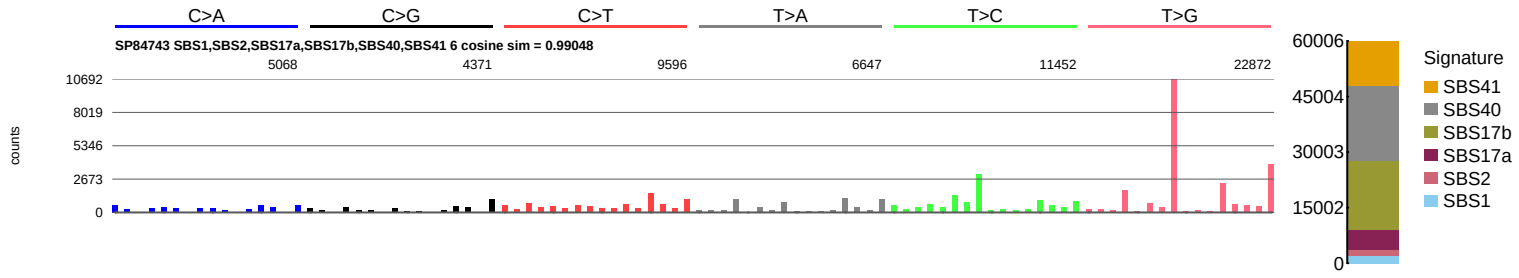
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
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SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99048
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4371
9596
6647
11452
22872
counts
10692
8019
5346
2673
0
60006
45004
30003
15002
0
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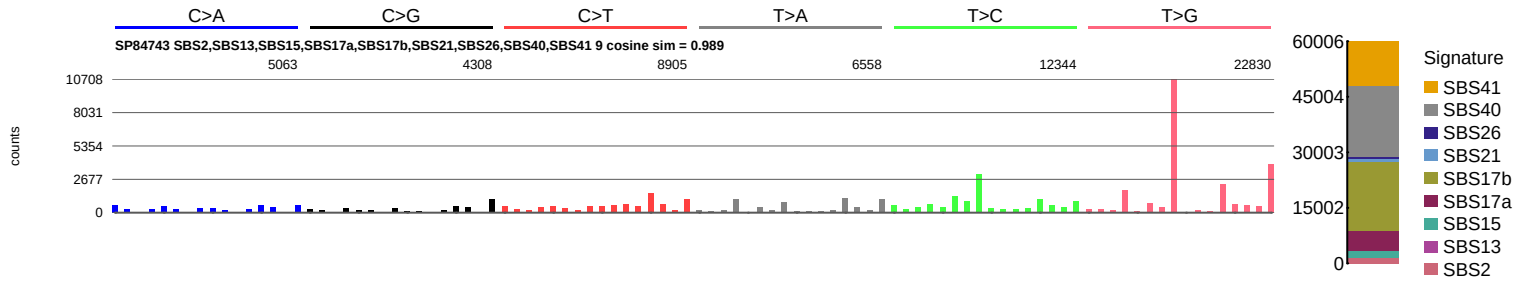
SBS40
SBS17b
SBS17a
SBS2
SBS1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T>C
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SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.989
5063
4308
8905
6558
12344
22830
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
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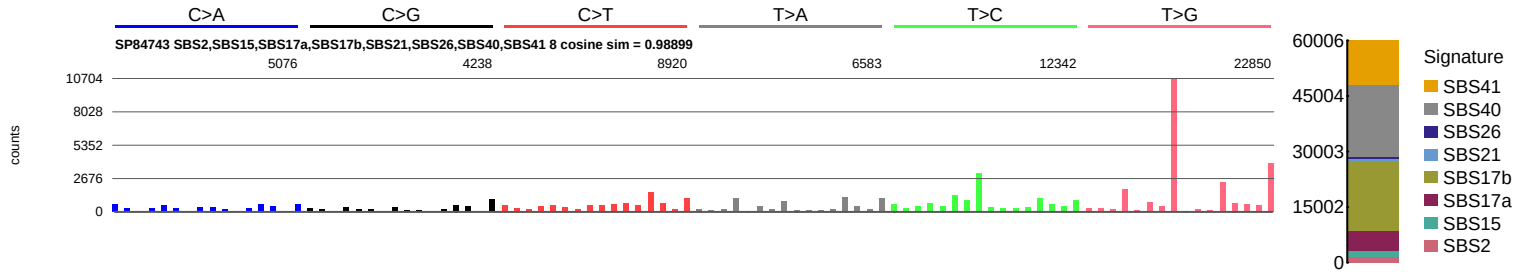
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98899
5076
4238
8920
6583
12342
22850
counts
10704
8028
5352
2676
0
60006
45004
30003
15002
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SBS40
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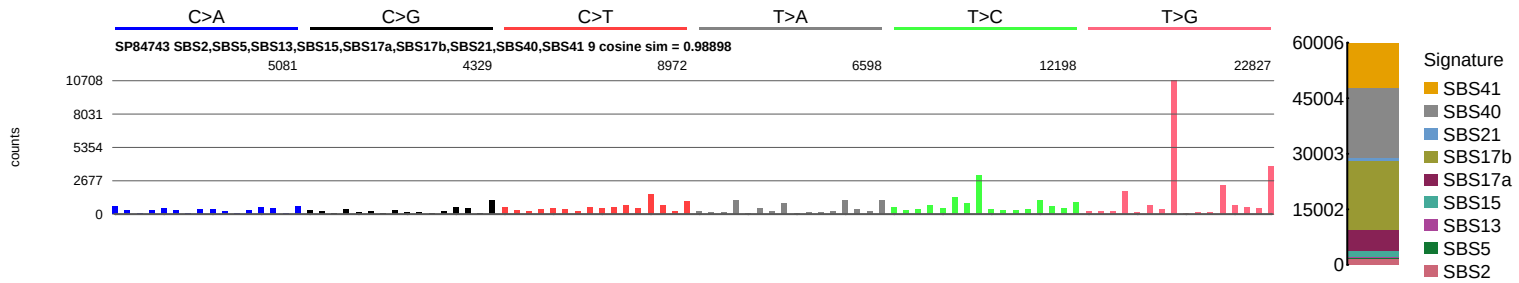
SBS21
SBS17b
SBS17a
SBS15
SBS2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C>G
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T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98898
5081
4329
8972
6598
12198
22827
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
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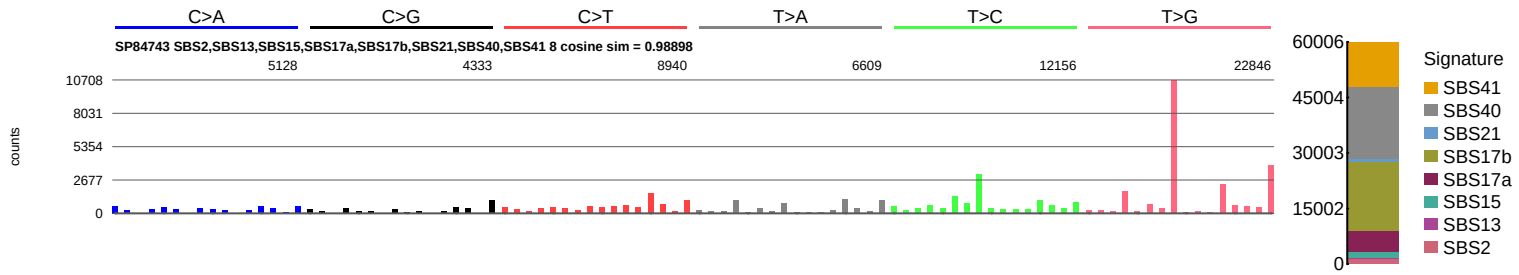
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98898
5128
4333
8940
6609
12156
22846
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
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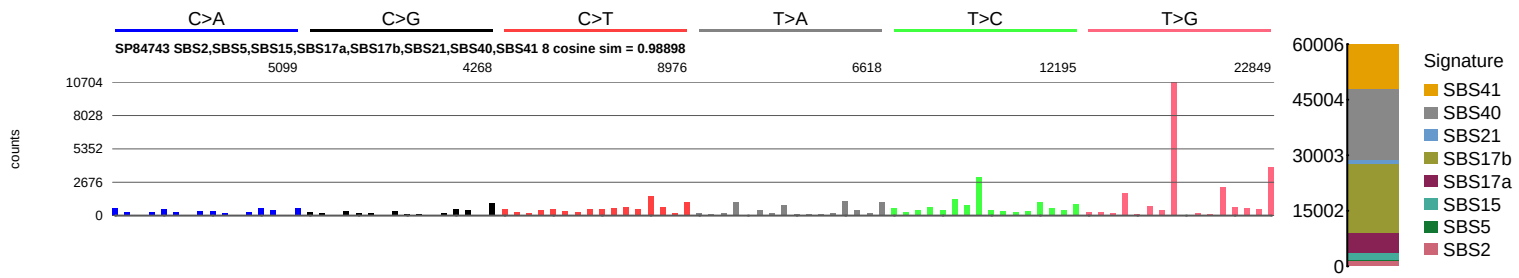
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2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SP84743 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98898
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4268
8976
6618
12195
22849
counts
10704
8028
5352
2676
0
60006
45004
30003
15002
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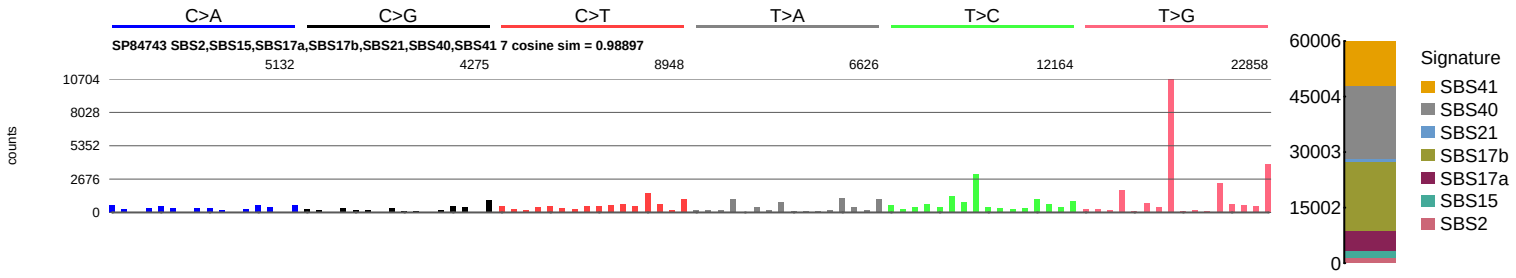
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2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SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98897
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4275
8948
6626
12164
22858
counts
10704
8028
5352
2676
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60006
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30003
15002
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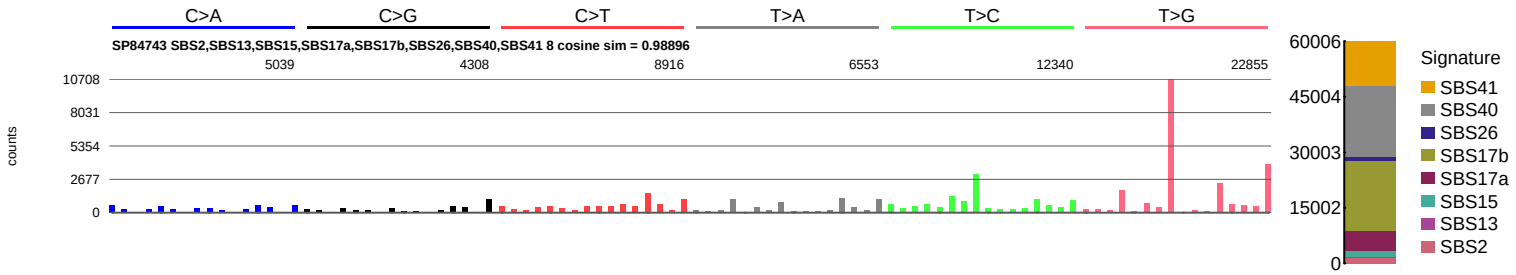
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2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C>T
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T>C
T>G
SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98896
5039
4308
8916
6553
12340
22855
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
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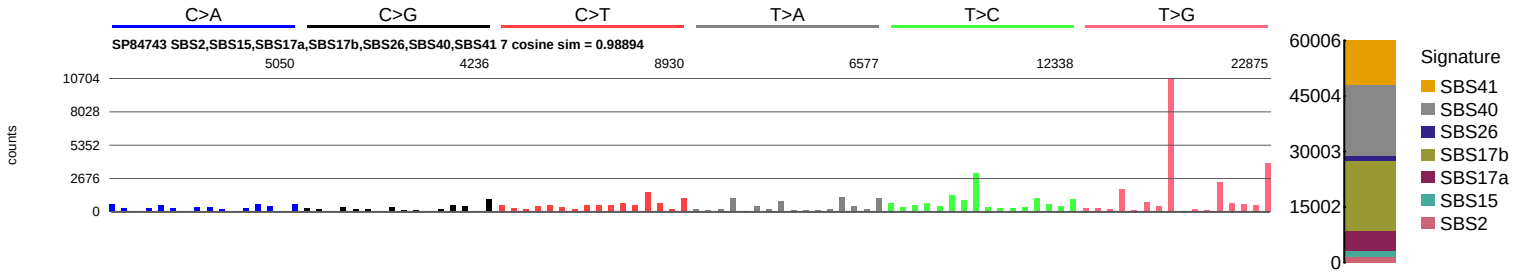
SBS13
SBS2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C>G
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SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98894
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4236
8930
6577
12338
22875
counts
10704
8028
5352
2676
0
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45004
30003
15002
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SBS17b
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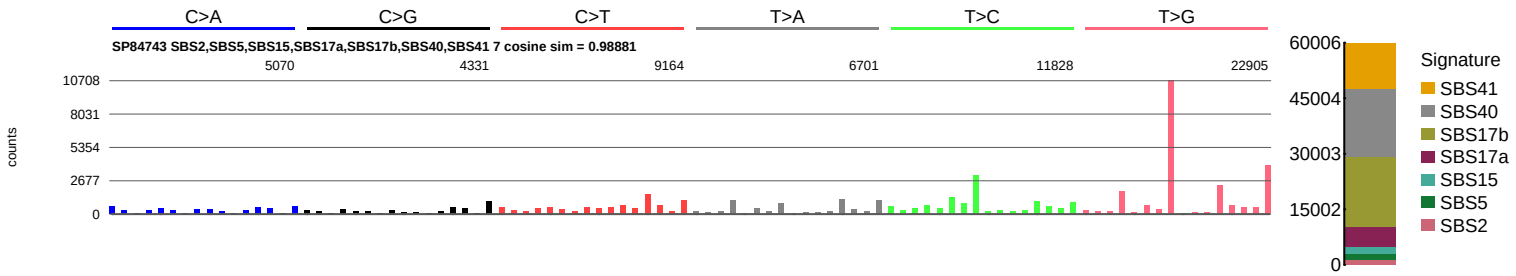
SBS15
SBS2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C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98881
5070
4331
9164
6701
11828
22905
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
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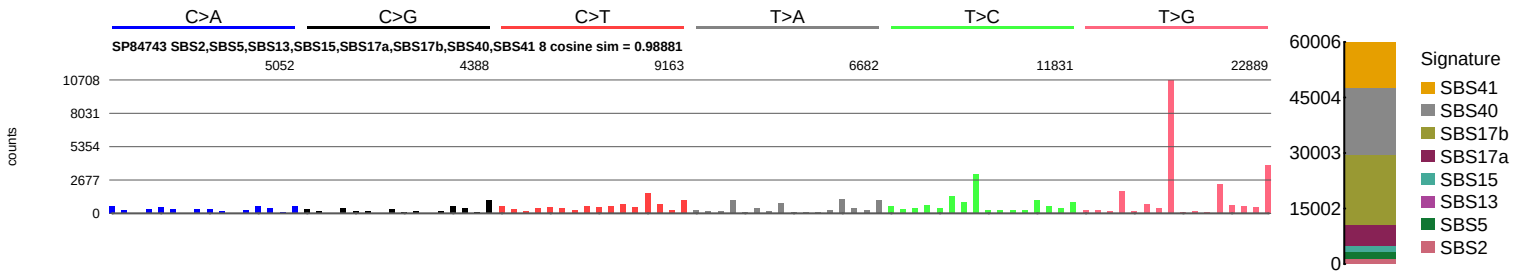
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2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C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98881
5052
4388
9163
6682
11831
22889
counts
10708
8031
5354
2677
0
60006
45004
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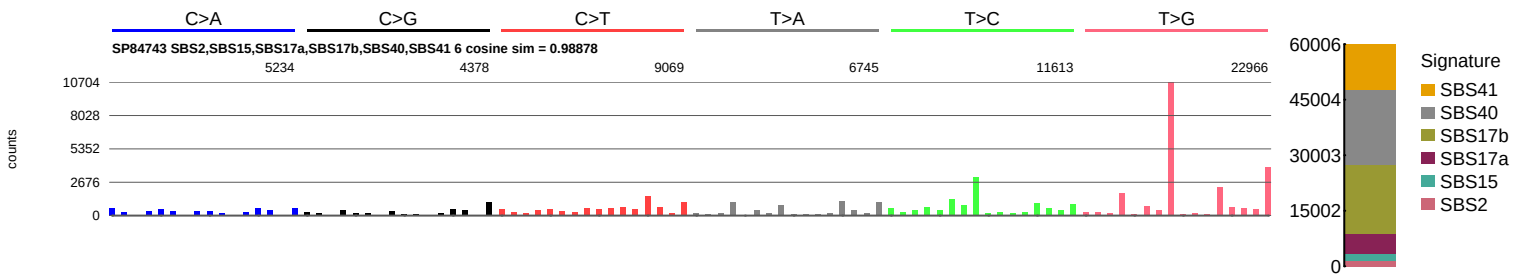
SBS5
SBS2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T>C
T>G
SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98878
5234
4378
9069
6745
11613
22966
counts
10704
8028
5352
2676
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60006
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30003
15002
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SBS40
SBS17b
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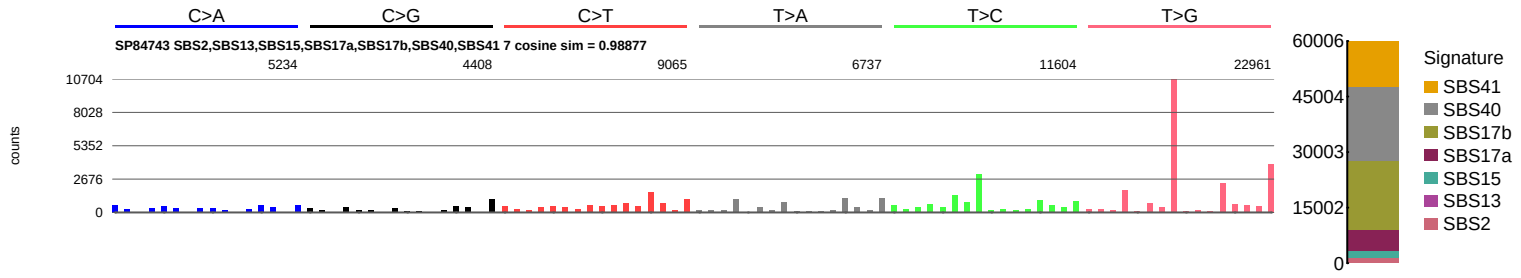
SBS2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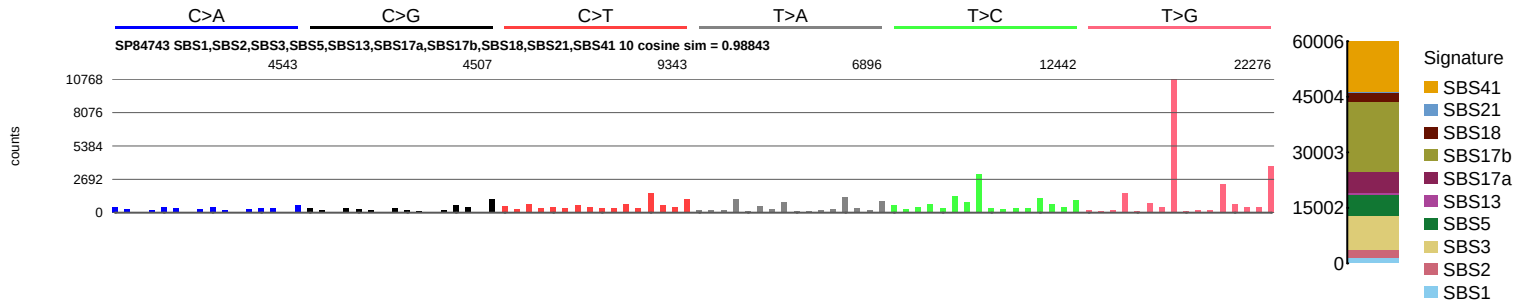
T>C
T>G
SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98877
5234
4408
9065
6737
11604
22961
counts
10704
8028
5352
2676
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2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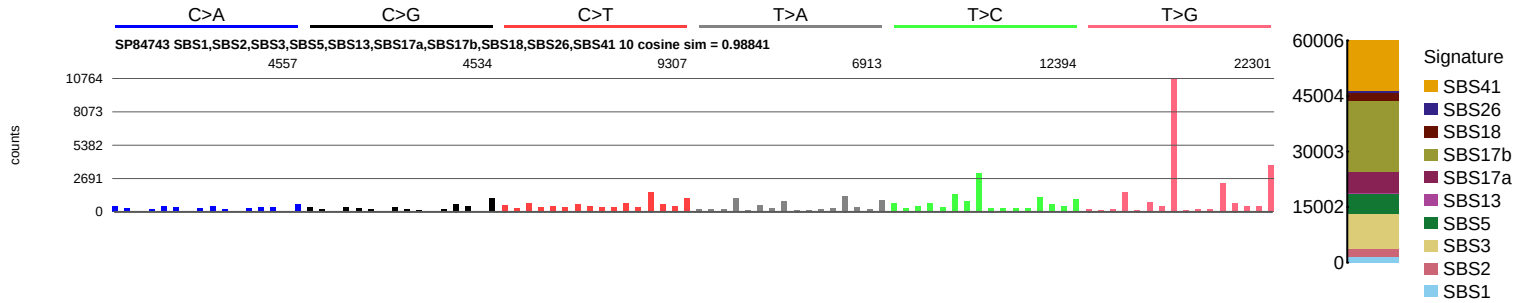
SP84743 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98843
4543
4507
9343
6896
12442
22276
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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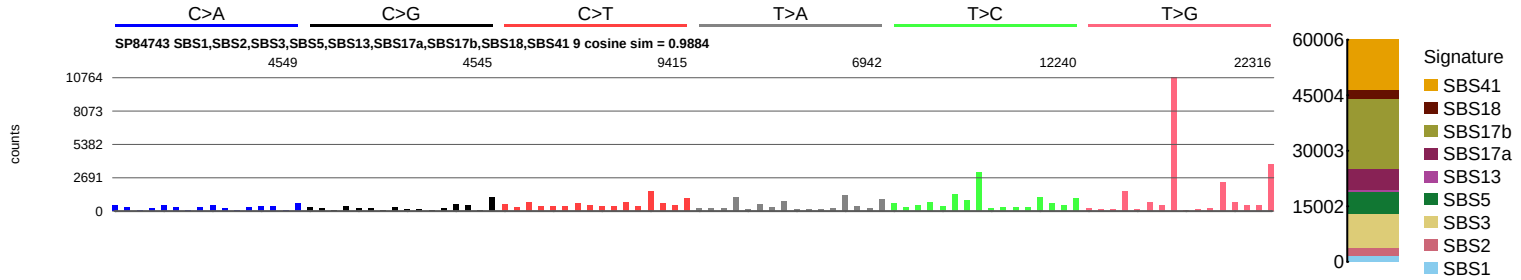
SP84743 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98841
4557
4534
9307
6913
12394
22301
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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SP84743 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9884
4549
4545
9415
6942
12240
22316
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
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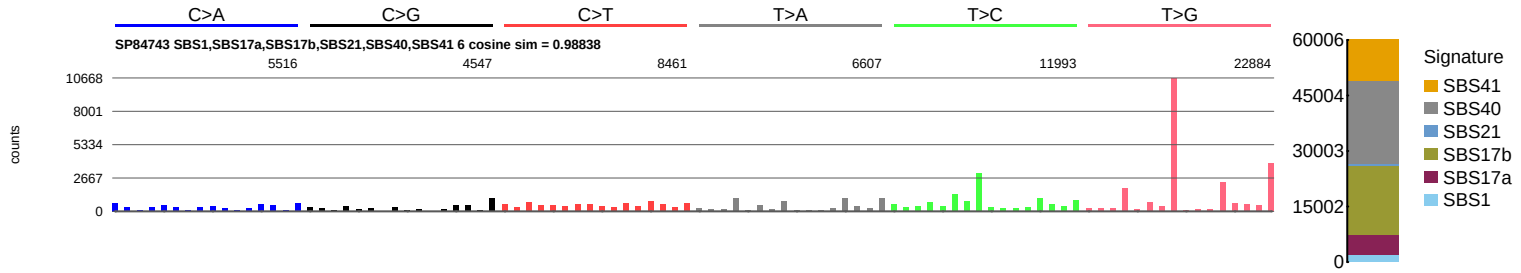
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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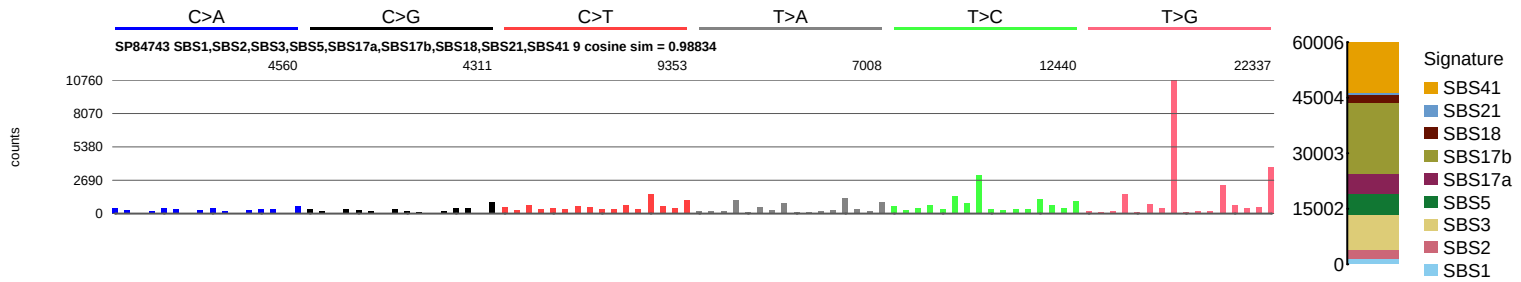
T>C
T>G
SP84743 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98838
5516
4547
8461
6607
11993
22884
counts
10668
8001
5334
2667
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1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C>G
C>T
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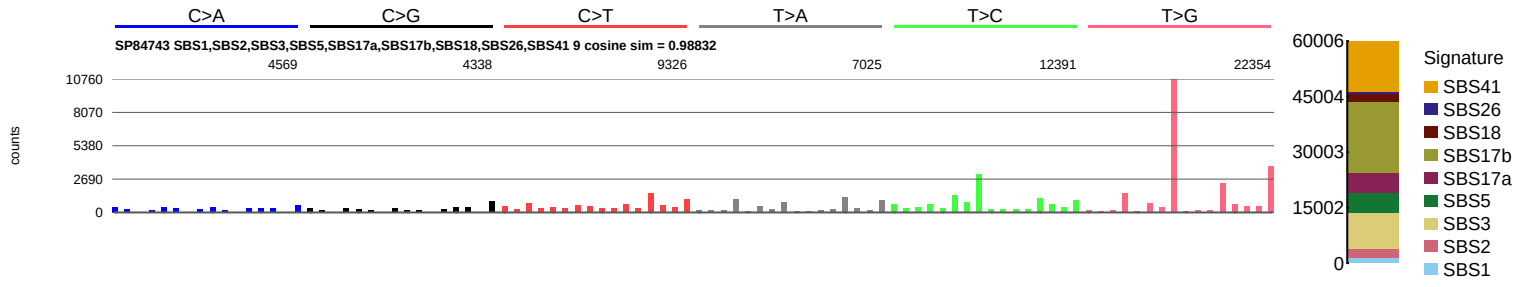
SP84743 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98834
4560
4311
9353
7008
12440
22337
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>T
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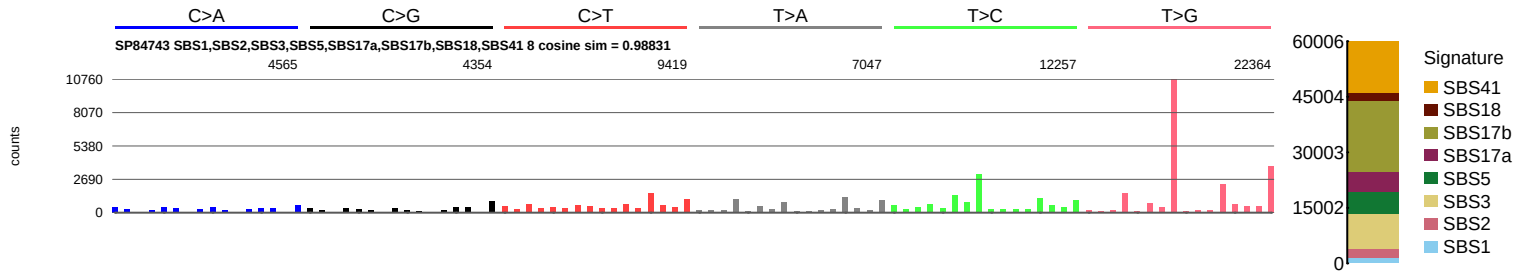
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98832
4569
4338
9326
7025
12391
22354
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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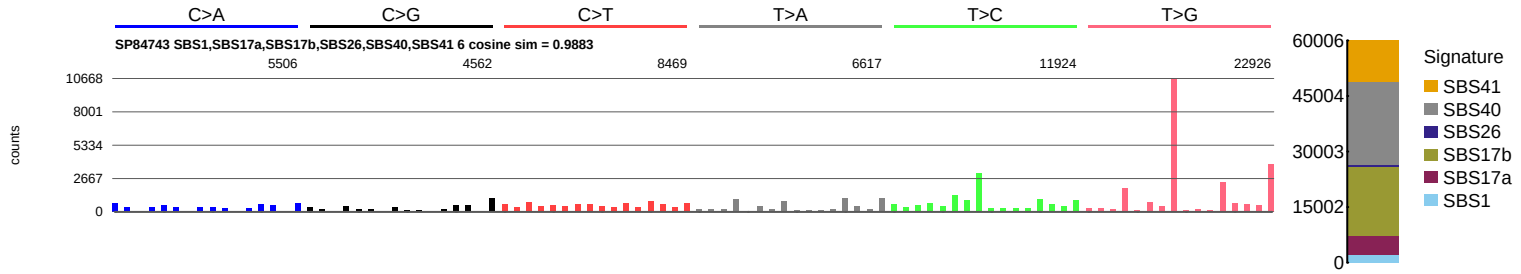
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98831
4565
4354
9419
7047
12257
22364
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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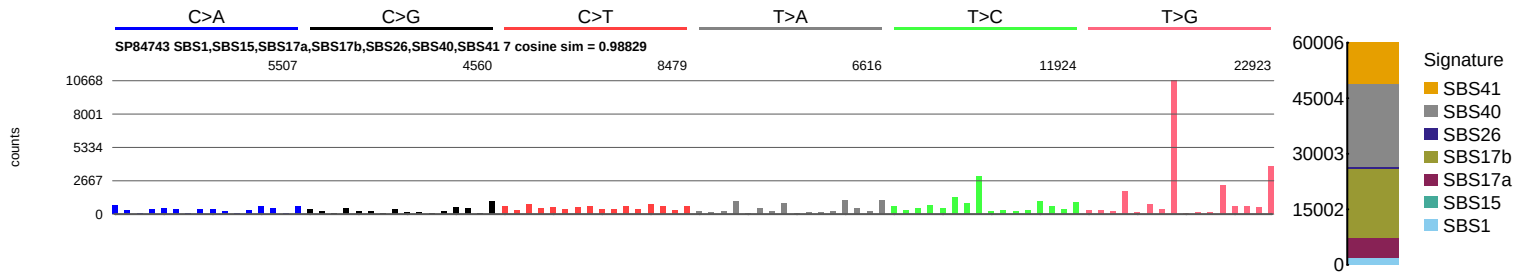
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.9883
5506
4562
8469
6617
11924
22926
counts
10668
8001
5334
2667
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1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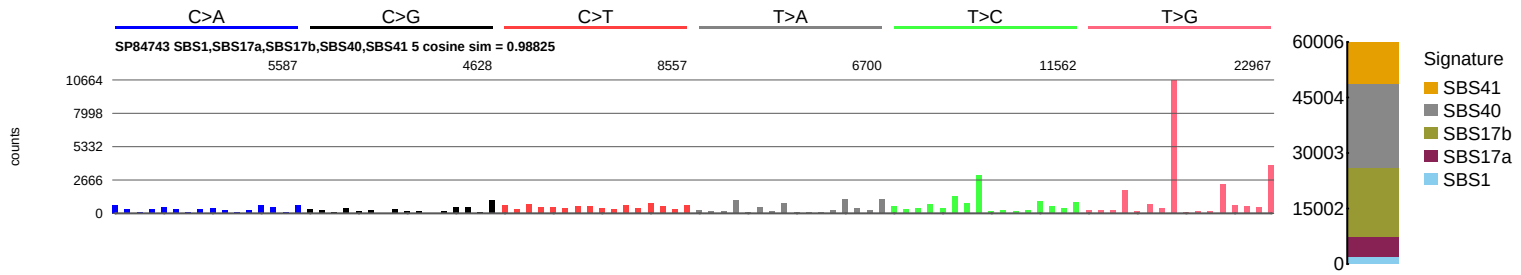
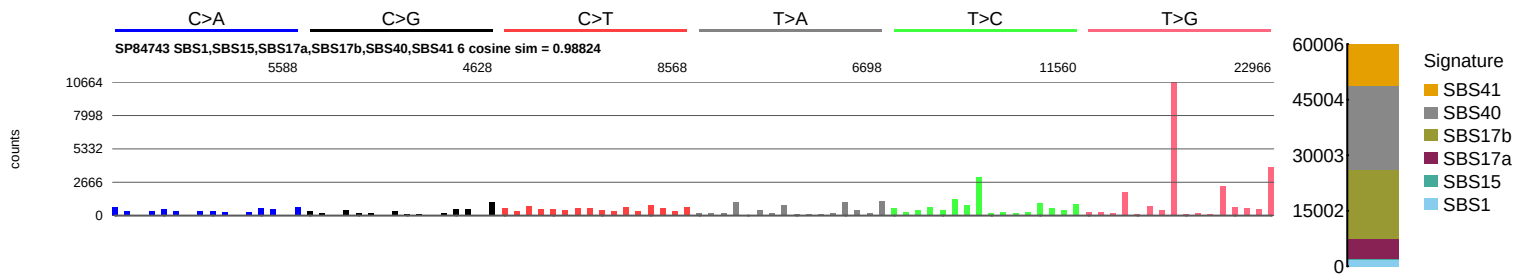
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98829
5507
4560
8479
6616
11924
22923
counts
10668
8001
5334
2667
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98825
5587
4628
8557
6700
11562
22967
counts
10664
7998
5332
2666
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1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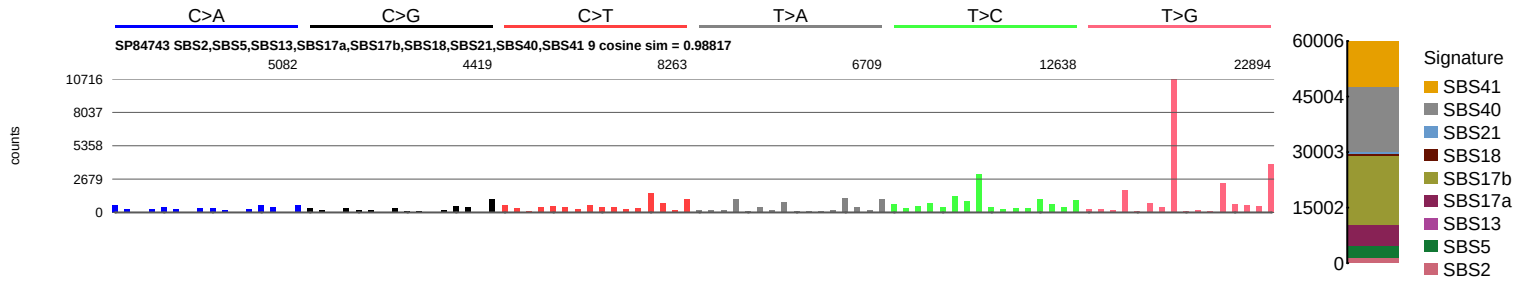
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98824
5588
4628
8568
6698
11560
22966
counts
10664
7998
5332
2666
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1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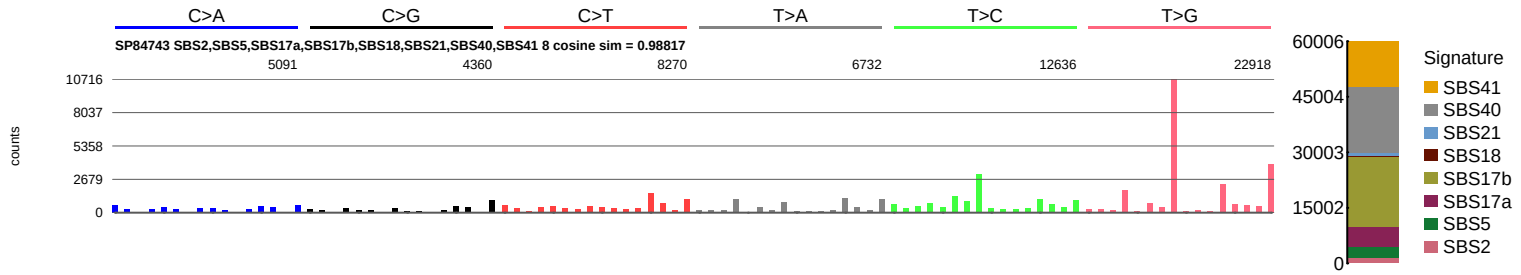
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98817
5082
4419
8263
6709
12638
22894
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2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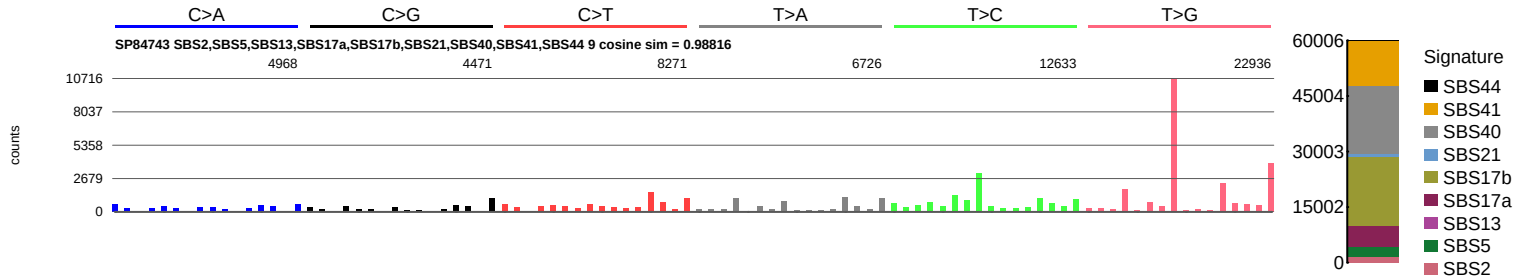
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98817
5091
4360
8270
6732
12636
22918
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98816
4968
4471
8271
6726
12633
22936
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
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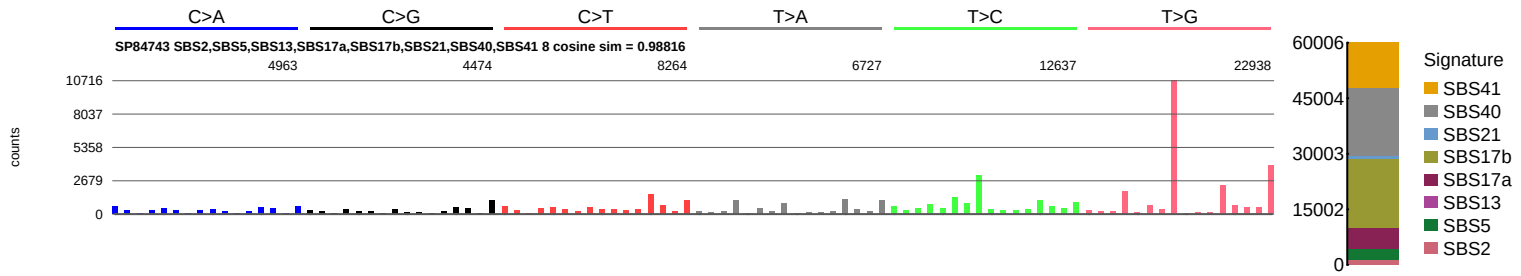
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98816
4963
4474
8264
6727
12637
22938
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
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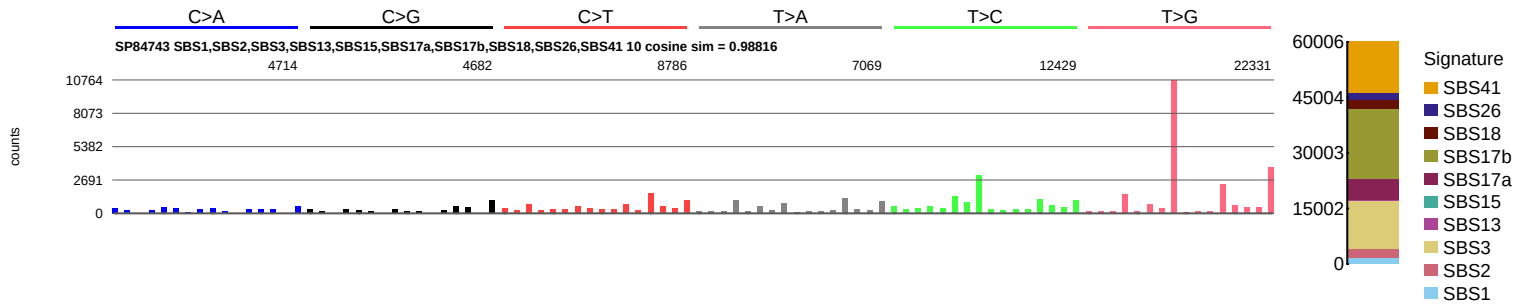
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2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C>G
C>T
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T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98816
4714
4682
8786
7069
12429
22331
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
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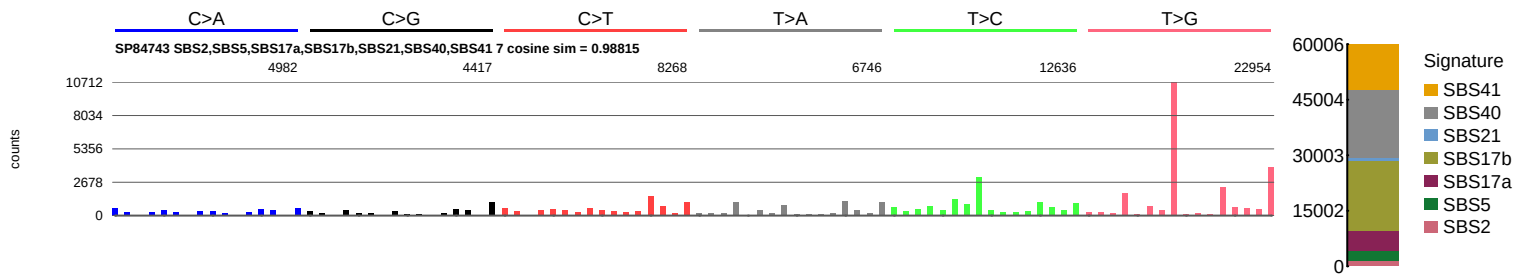
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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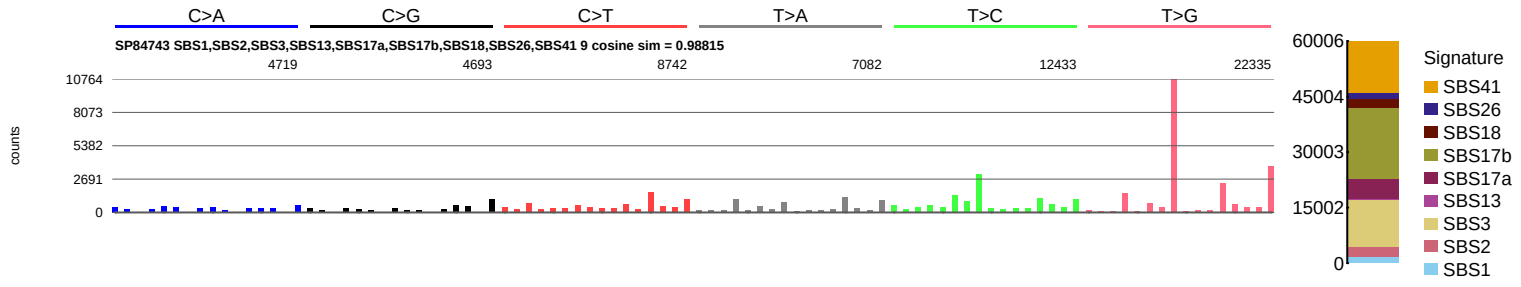
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98815
4982
4417
8268
6746
12636
22954
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2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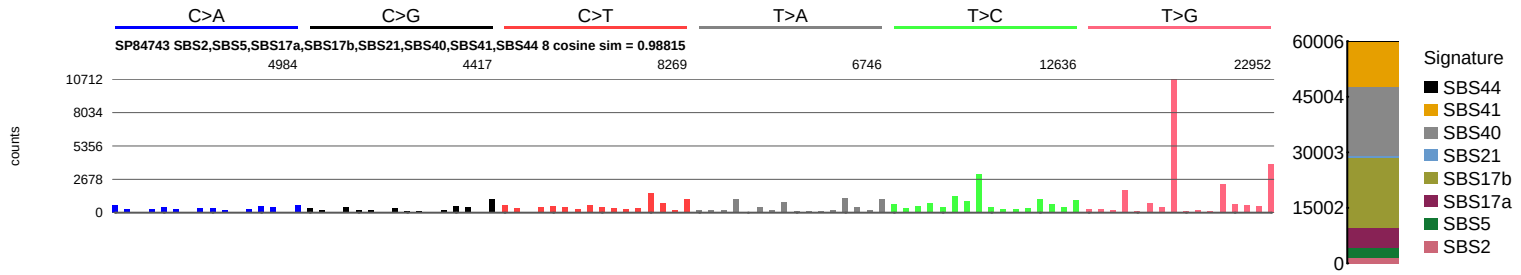
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98815
4719
4693
8742
7082
12433
22335
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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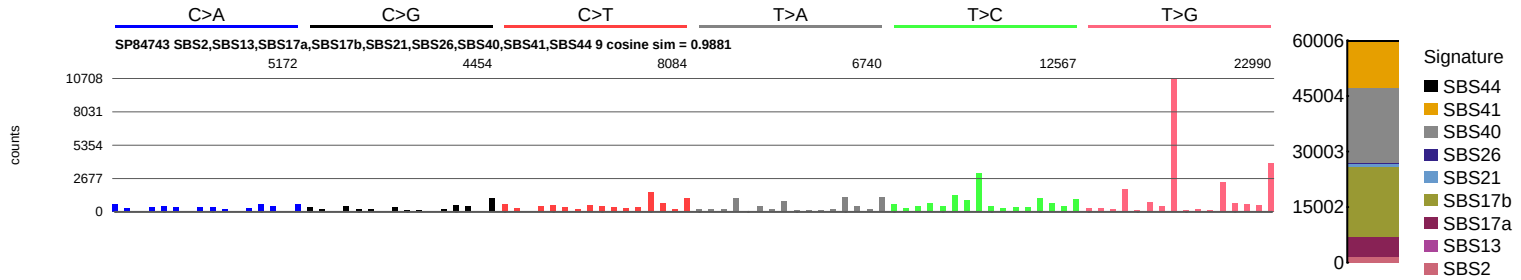
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98815
4984
4417
8269
6746
12636
22952
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2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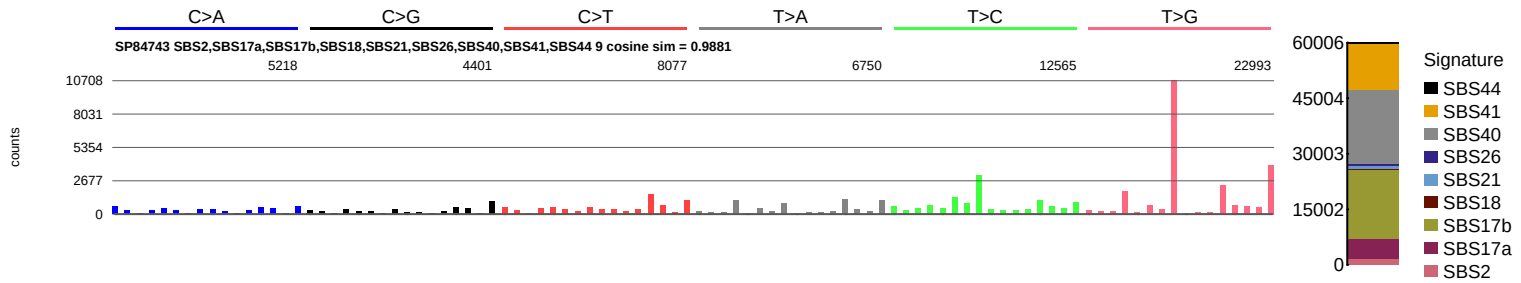
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9881
5172
4454
8084
6740
12567
22990
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2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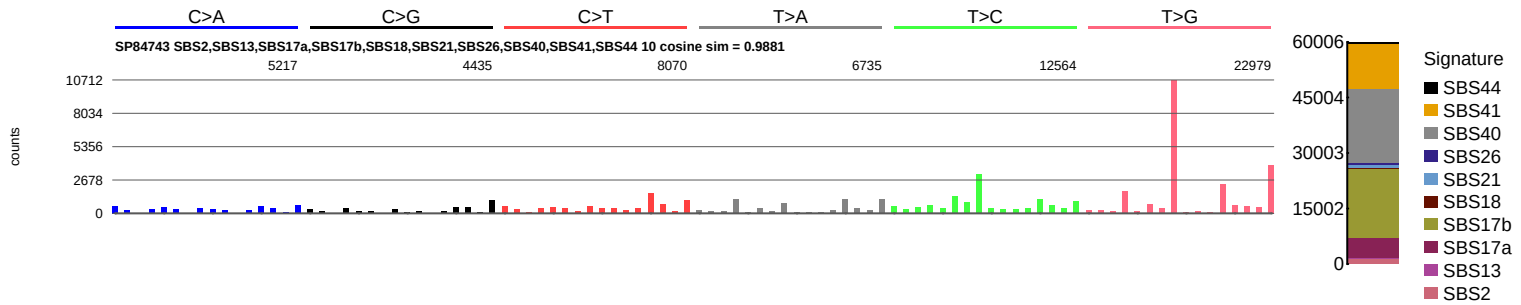
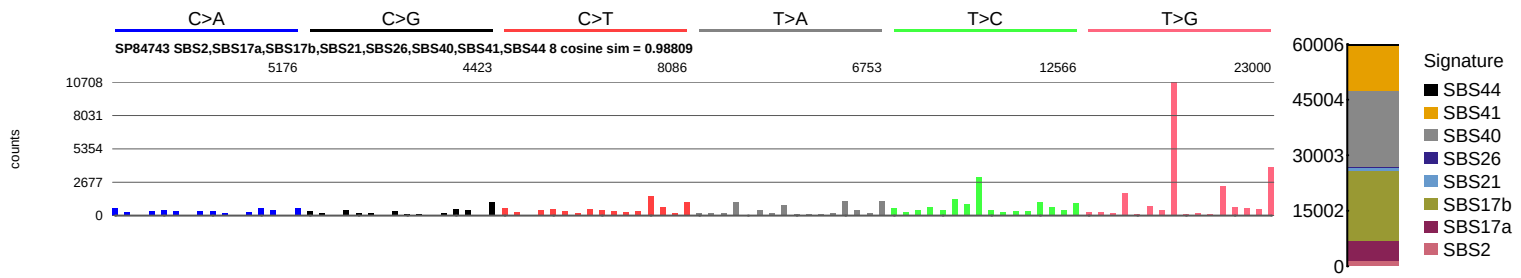
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9881
5218
4401
8077
6750
12565
22993
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9881
5217
4435
8070
6735
12564
22979
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2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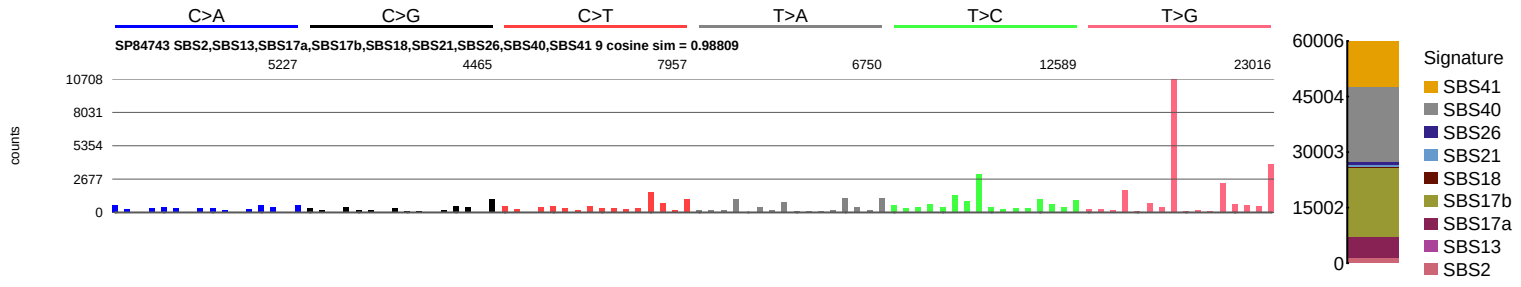
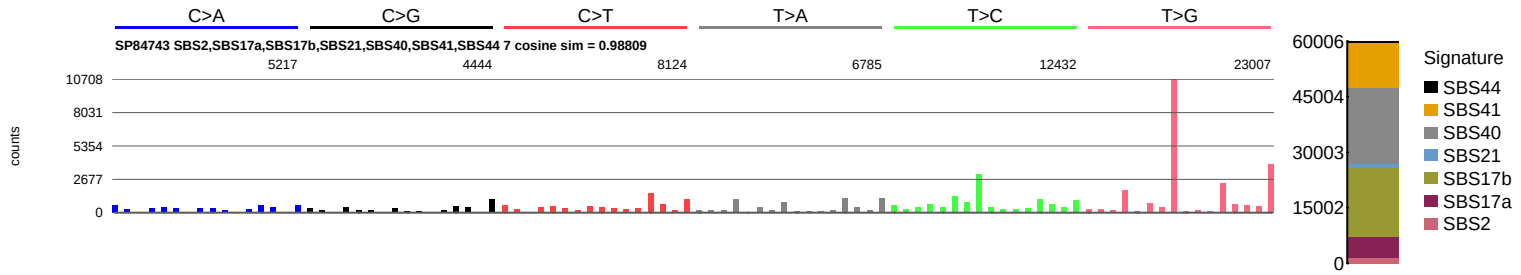
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98809
5176
4423
8086
6753
12566
23000
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98809
5227
4465
7957
6750
12589
23016
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
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T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98809
5217
4444
8124
6785
12432
23007
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
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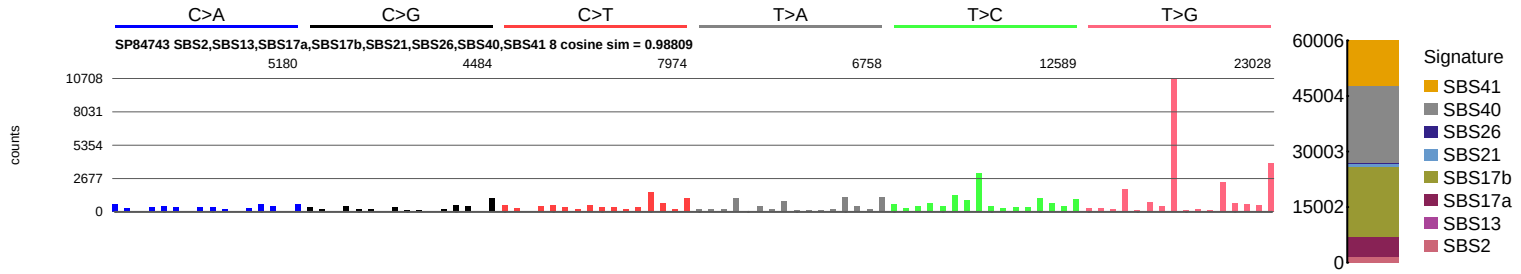
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2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C>G
C>T
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T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98809
5180
4484
7974
6758
12589
23028
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
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SBS40
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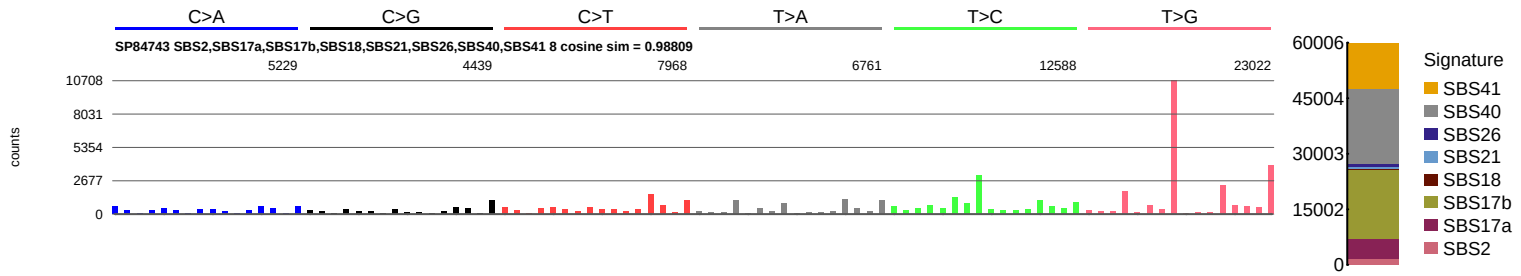
SBS17b
SBS17a
SBS13
SBS2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C>G
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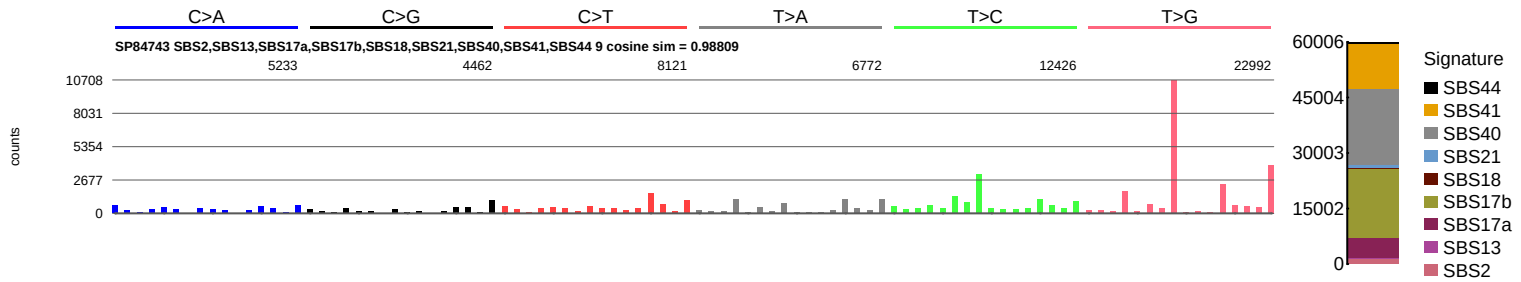
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98809
5229
4439
7968
6761
12588
23022
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2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C>G
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T>A
T>C
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SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98809
5233
4462
8121
6772
12426
22992
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
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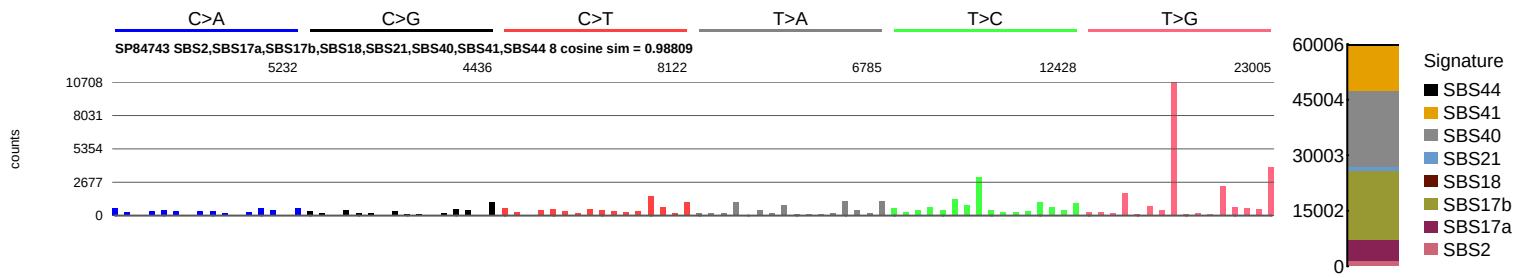
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2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C>G
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T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98809
5232
4436
8122
6785
12428
23005
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
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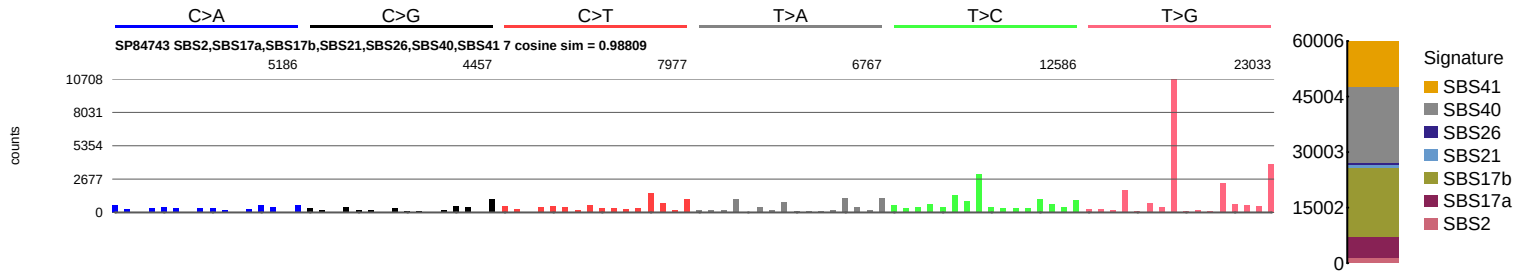
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98809
5186
4457
7977
6767
12586
23033
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
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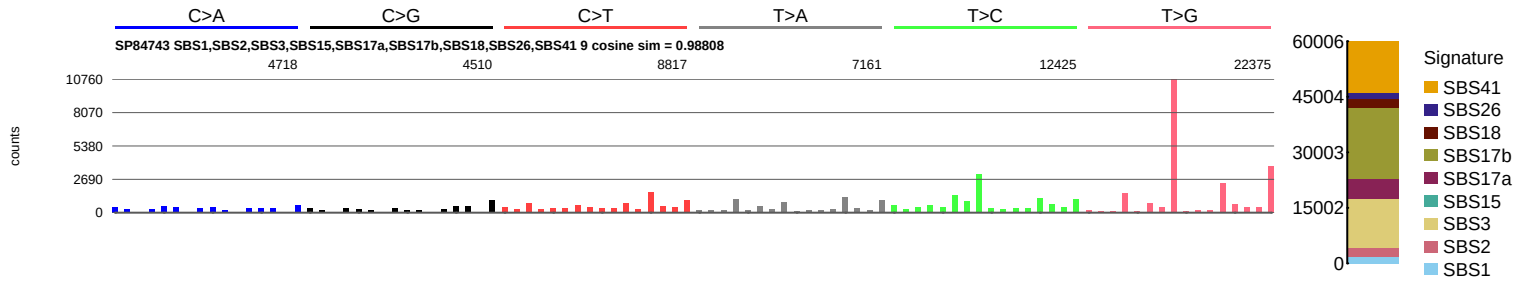
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98808
4718
4510
8817
7161
12425
22375
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
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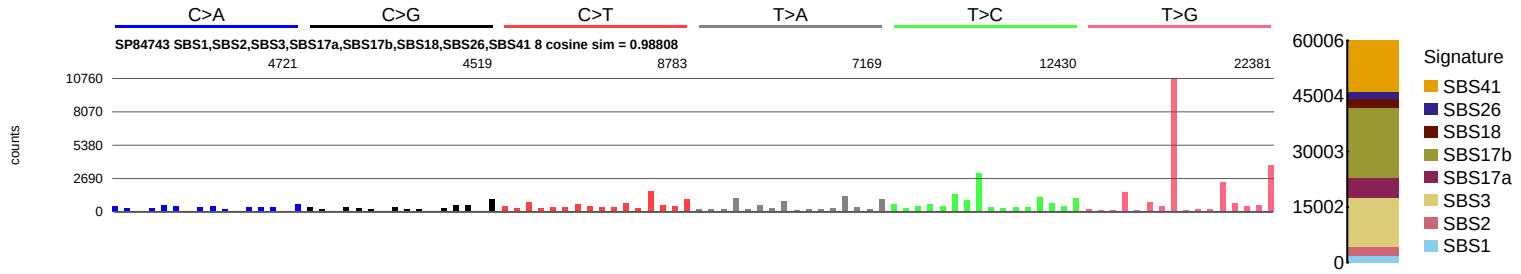
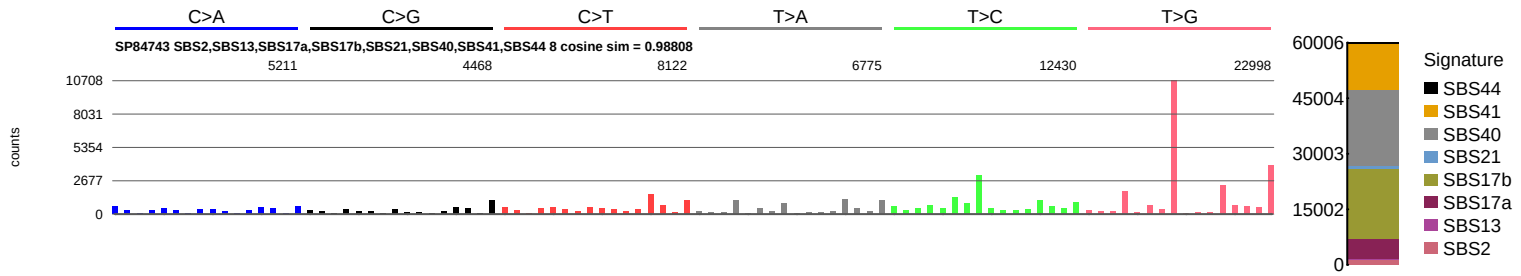
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98808
4721
4519
8783
7169
12430
22381
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98808
5211
4468
8122
6775
12430
22998
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
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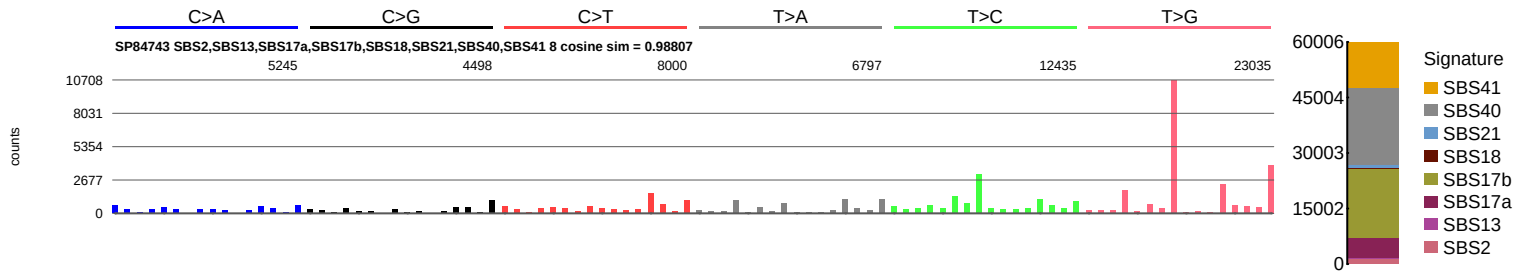
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98807
5245
4498
8000
6797
12435
23035
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
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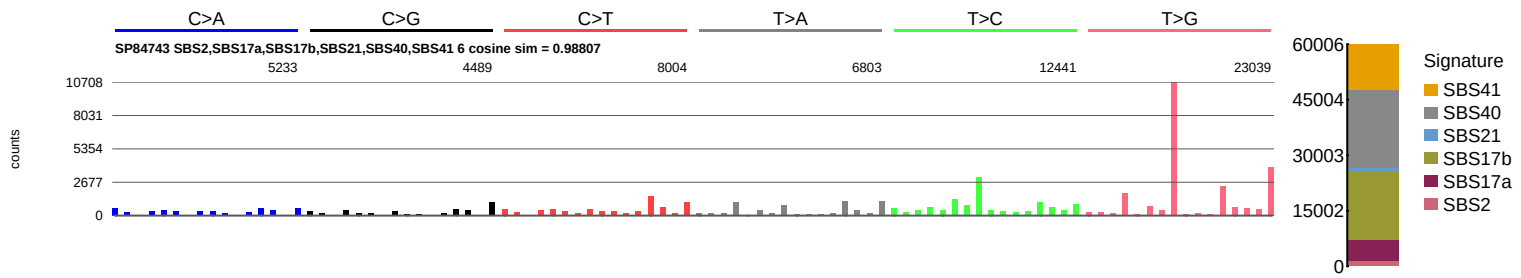
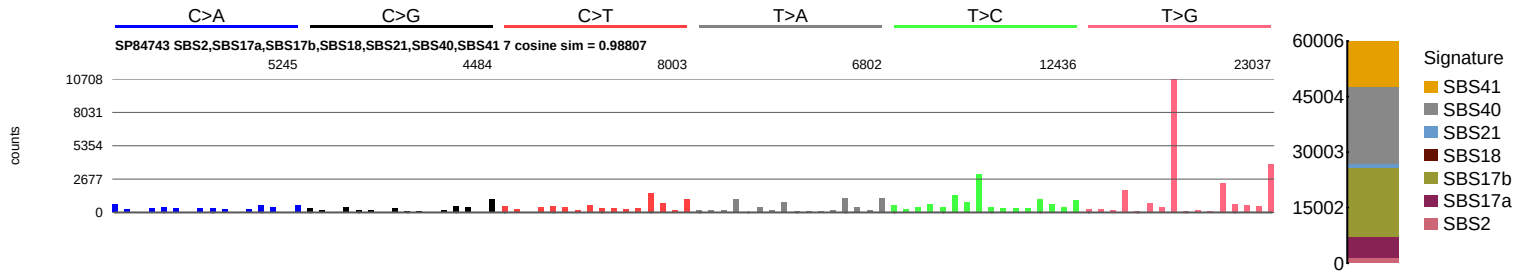
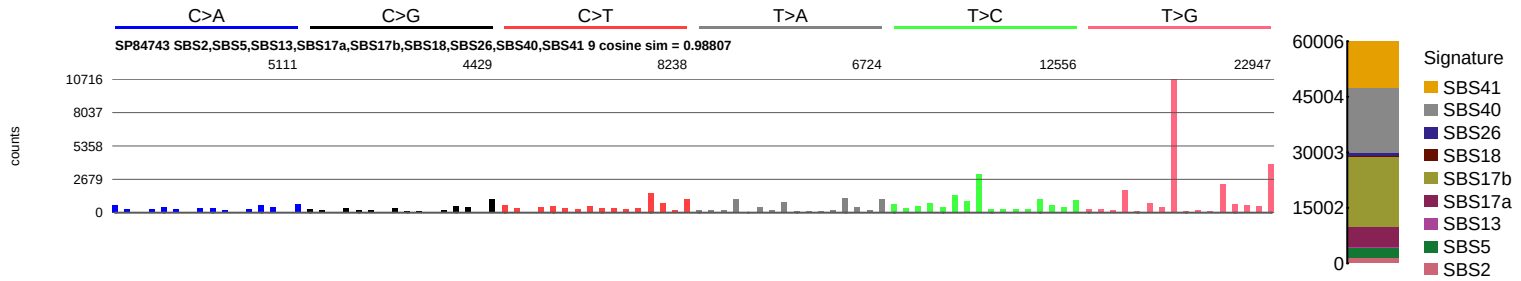
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98807
5233
4489
8004
6803
12441
23039
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98807
5245
4484
8003
6802
12436
23037
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98807
5111
4429
8238
6724
12556
22947
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
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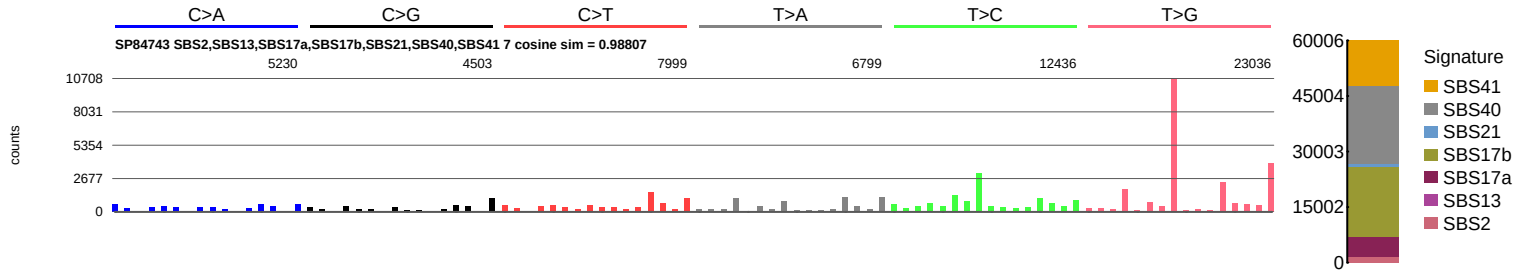
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98807
5230
4503
7999
6799
12436
23036
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
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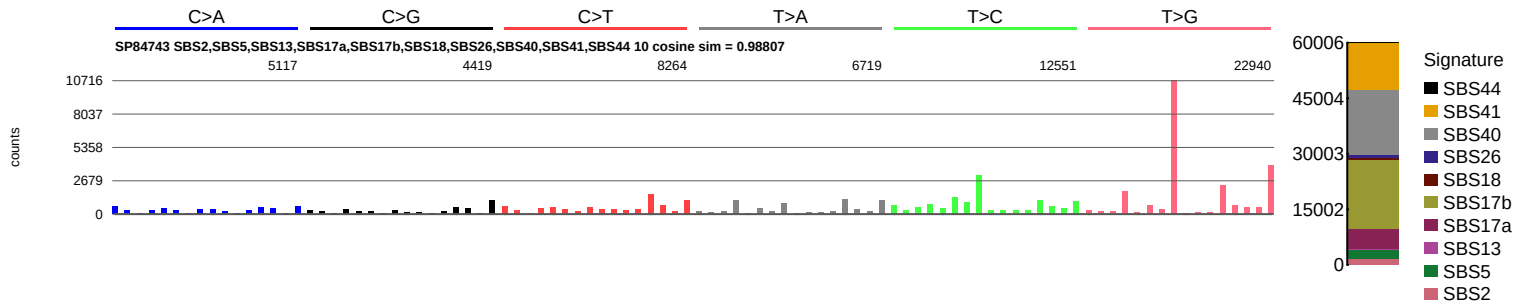
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2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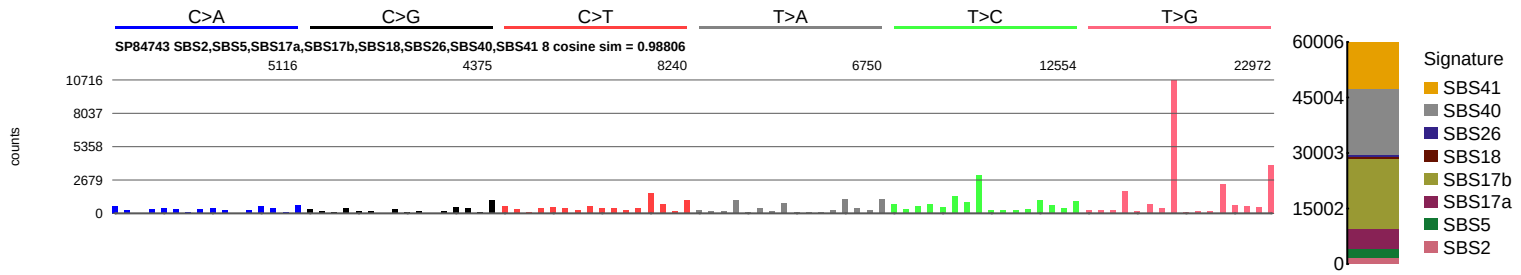
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98807
5117
4419
8264
6719
12551
22940
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98806
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4375
8240
6750
12554
22972
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
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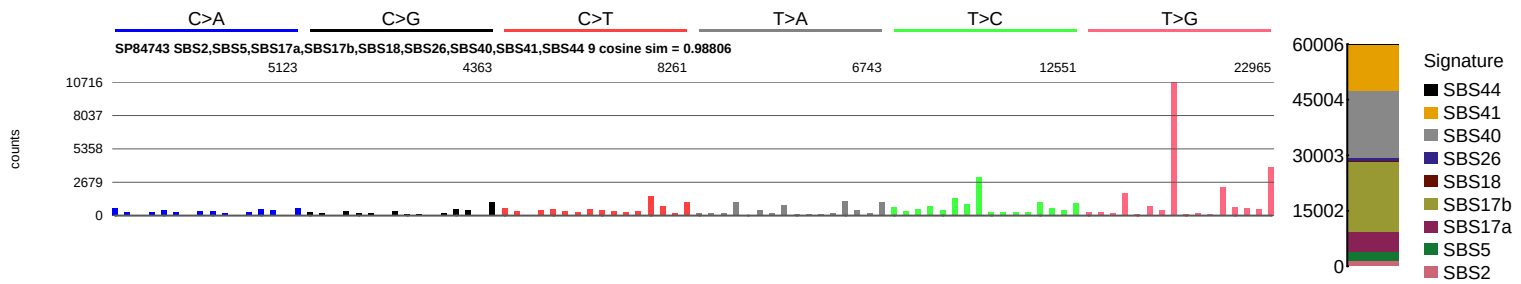
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98806
5123
4363
8261
6743
12551
22965
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
0
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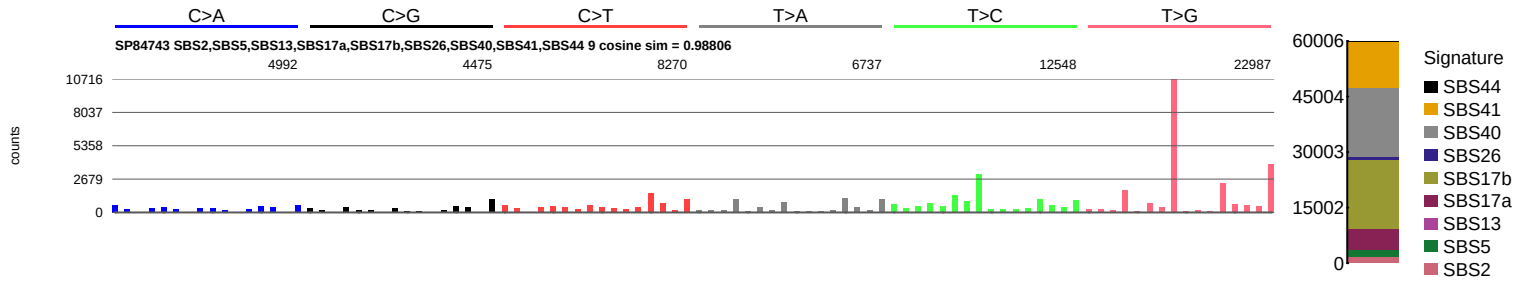
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2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C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98806
4992
4475
8270
6737
12548
22987
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
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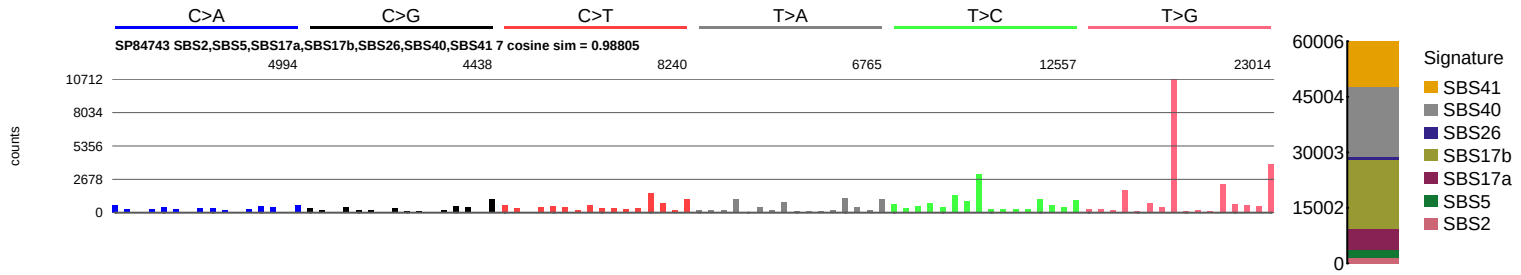
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98805
4994
4438
8240
6765
12557
23014
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
0
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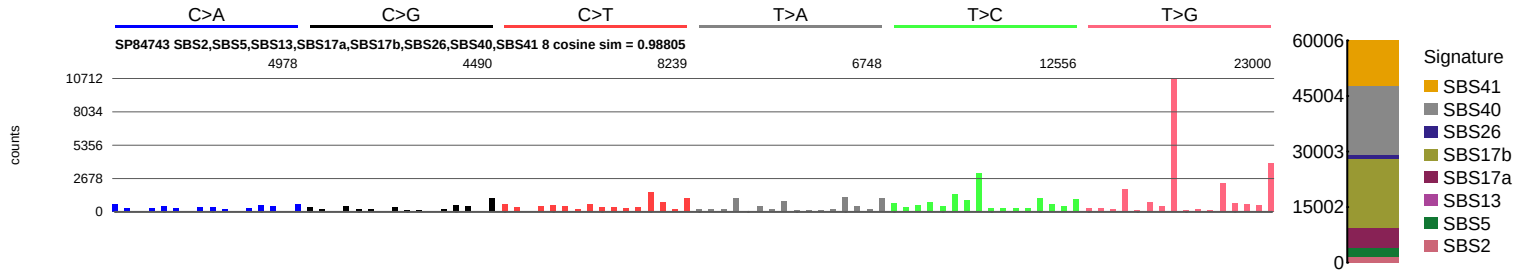
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2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C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98805
4978
4490
8239
6748
12556
23000
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
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Signature
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SBS26
SBS17b
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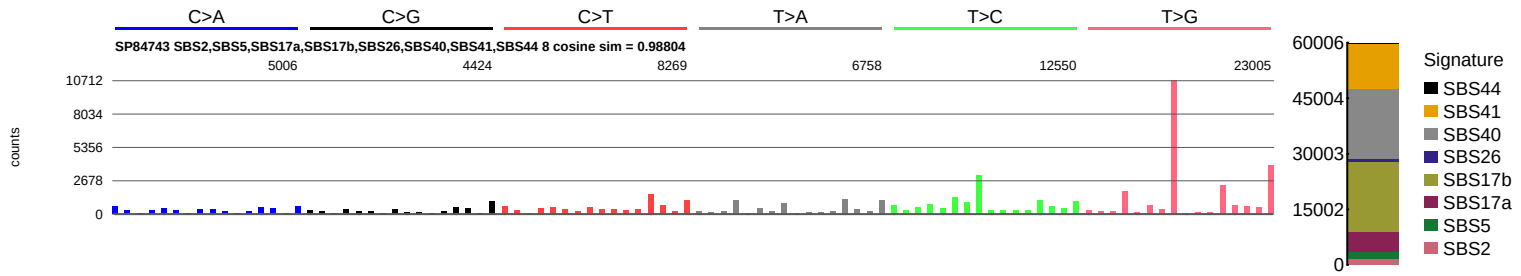
SBS13
SBS5
SBS2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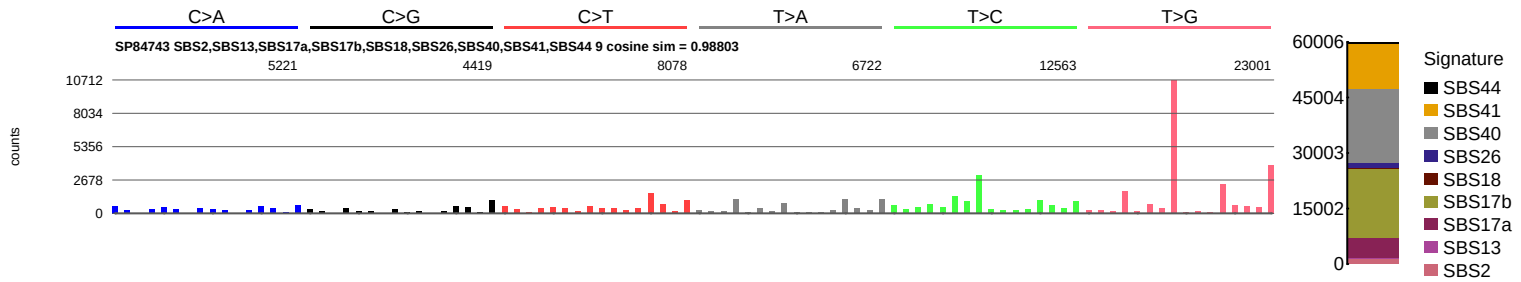
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98804
5006
4424
8269
6758
12550
23005
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98803
5221
4419
8078
6722
12563
23001
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
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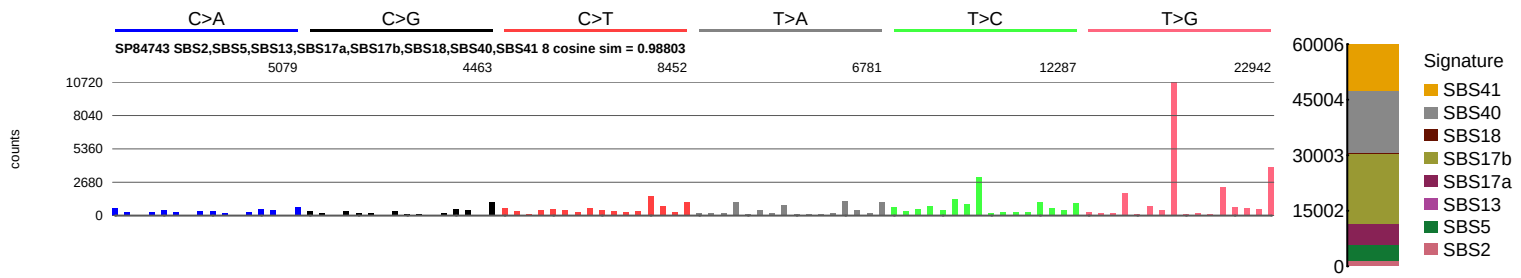
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2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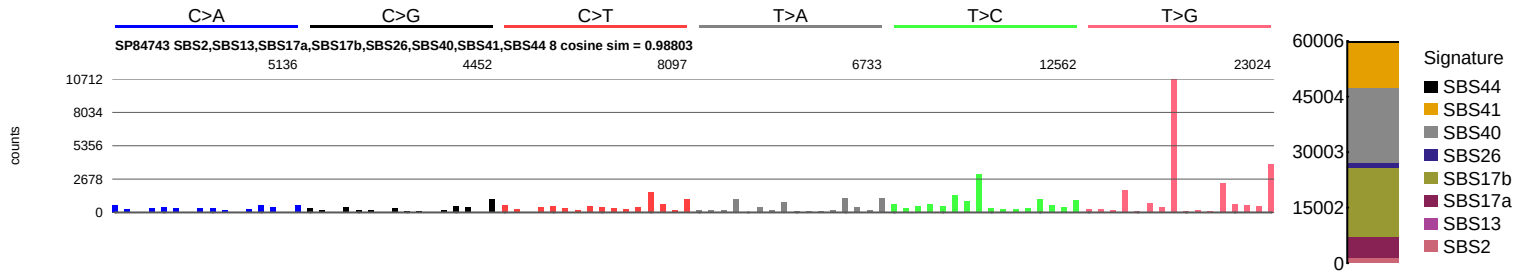
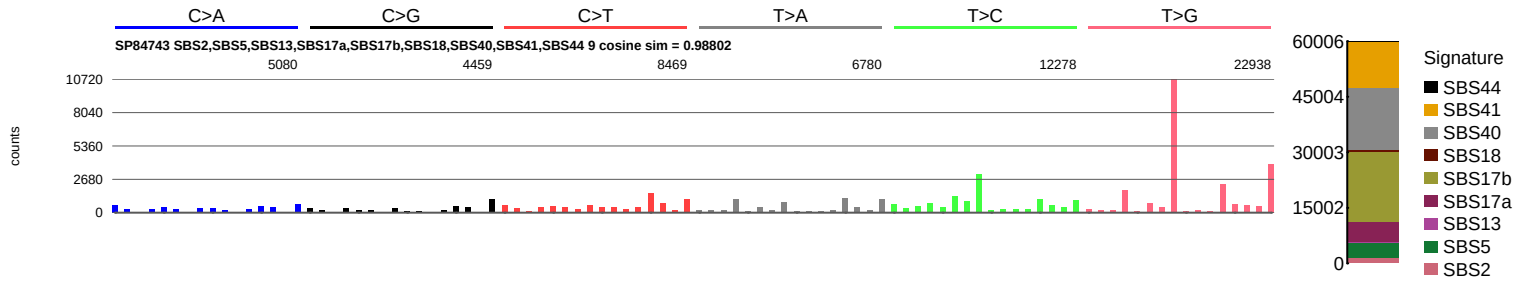
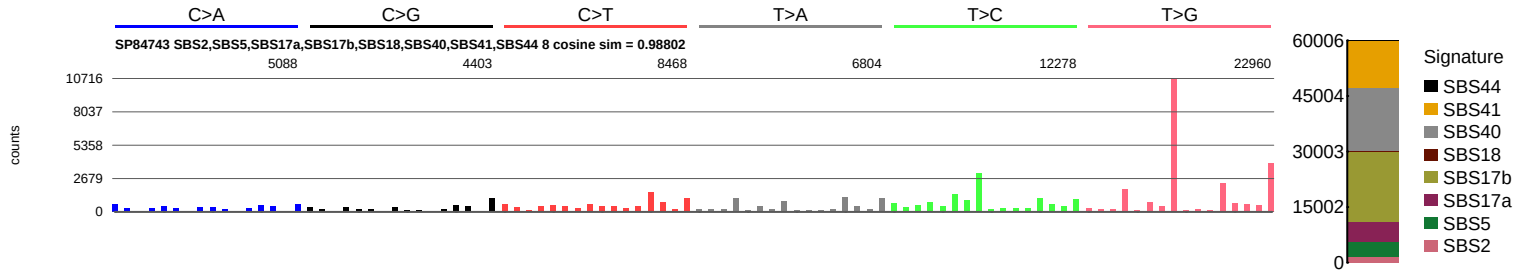
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98803
5079
4463
8452
6781
12287
22942
counts
10720
8040
5360
2680
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98803
5136
4452
8097
6733
12562
23024
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98802
5080
4459
8469
6780
12278
22938
counts
10720
8040
5360
2680
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2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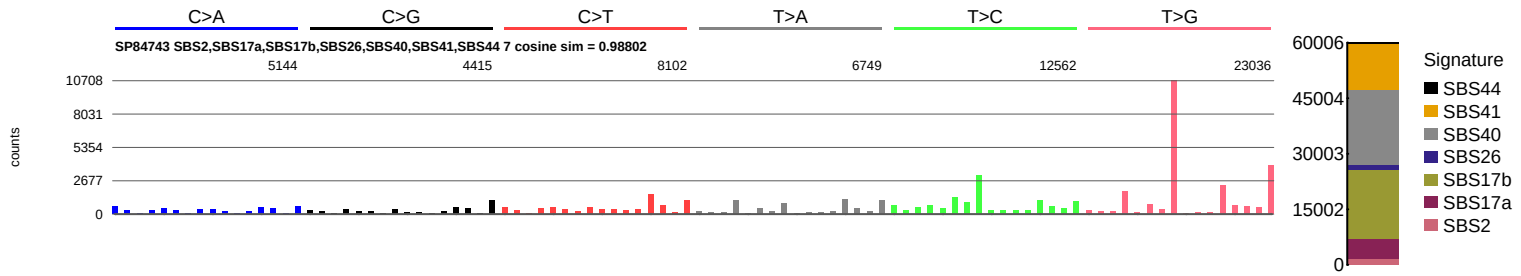
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98802
5088
4403
8468
6804
12278
22960
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2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SP84743 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98802
5144
4415
8102
6749
12562
23036
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
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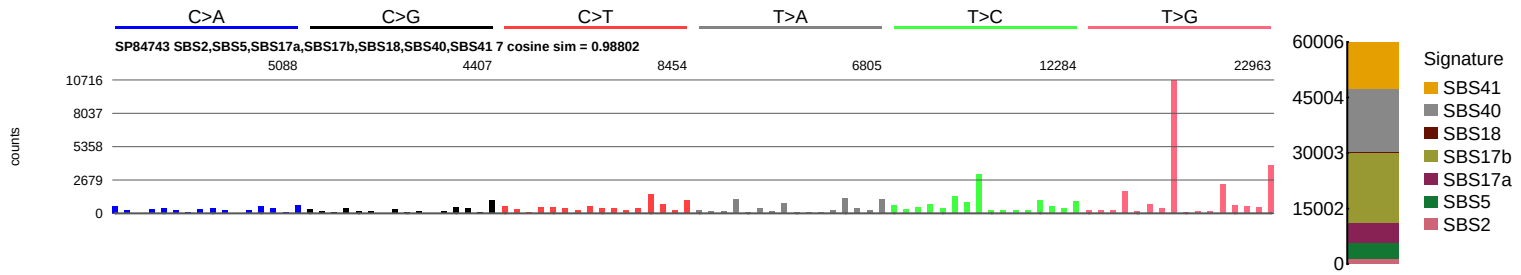
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2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SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98802
5088
4407
8454
6805
12284
22963
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
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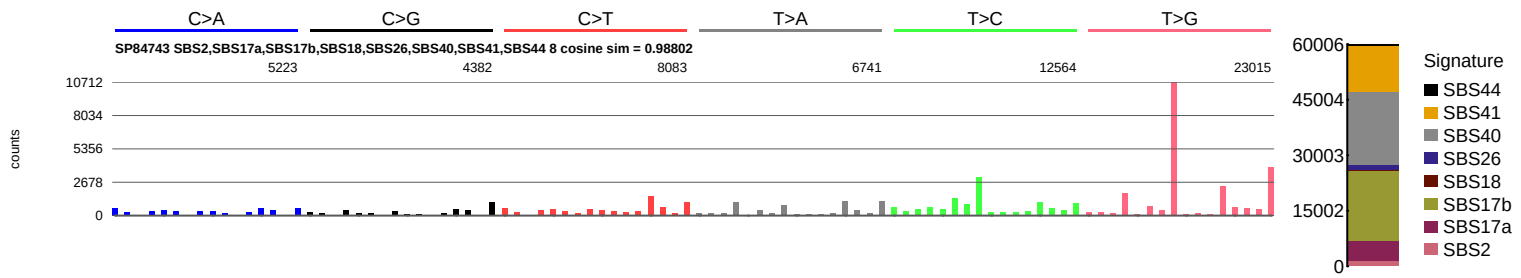
SBS17b
SBS17a
SBS5
SBS2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SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98802
5223
4382
8083
6741
12564
23015
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
0
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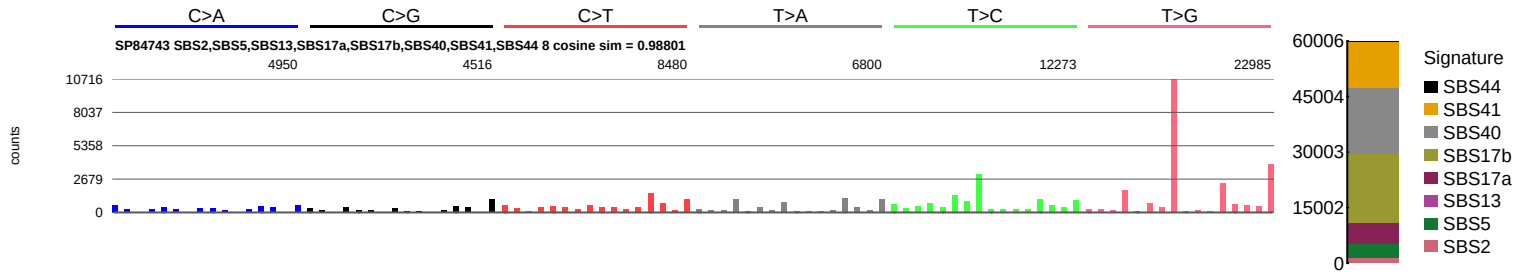
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2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SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98801
4950
4516
8480
6800
12273
22985
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
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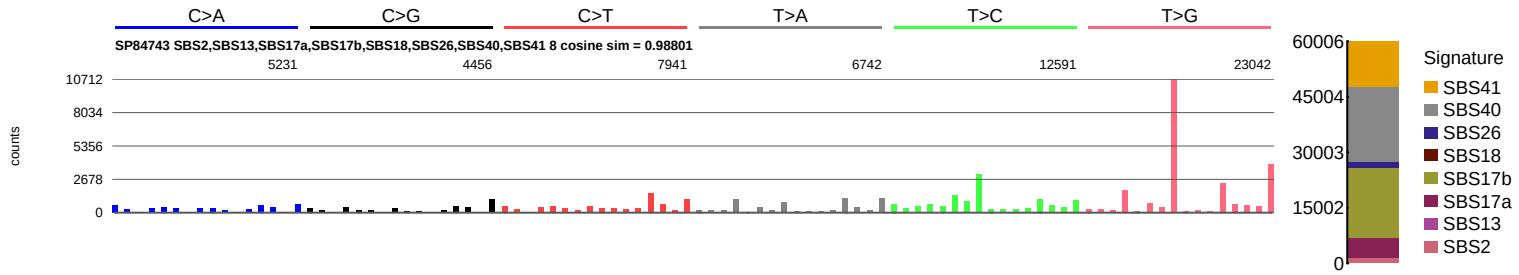
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2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SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98801
5231
4456
7941
6742
12591
23042
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
0
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SBS40
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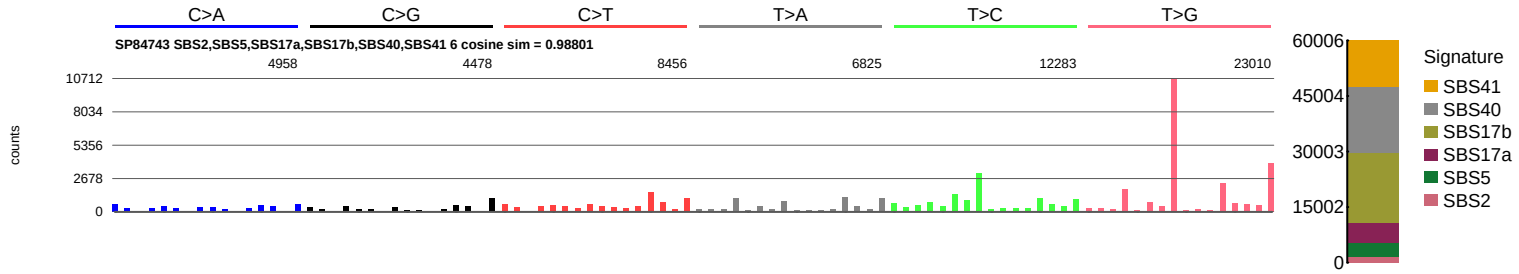
SBS18
SBS17b
SBS17a
SBS13
SBS2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SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98801
4958
4478
8456
6825
12283
23010
counts
10712
8034
5356
2678
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60006
45004
30003
15002
0
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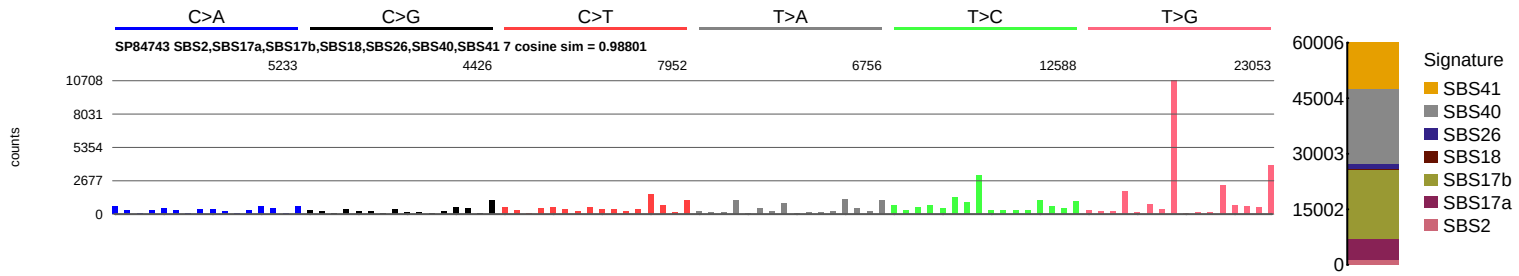
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2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C>G
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T>C
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SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98801
5233
4426
7952
6756
12588
23053
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
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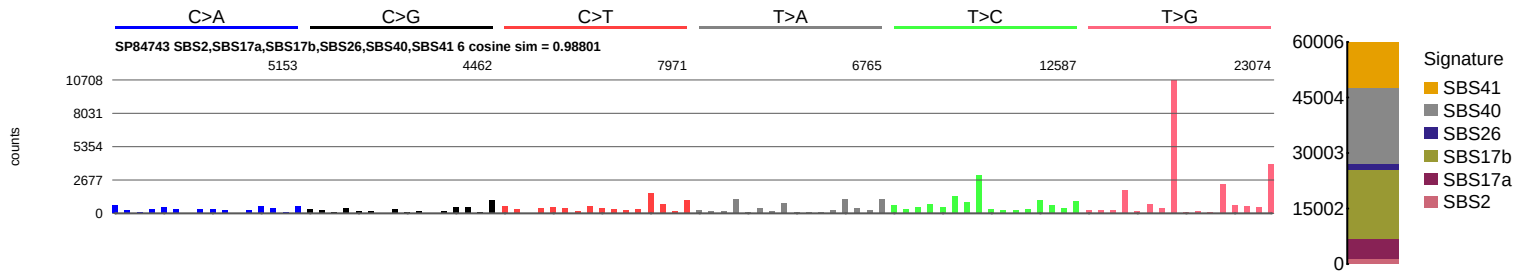
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2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T>G
SP84743 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98801
5153
4462
7971
6765
12587
23074
counts
10708
8031
5354
2677
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30003
15002
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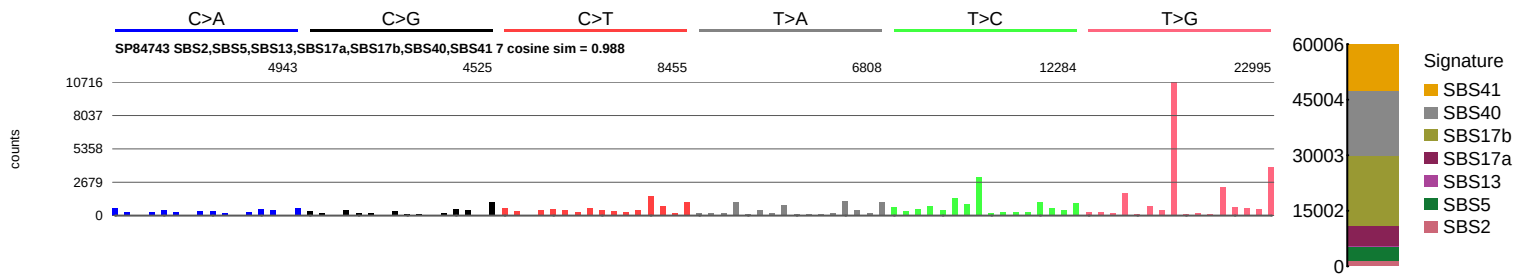
SBS17a
SBS2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C>G
C>T
T>A
T>C
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SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.988
4943
4525
8455
6808
12284
22995
counts
10716
8037
5358
2679
0
60006
45004
30003
15002
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SBS40
SBS17b
SBS17a
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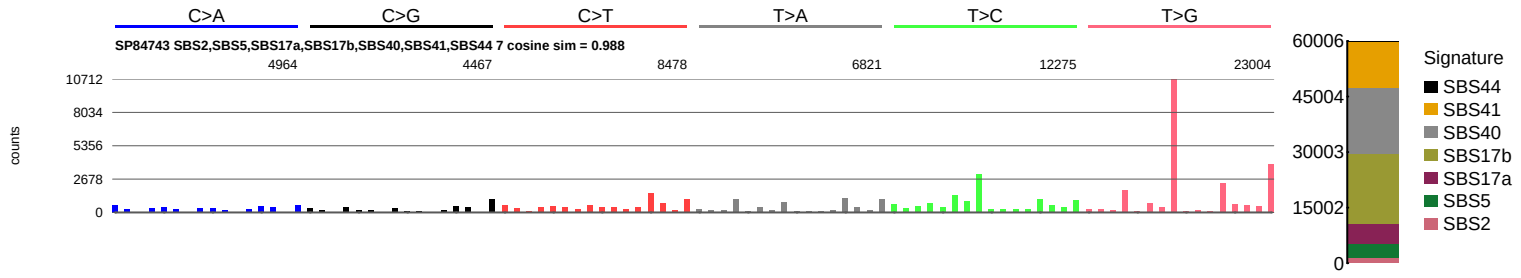
SBS5
SBS2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C>G
C>T
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T>C
T>G
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.988
4964
4467
8478
6821
12275
23004
counts
10712
8034
5356
2678
0
60006
45004
30003
15002
0
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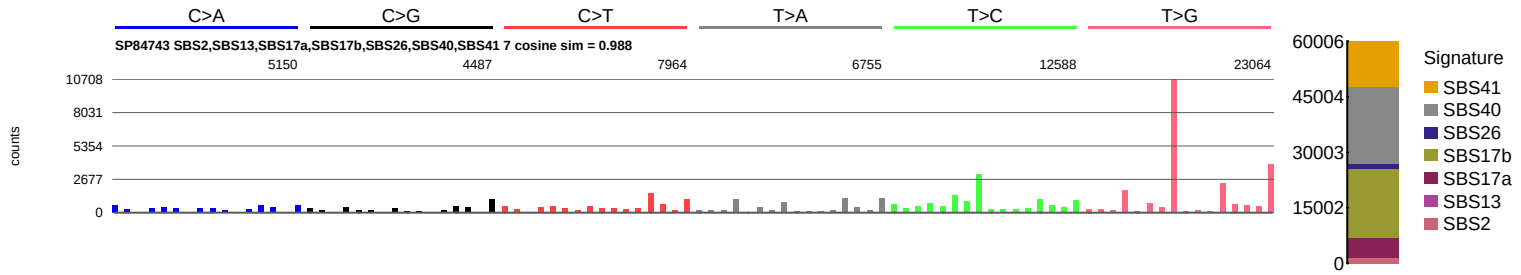
SBS17b
SBS17a
SBS5
SBS2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C>G
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T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.988
5150
4487
7964
6755
12588
23064
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
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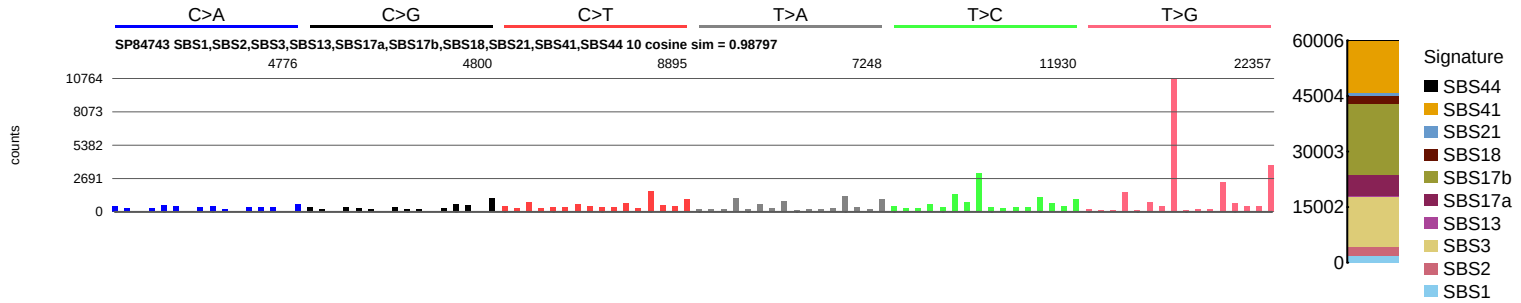
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2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SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98797
4776
4800
8895
7248
11930
22357
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
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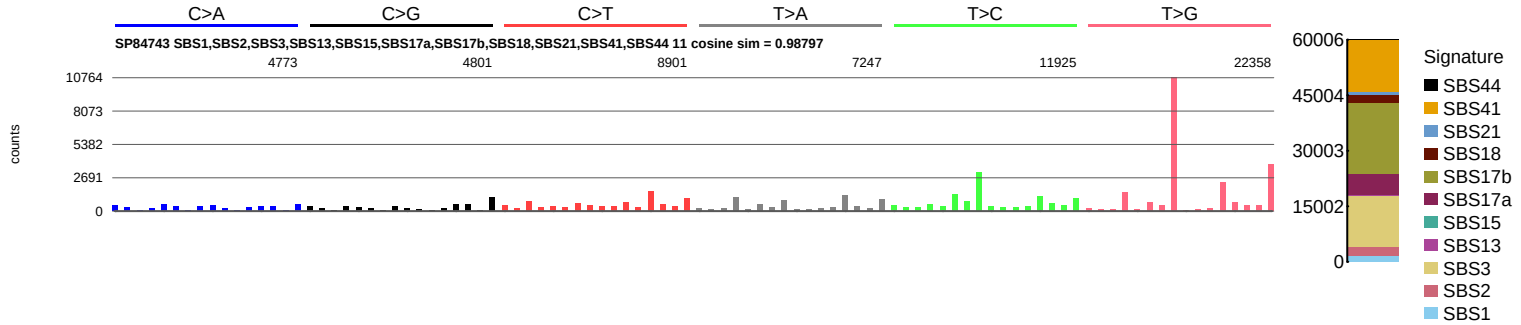
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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T>C
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SP84743 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98797
4773
4801
8901
7247
11925
22358
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
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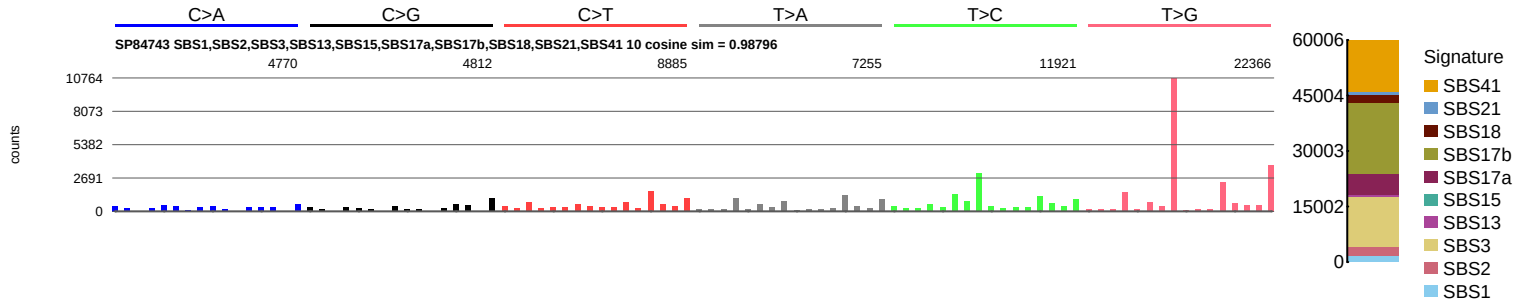
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
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T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98796
4770
4812
8885
7255
11921
22366
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
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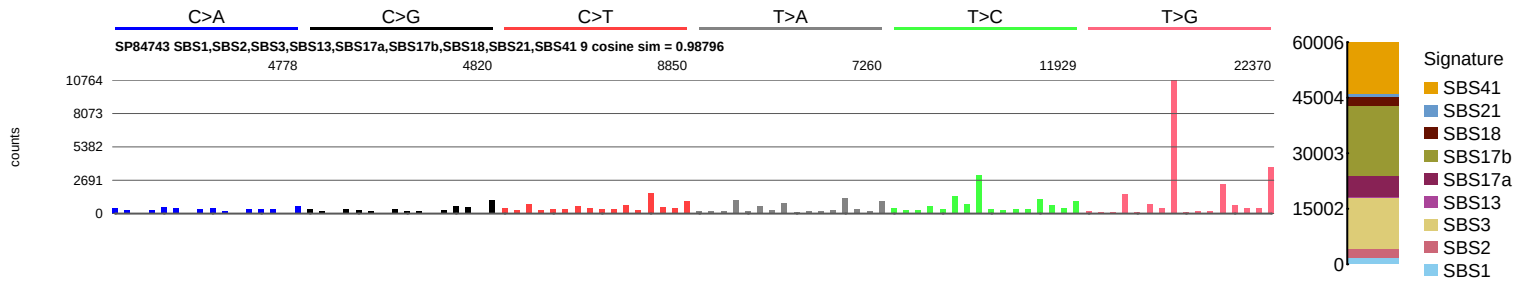
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98796
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4820
8850
7260
11929
22370
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
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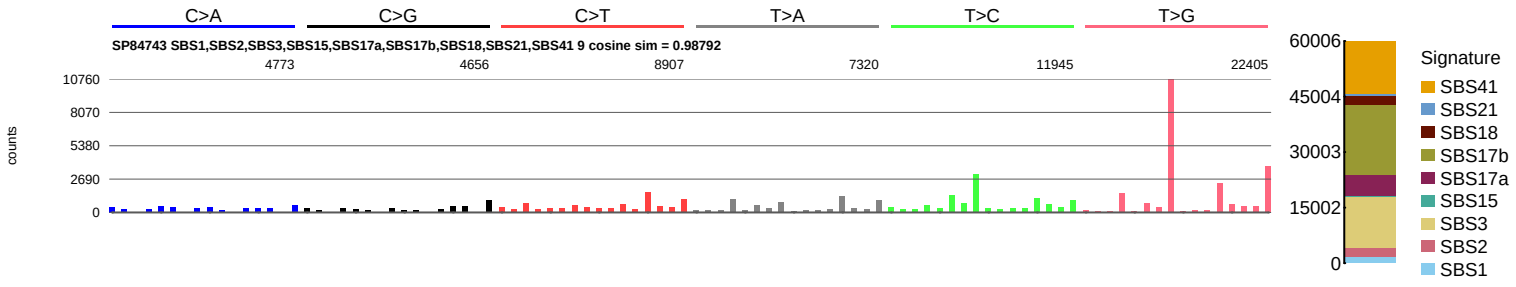
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98792
4773
4656
8907
7320
11945
22405
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
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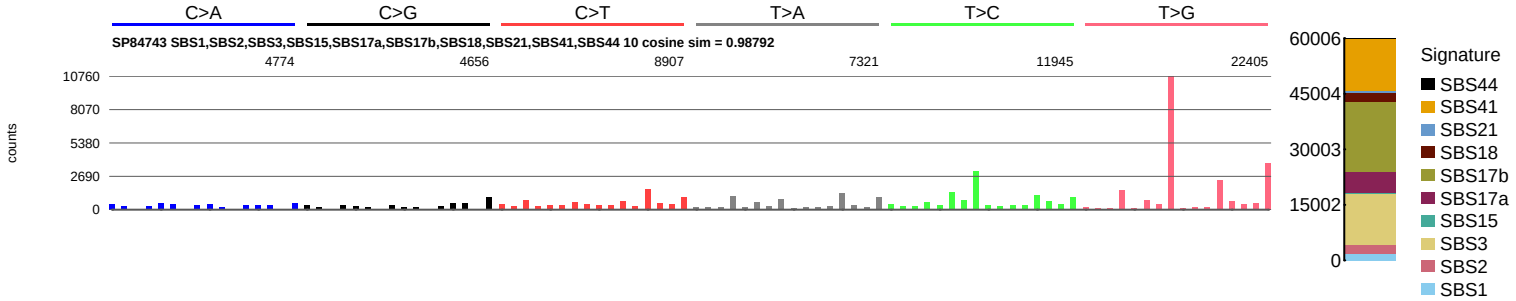
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98792
4774
4656
8907
7321
11945
22405
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
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Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
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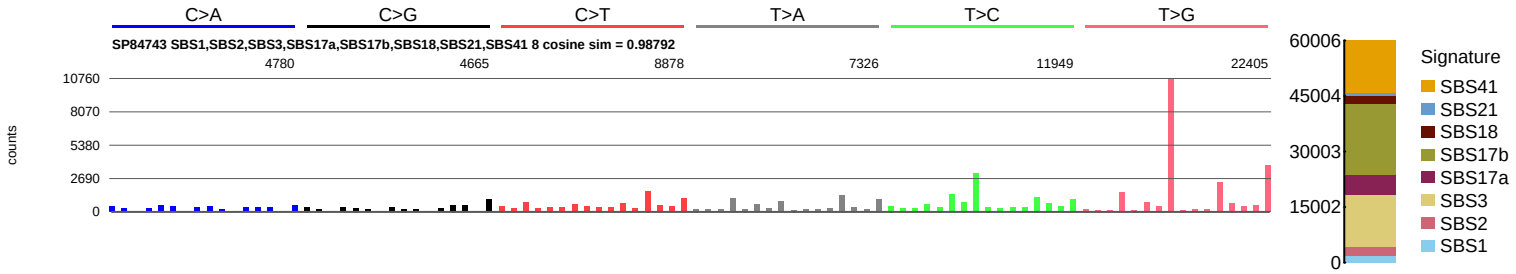
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98792
4780
4665
8878
7326
11949
22405
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
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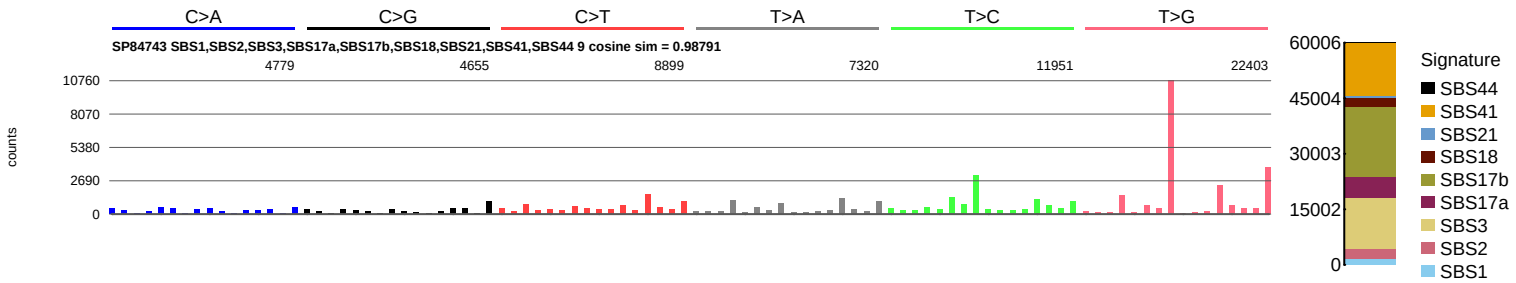
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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T>G
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98791
4779
4655
8899
7320
11951
22403
counts
10760
8070
5380
2690
0
60006
45004
30003
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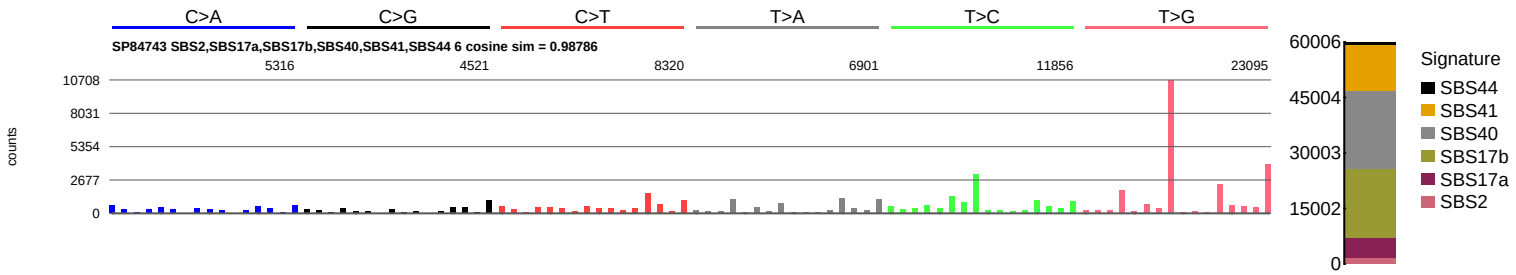
SBS2
SBS1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C>T
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T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98786
5316
4521
8320
6901
11856
23095
counts
10708
8031
5354
2677
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60006
45004
30003
15002
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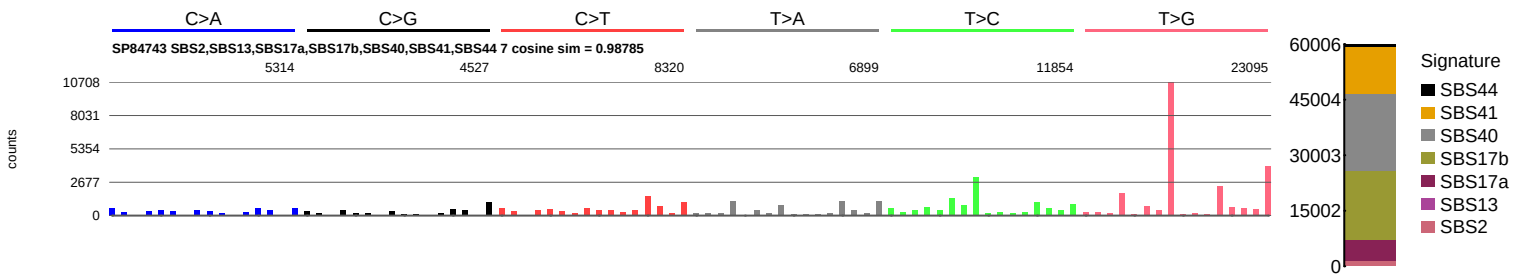
SBS40
SBS17b
SBS17a
SBS2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SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98785
5314
4527
8320
6899
11854
23095
counts
10708
8031
5354
2677
0
60006
45004
30003
15002
0
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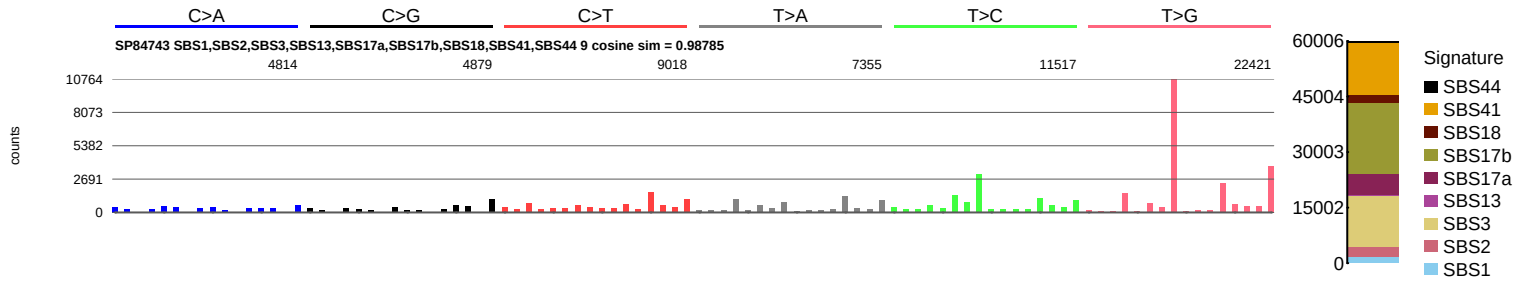
SBS17b
SBS17a
SBS13
SBS2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SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98785
4814
4879
9018
7355
11517
22421
counts
10764
8073
5382
2691
0
60006
45004
30003
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SBS41
SBS18
SBS17b
SBS17a
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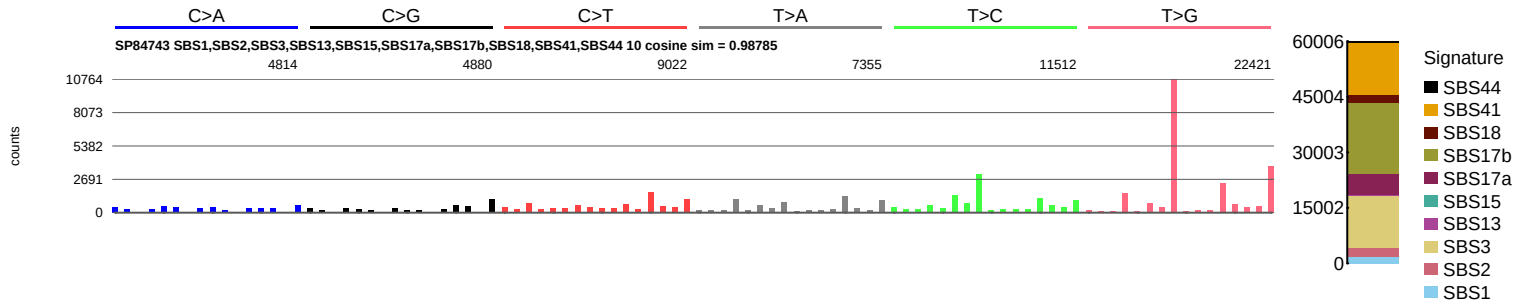
SBS3
SBS2
SBS1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T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98785
4814
4880
9022
7355
11512
22421
counts
10764
8073
5382
2691
0
60006
45004
30003
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Signature
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SBS18
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SBS15
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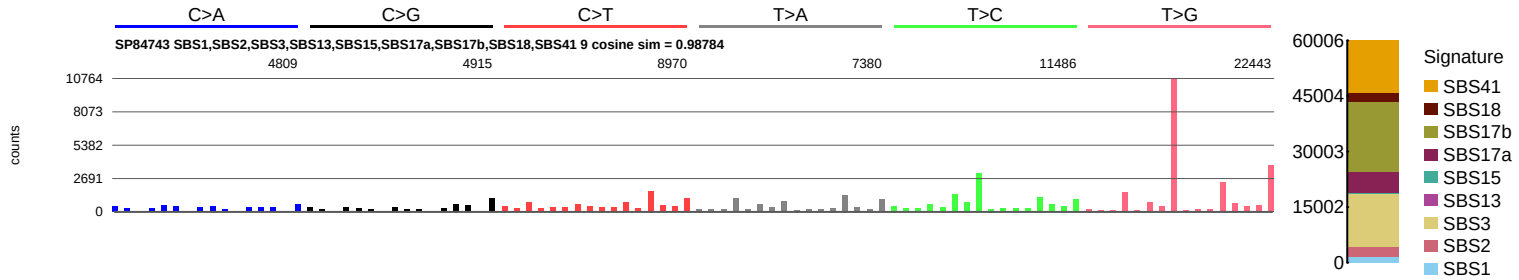
SBS3
SBS2
SBS1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C>G
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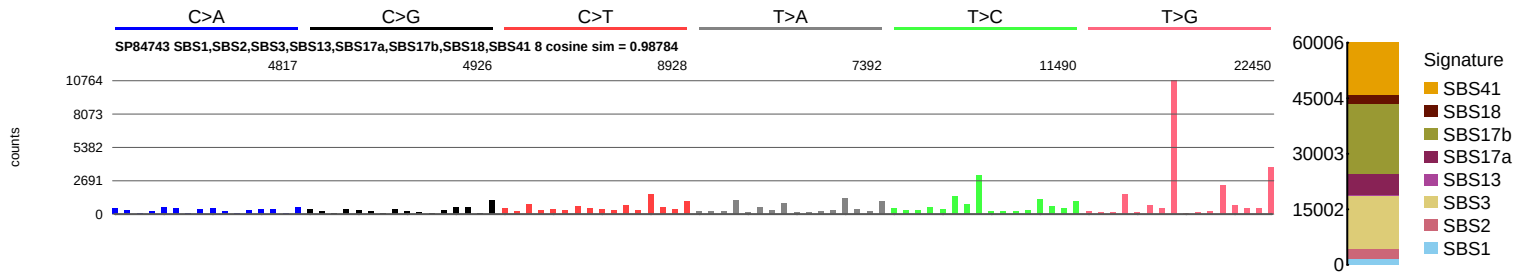
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98784
4809
4915
8970
7380
11486
22443
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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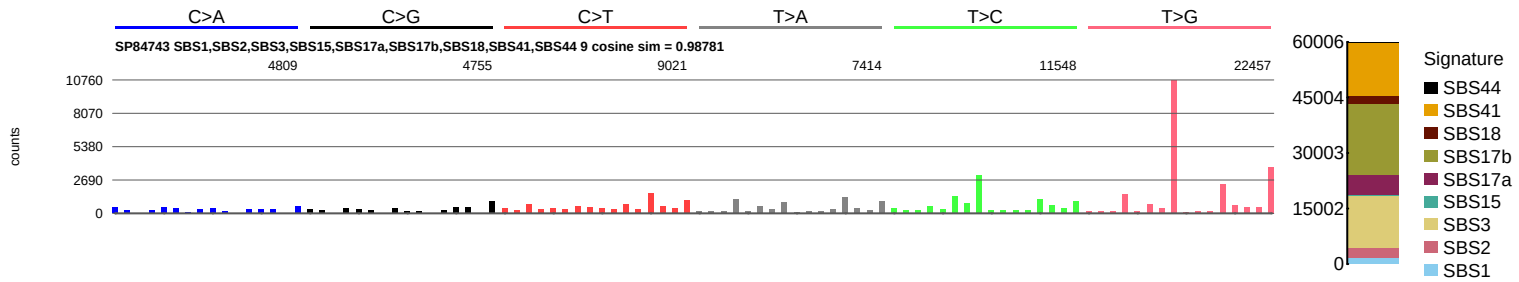
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98784
4817
4926
8928
7392
11490
22450
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98781
4809
4755
9021
7414
11548
22457
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
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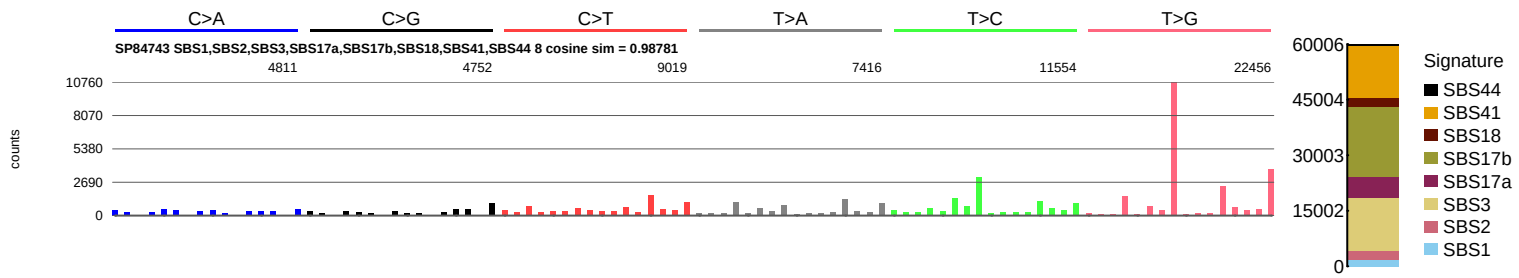
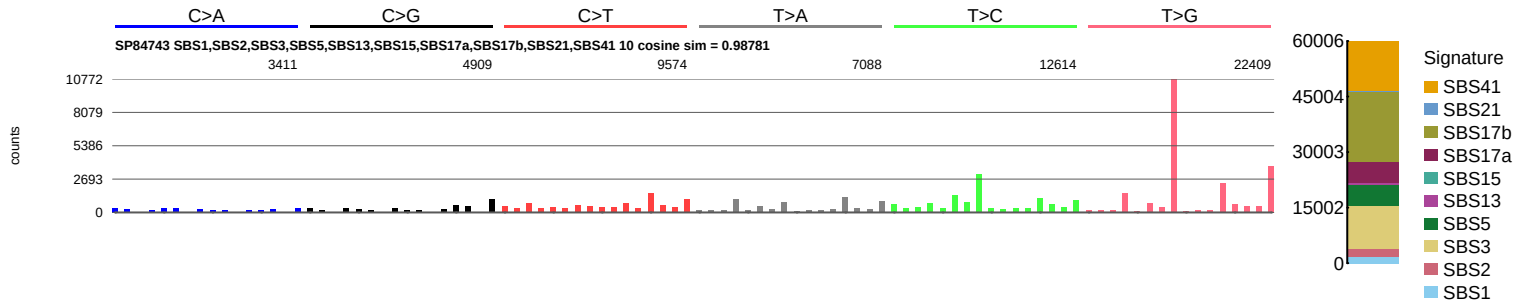
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98781
4811
4752
9019
7416
11554
22456
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 10 cosine sim = 0.98781
3411
4909
9574
7088
12614
22409
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
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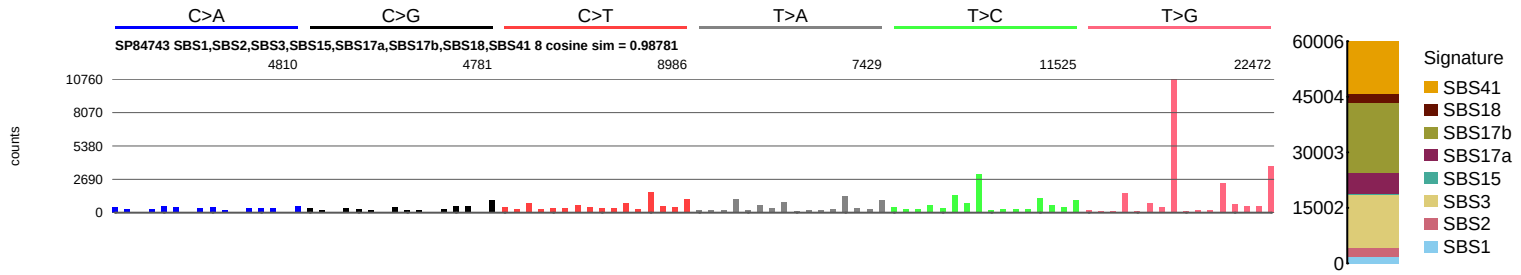
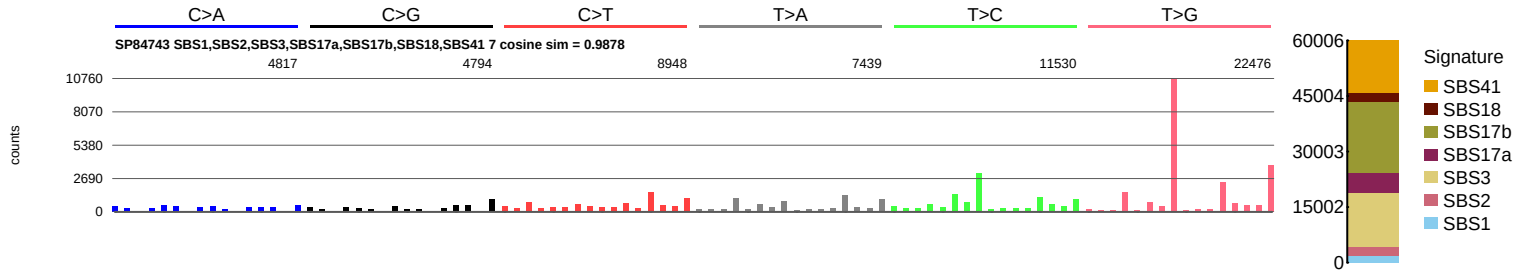
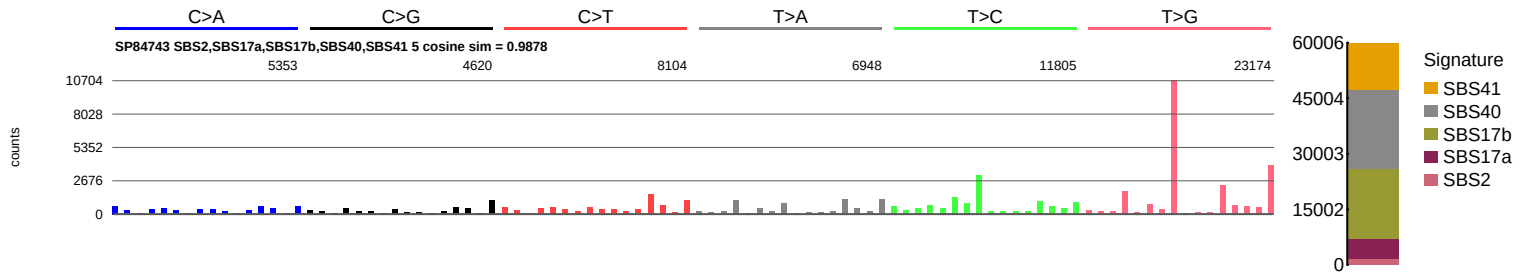
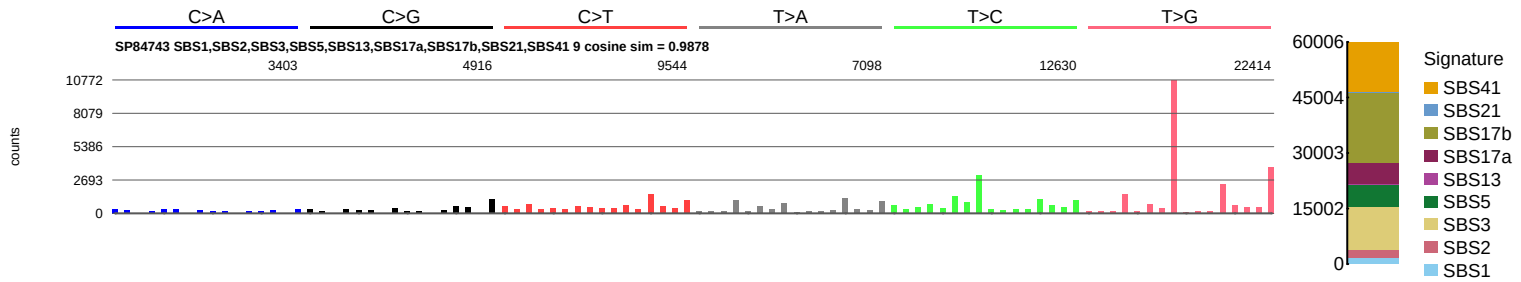
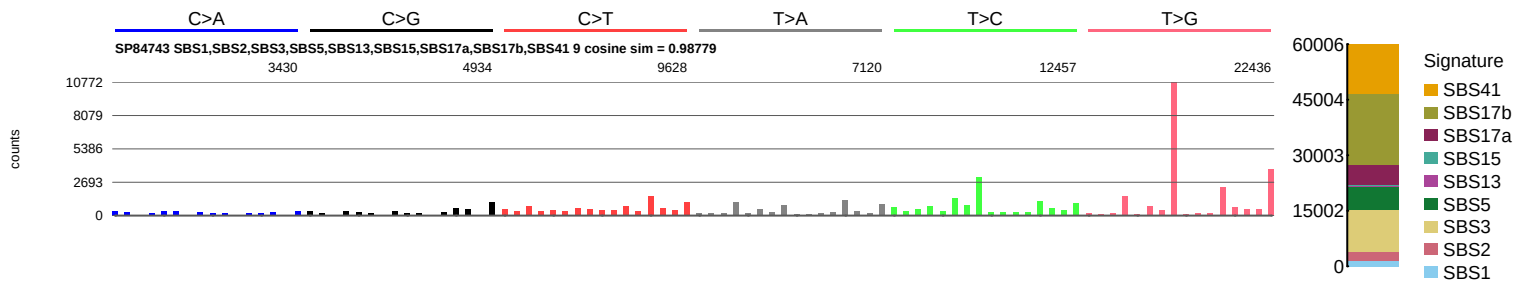
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98781
4810
4781
8986
7429
11525
22472
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9878
4817
4794
8948
7439
11530
22476
counts
10760
8070
5380
2690
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.9878
5353
4620
8104
6948
11805
23174
counts
10704
8028
5352
2676
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.9878
3403
4916
9544
7098
12630
22414
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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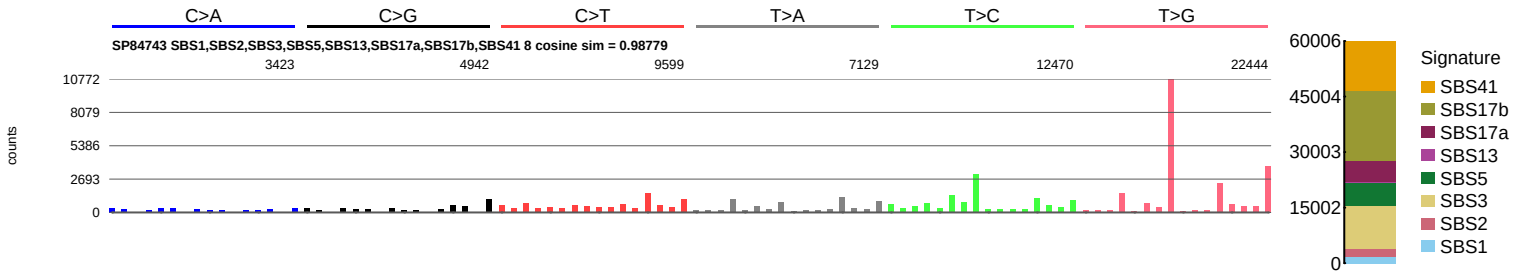
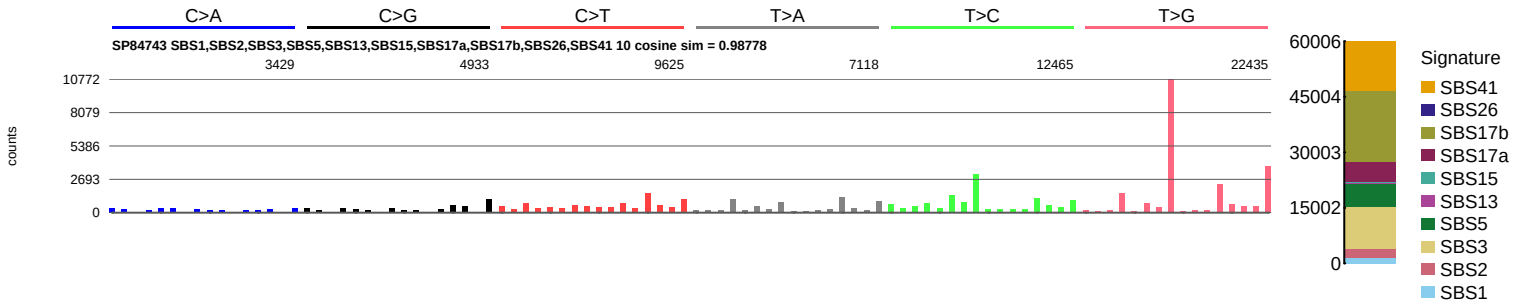
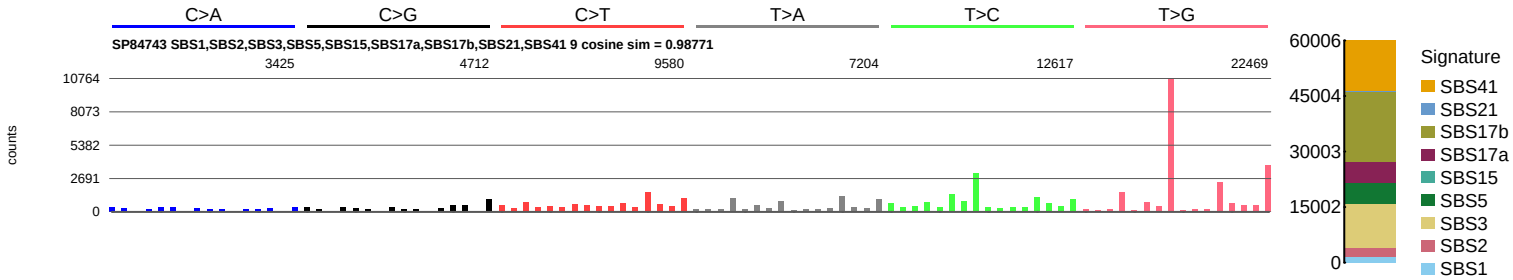
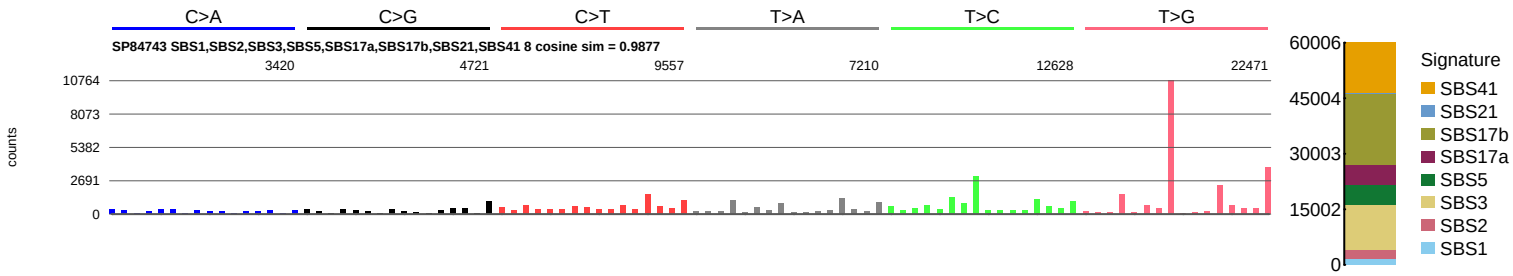
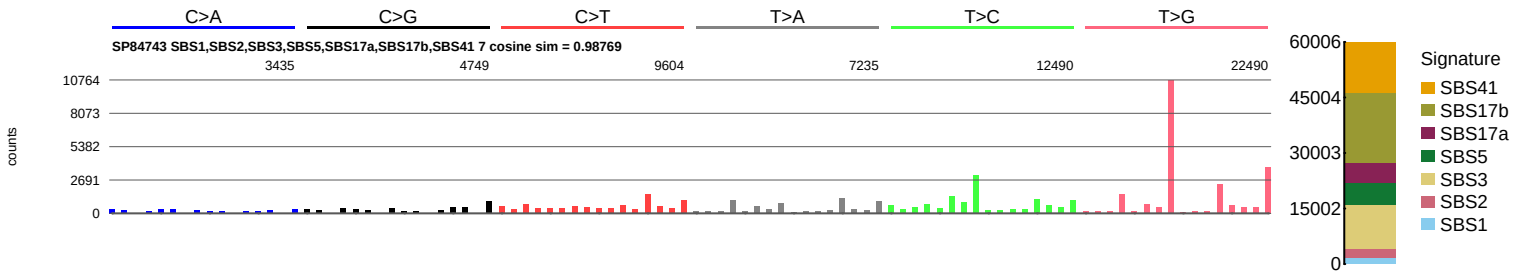
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 9 cosine sim = 0.98779
3430
4934
9628
7120
12457
22436
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98779
3423
4942
9599
7129
12470
22444
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 10 cosine sim = 0.98778
3429
4933
9625
7118
12465
22435
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98771
3425
4712
9580
7204
12617
22469
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9877
3420
4721
9557
7210
12628
22471
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98769
3435
4749
9604
7235
12490
22490
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
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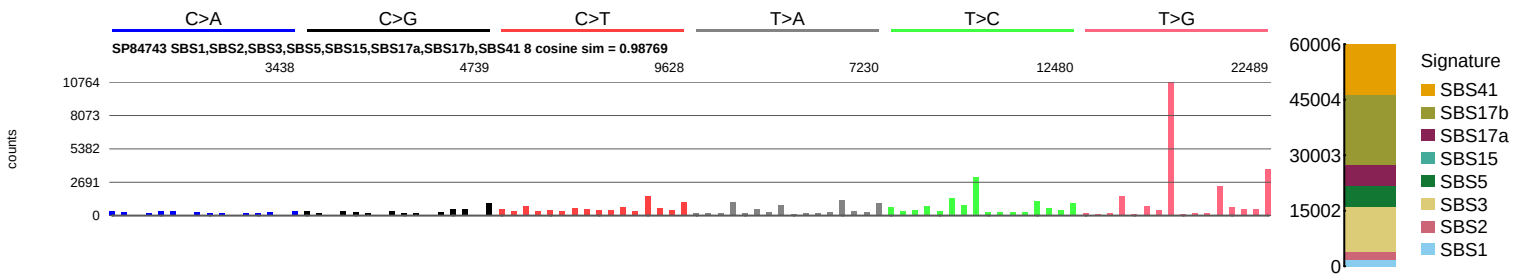
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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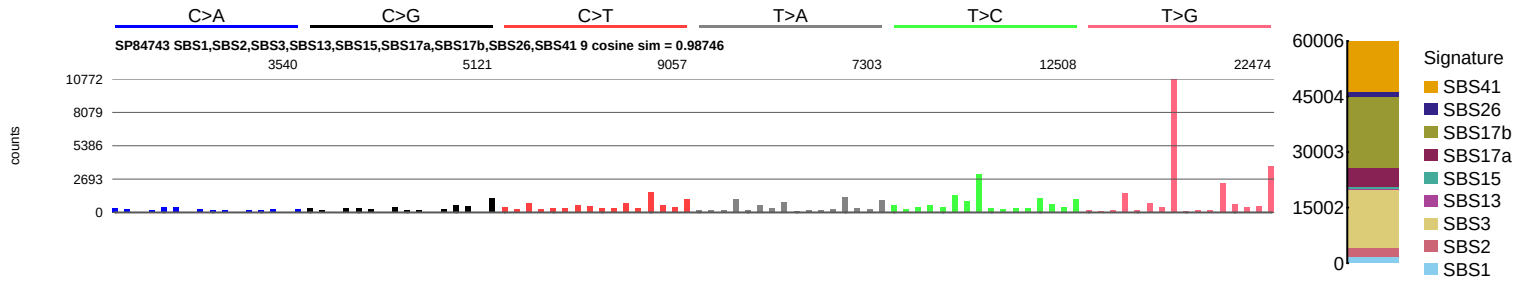
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98769
3438
4739
9628
7230
12480
22489
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.98746
3540
5121
9057
7303
12508
22474
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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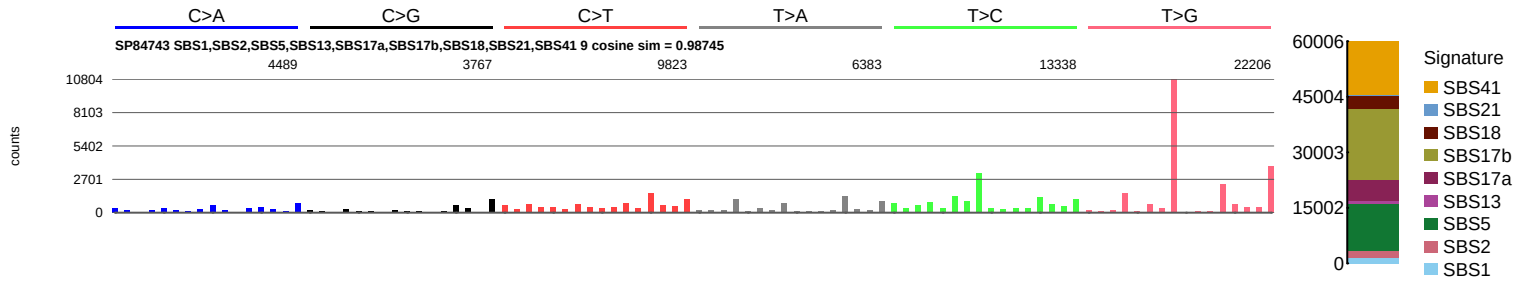
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98745
4489
3767
9823
6383
13338
22206
counts
10804
8103
5402
2701
0
60006
45004
30003
15002
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Signature
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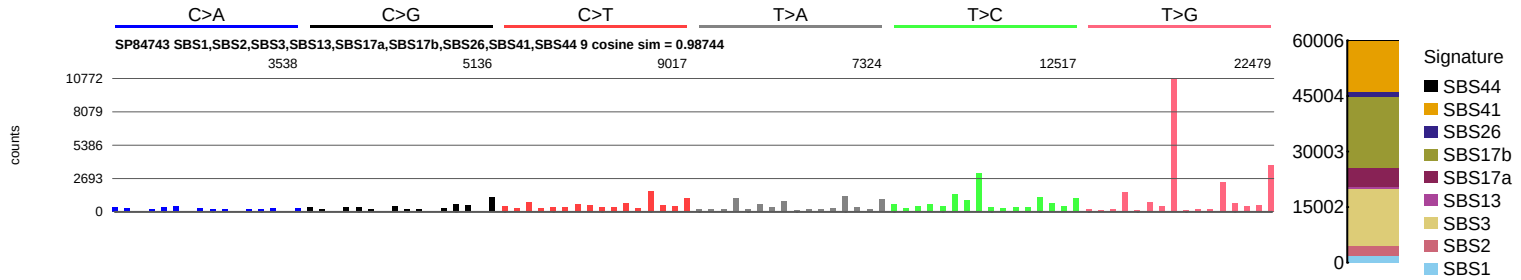
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98744
3538
5136
9017
7324
12517
22479
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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Signature
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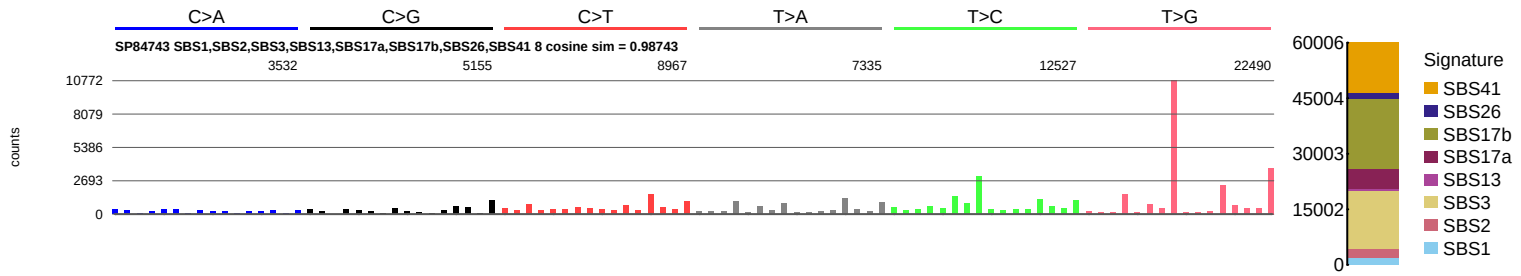
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98743
3532
5155
8967
7335
12527
22490
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
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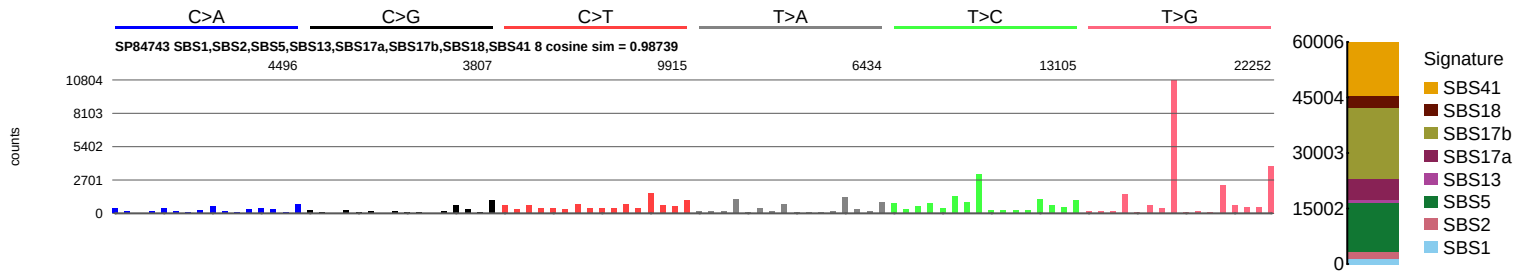
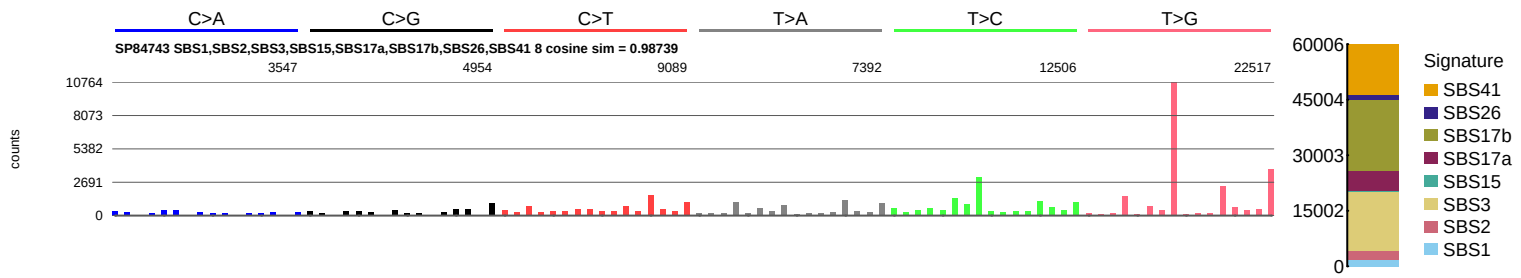
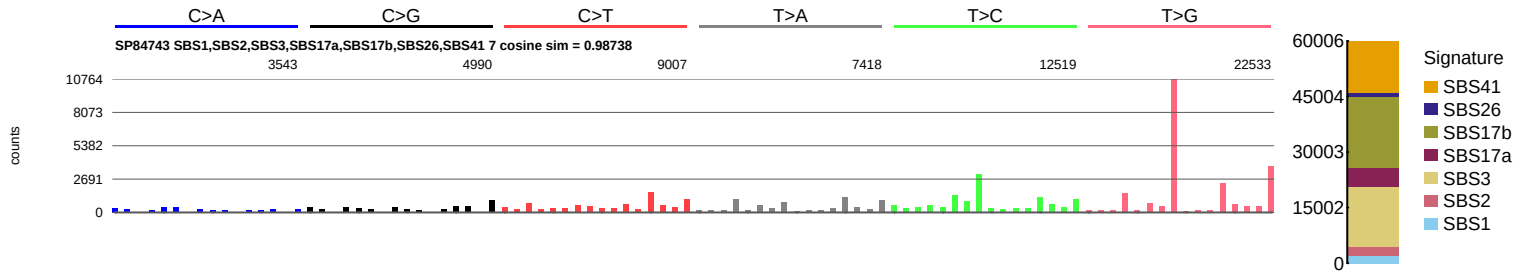
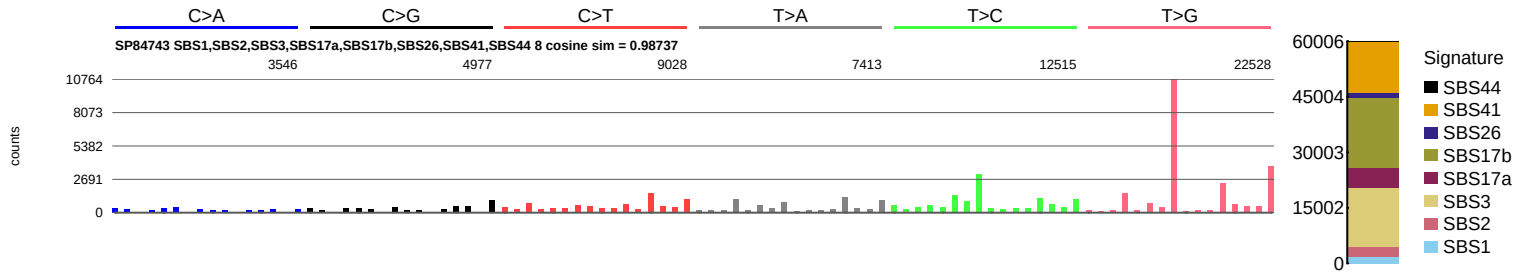
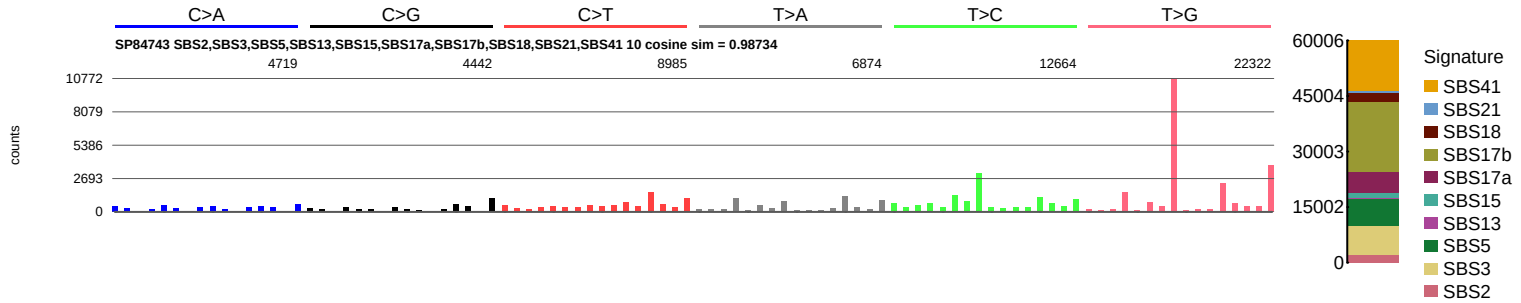
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98739
4496
3807
9915
6434
13105
22252
counts
10804
8103
5402
2701
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98739
3547
4954
9089
7392
12506
22517
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98738
3543
4990
9007
7418
12519
22533
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98737
3546
4977
9028
7413
12515
22528
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
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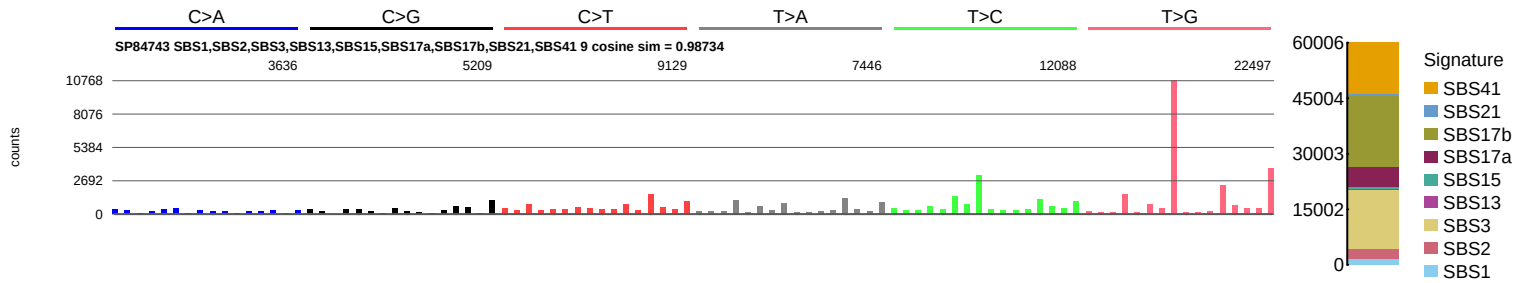
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98734
4719
4442
8985
6874
12664
22322
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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3636
5209
9129
7446
12088
22497
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
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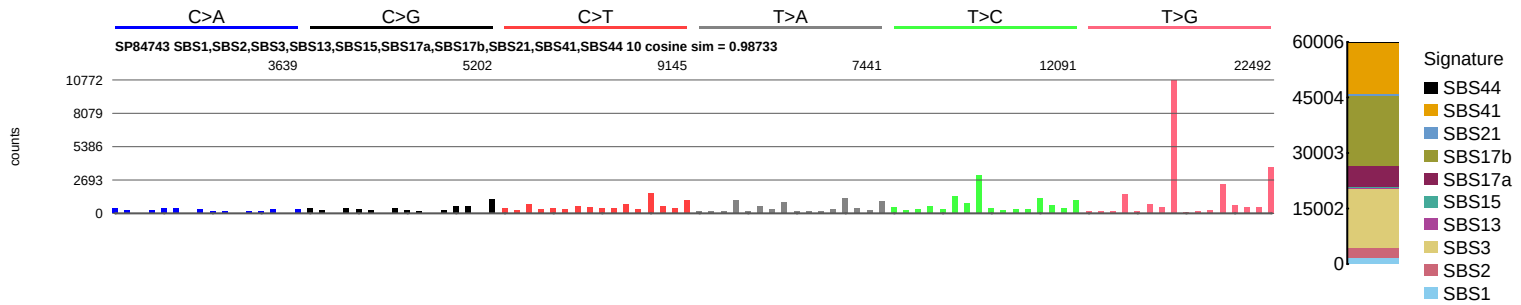
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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SP84743 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 10 cosine sim = 0.98733
3639
5202
9145
7441
12091
22492
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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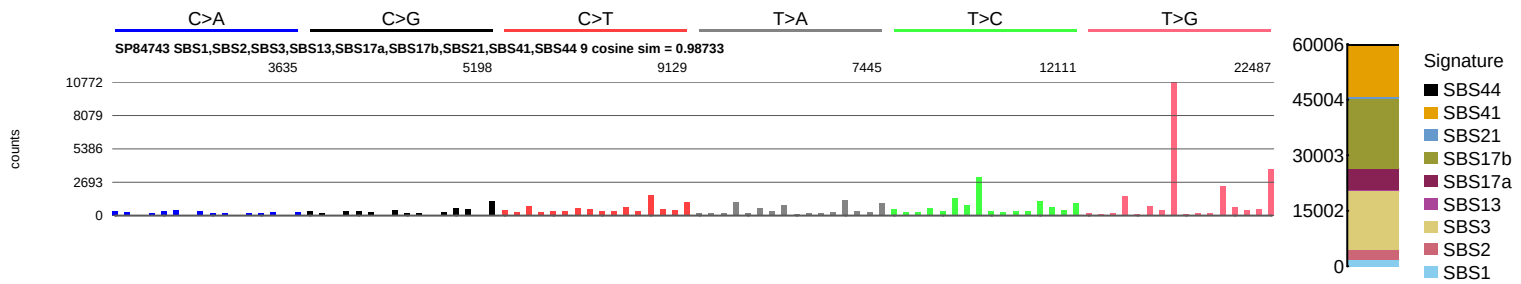
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98733
3635
5198
9129
7445
12111
22487
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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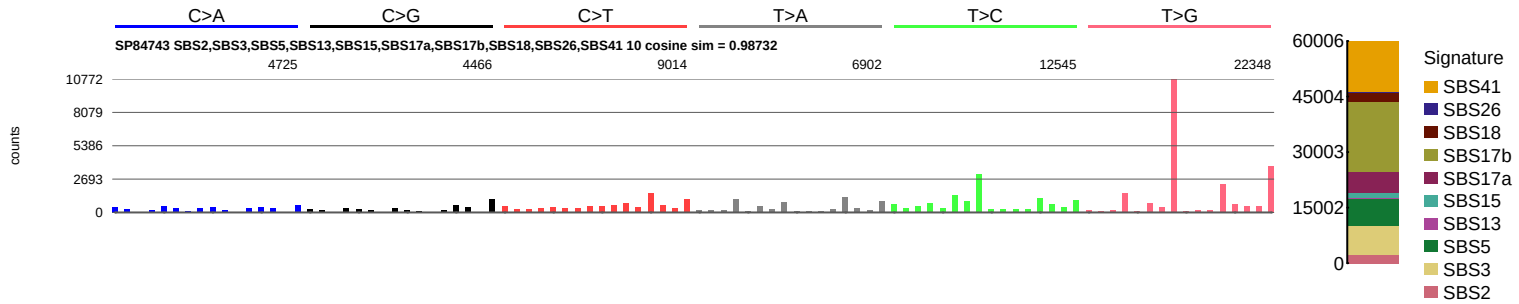
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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SP84743 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98732
4725
4466
9014
6902
12545
22348
10772
8079
5386
2693
0
counts
60006
45004
30003
15002
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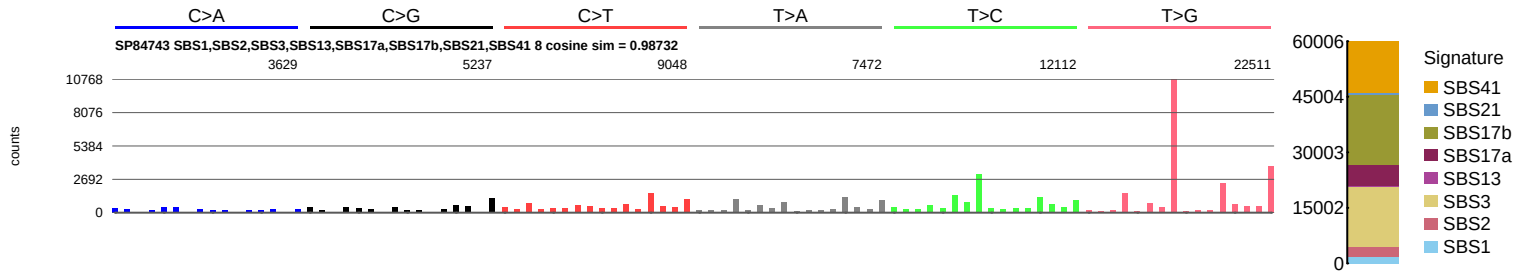
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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C>G
C>T
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SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98732
3629
5237
9048
7472
12112
22511
10768
8076
5384
2692
0
counts
60006
45004
30003
15002
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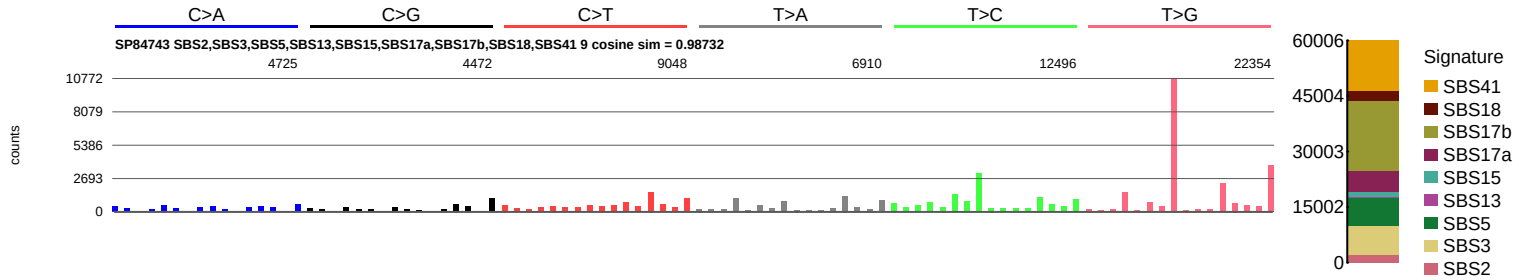
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98732
4725
4472
9048
6910
12496
22354
10772
8079
5386
2693
0
counts
60006
45004
30003
15002
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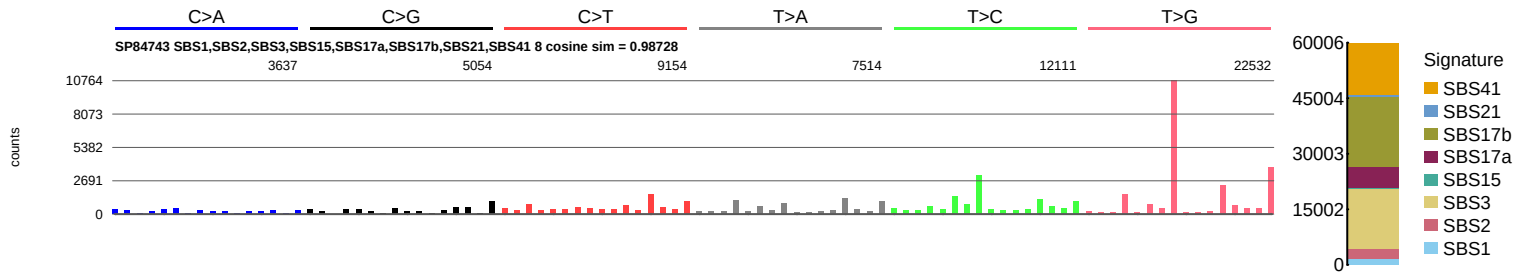
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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C>T
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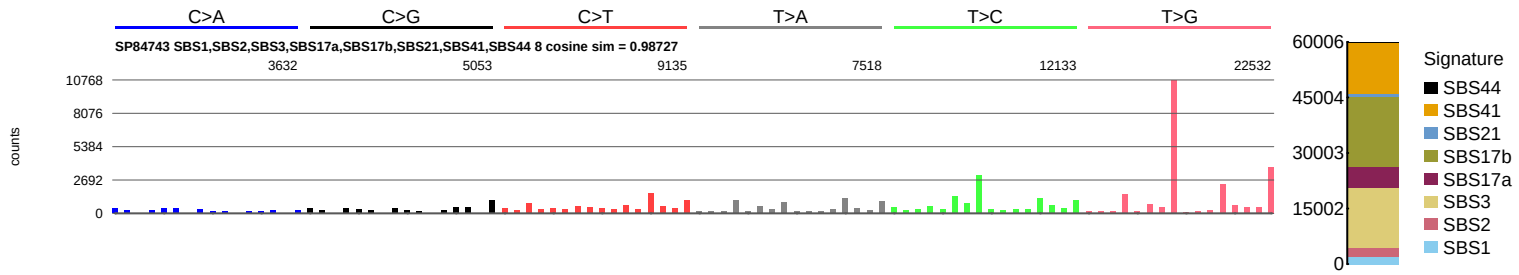
T>G
SP84743 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98728
3637
5054
9154
7514
12111
22532
10764
8073
5382
2691
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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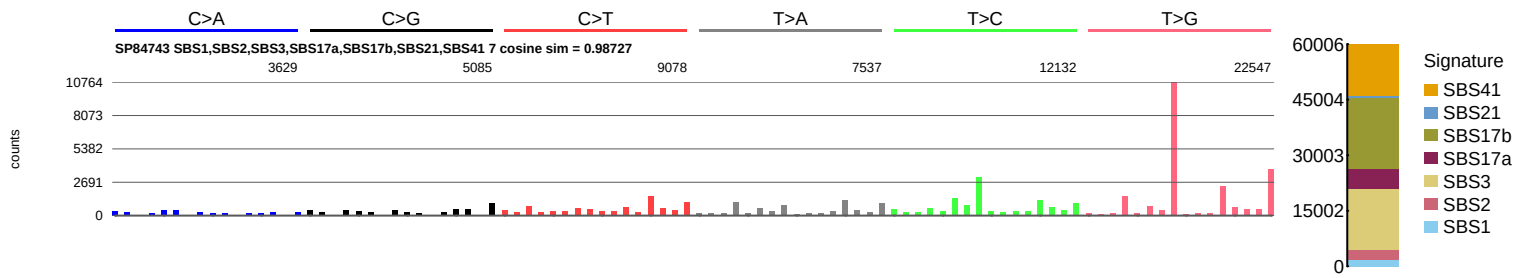
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98727
3632
5053
9135
7518
12133
22532
10768
8076
5384
2692
0
counts
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98727
3629
5085
9078
7537
12132
22547
10764
8073
5382
2691
0
counts
60006
45004
30003
15002
0
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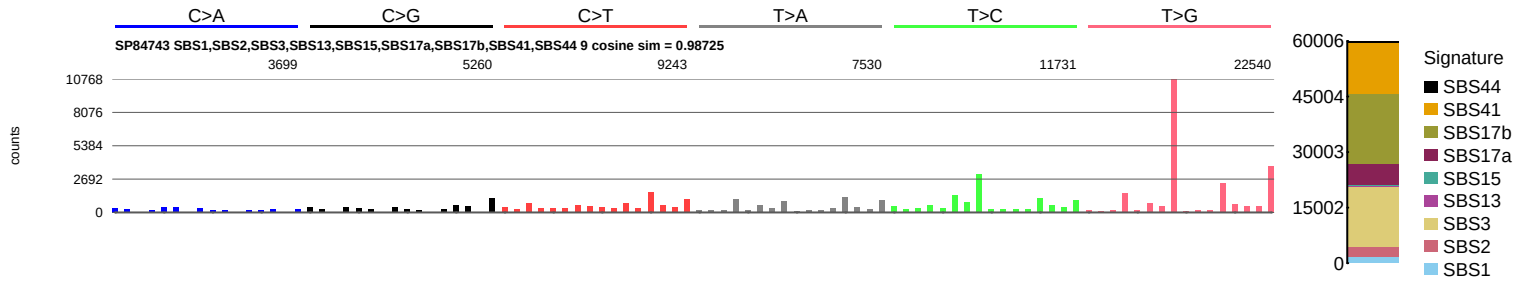
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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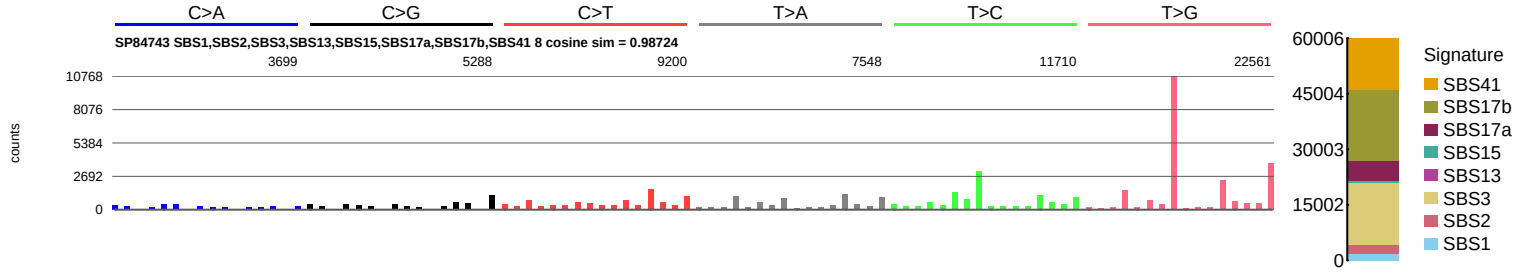
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.98725
3699
5260
9243
7530
11731
22540
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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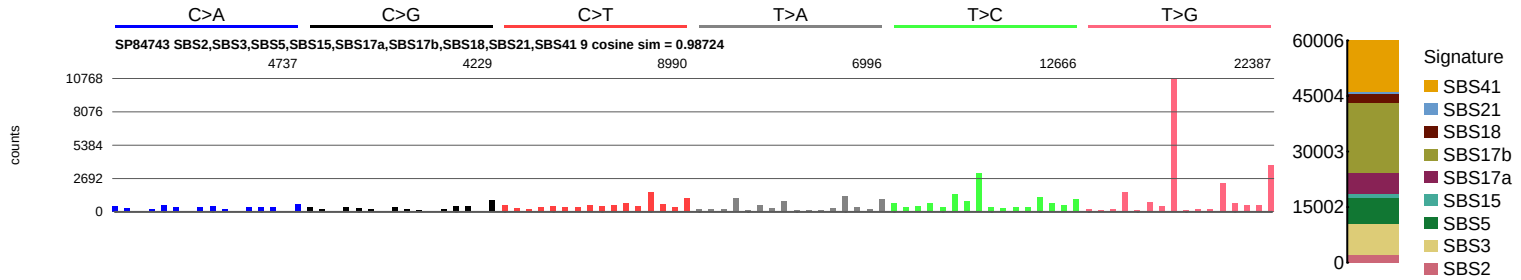
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98724
3699
5288
9200
7548
11710
22561
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98724
4737
4229
8990
6996
12666
22387
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
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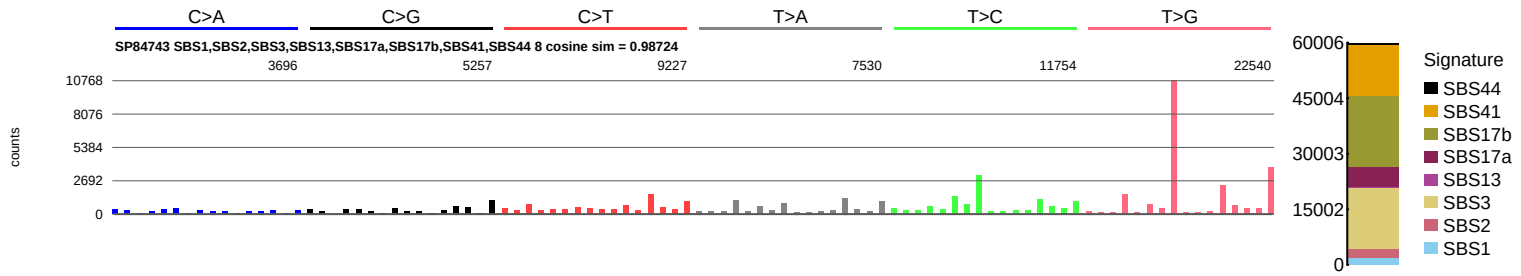
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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C>G
C>T
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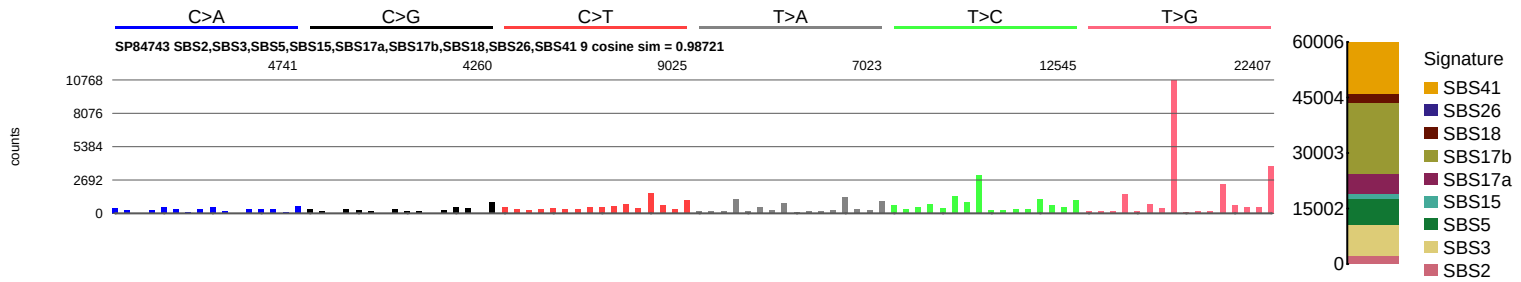
T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98724
3696
5257
9227
7530
11754
22540
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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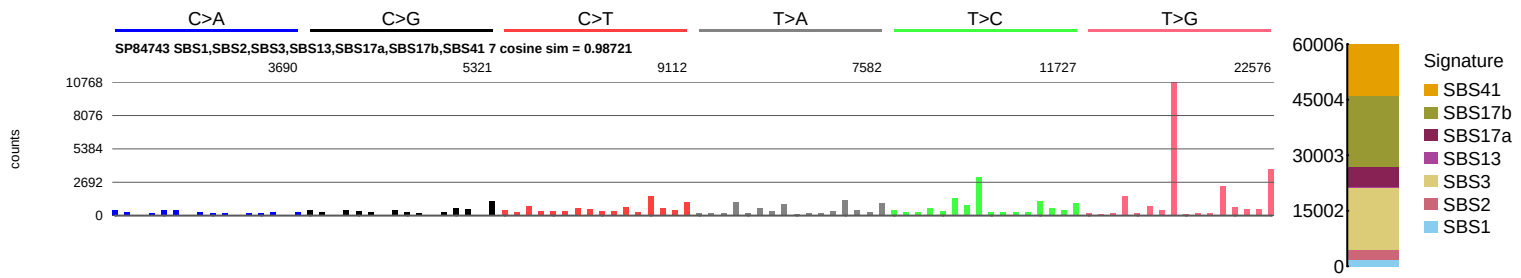
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98721
4741
4260
9025
7023
12545
22407
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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C>G
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T>C
T>G
SP84743 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98721
3690
5321
9112
7582
11727
22576
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
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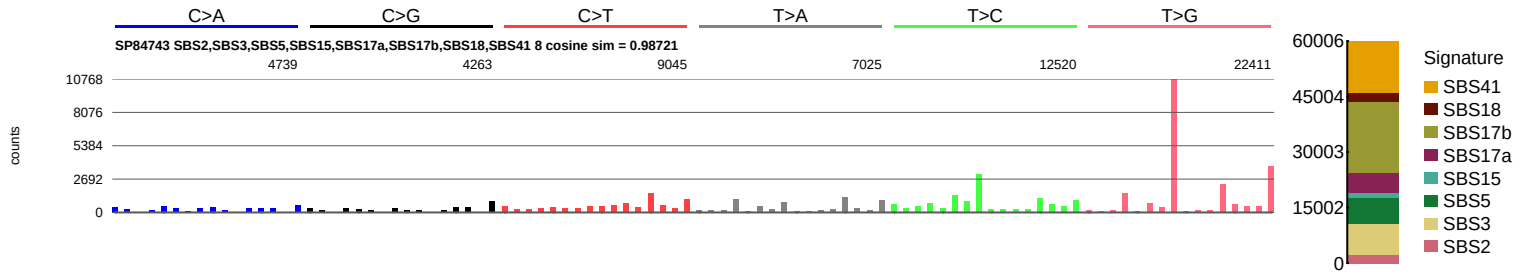
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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SP84743 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98721
4739
4263
9045
7025
12520
22411
counts
10768
8076
5384
2692
0
60006
45004
30003
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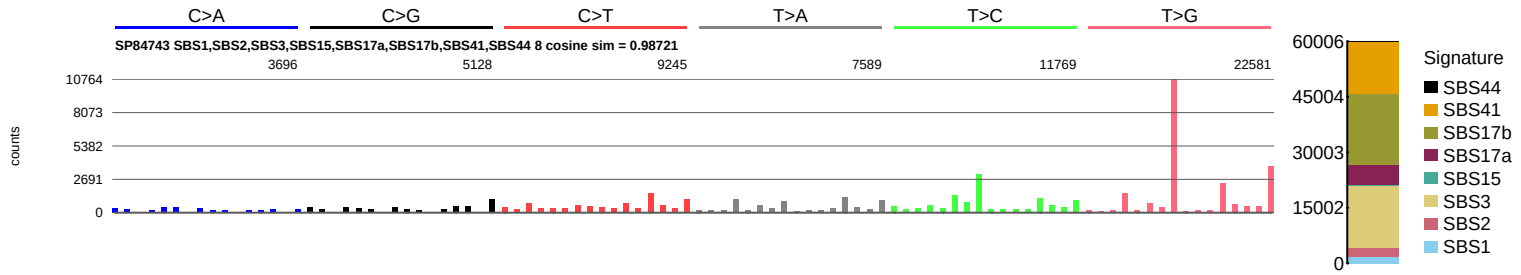
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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SP84743 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98721
3696
5128
9245
7589
11769
22581
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
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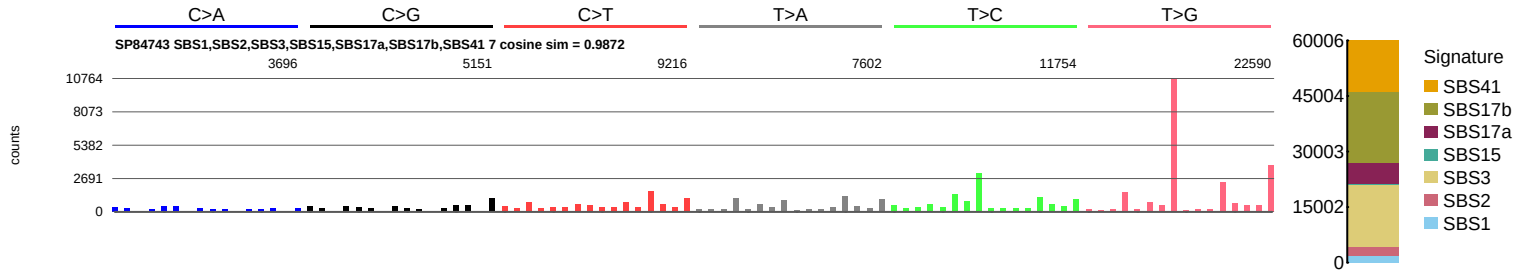
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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T>C
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SP84743 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.9872
3696
5151
9216
7602
11754
22590
counts
10764
8073
5382
2691
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60006
45004
30003
15002
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SBS17b
SBS17a
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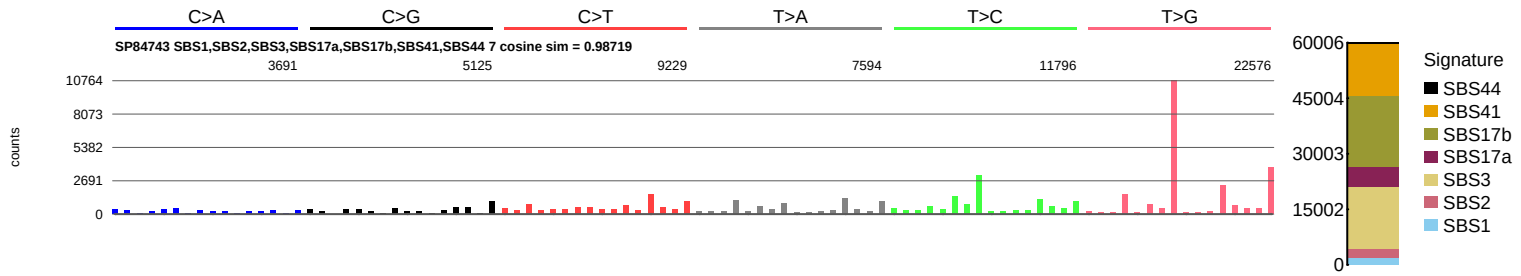
SBS3
SBS2
SBS1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SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98719
3691
5125
9229
7594
11796
22576
counts
10764
8073
5382
2691
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60006
45004
30003
15002
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SBS41
SBS17b
SBS17a
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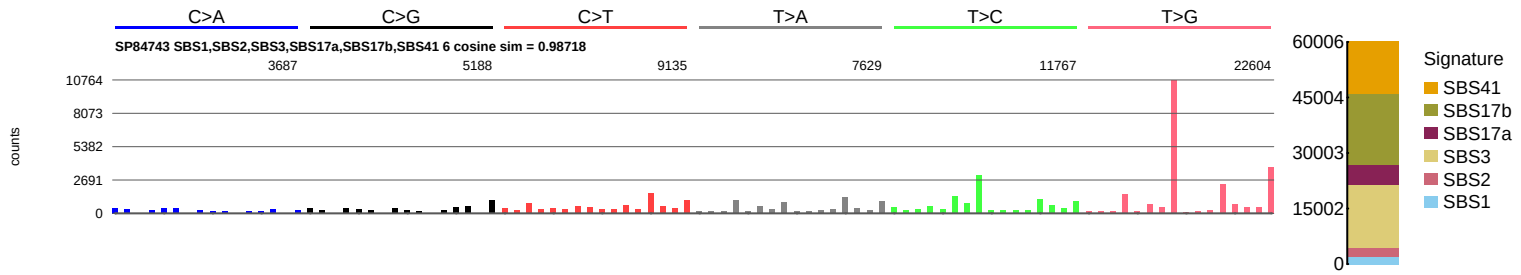
SBS2
SBS1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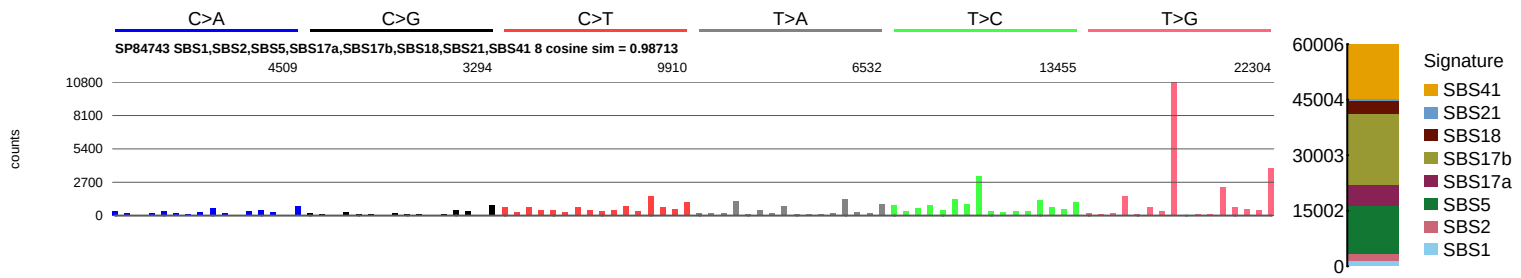
SP84743 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98718
3687
5188
9135
7629
11767
22604
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1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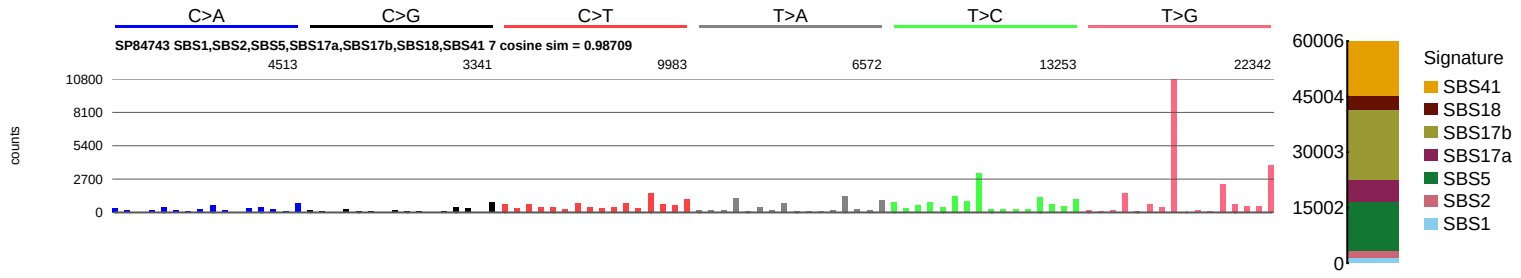
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98713
4509
3294
9910
6532
13455
22304
counts
10800
8100
5400
2700
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98709
4513
3341
9983
6572
13253
22342
counts
10800
8100
5400
2700
0
60006
45004
30003
15002
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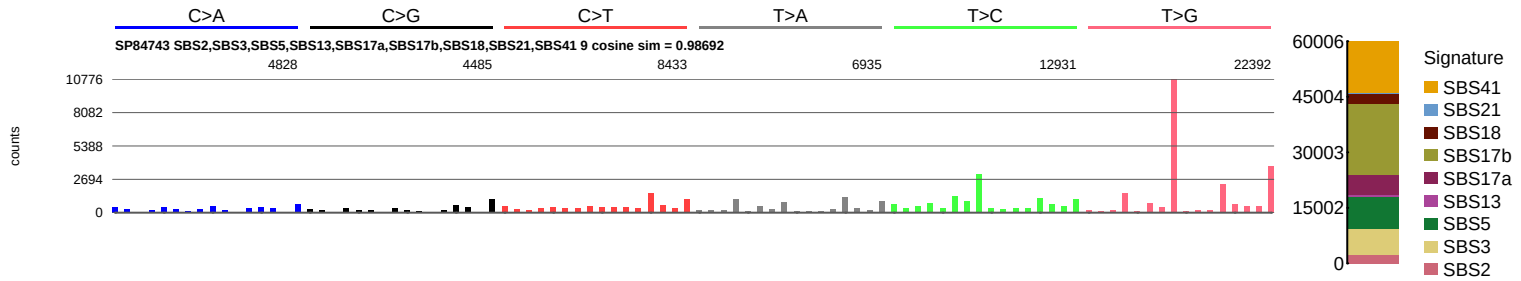
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>T
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T>C
T>G
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4828
4485
8433
6935
12931
22392
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
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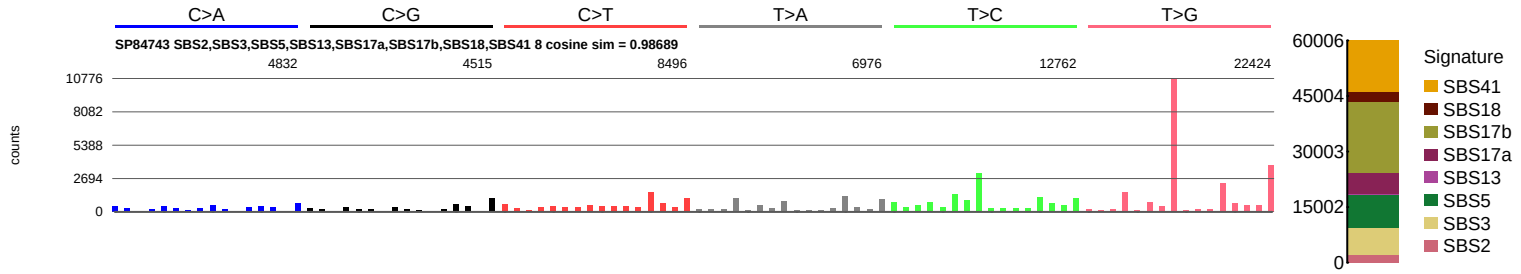
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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C>G
C>T
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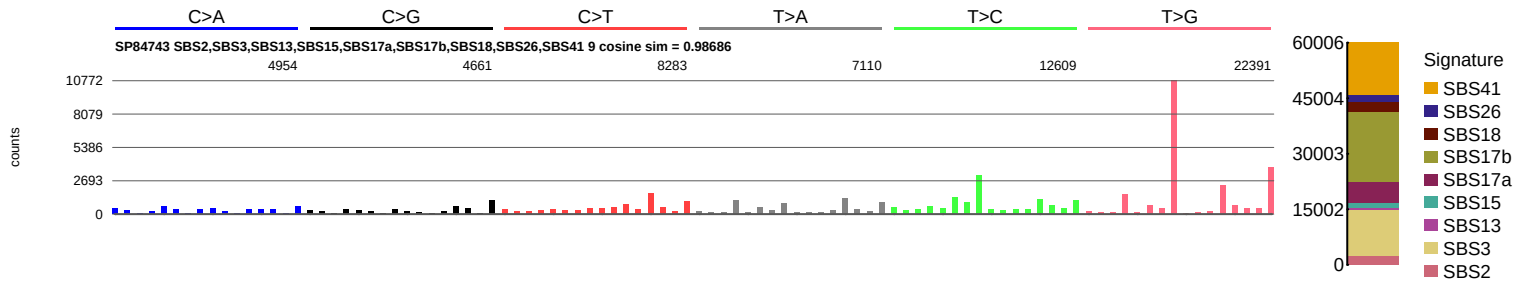
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98689
4832
4515
8496
6976
12762
22424
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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C>G
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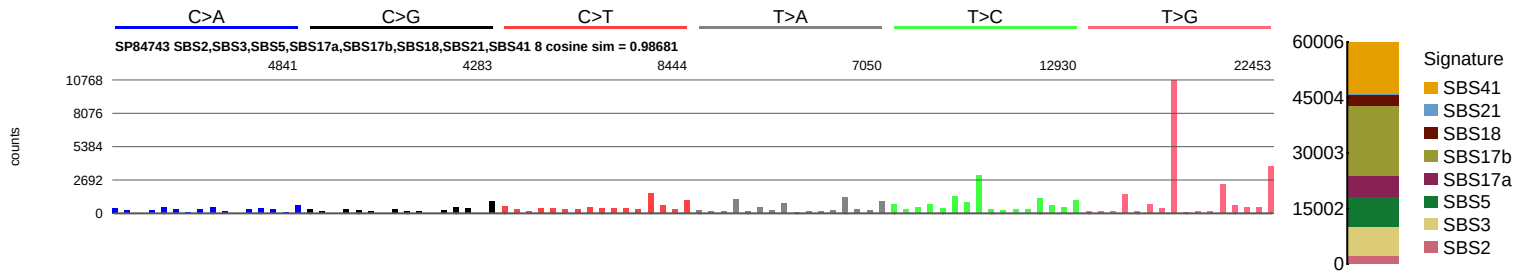
T>G
SP84743 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98686
4954
4661
8283
7110
12609
22391
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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C>G
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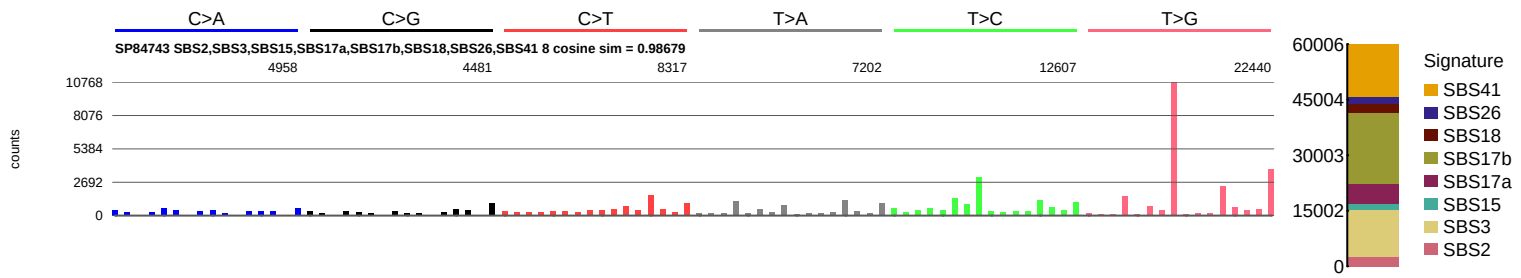
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98681
4841
4283
8444
7050
12930
22453
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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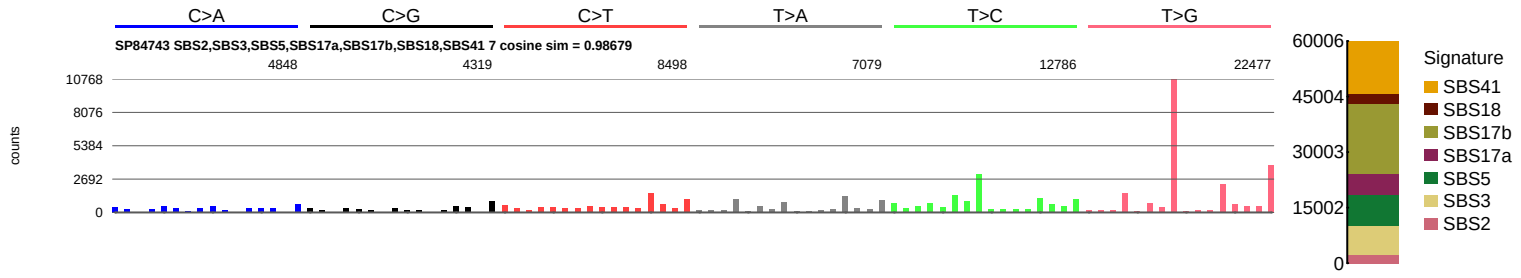
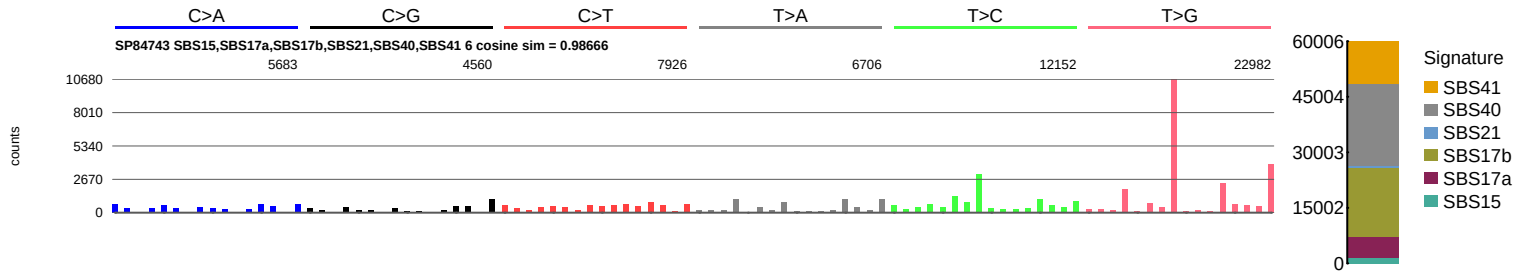
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98679
4958
4481
8317
7202
12607
22440
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98679
4848
4319
8498
7079
12786
22477
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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C>G
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T>A
T>C
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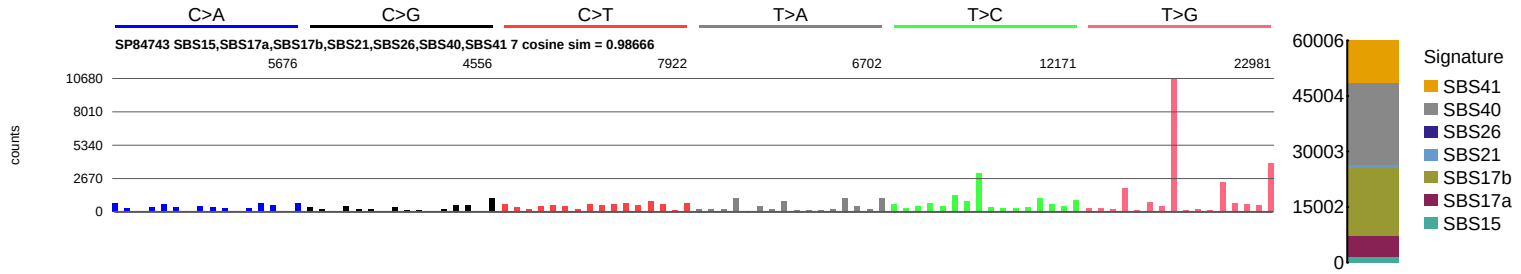
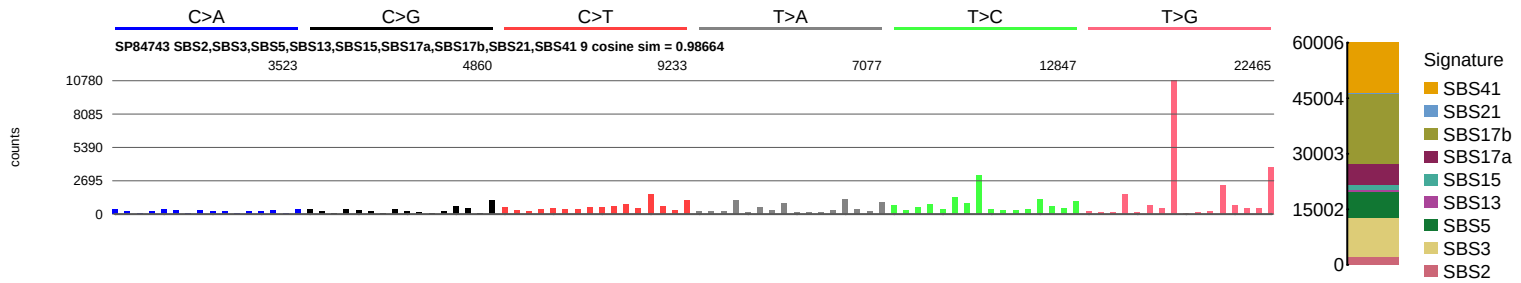
SP84743 SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98666
5683
4560
7926
6706
12152
22982
counts
10680
8010
5340
2670
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98666
5676
4556
7922
6702
12171
22981
counts
10680
8010
5340
2670
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15


C>A
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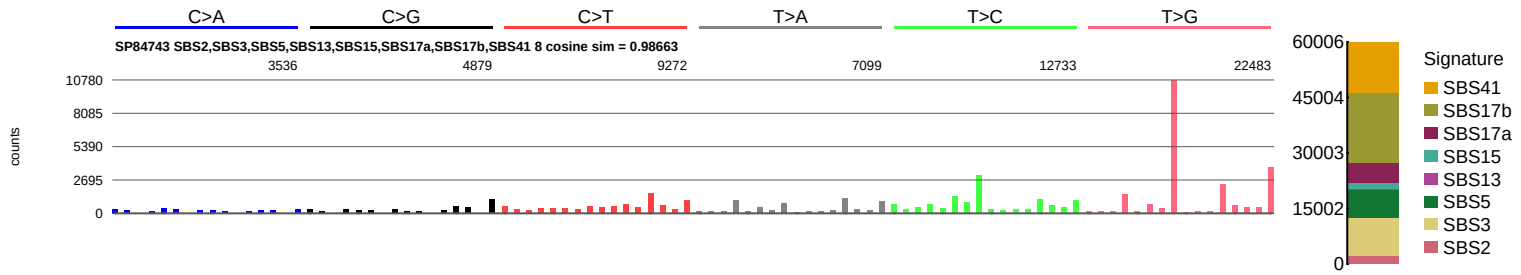
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98664
3523
4860
9233
7077
12847
22465
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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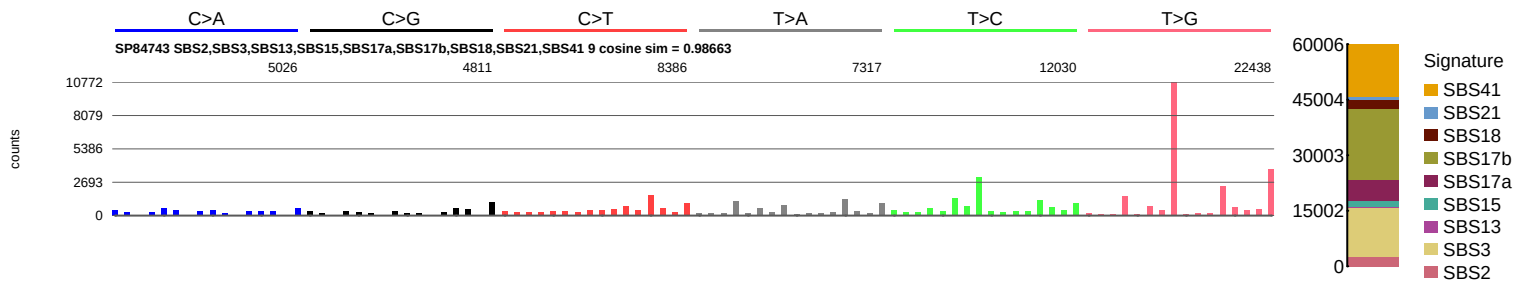
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98663
3536
4879
9272
7099
12733
22483
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
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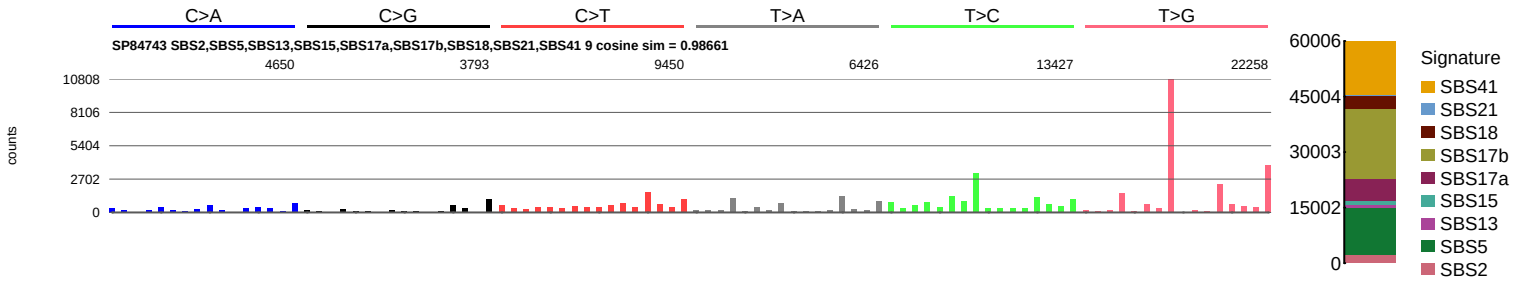
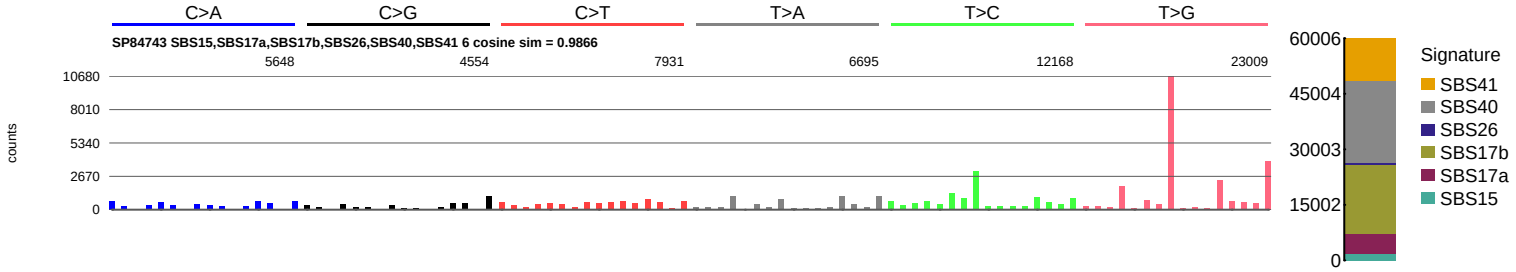
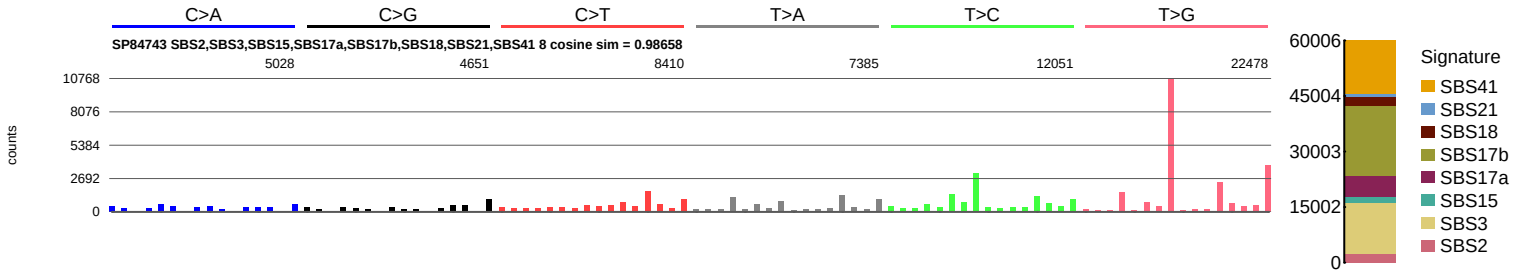
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98663
5026
4811
8386
7317
12030
22438
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98661
4650
3793
9450
6426
13427
22258
counts
10808
8106
5404
2702
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.9866
5648
4554
7931
6695
12168
23009
counts
10680
8010
5340
2670
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15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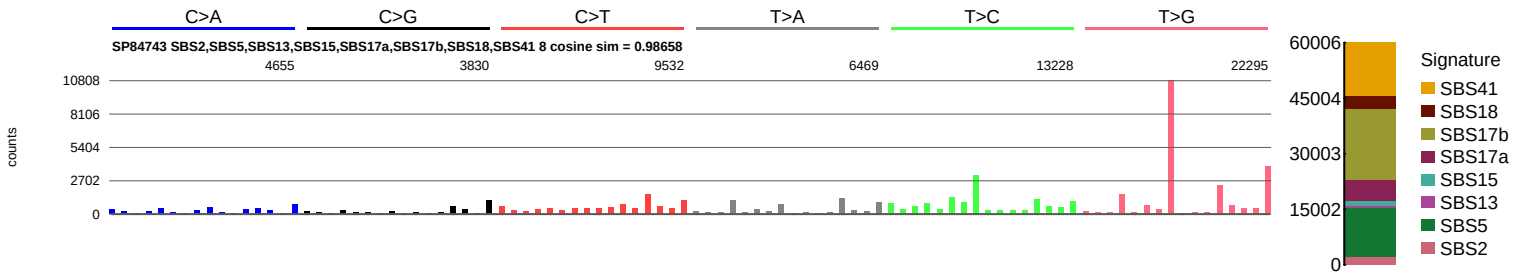
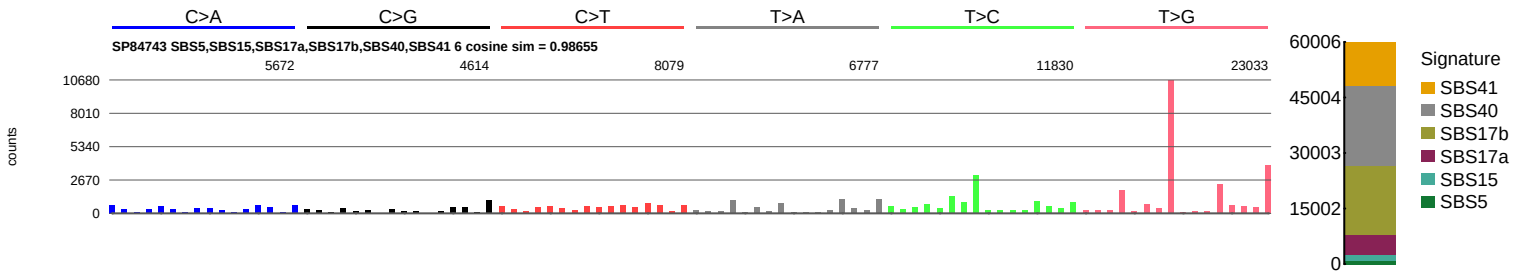
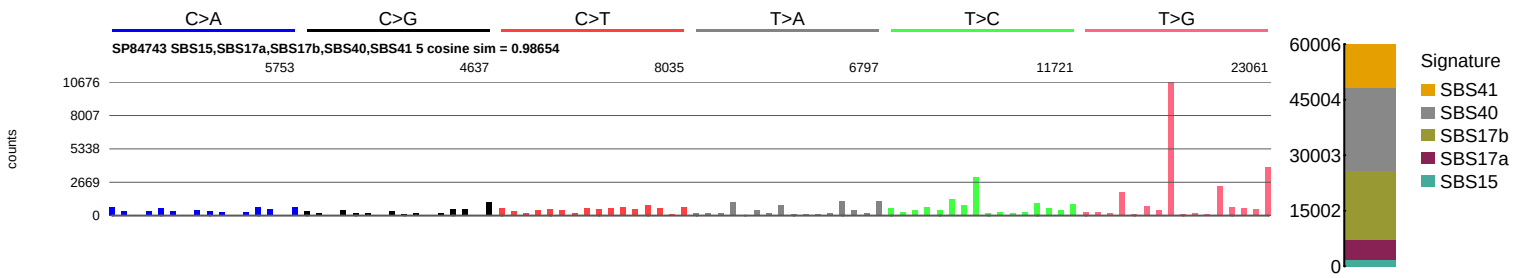
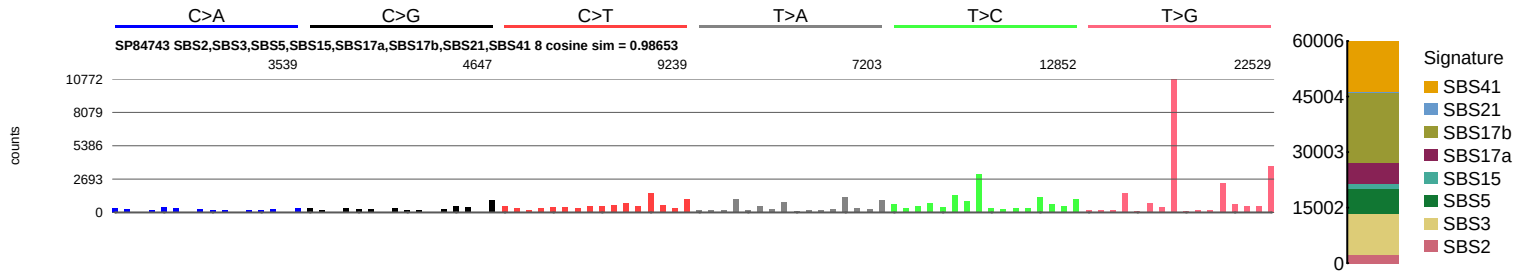
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98658
5028
4651
8410
7385
12051
22478
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98658
4655
3830
9532
6469
13228
22295
counts
10808
8106
5404
2702
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98655
5672
4614
8079
6777
11830
23033
counts
10680
8010
5340
2670
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS15,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98654
5753
4637
8035
6797
11721
23061
counts
10676
8007
5338
2669
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15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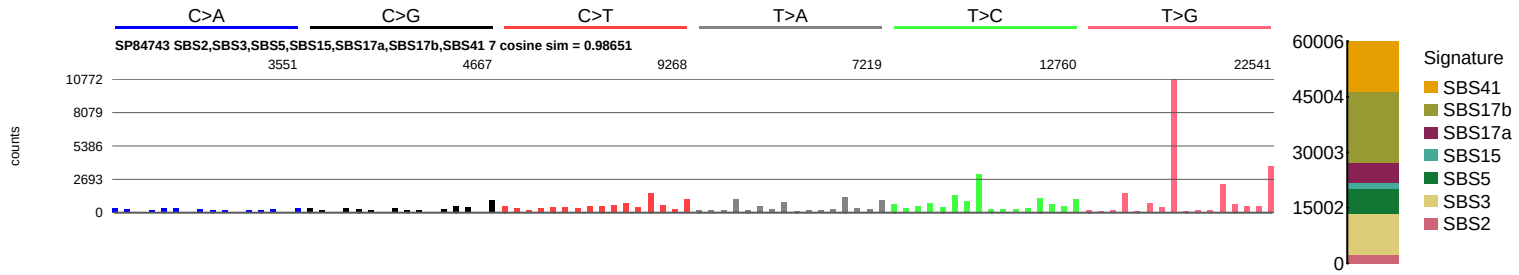
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98653
3539
4647
9239
7203
12852
22529
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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C>G
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T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98651
3551
4667
9268
7219
12760
22541
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
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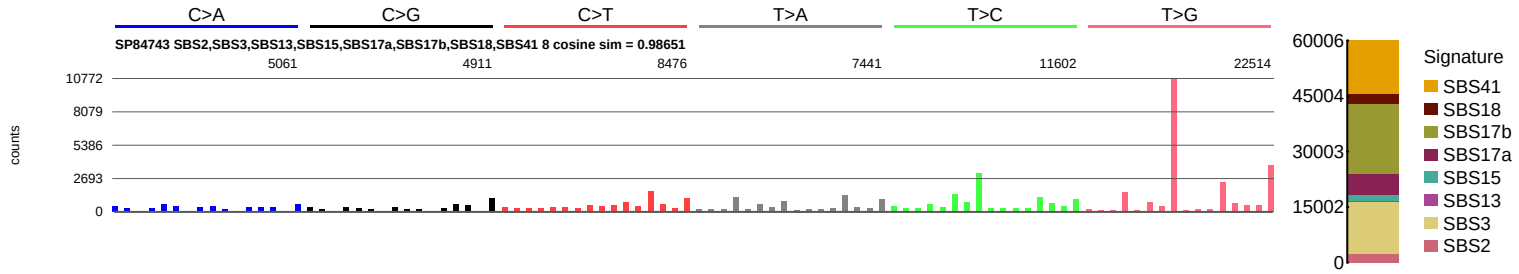
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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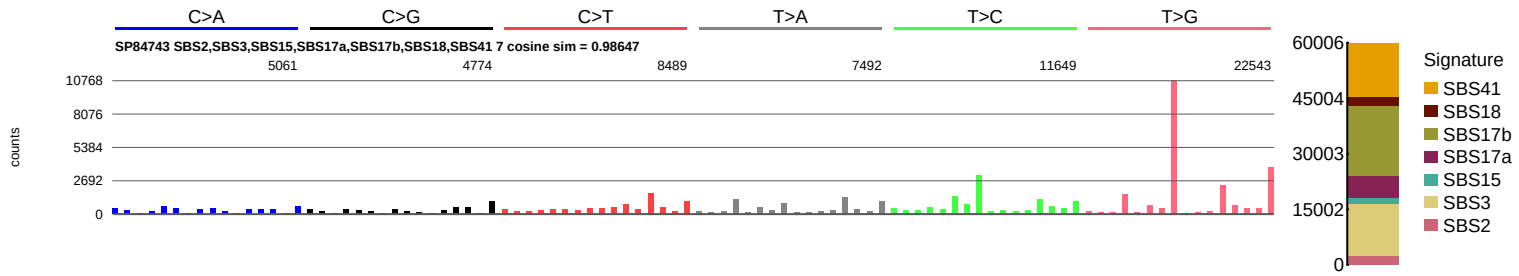
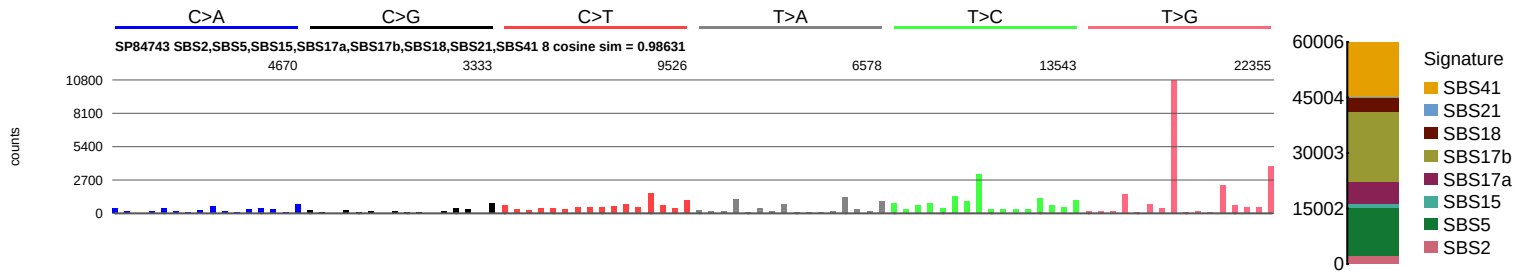
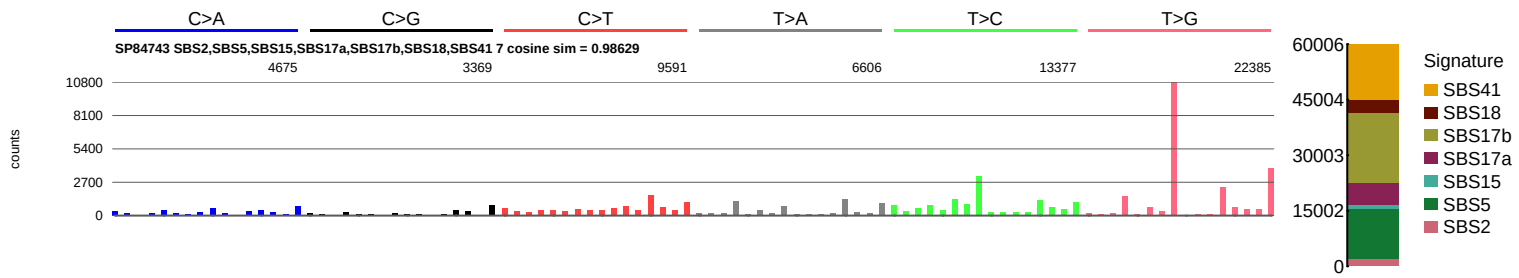
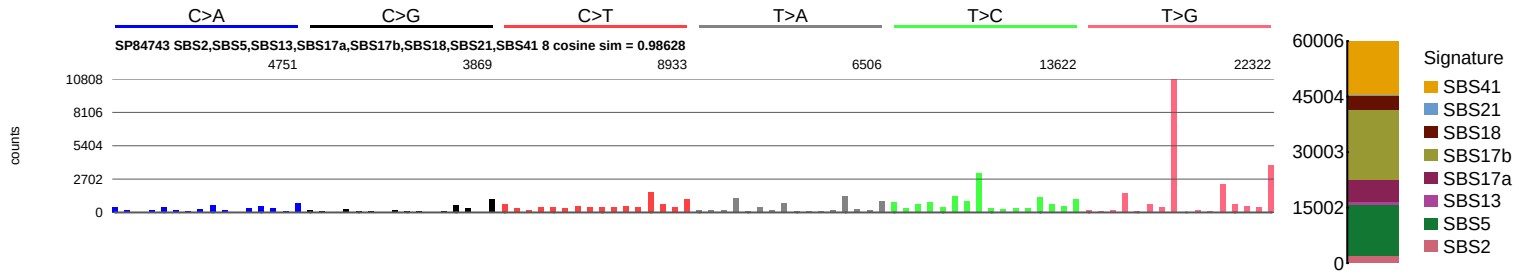
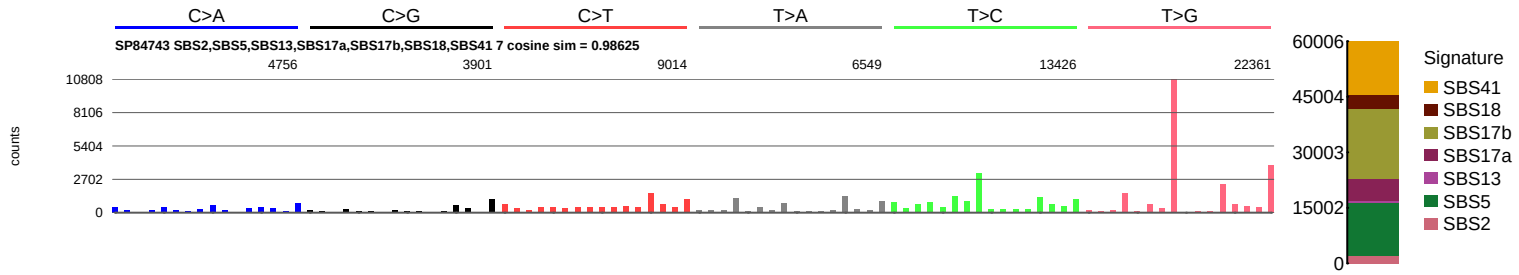
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98651
5061
4911
8476
7441
11602
22514
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

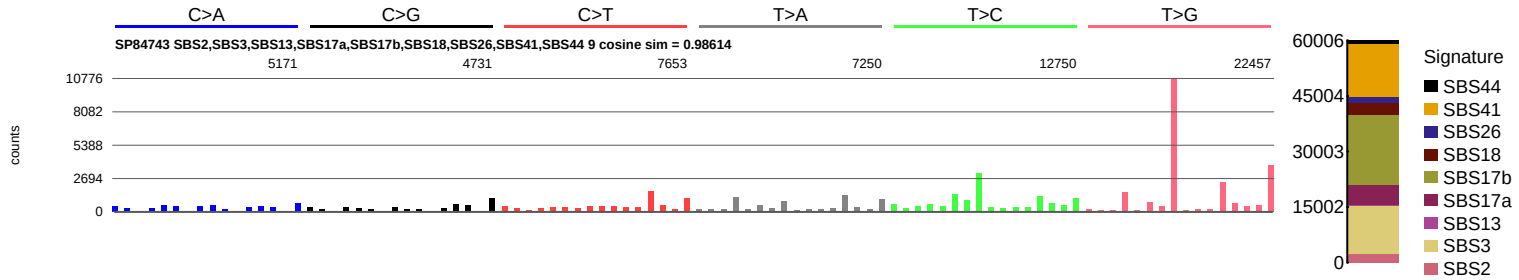

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98647
5061
4774
8489
7492
11649
22543
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98631
4670
3333
9526
6578
13543
22355
counts
10800
8100
5400
2700
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98629
4675
3369
9591
6606
13377
22385
counts
10800
8100
5400
2700
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98628
4751
3869
8933
6506
13622
22322
counts
10808
8106
5404
2702
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98625
4756
3901
9014
6549
13426
22361
counts
10808
8106
5404
2702
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98614
5171
4731
7653
7250
12750
22457
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
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Signature
SBS44
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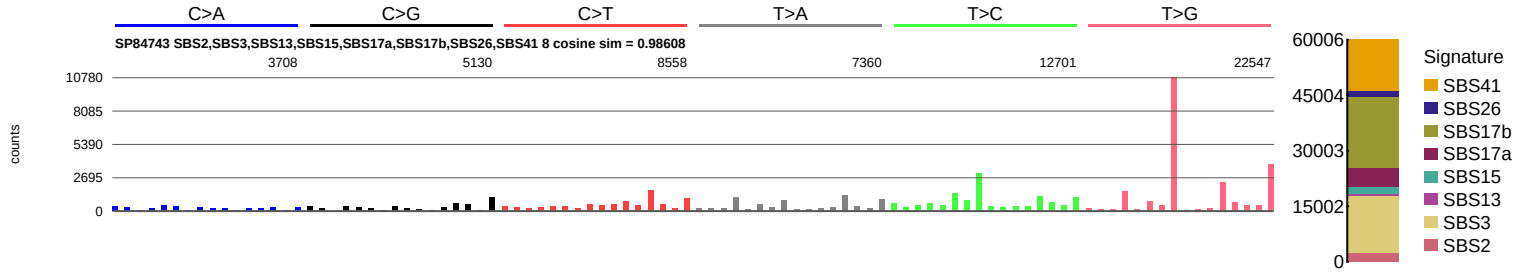
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98608
3708
5130
8558
7360
12701
22547
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
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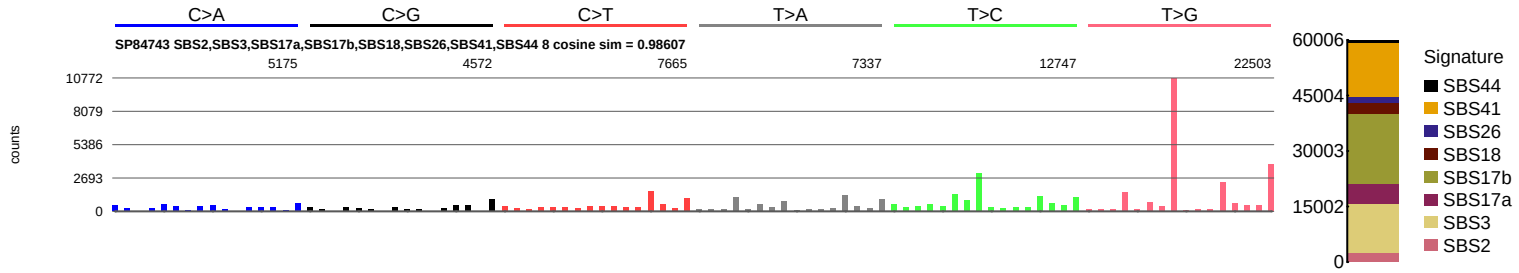
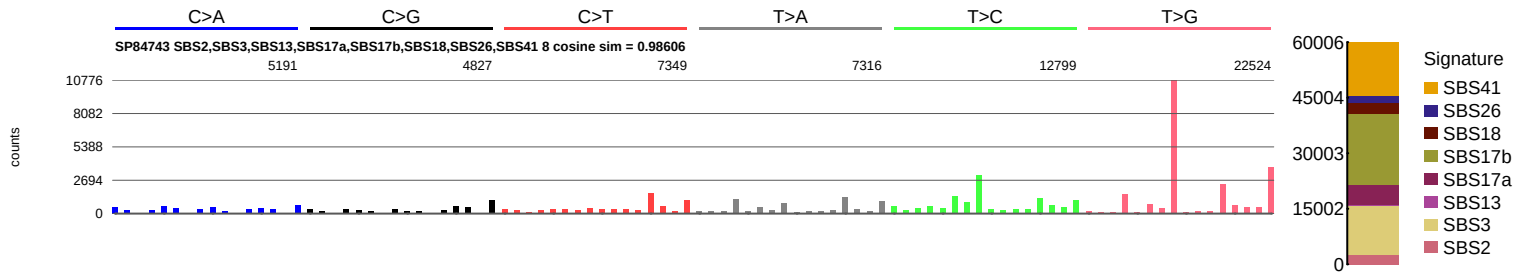
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98607
5175
4572
7665
7337
12747
22503
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2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C>G
C>T
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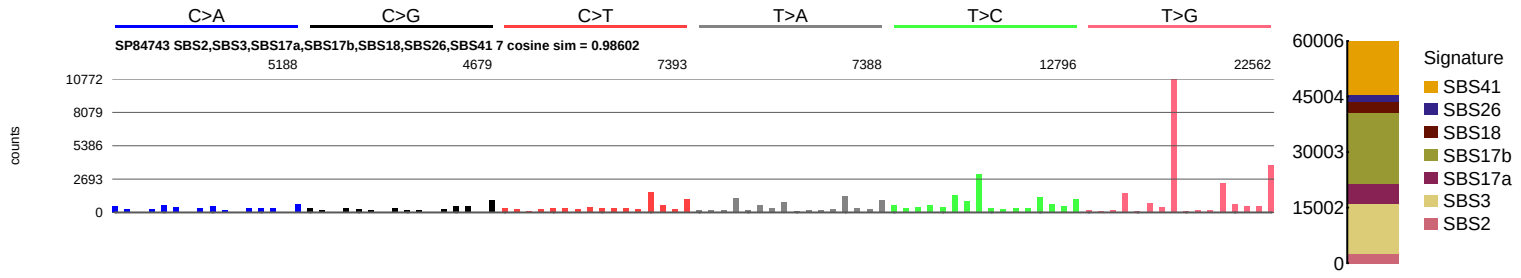
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98606
5191
4827
7349
7316
12799
22524
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2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C>G
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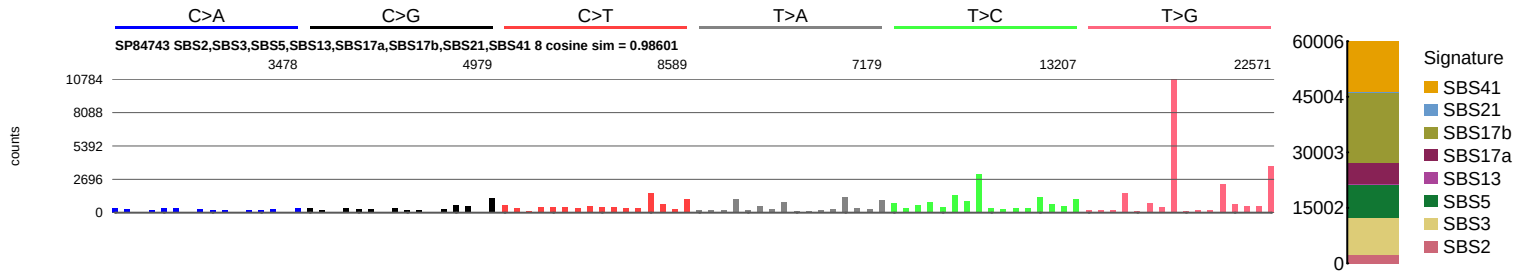
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98602
5188
4679
7393
7388
12796
22562
10772
8079
5386
2693
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2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C>T
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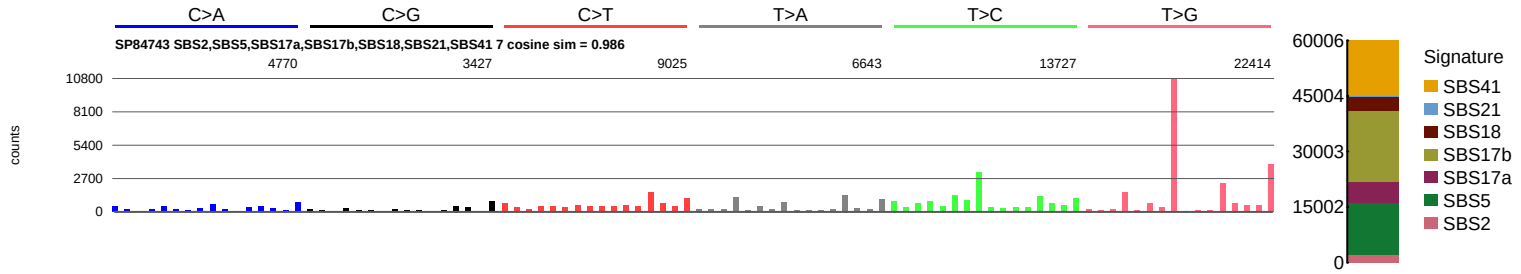
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98601
3478
4979
8589
7179
13207
22571
10784
8088
5392
2696
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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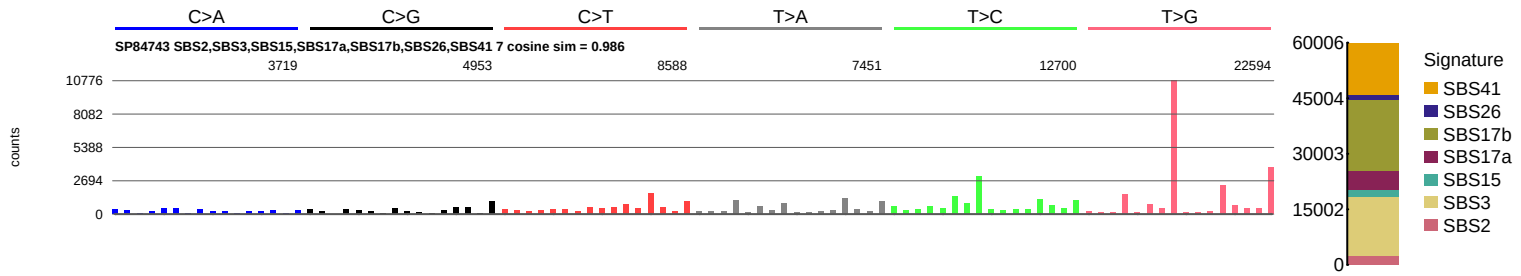
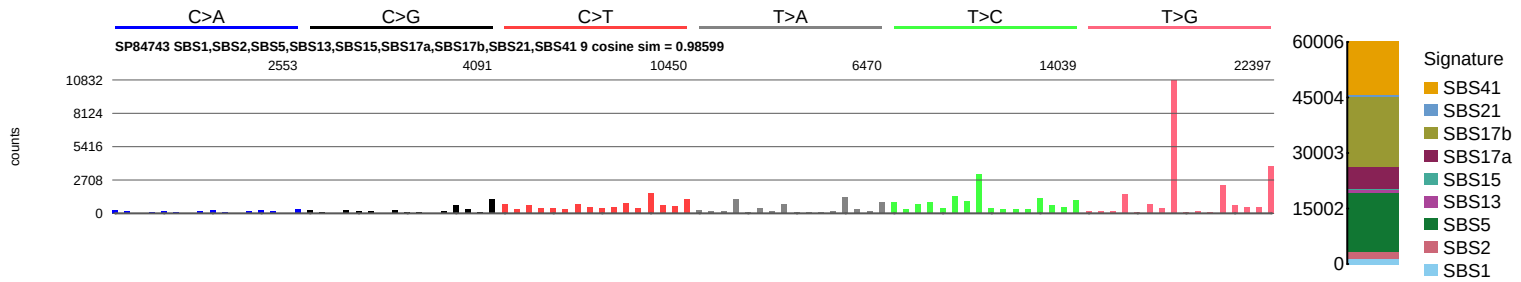
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.986
4770
3427
9025
6643
13727
22414
10800
8100
5400
2700
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.986
3719
4953
8588
7451
12700
22594
10776
8082
5388
2694
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98599
2553
4091
10450
6470
14039
22397
10832
8124
5416
2708
0
counts
60006
45004
30003
15002
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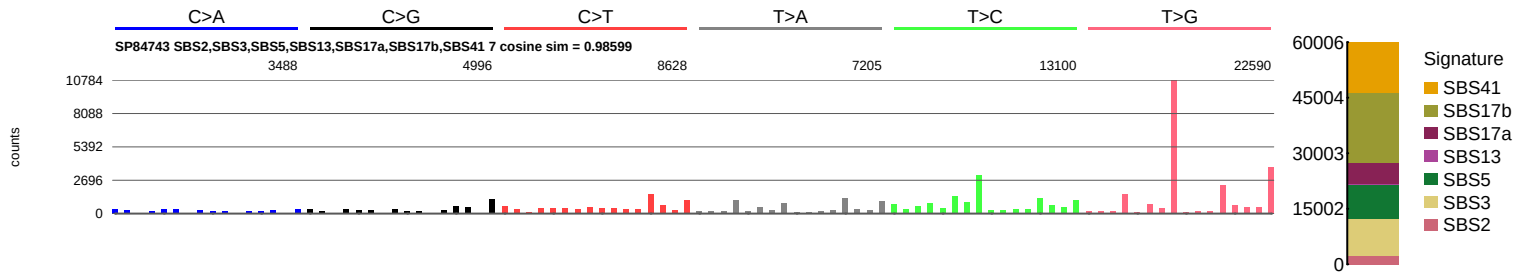
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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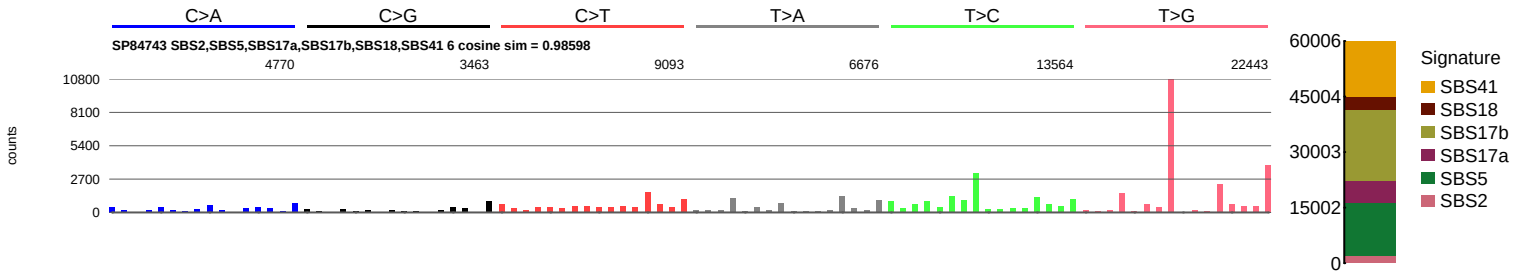
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98599
3488
4996
8628
7205
13100
22590
10784
8088
5392
2696
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98598
4770
3463
9093
6676
13564
22443
counts
10800
8100
5400
2700
0
60006
45004
30003
15002
0
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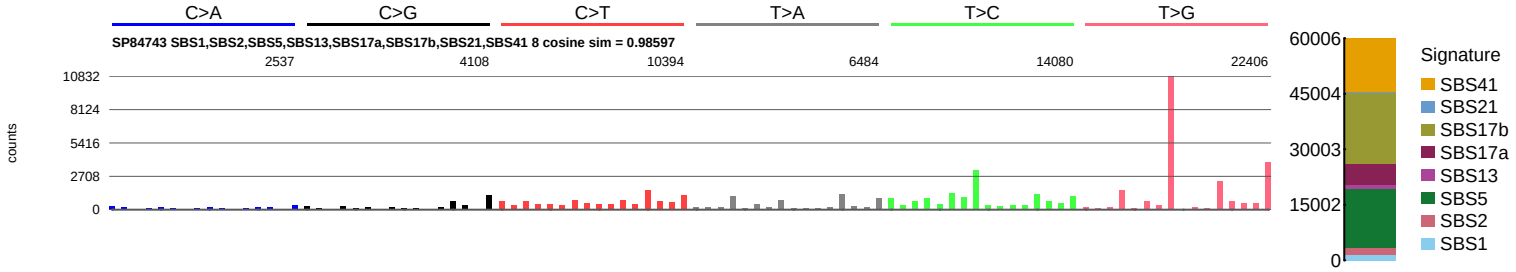
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2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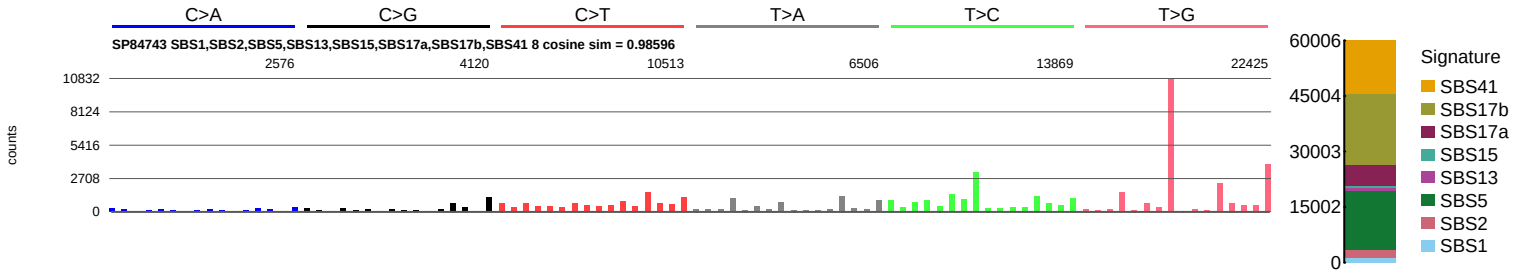
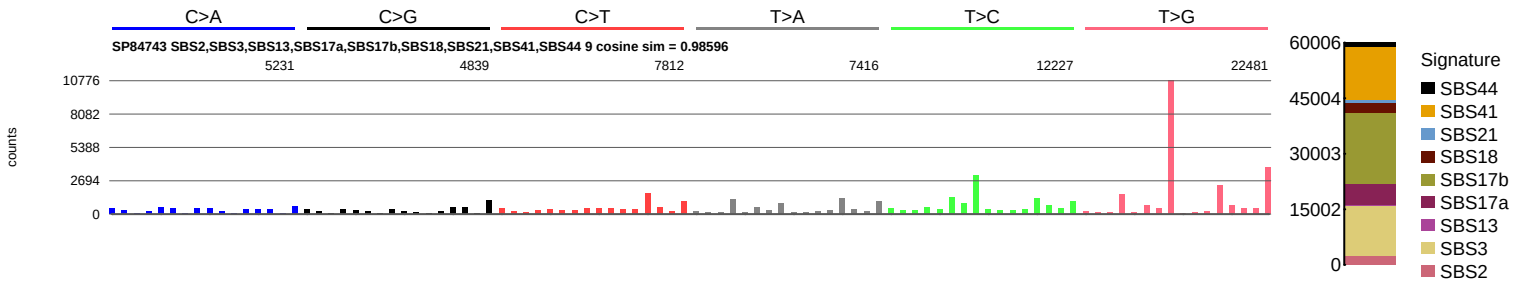
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98597
2537
4108
10394
6484
14080
22406
counts
10832
8124
5416
2708
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98596
2576
4120
10513
6506
13869
22425
counts
10832
8124
5416
2708
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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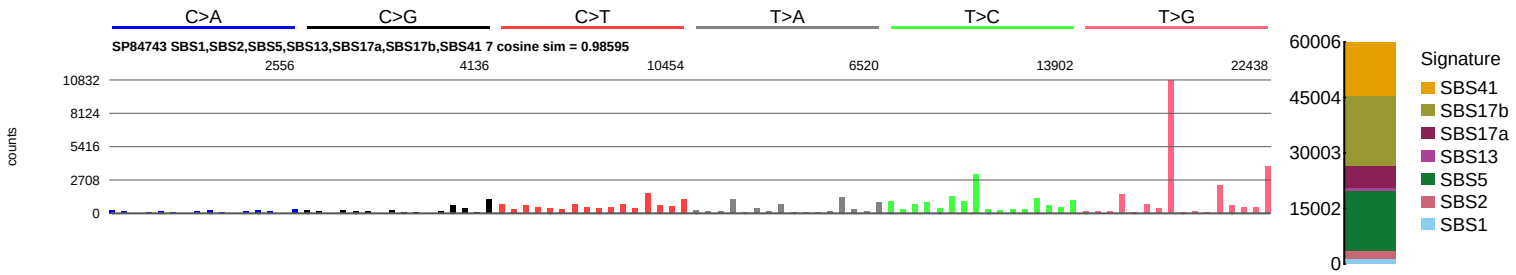
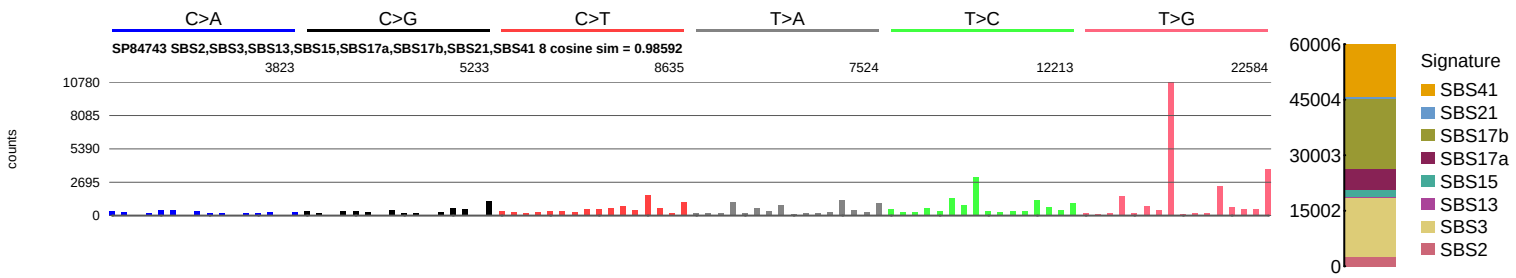
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98596
5231
4839
7812
7416
12227
22481
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98595
2556
4136
10454
6520
13902
22438
counts
10832
8124
5416
2708
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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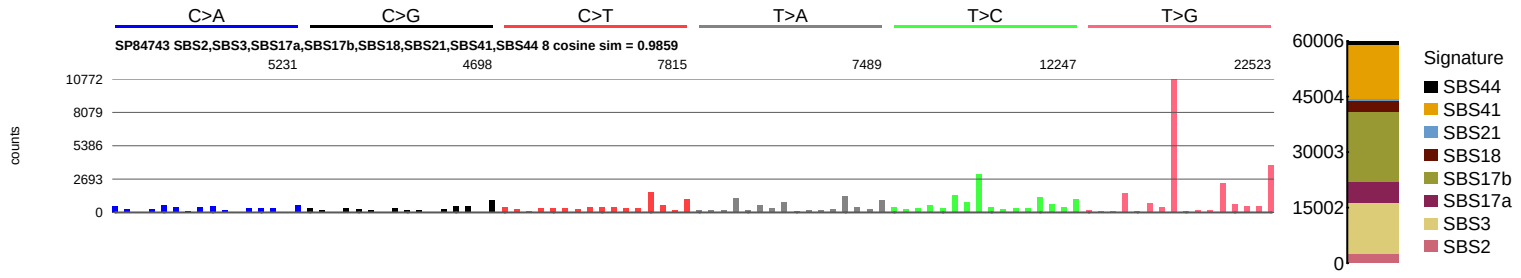
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98592
3823
5233
8635
7524
12213
22584
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.9859
5231
4698
7815
7489
12247
22523
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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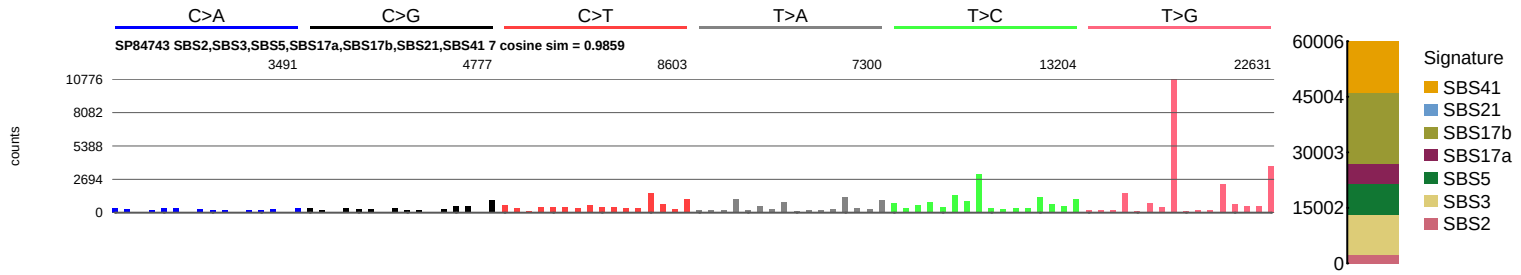
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2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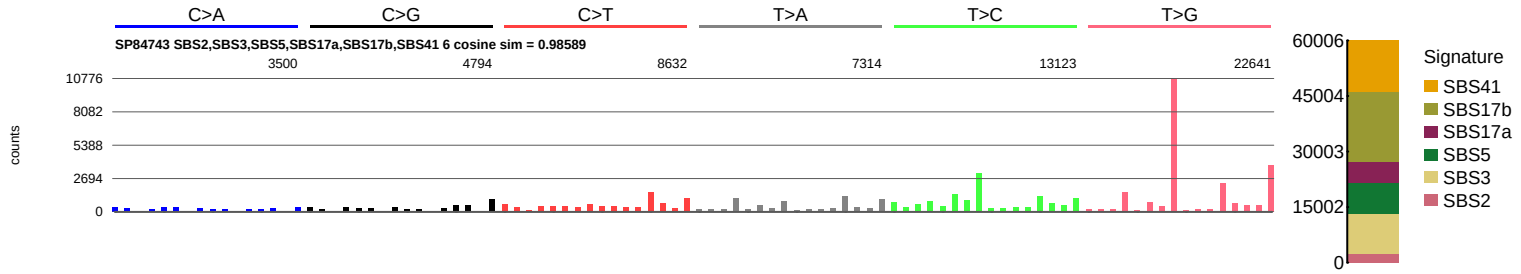
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9859
3491
4777
8603
7300
13204
22631
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98589
3500
4794
8632
7314
13123
22641
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
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Signature
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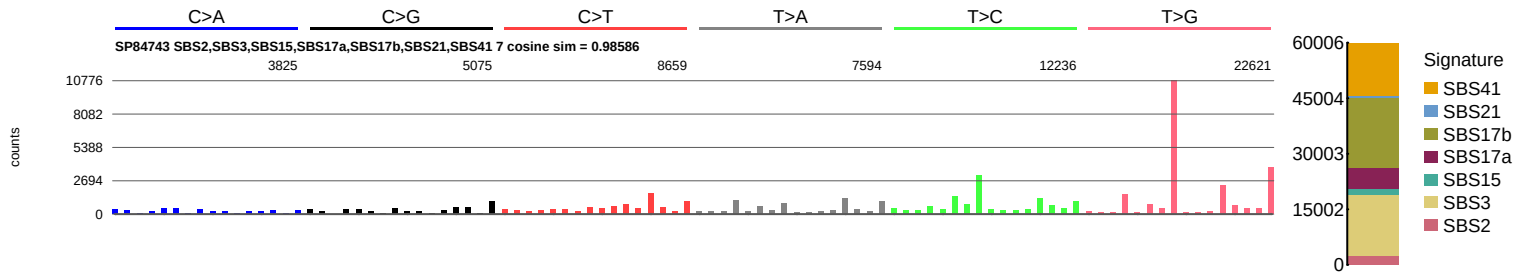
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98586
3825
5075
8659
7594
12236
22621
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
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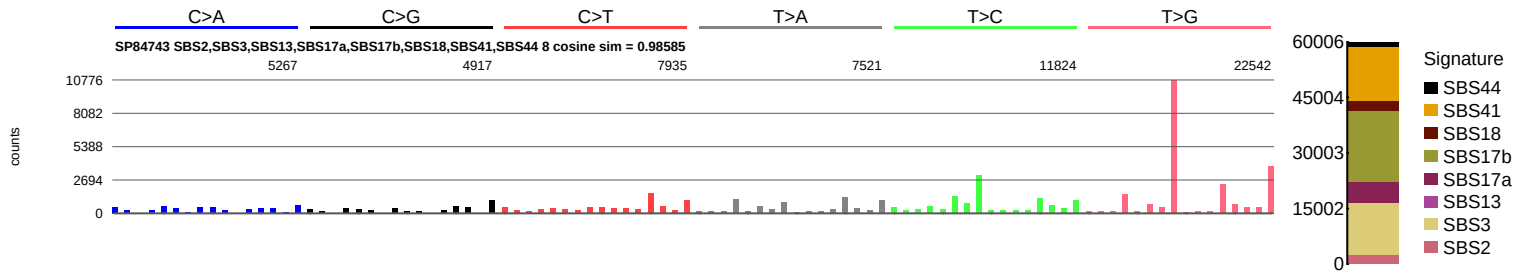
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
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T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98585
5267
4917
7935
7521
11824
22542
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
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Signature
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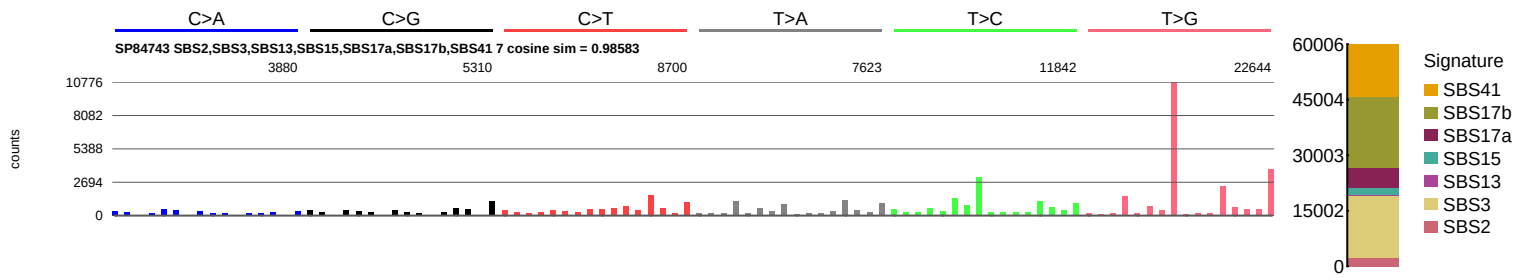
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
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T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98583
3880
5310
8700
7623
11842
22644
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
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Signature
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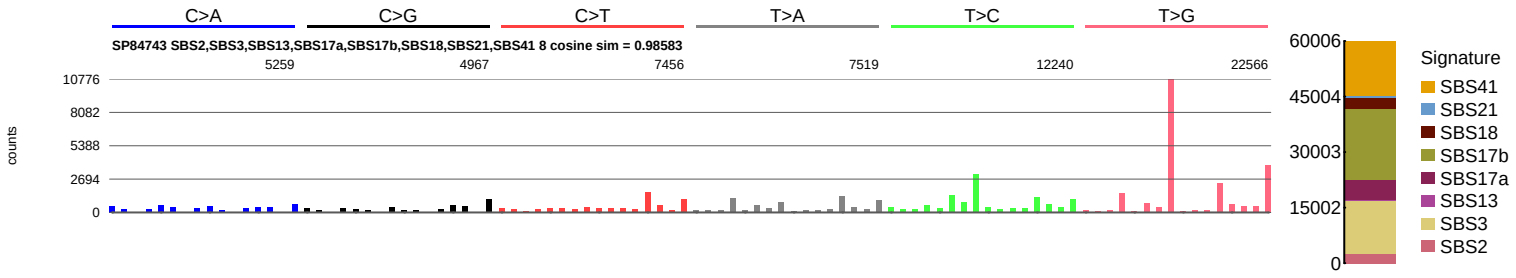
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98583
5259
4967
7456
7519
12240
22566
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
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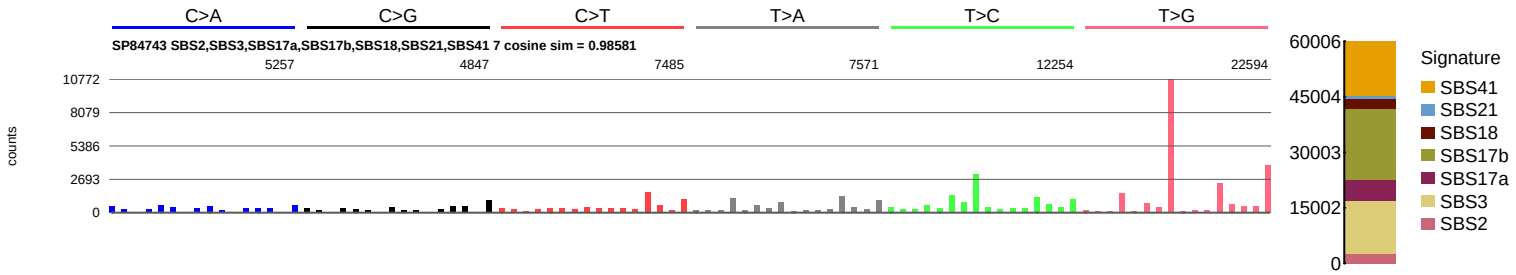
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98581
5257
4847
7485
7571
12254
22594
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
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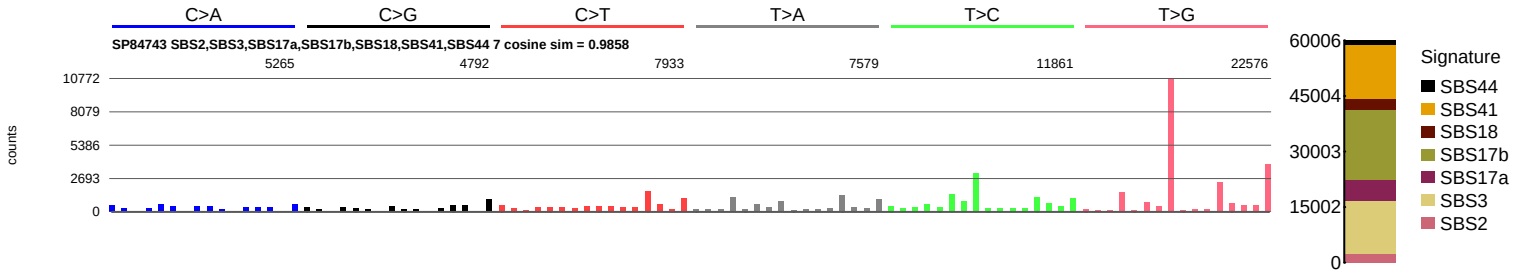
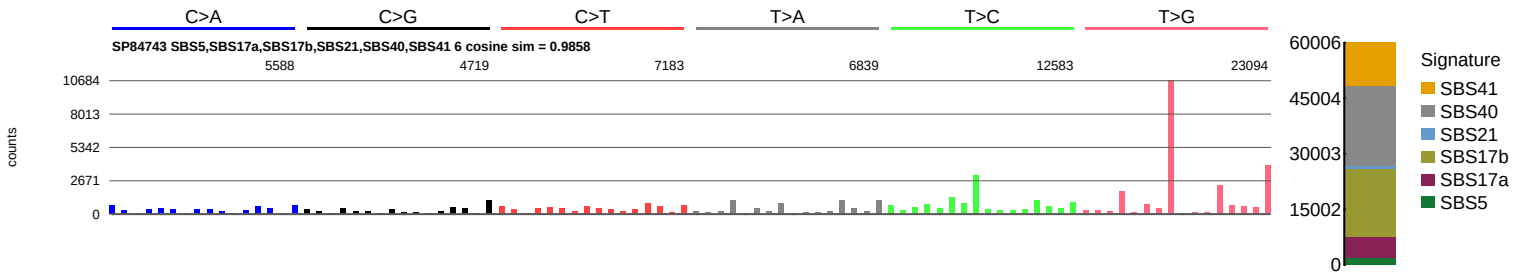
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.9858
5265
4792
7933
7579
11861
22576
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2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C>T
T>A
T>C
T>G
SP84743 SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.9858
5588
4719
7183
6839
12583
23094
counts
10684
8013
5342
2671
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60006
45004
30003
15002
0
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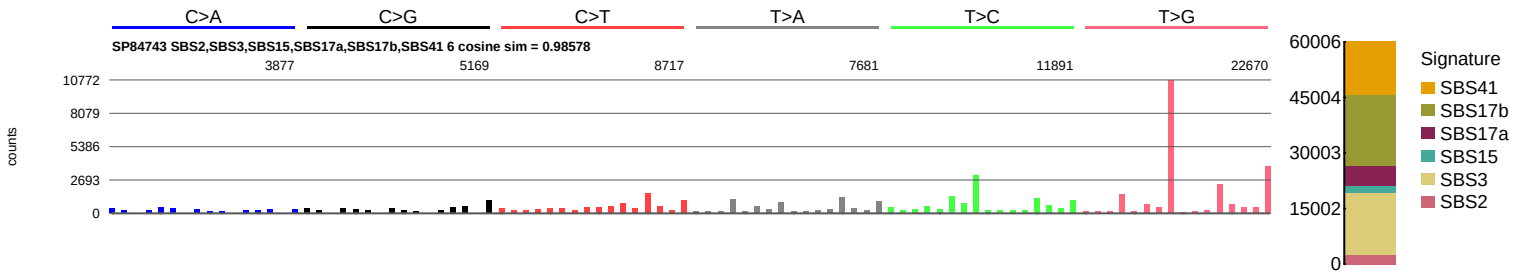
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
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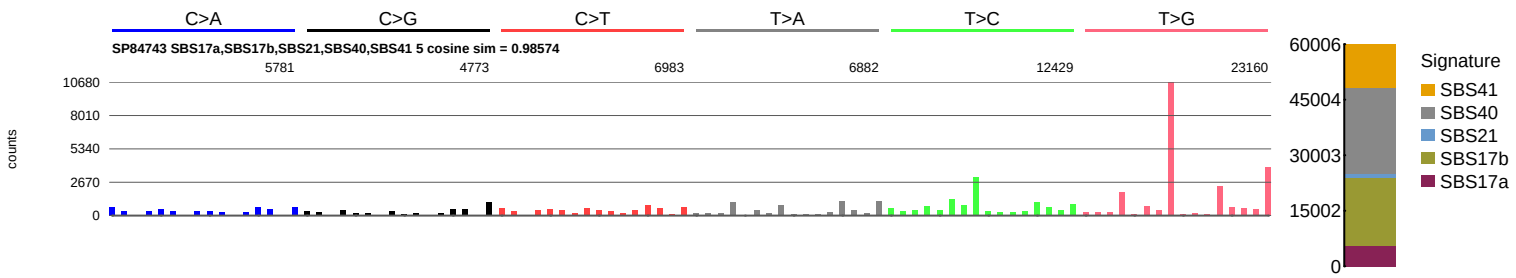
T>C
T>G
SP84743 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98578
3877
5169
8717
7681
11891
22670
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
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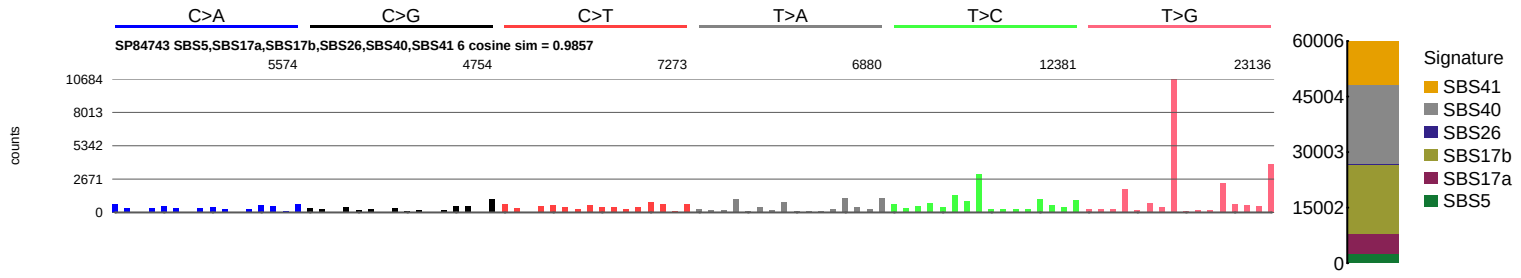
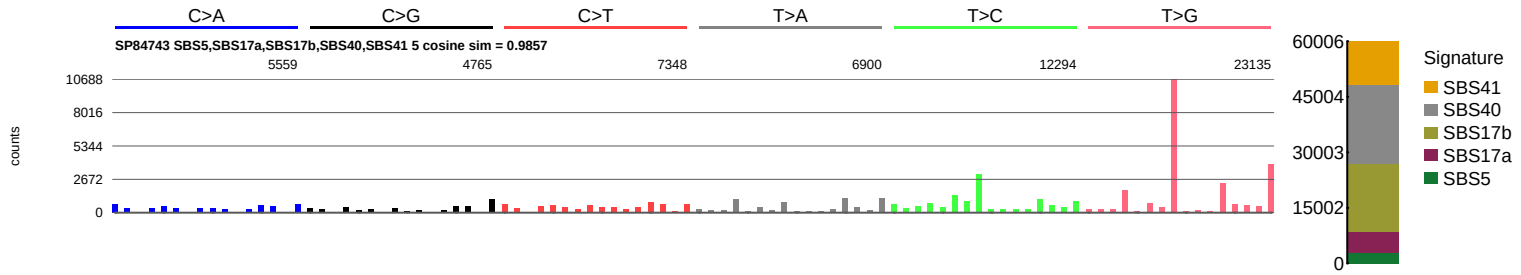
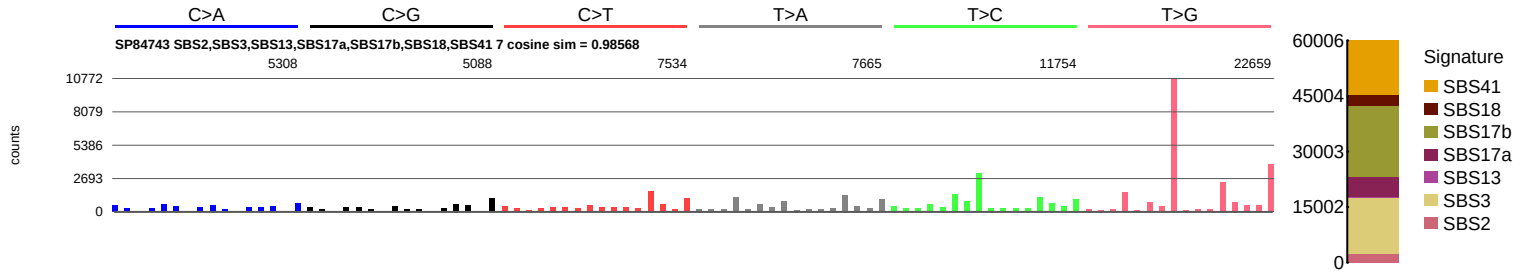
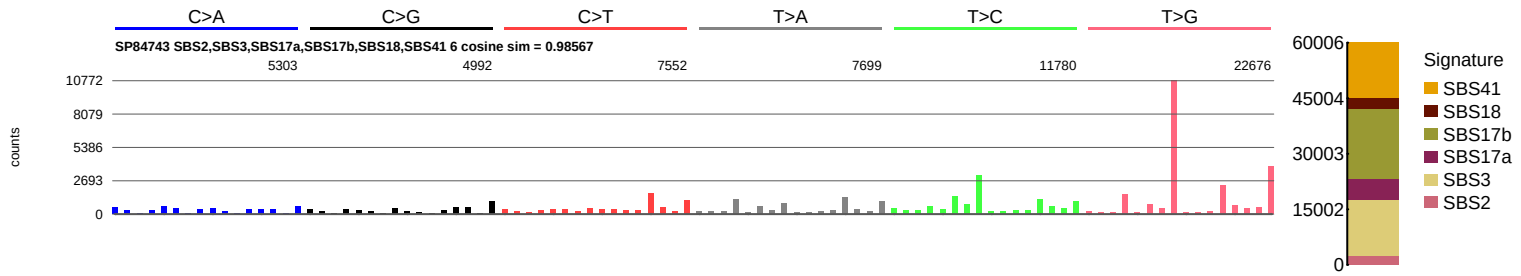
T>C
T>G
SP84743 SBS17a,SBS17b,SBS21,SBS40,SBS41 5 cosine sim = 0.98574
5781
4773
6983
6882
12429
23160
counts
10680
8010
5340
2670
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.9857
5574
4754
7273
6880
12381
23136
counts
10684
8013
5342
2671
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.9857
5559
4765
7348
6900
12294
23135
counts
10688
8016
5344
2672
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98568
5308
5088
7534
7665
11754
22659
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98567
5303
4992
7552
7699
11780
22676
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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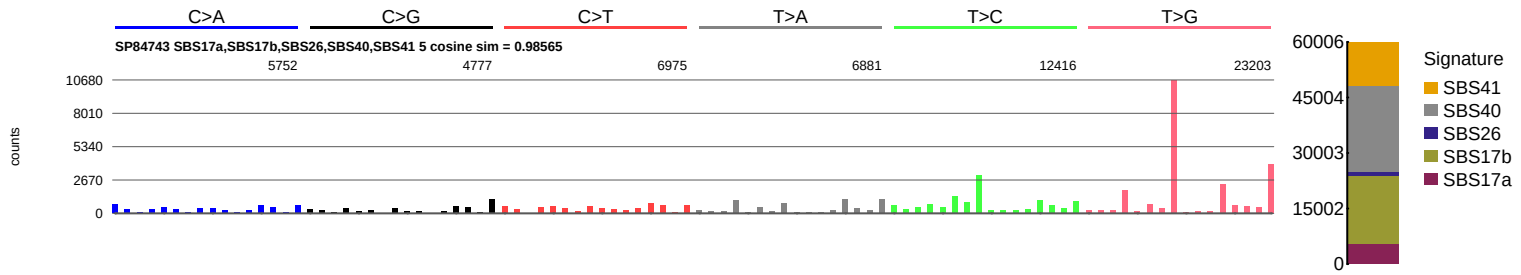
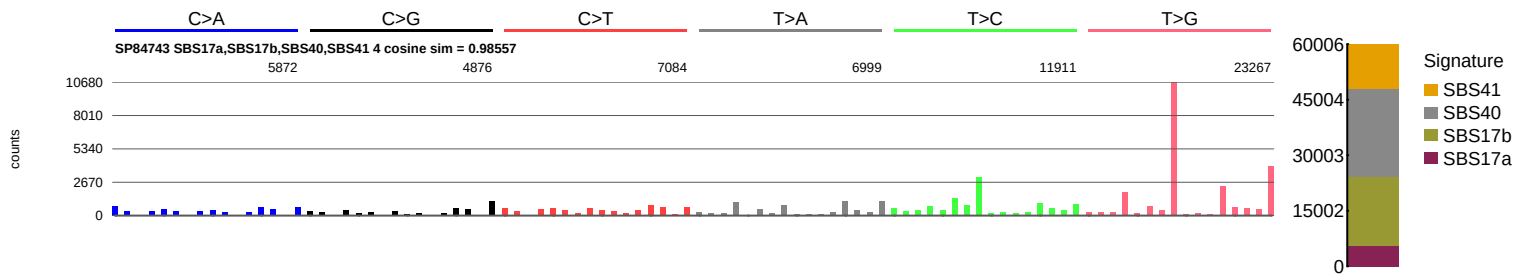
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS17a,SBS17b,SBS26,SBS40,SBS41 5 cosine sim = 0.98565
5752
4777
6975
6881
12416
23203
counts
10680
8010
5340
2670
0
60006
45004
30003
15002
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS17a,SBS17b,SBS40,SBS41 4 cosine sim = 0.98557
5872
4876
7084
6999
11911
23267
counts
10680
8010
5340
2670
0
60006
45004
30003
15002
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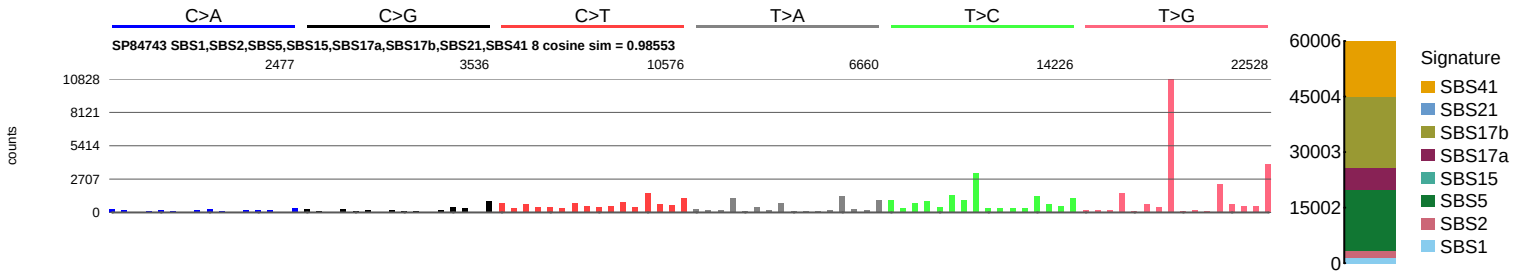
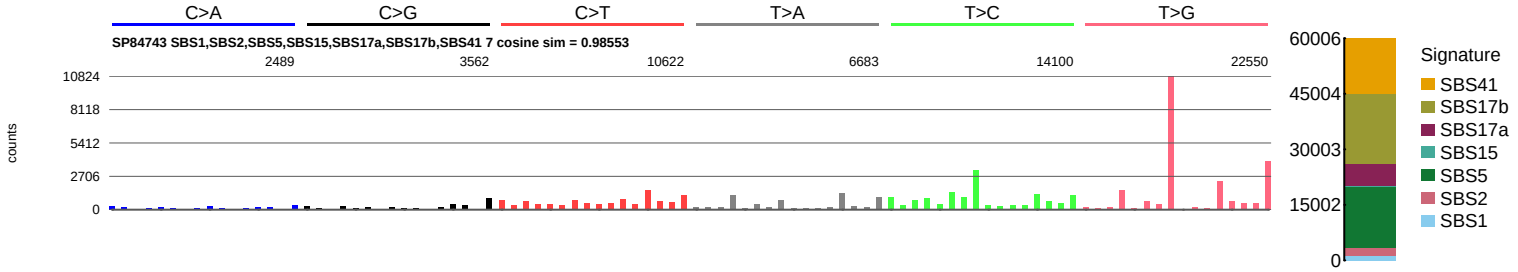
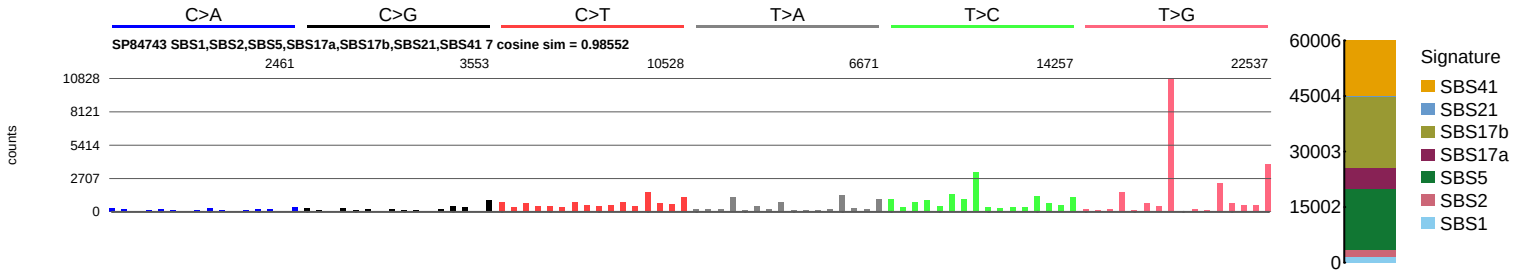
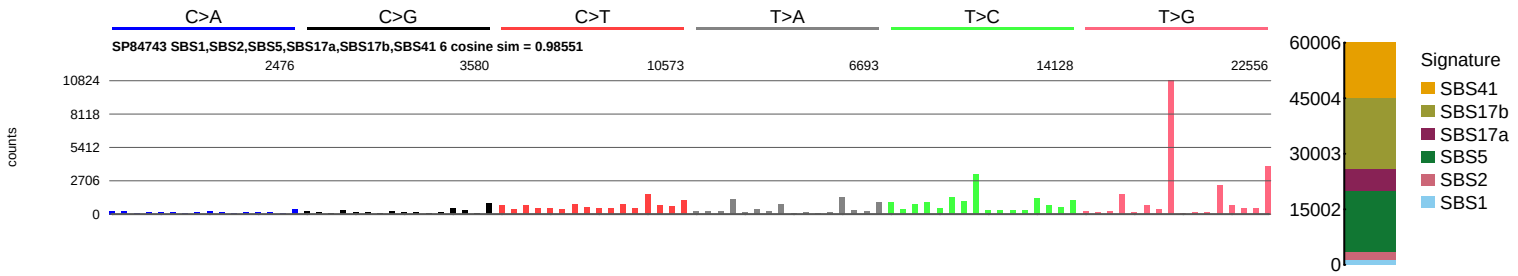
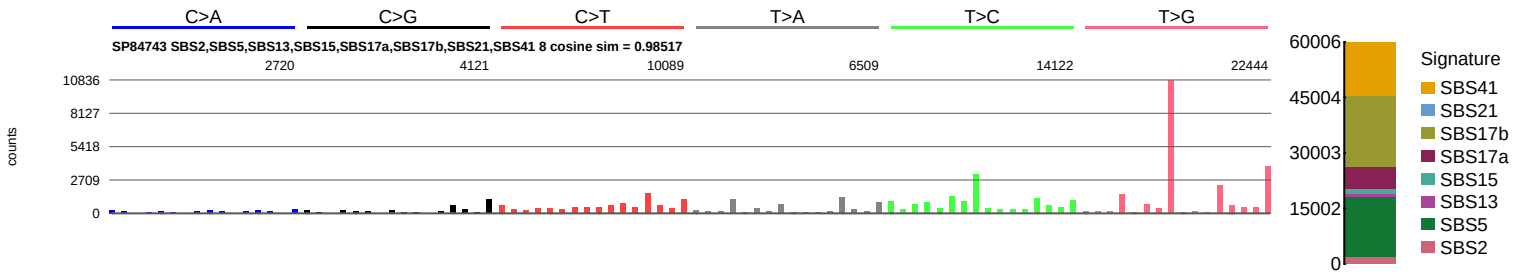
Signature
SBS41
SBS40
SBS17b
SBS17a

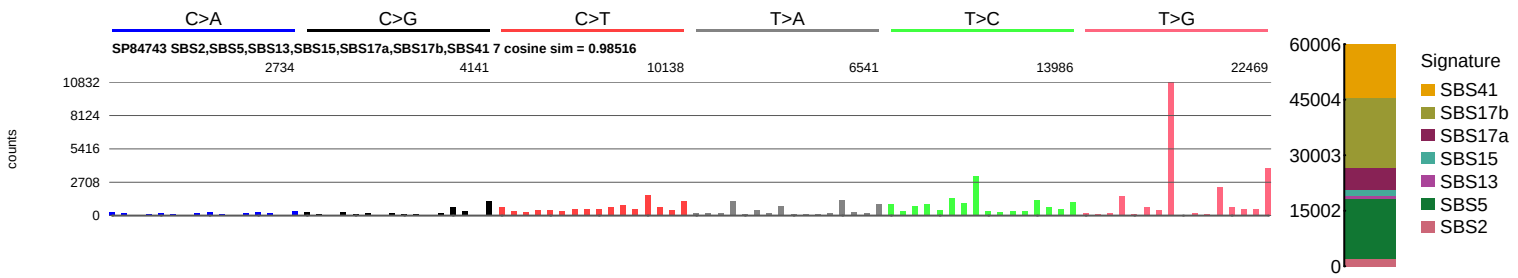

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98553
2477
3536
10576
6660
14226
22528
counts
10828
8121
5414
2707
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

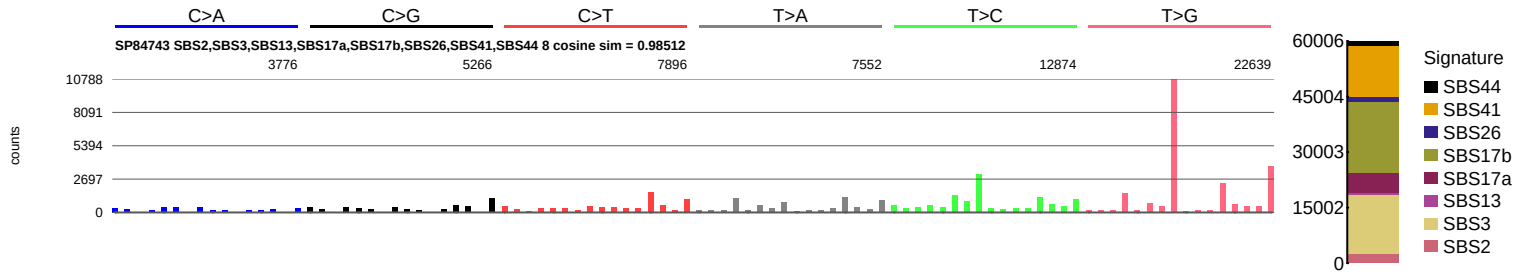

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98553
2489
3562
10622
6683
14100
22550
counts
10824
8118
5412
2706
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

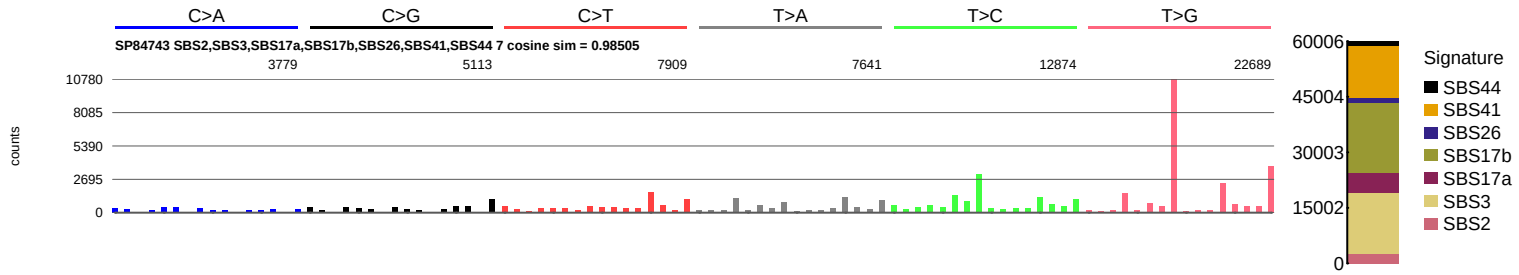

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98552
2461
3553
10528
6671
14257
22537
counts
10828
8121
5414
2707
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

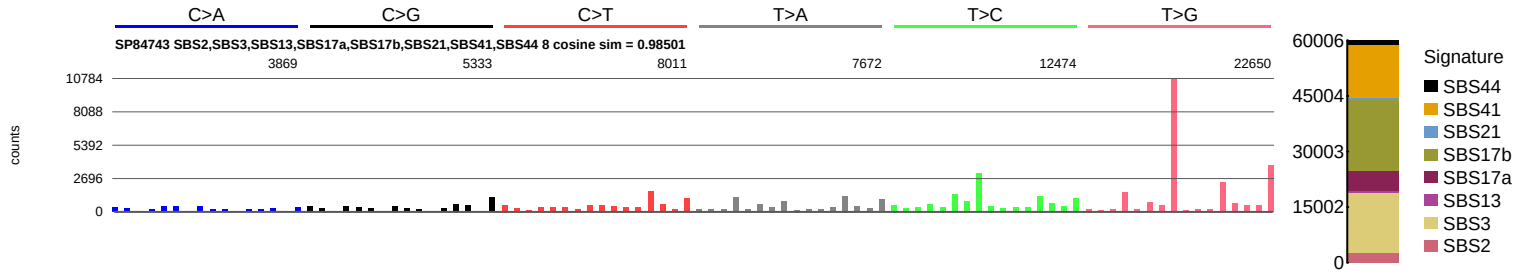

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98551
2476
3580
10573
6693
14128
22556
counts
10824
8118
5412
2706
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98517
2720
4121
10089
6509
14122
22444
counts
10836
8127
5418
2709
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98516
2734
4141
10138
6541
13986
22469
counts
10832
8124
5416
2708
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98512
3776
5266
7896
7552
12874
22639
counts
10788
8091
5394
2697
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98505
3779
5113
7909
7641
12874
22689
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98501
3869
5333
8011
7672
12474
22650
counts
10784
8088
5392
2696
0
60006
45004
30003
15002
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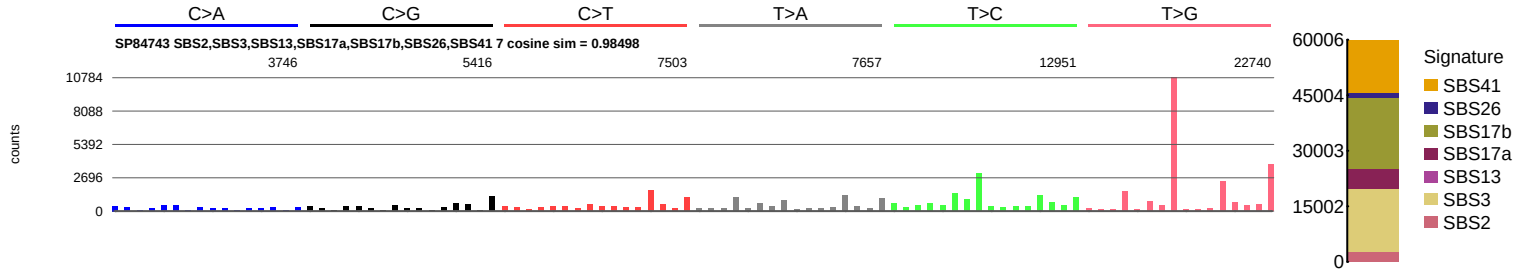
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98498
3746
5416
7503
7657
12951
22740
counts
10784
8088
5392
2696
0
60006
45004
30003
15002
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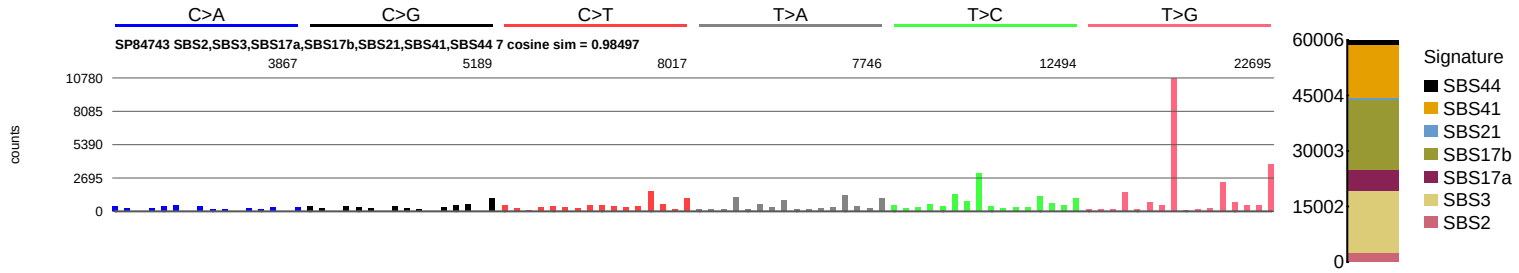
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98497
3867
5189
8017
7746
12494
22695
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
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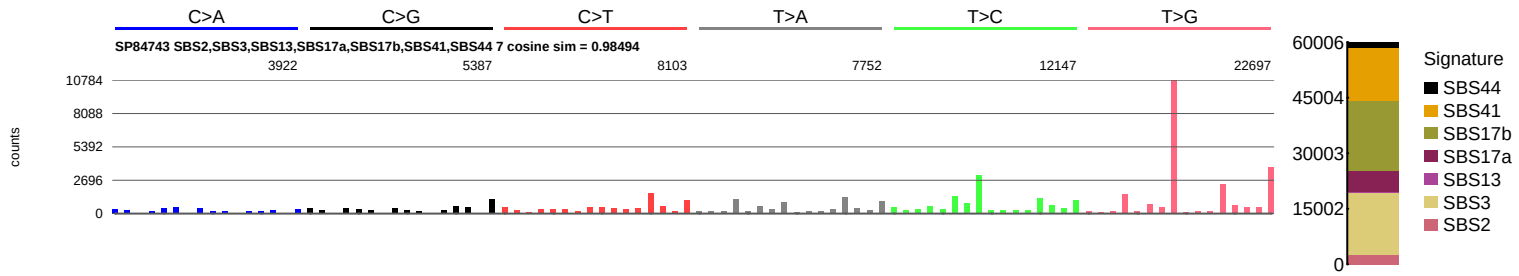
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98494
3922
5387
8103
7752
12147
22697
counts
10784
8088
5392
2696
0
60006
45004
30003
15002
0
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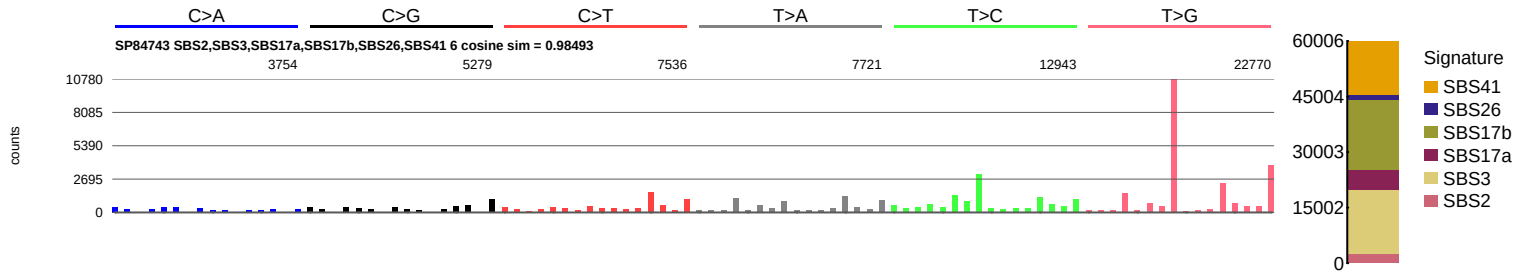
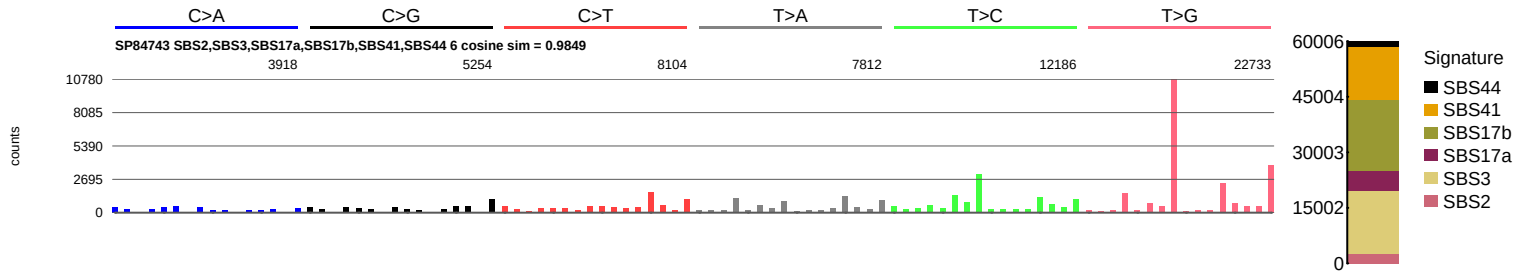
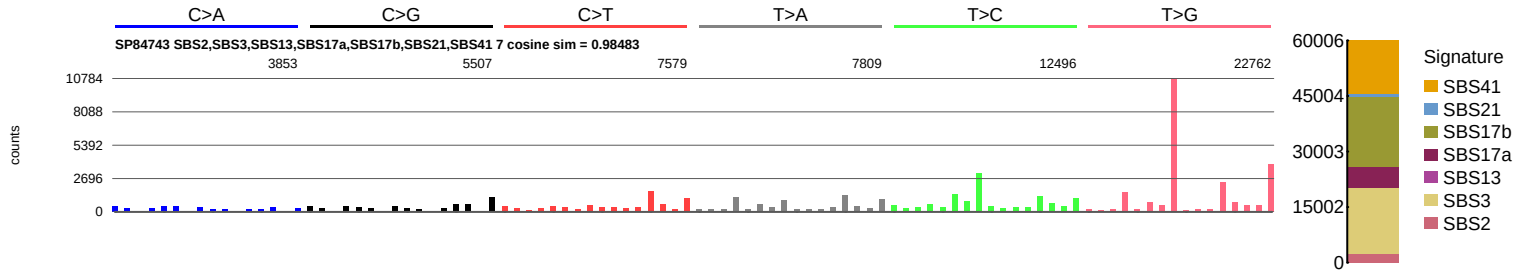
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98493
3754
5279
7536
7721
12943
22770
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.9849
3918
5254
8104
7812
12186
22733
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98483
3853
5507
7579
7809
12496
22762
counts
10784
8088
5392
2696
0
60006
45004
30003
15002
0
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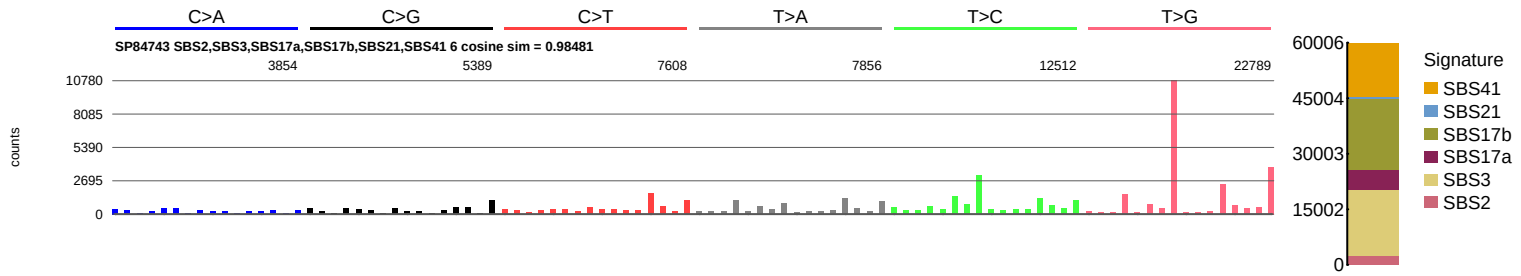
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98481
3854
5389
7608
7856
12512
22789
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
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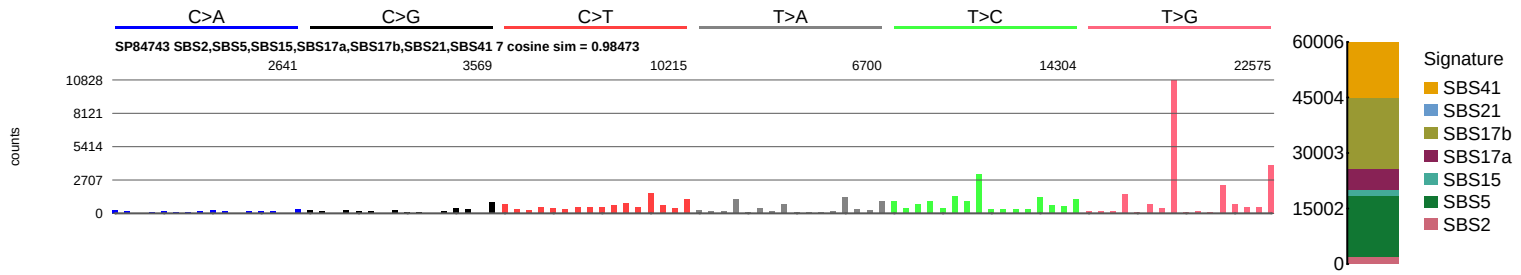
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98473
2641
3569
10215
6700
14304
22575
counts
10828
8121
5414
2707
0
60006
45004
30003
15002
0
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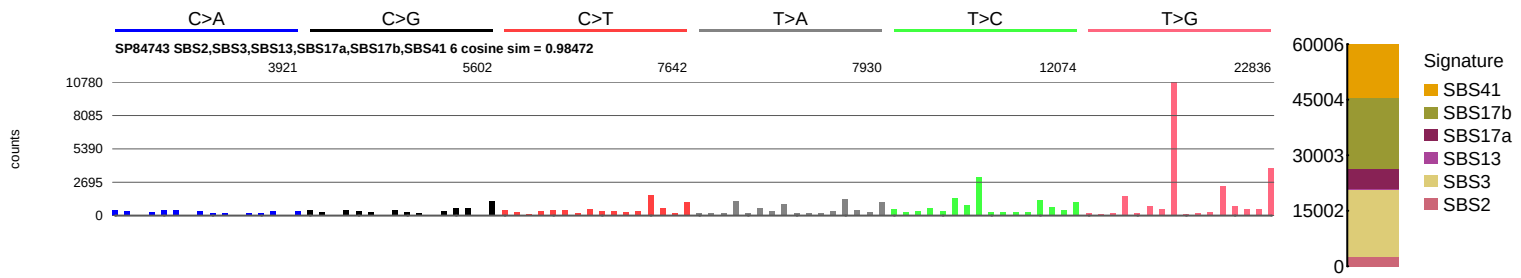
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98472
3921
5602
7642
7930
12074
22836
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
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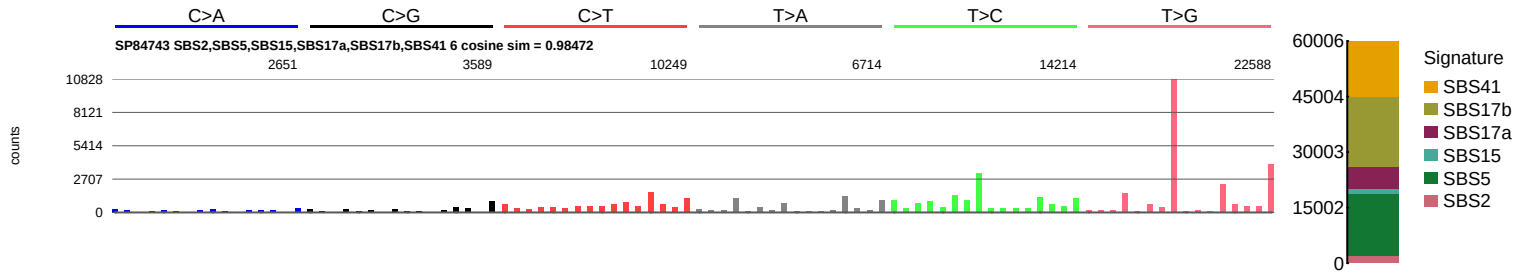
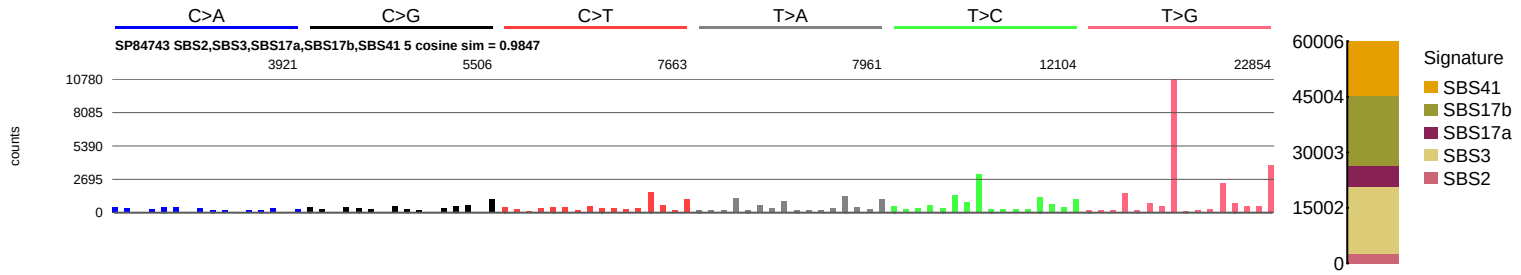
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98472
2651
3589
10249
6714
14214
22588
counts
10828
8121
5414
2707
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2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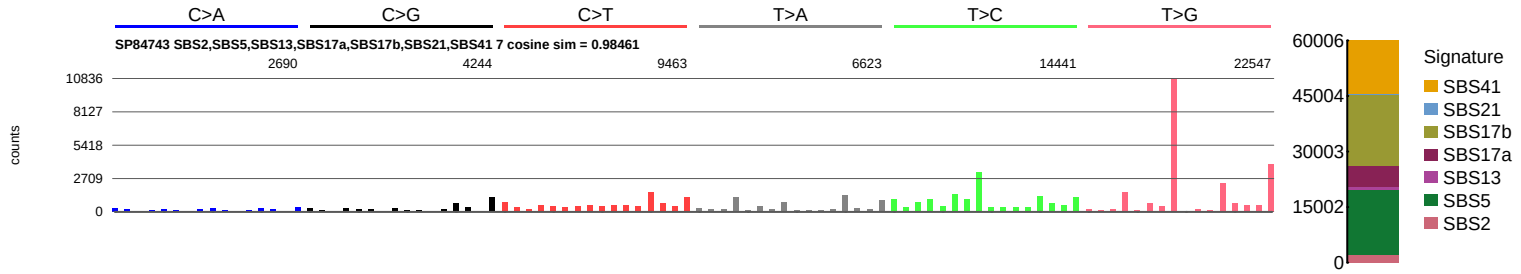
SP84743 SBS2,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.9847
3921
5506
7663
7961
12104
22854
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2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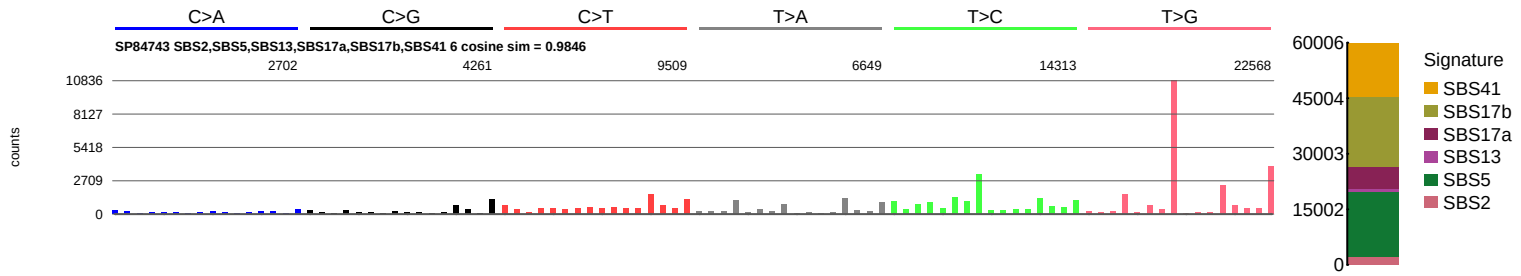
SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98461
2690
4244
9463
6623
14441
22547
counts
10836
8127
5418
2709
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2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SP84743 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9846
2702
4261
9509
6649
14313
22568
counts
10836
8127
5418
2709
0
60006
45004
30003
15002
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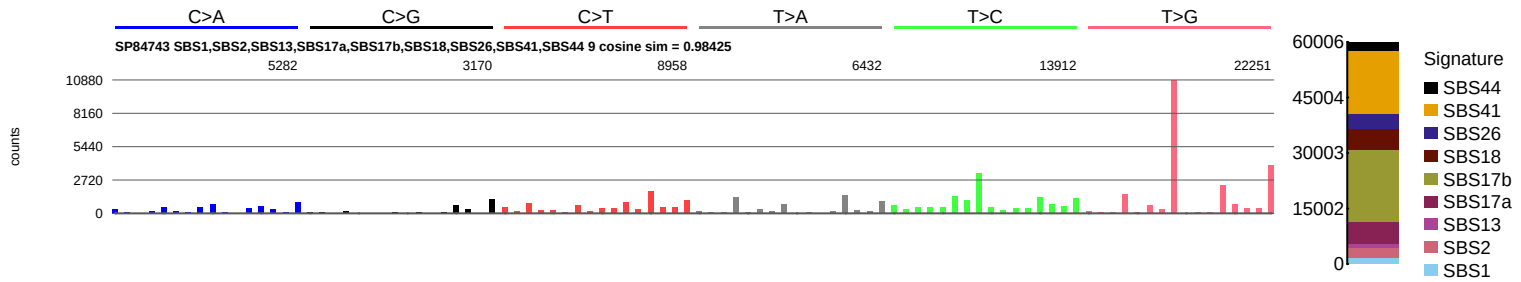
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2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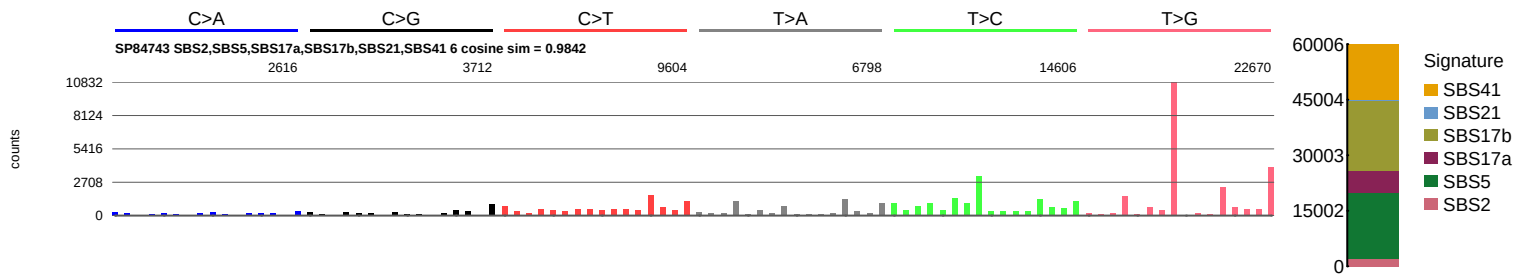
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98425
5282
3170
8958
6432
13912
22251
counts
10880
8160
5440
2720
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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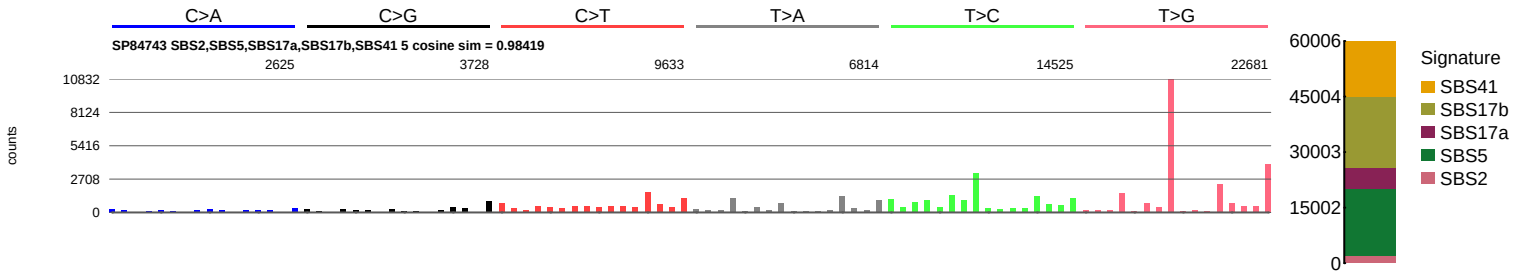
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.9842
2616
3712
9604
6798
14606
22670
counts
10832
8124
5416
2708
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2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T>C
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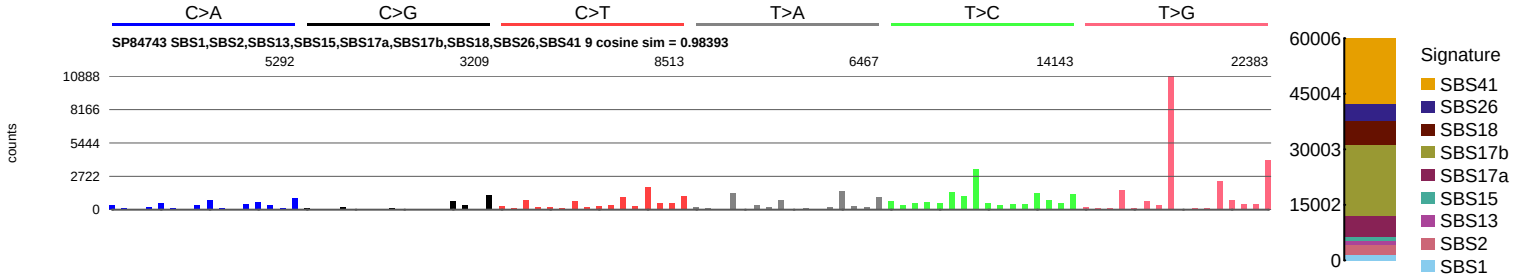
SP84743 SBS2,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98419
2625
3728
9633
6814
14525
22681
counts
10832
8124
5416
2708
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2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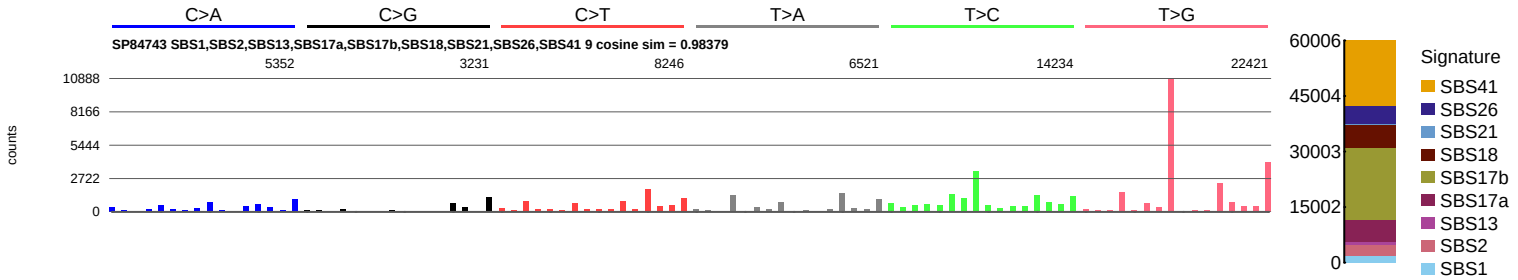
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98393
5292
3209
8513
6467
14143
22383
counts
10888
8166
5444
2722
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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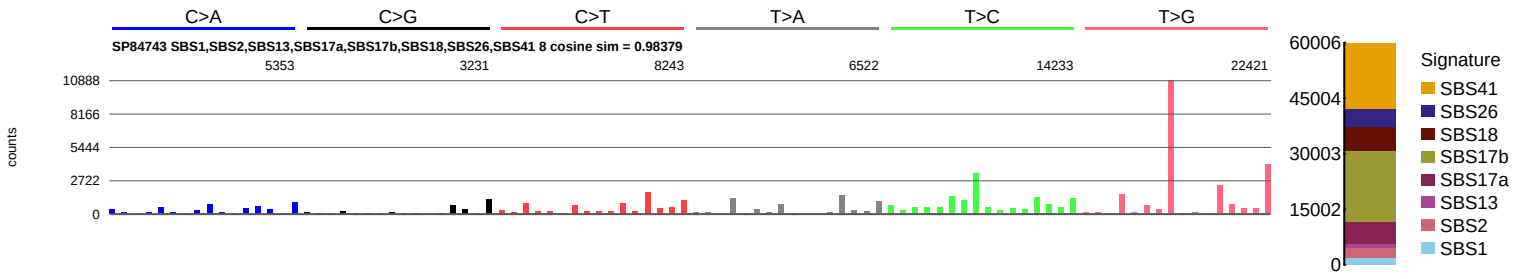
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98379
5352
3231
8246
6521
14234
22421
counts
10888
8166
5444
2722
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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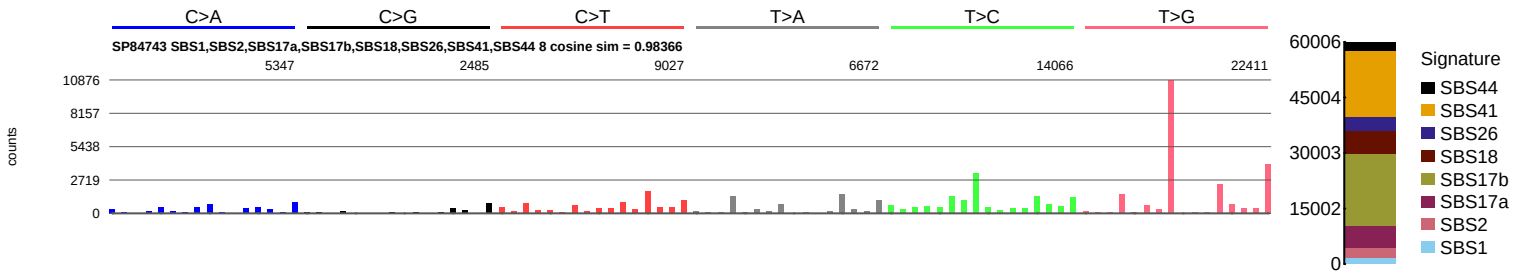
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98379
5353
3231
8243
6522
14233
22421
counts
10888
8166
5444
2722
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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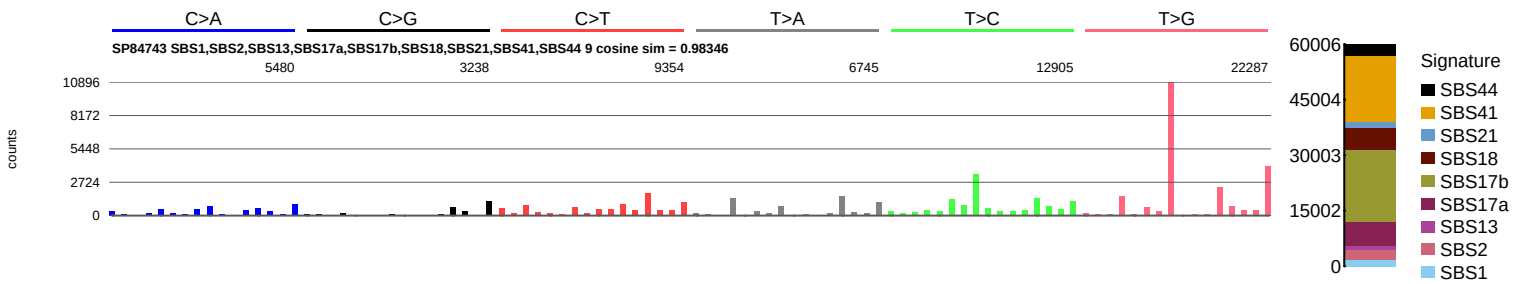
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98366
5347
2485
9027
6672
14066
22411
counts
10876
8157
5438
2719
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98346
5480
3238
9354
6745
12905
22287
counts
10896
8172
5448
2724
0
60006
45004
30003
15002
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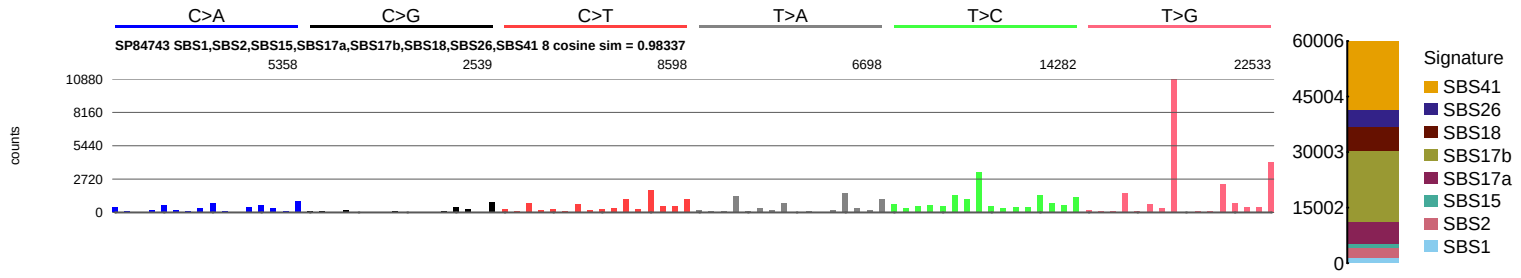
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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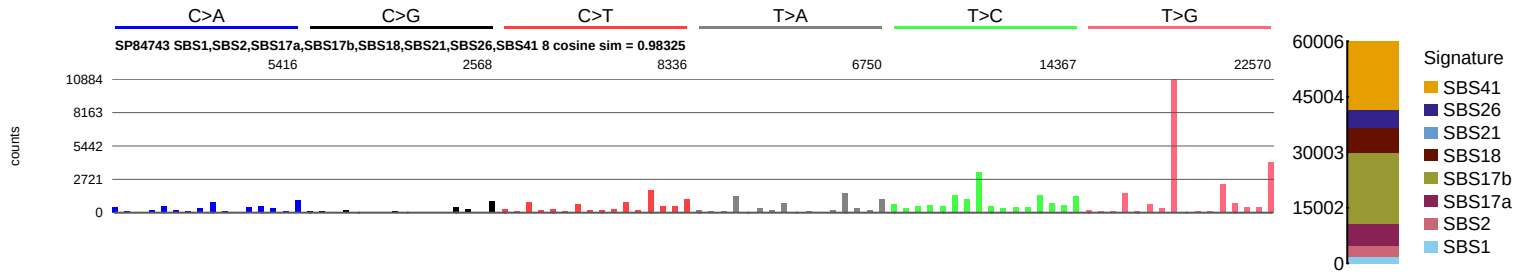
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98337
5358
2539
8598
6698
14282
22533
counts
10880
8160
5440
2720
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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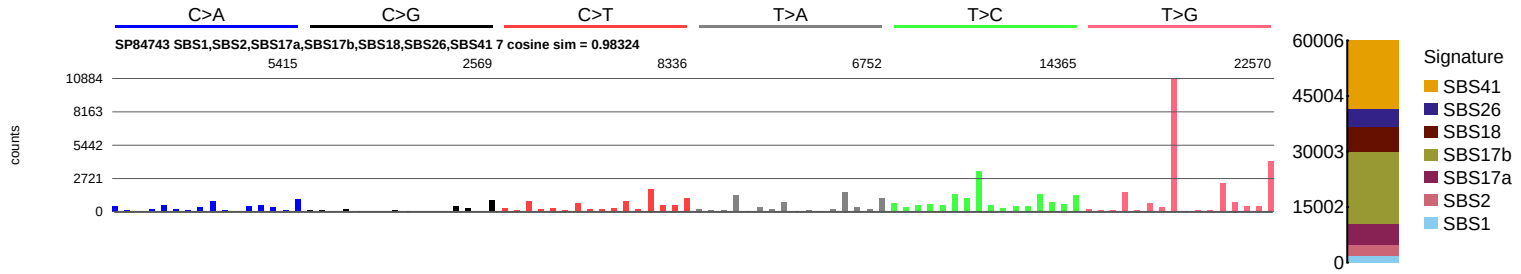
T>C
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98325
5416
2568
8336
6750
14367
22570
counts
10884
8163
5442
2721
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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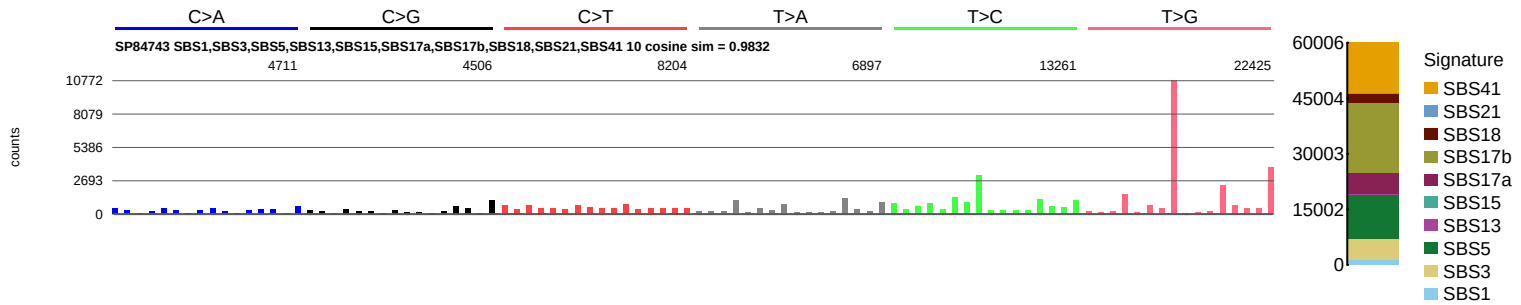
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98324
5415
2569
8336
6752
14365
22570
counts
10884
8163
5442
2721
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84743 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.9832
4711
4506
8204
6897
13261
22425
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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Signature
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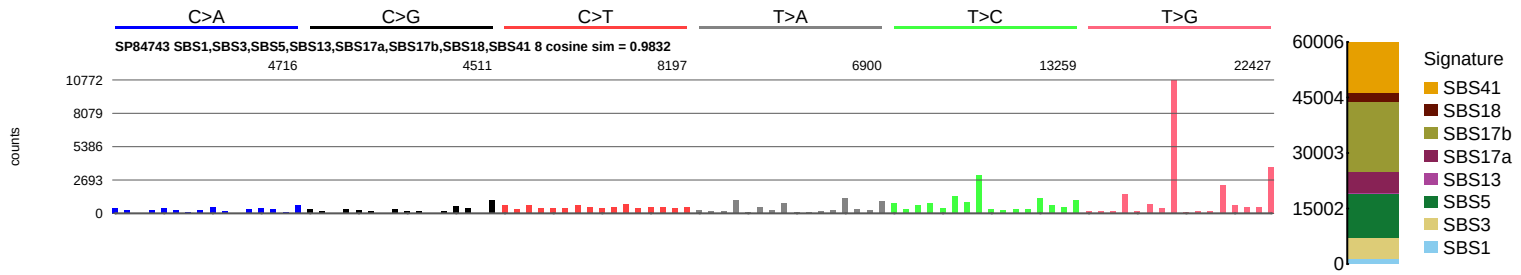
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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T>C
T>G
SP84743 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9832
4716
4511
8197
6900
13259
22427
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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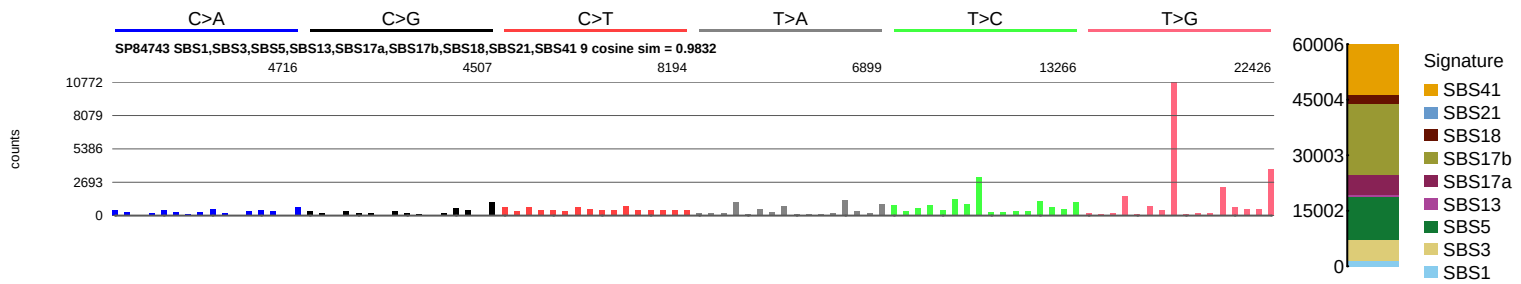
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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T>G
SP84743 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9832
4716
4507
8194
6899
13266
22426
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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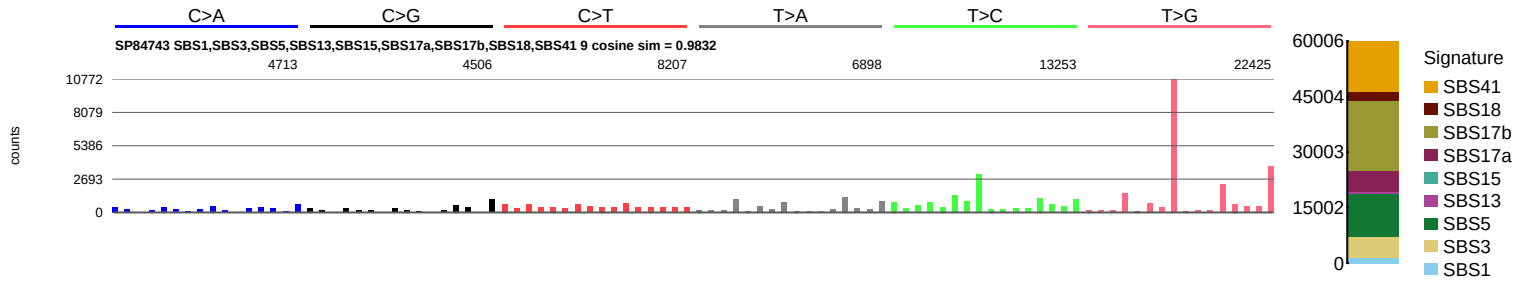
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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SP84743 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9832
4713
4506
8207
6898
13253
22425
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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Signature
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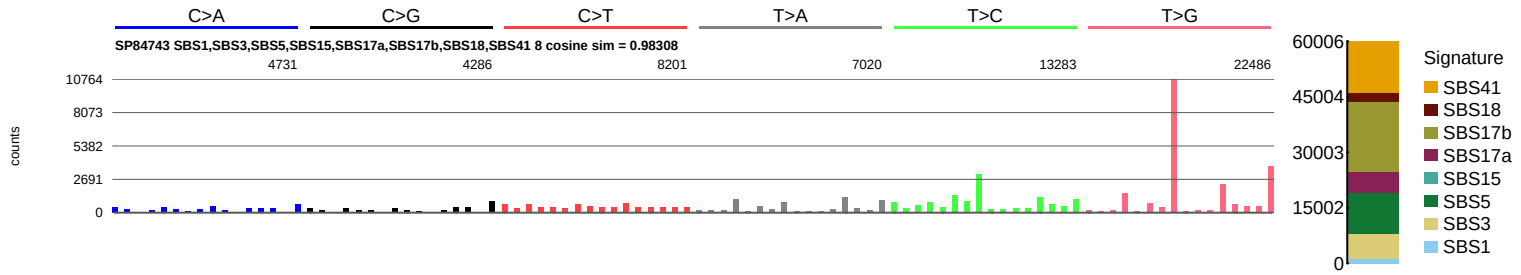
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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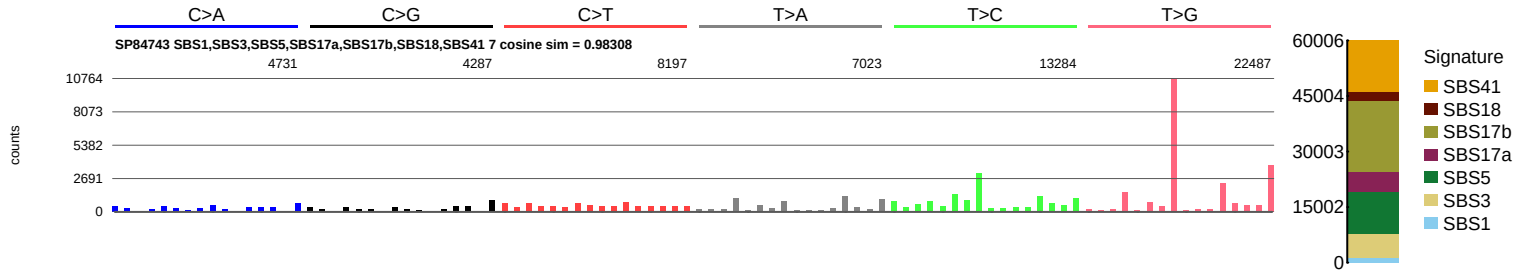
T>G
SP84743 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98308
4731
4286
8201
7020
13283
22486
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>T
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T>C
T>G
SP84743 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98308
4731
4287
8197
7023
13284
22487
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
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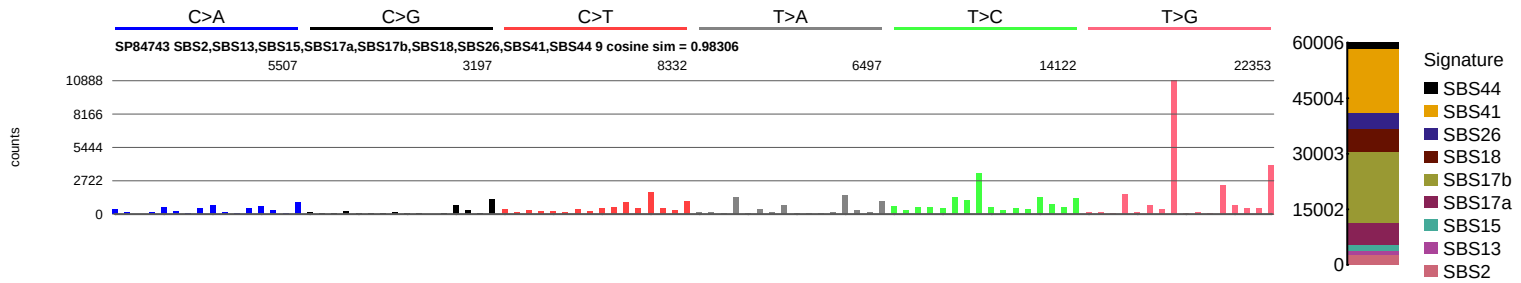
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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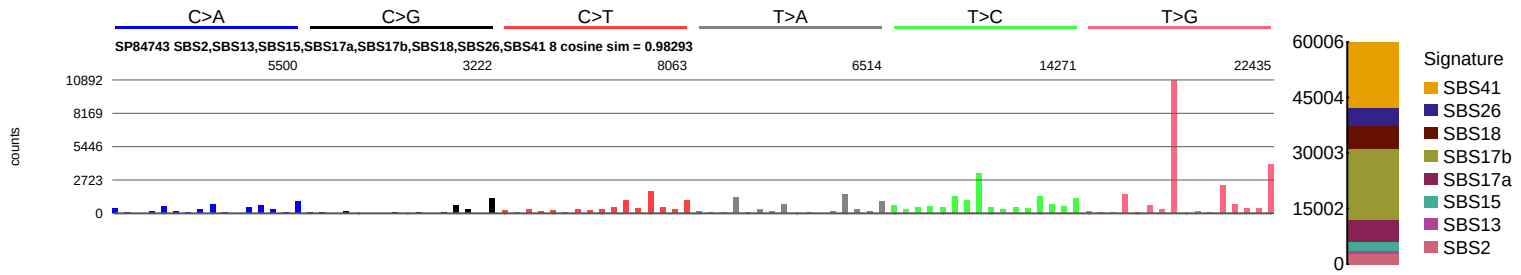
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98306
5507
3197
8332
6497
14122
22353
counts
10888
8166
5444
2722
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98293
5500
3222
8063
6514
14271
22435
counts
10892
8169
5446
2723
0
60006
45004
30003
15002
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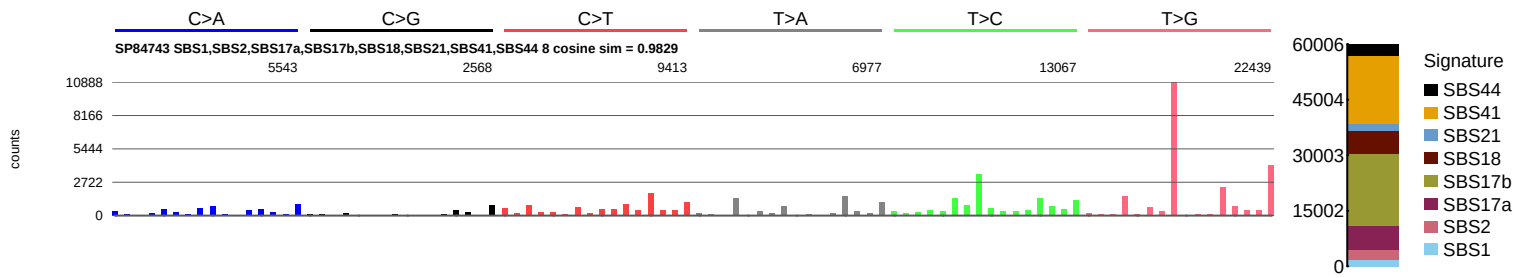
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2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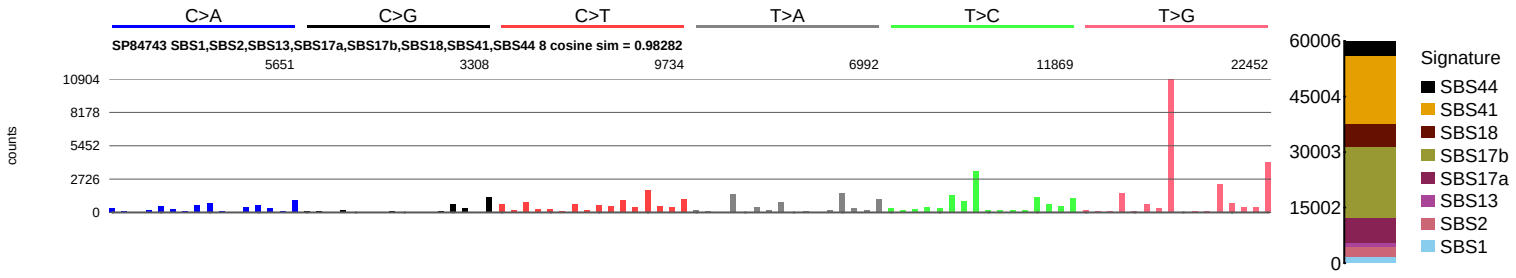
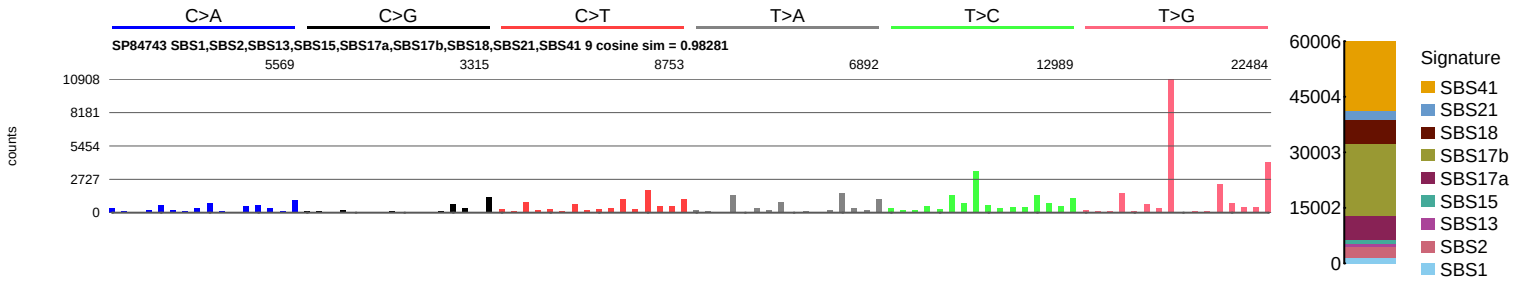
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.9829
5543
2568
9413
6977
13067
22439
counts
10888
8166
5444
2722
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98282
5651
3308
9734
6992
11869
22452
counts
10904
8178
5452
2726
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98281
5569
3315
8753
6892
12989
22484
counts
10908
8181
5454
2727
0
60006
45004
30003
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SBS41
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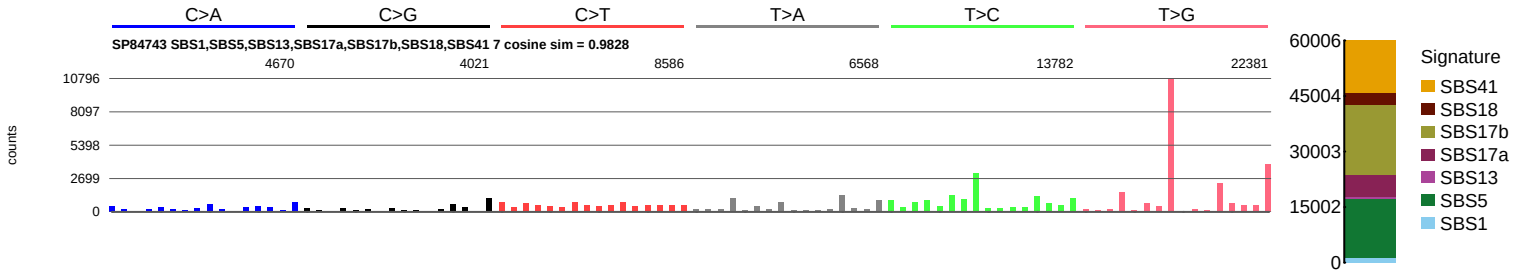
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SP84743 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9828
4670
4021
8586
6568
13782
22381
counts
10796
8097
5398
2699
0
60006
45004
30003
15002
0
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SBS41
SBS18
SBS17b
SBS17a
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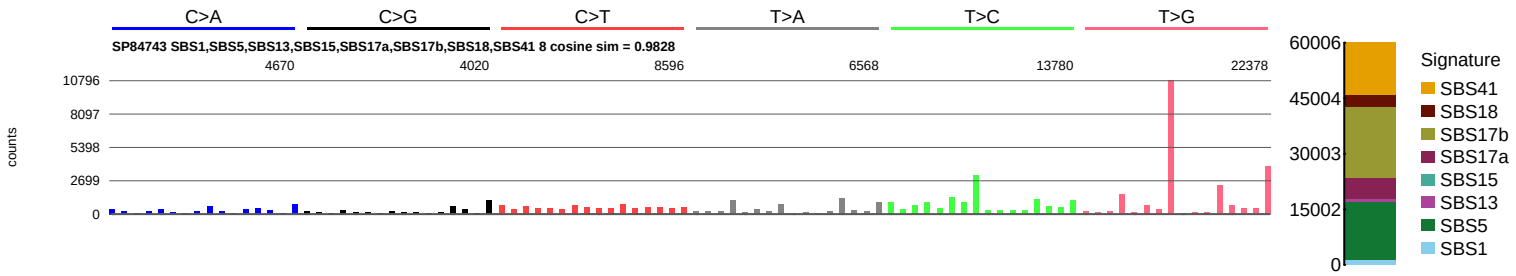
SBS5
SBS1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SP84743 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9828
4670
4020
8596
6568
13780
22378
counts
10796
8097
5398
2699
0
60006
45004
30003
15002
0
Signature
SBS41
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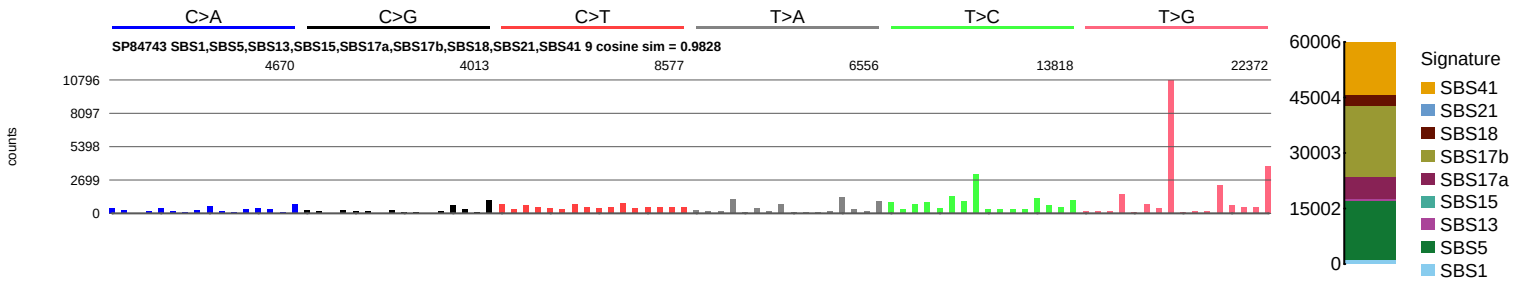
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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SP84743 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9828
4670
4013
8577
6556
13818
22372
counts
10796
8097
5398
2699
0
60006
45004
30003
15002
0
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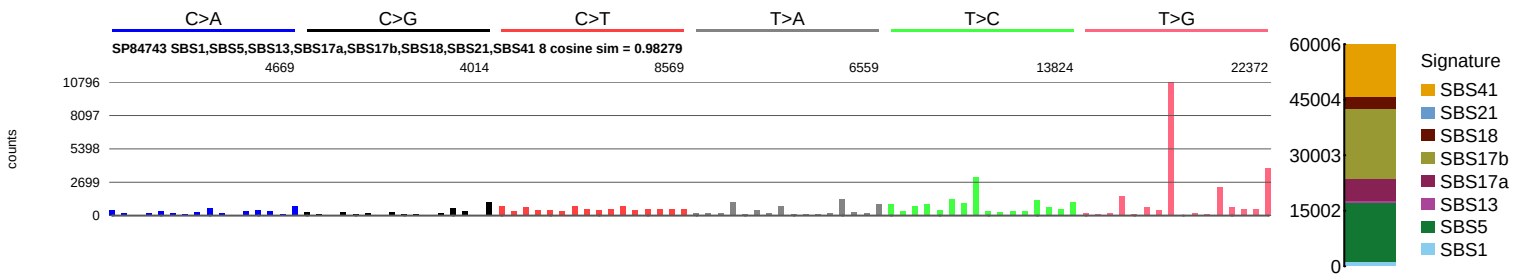
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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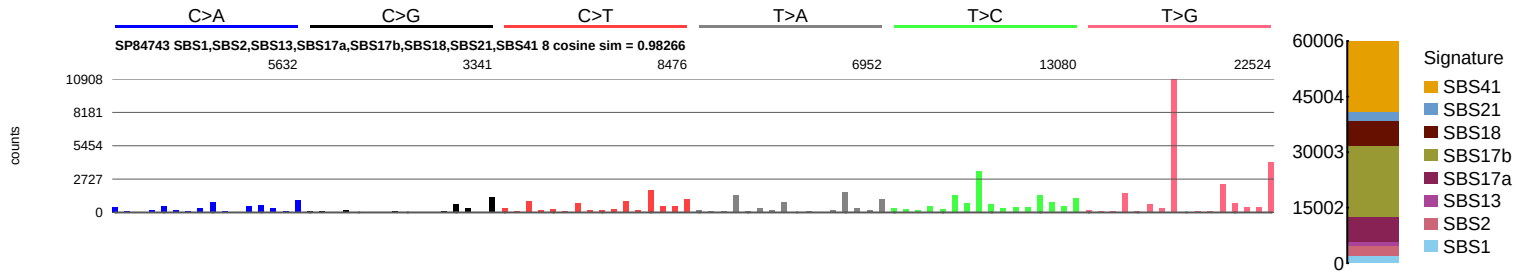
SP84743 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98279
4669
4014
8569
6559
13824
22372
counts
10796
8097
5398
2699
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98266
5632
3341
8476
6952
13080
22524
counts
10908
8181
5454
2727
0
60006
45004
30003
15002
0
Signature
SBS41
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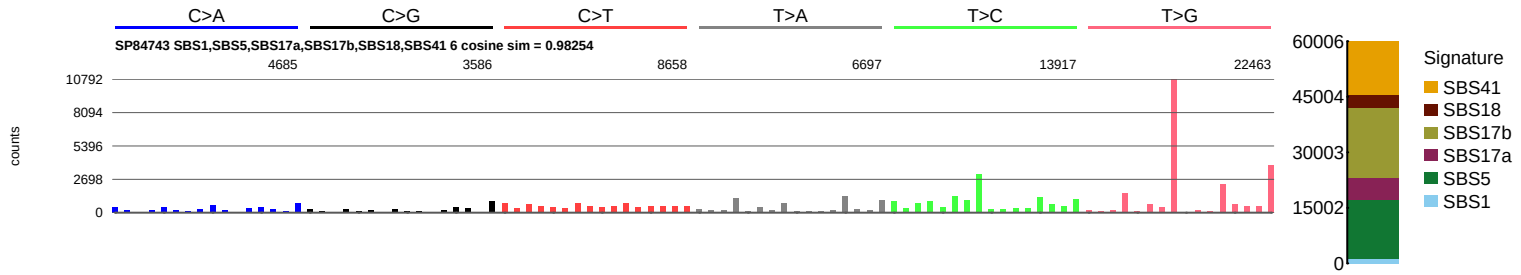
SBS17b
SBS17a
SBS13
SBS2
SBS1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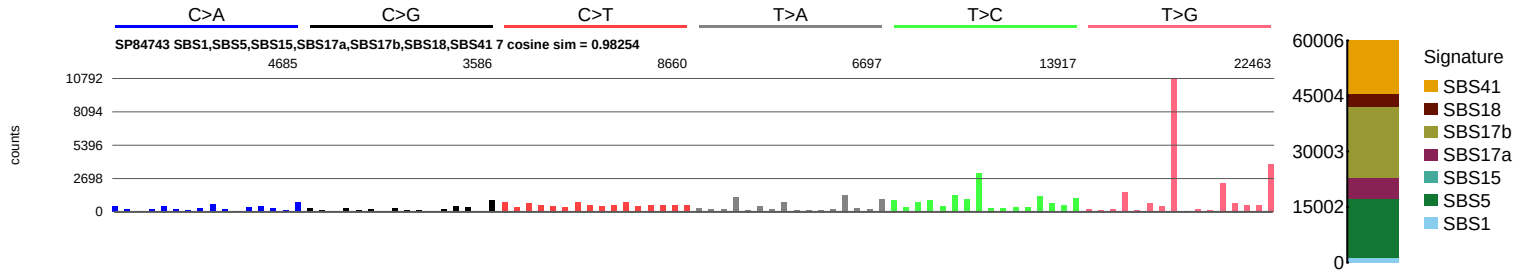
SP84743 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98254
4685
3586
8658
6697
13917
22463
counts
10792
8094
5396
2698
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1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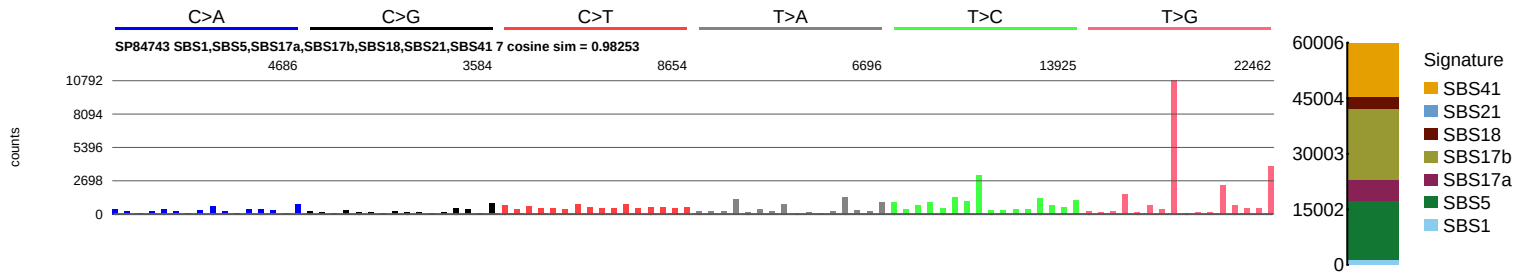
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98254
4685
3586
8660
6697
13917
22463
counts
10792
8094
5396
2698
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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T>C
T>G
SP84743 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98253
4686
3584
8654
6696
13925
22462
counts
10792
8094
5396
2698
0
60006
45004
30003
15002
0
Signature
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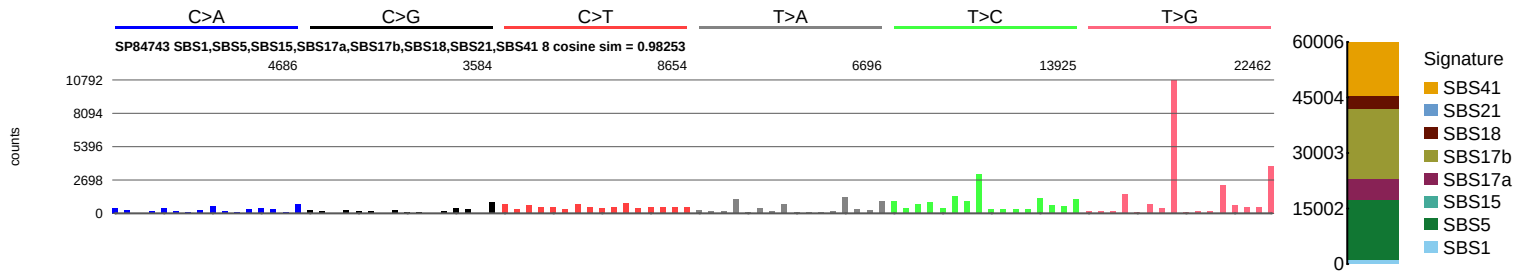
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>T
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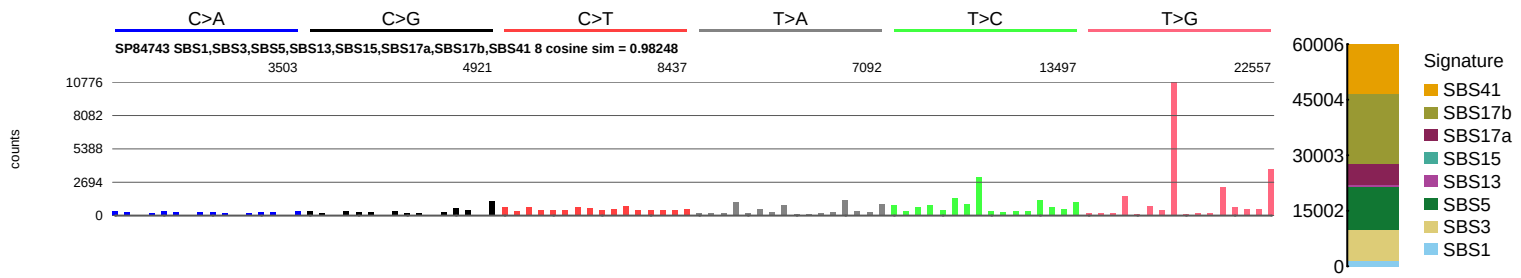
T>C
T>G
SP84743 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98253
4686
3584
8654
6696
13925
22462
counts
10792
8094
5396
2698
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
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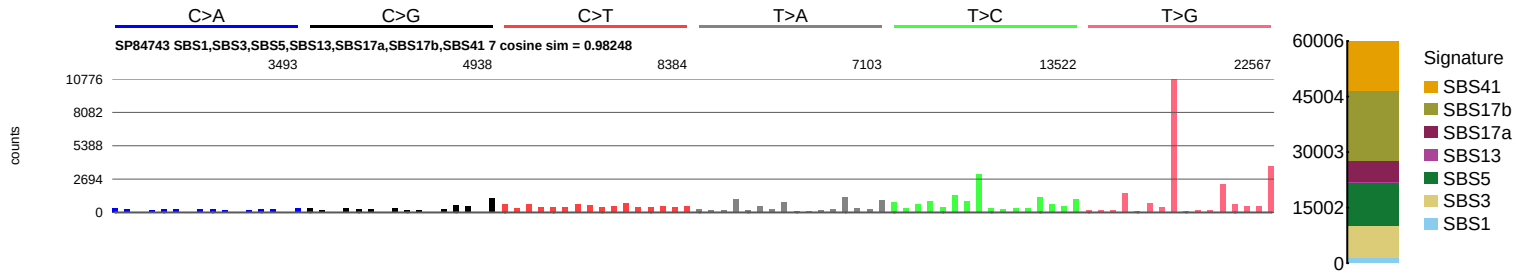
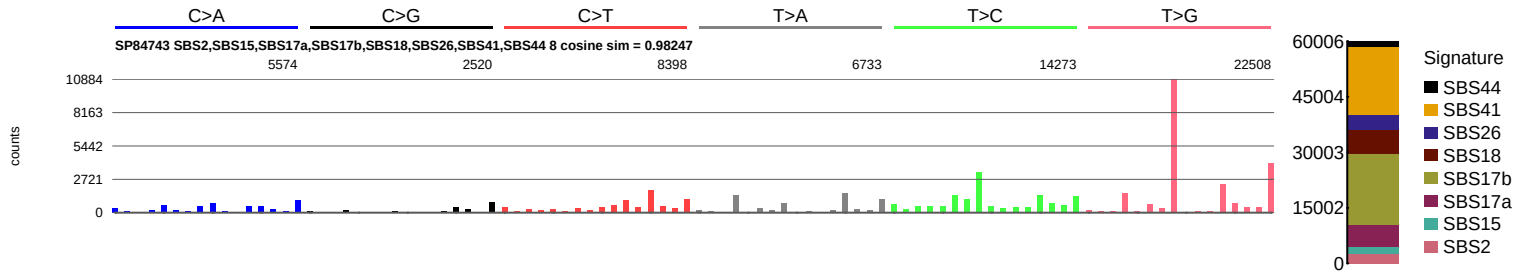
T>C
T>G
SP84743 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98248
3503
4921
8437
7092
13497
22557
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98248
3493
4938
8384
7103
13522
22567
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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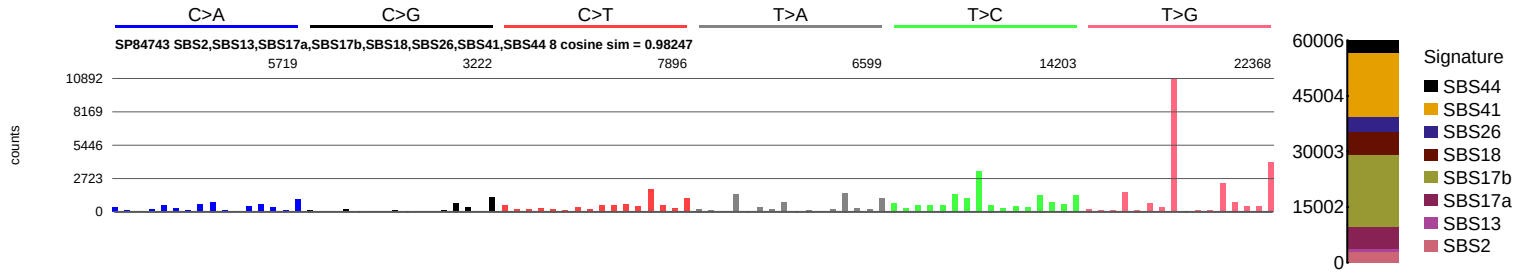
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98247
5574
2520
8398
6733
14273
22508
counts
10884
8163
5442
2721
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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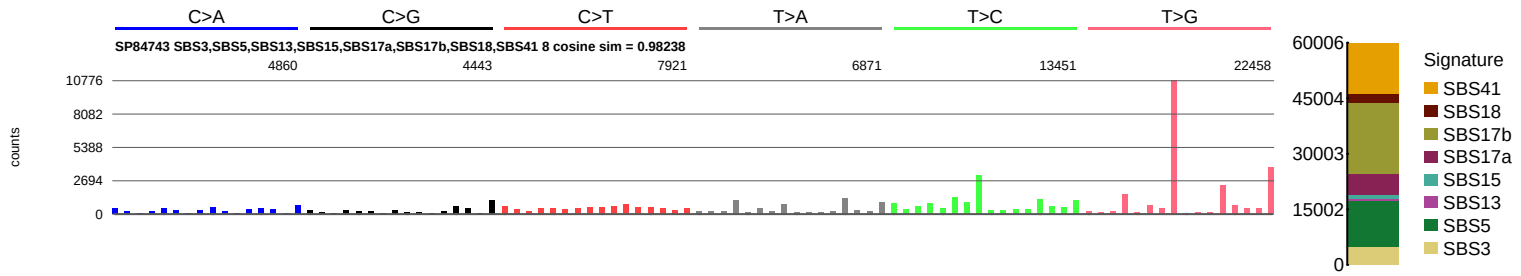
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98247
5719
3222
7896
6599
14203
22368
counts
10892
8169
5446
2723
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2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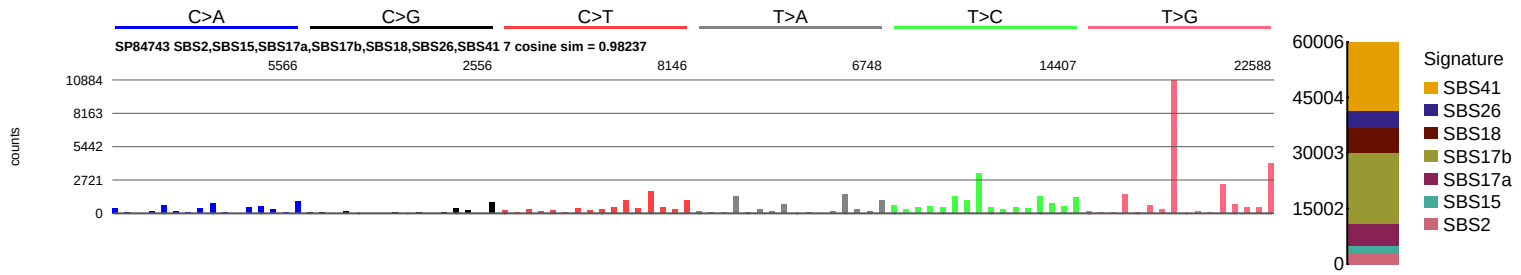
C>T
T>A
T>C
T>G
SP84743 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98238
4860
4443
7921
6871
13451
22458
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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C>G
C>T
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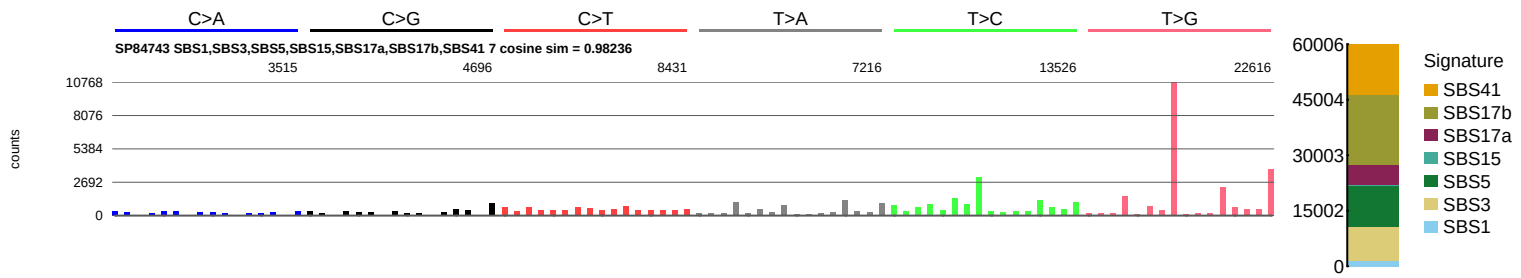
T>C
T>G
SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98237
5566
2556
8146
6748
14407
22588
counts
10884
8163
5442
2721
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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C>G
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T>C
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SP84743 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98236
3515
4696
8431
7216
13526
22616
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
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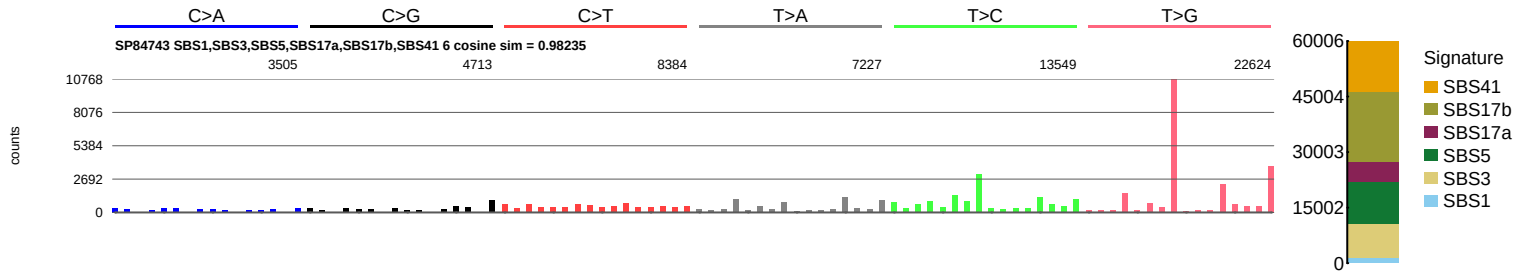
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98235
3505
4713
8384
7227
13549
22624
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
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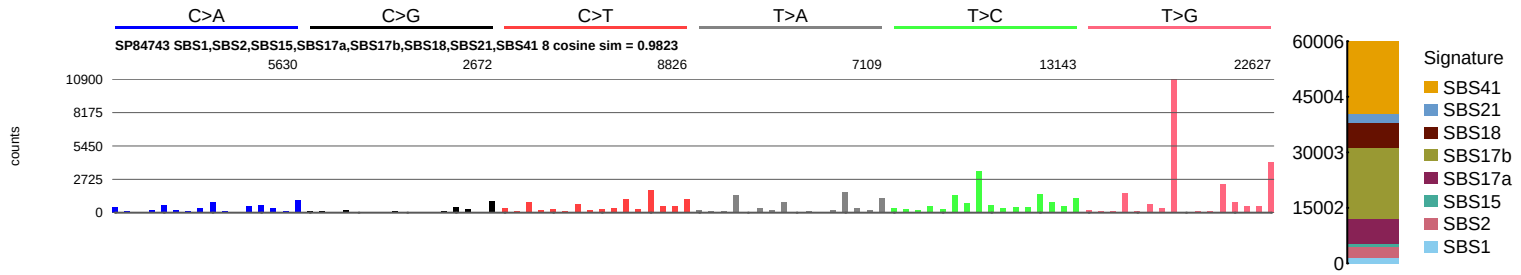
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9823
5630
2672
8826
7109
13143
22627
counts
10900
8175
5450
2725
0
60006
45004
30003
15002
0
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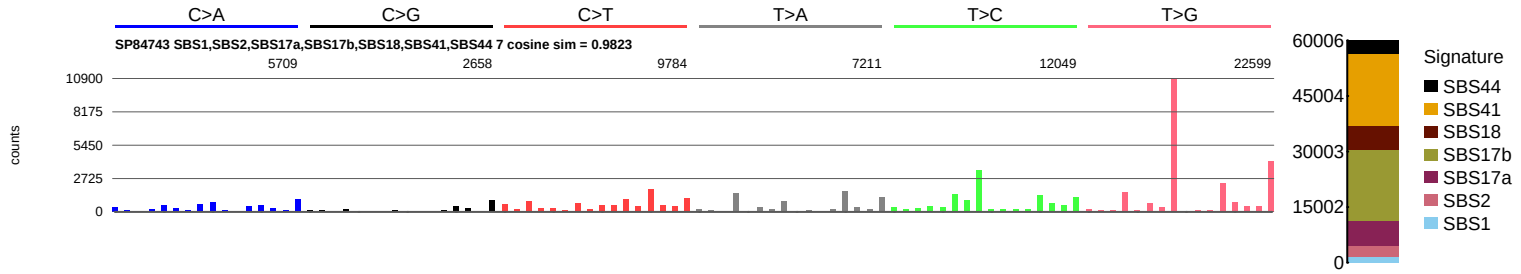
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.9823
5709
2658
9784
7211
12049
22599
counts
10900
8175
5450
2725
0
60006
45004
30003
15002
0
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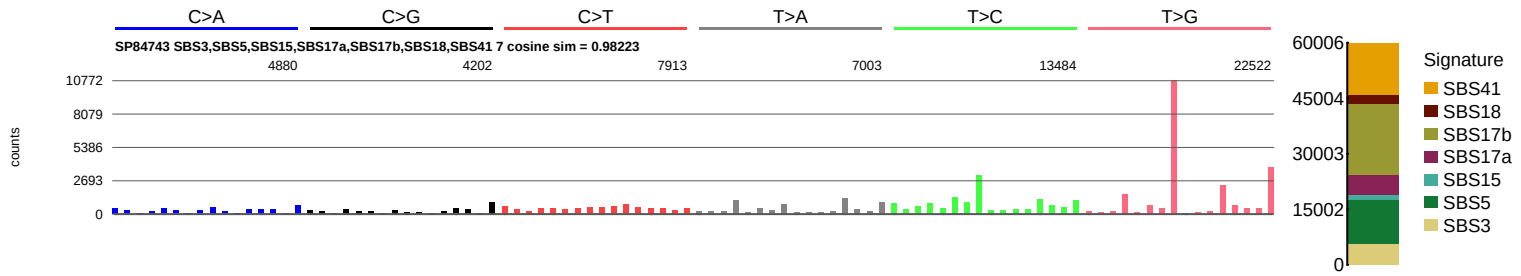
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98223
4880
4202
7913
7003
13484
22522
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
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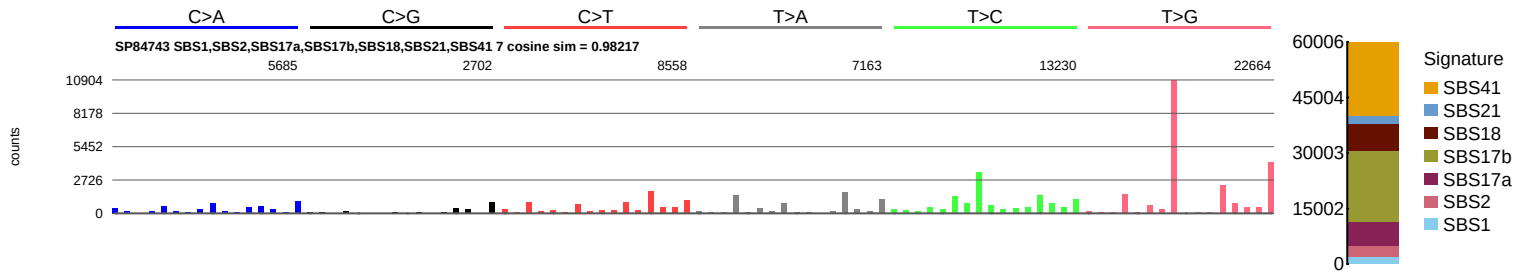
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98217
5685
2702
8558
7163
13230
22664
counts
10904
8178
5452
2726
0
60006
45004
30003
15002
0
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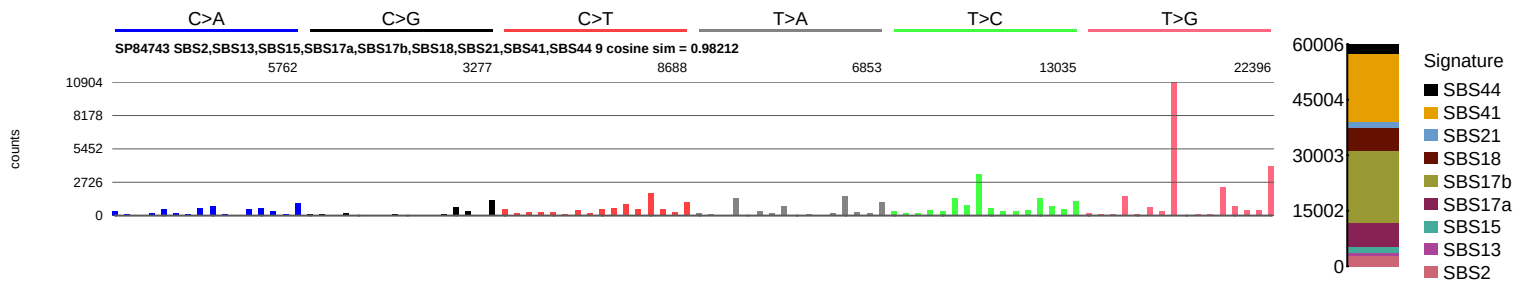
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98212
5762
3277
8688
6853
13035
22396
counts
10904
8178
5452
2726
0
60006
45004
30003
15002
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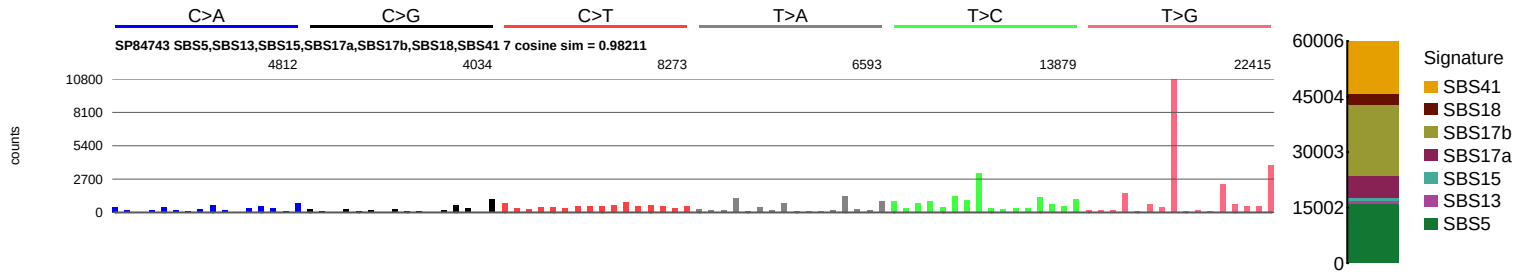
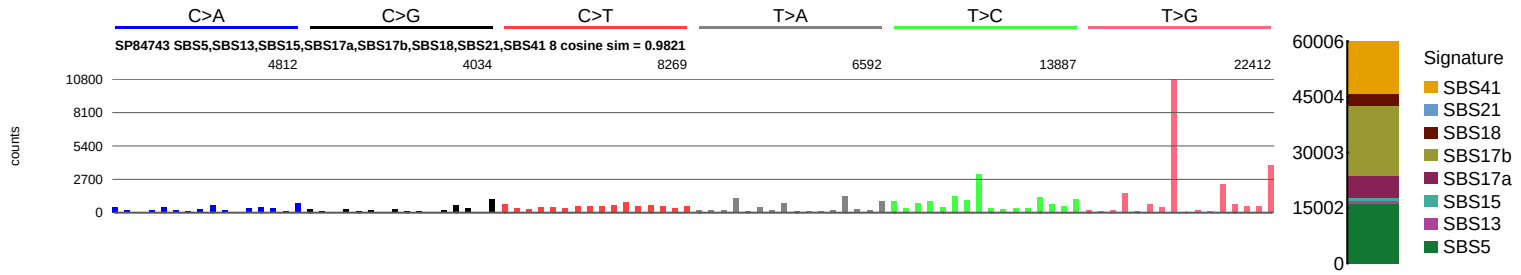
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98211
4812
4034
8273
6593
13879
22415
counts
10800
8100
5400
2700
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9821
4812
4034
8269
6592
13887
22412
counts
10800
8100
5400
2700
0
60006
45004
30003
15002
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Signature
SBS41
SBS21
SBS18
SBS17b
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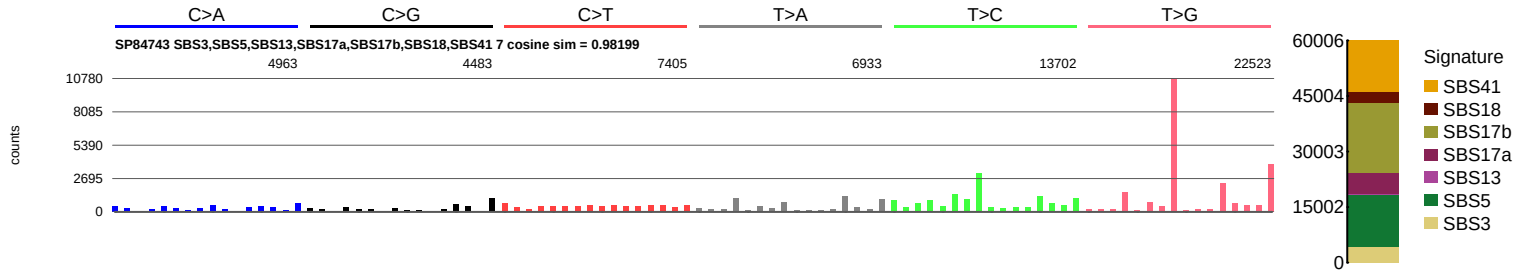
SBS15
SBS13
SBS5


C>A
C>G
C>T
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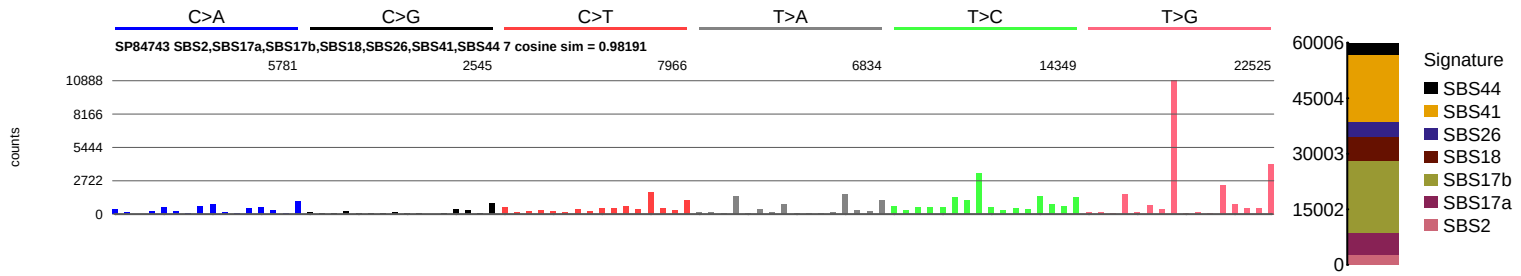
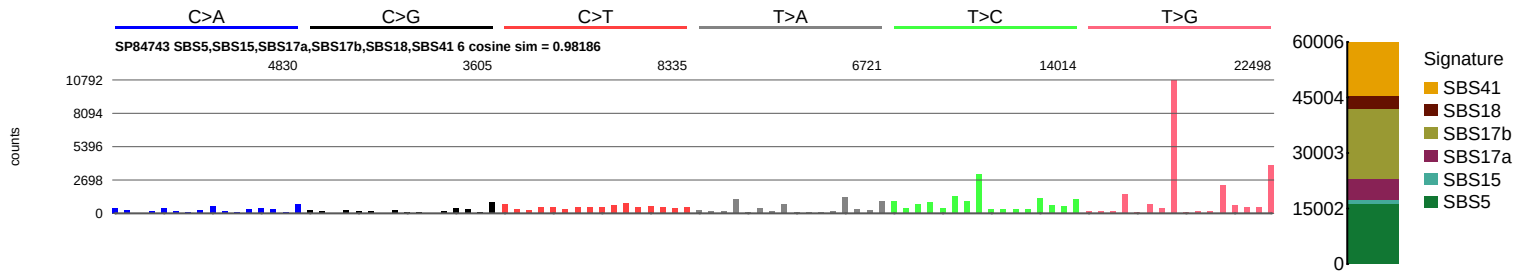
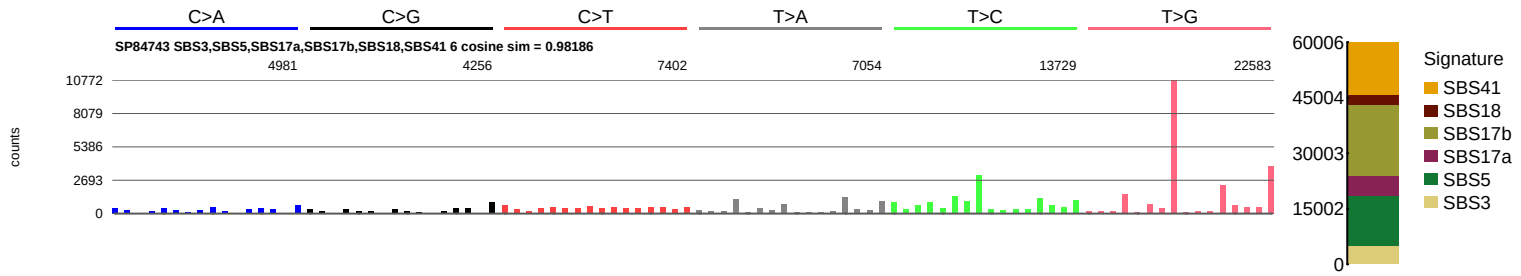
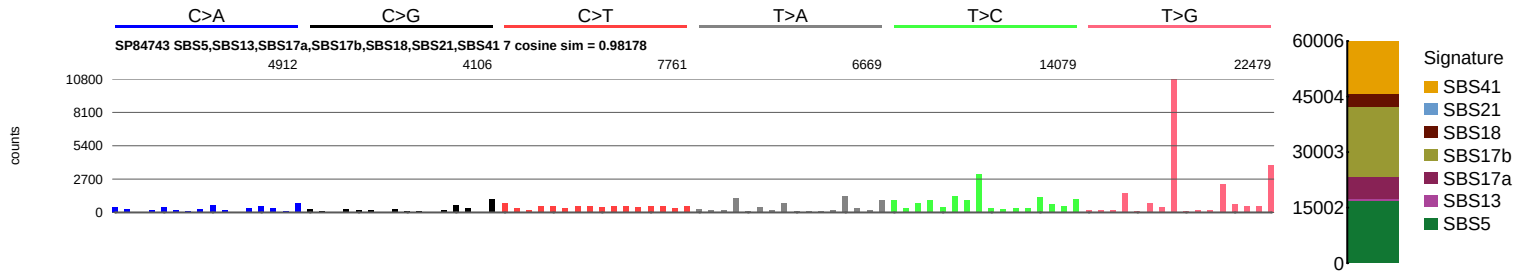
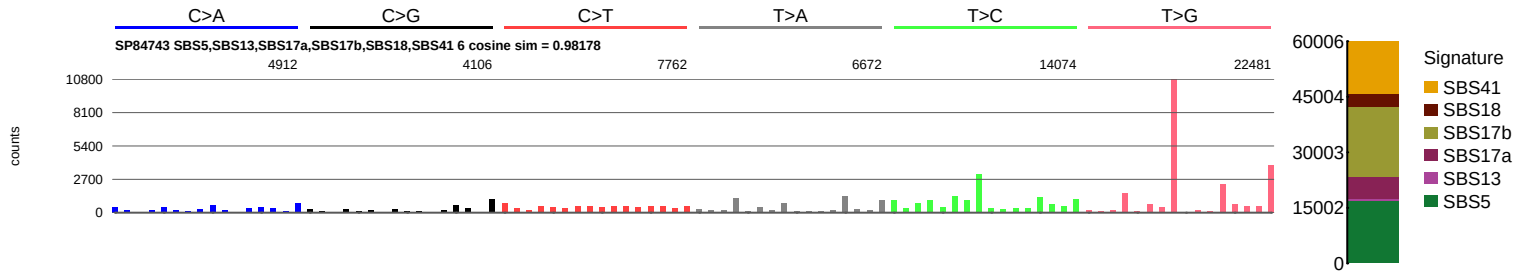
T>C
T>G
SP84743 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98199
4963
4483
7405
6933
13702
22523
counts
10780
8085
5390
2695
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3

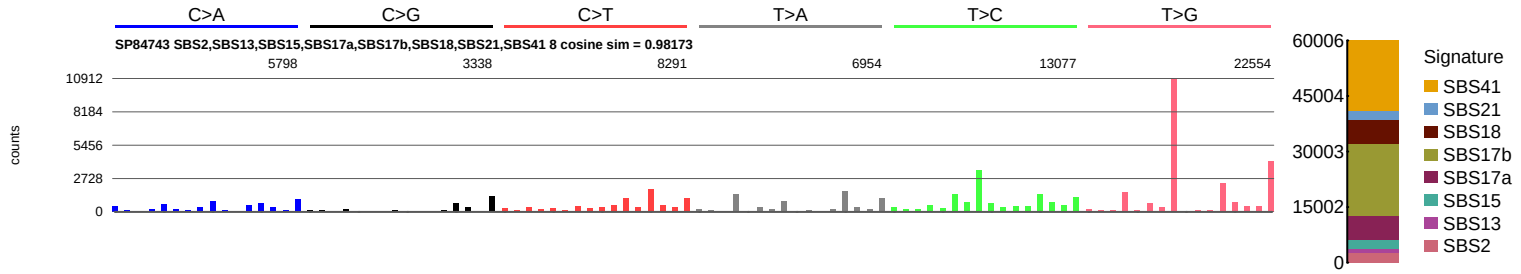

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98191
5781
2545
7966
6834
14349
22525
counts
10888
8166
5444
2722
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2

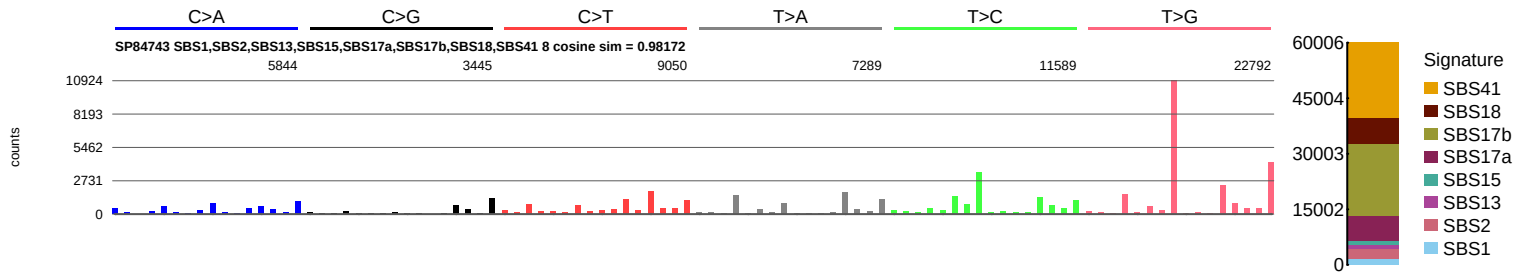

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98186
4830
3605
8335
6721
14014
22498
counts
10792
8094
5396
2698
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5

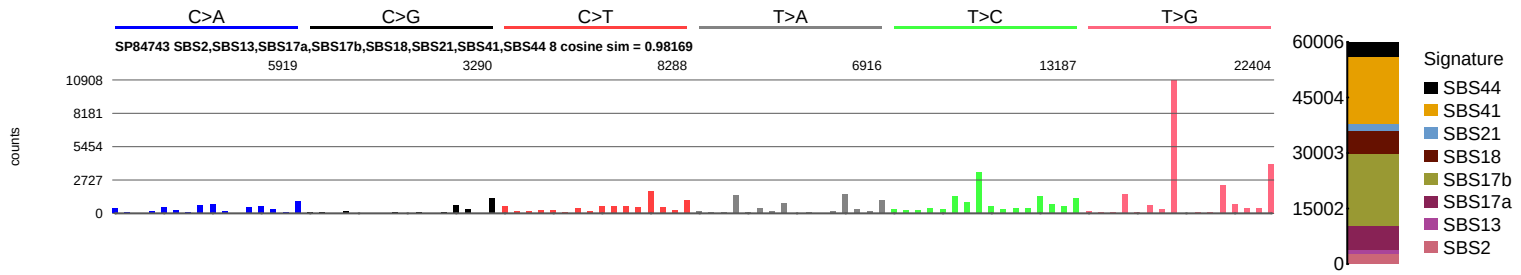

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98186
4981
4256
7402
7054
13729
22583
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3

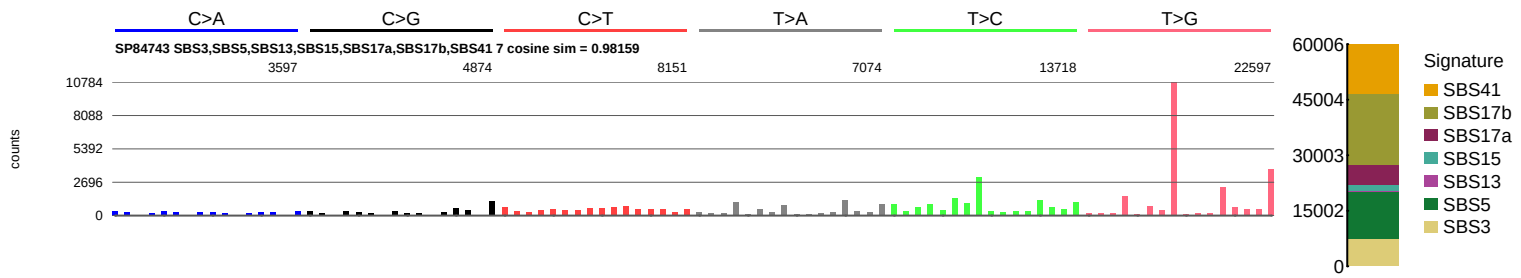

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98178
4912
4106
7761
6669
14079
22479
counts
10800
8100
5400
2700
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5

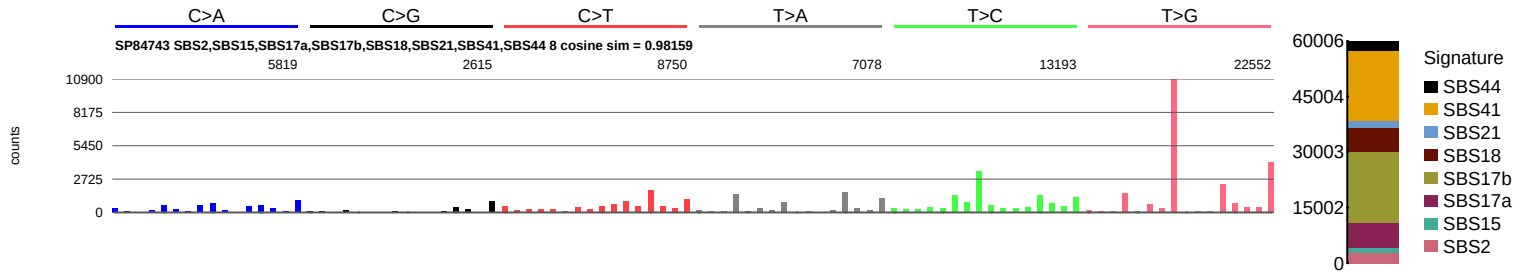

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98178
4912
4106
7762
6672
14074
22481
counts
10800
8100
5400
2700
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5

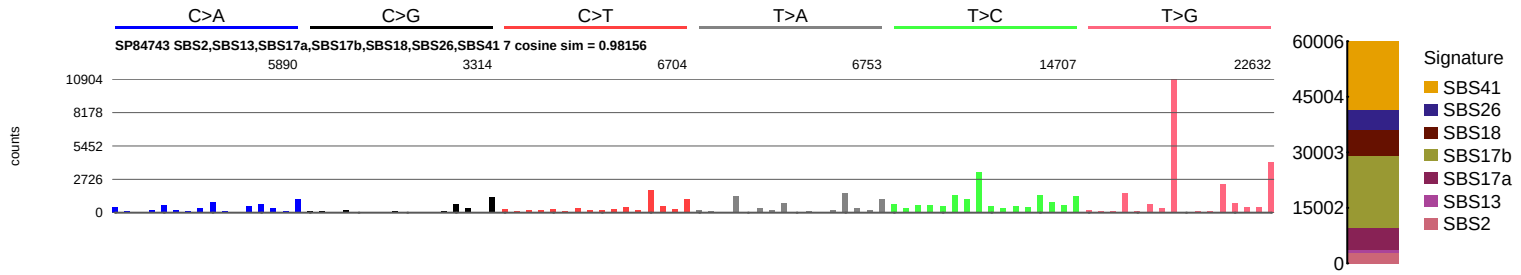

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98173
5798
3338
8291
6954
13077
22554
counts
10912
8184
5456
2728
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

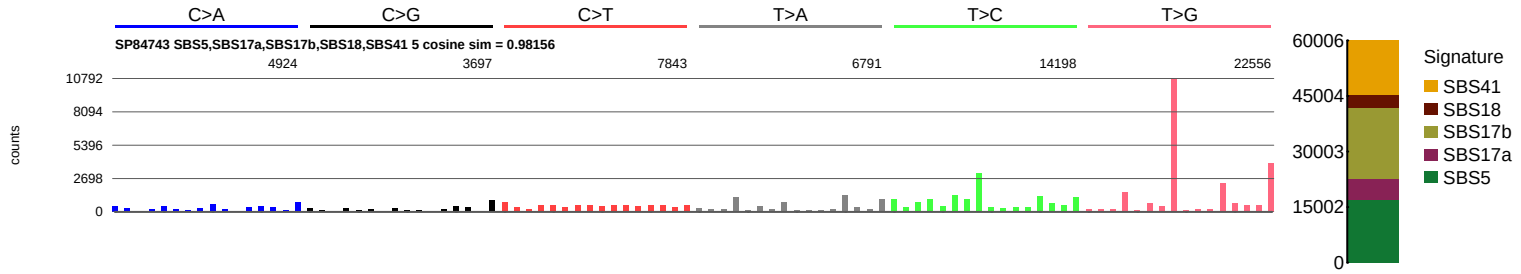

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98172
5844
3445
9050
7289
11589
22792
counts
10924
8193
5462
2731
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

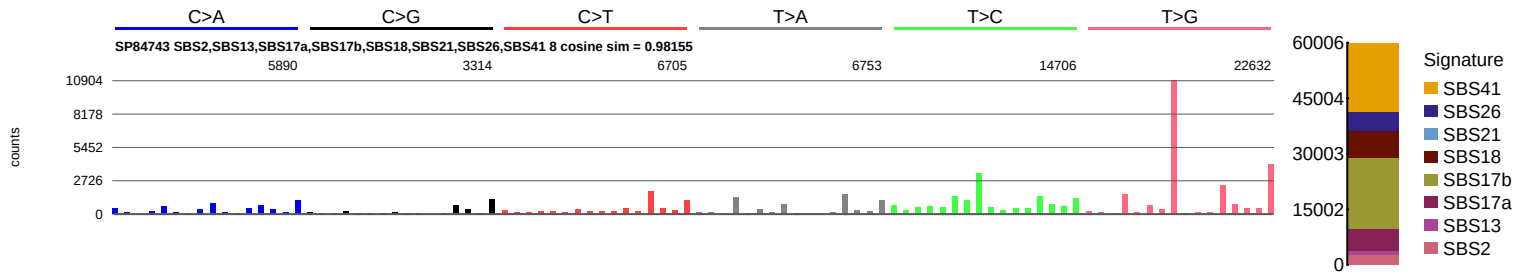

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98169
5919
3290
8288
6916
13187
22404
counts
10908
8181
5454
2727
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2

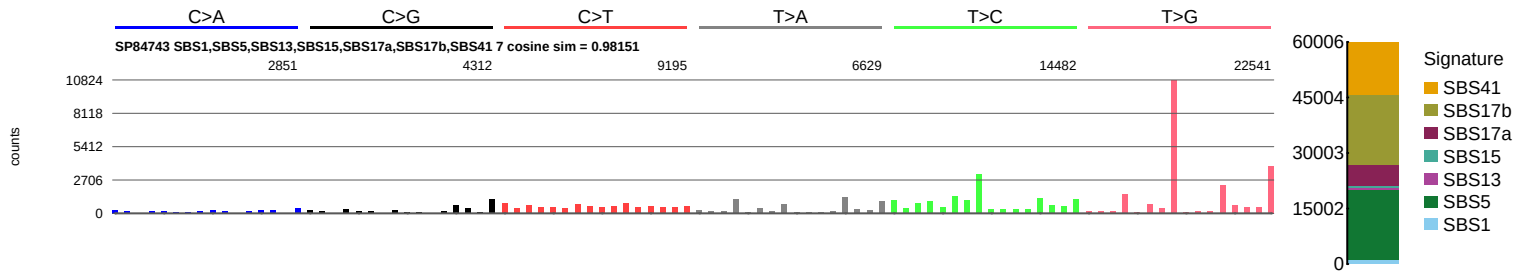

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98159
3597
4874
8151
7074
13718
22597
counts
10784
8088
5392
2696
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

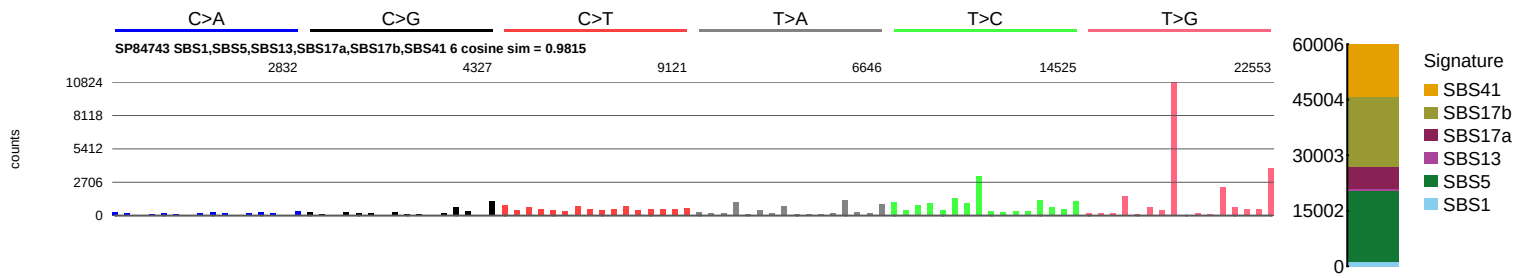

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98159
5819
2615
8750
7078
13193
22552
counts
10900
8175
5450
2725
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2

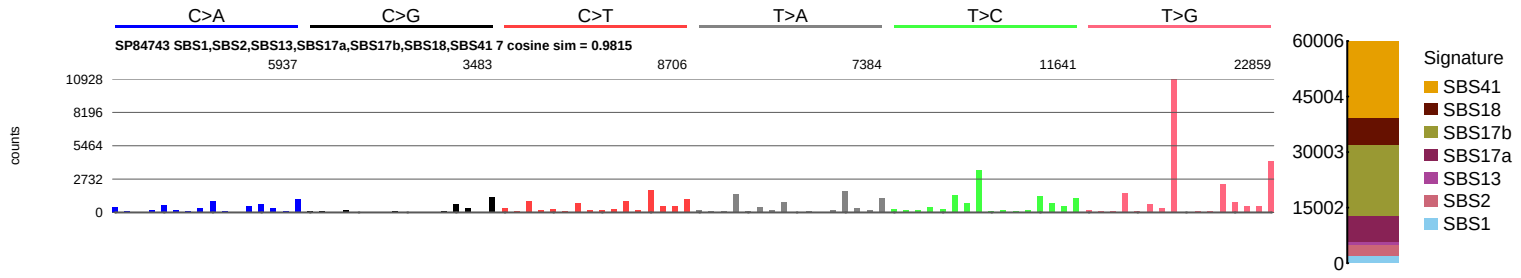

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98156
5890
3314
6704
6753
14707
22632
counts
10904
8178
5452
2726
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2

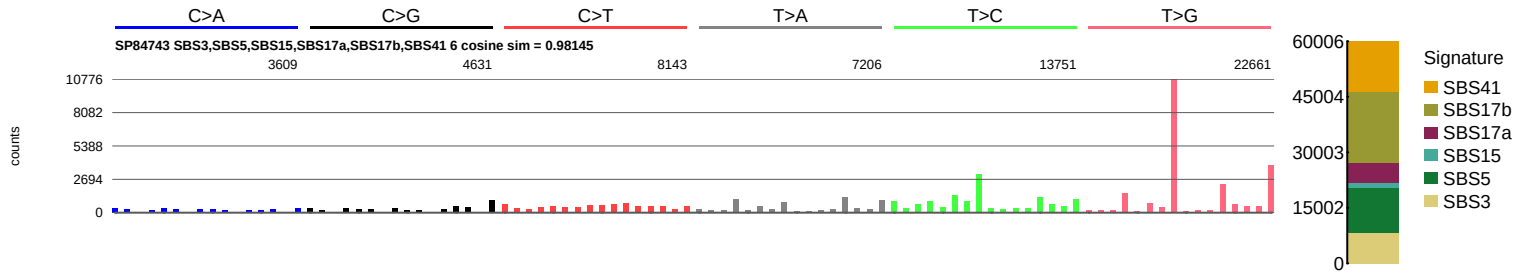

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98156
4924
3697
7843
6791
14198
22556
counts
10792
8094
5396
2698
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5

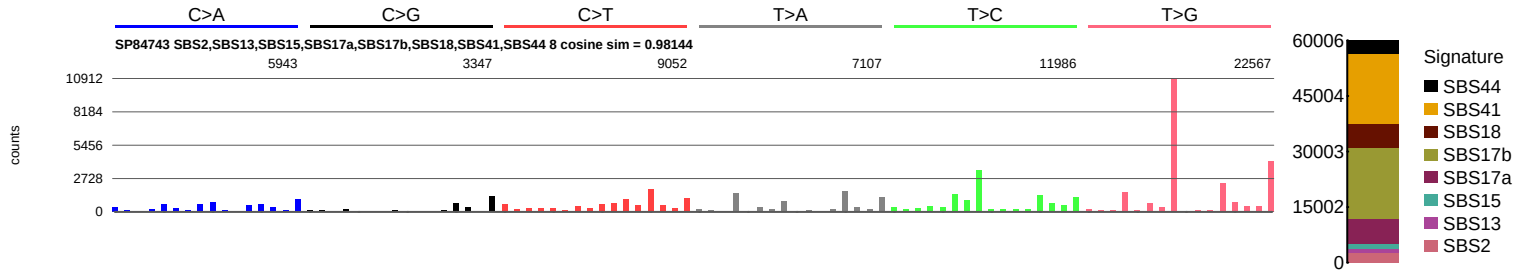

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98155
5890
3314
6705
6753
14706
22632
counts
10904
8178
5452
2726
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2

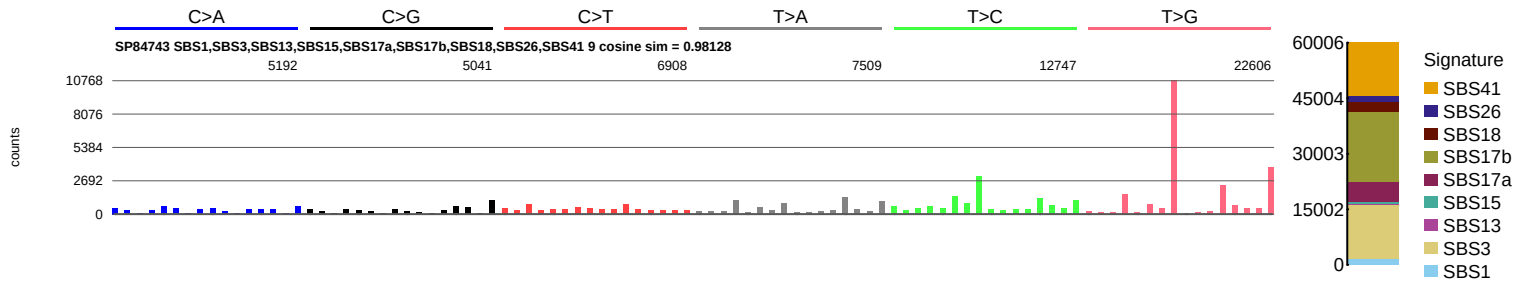

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98151
2851
4312
9195
6629
14482
22541
counts
10824
8118
5412
2706
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

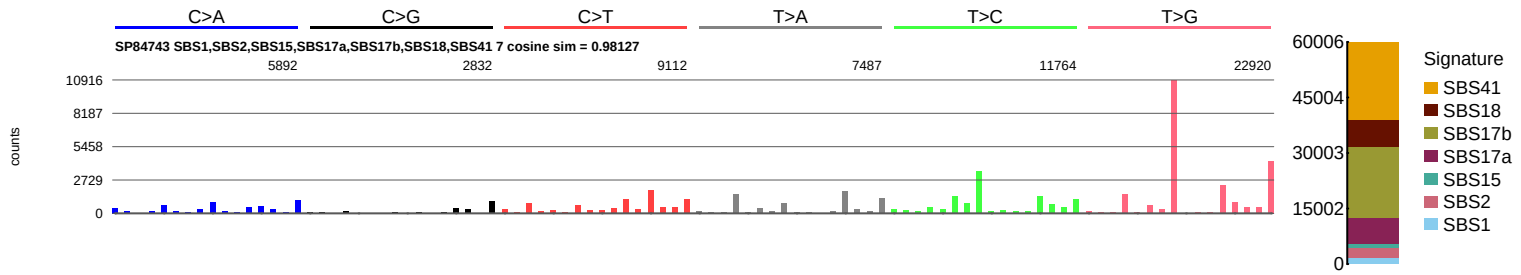

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9815
2832
4327
9121
6646
14525
22553
counts
10824
8118
5412
2706
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS1

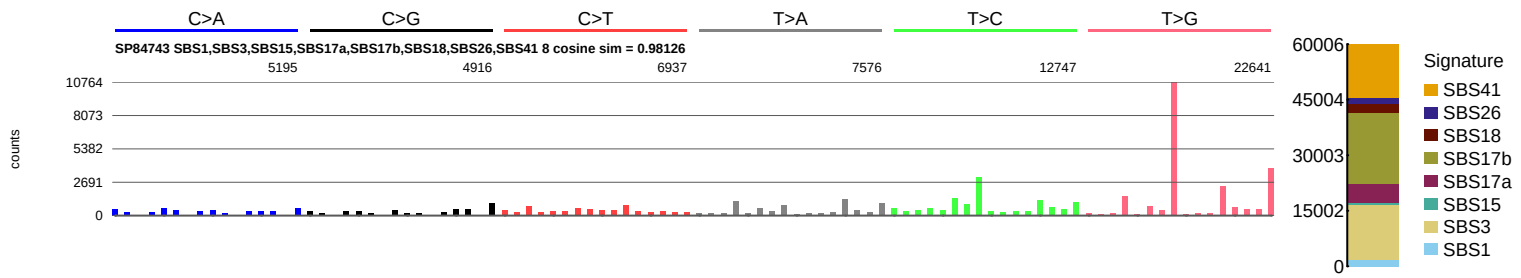

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9815
5937
3483
8706
7384
11641
22859
counts
10928
8196
5464
2732
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

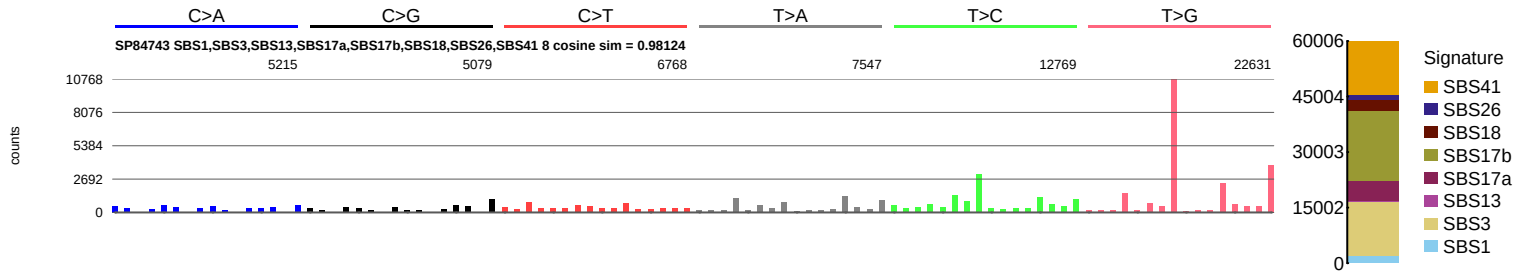

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98145
3609
4631
8143
7206
13751
22661
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3

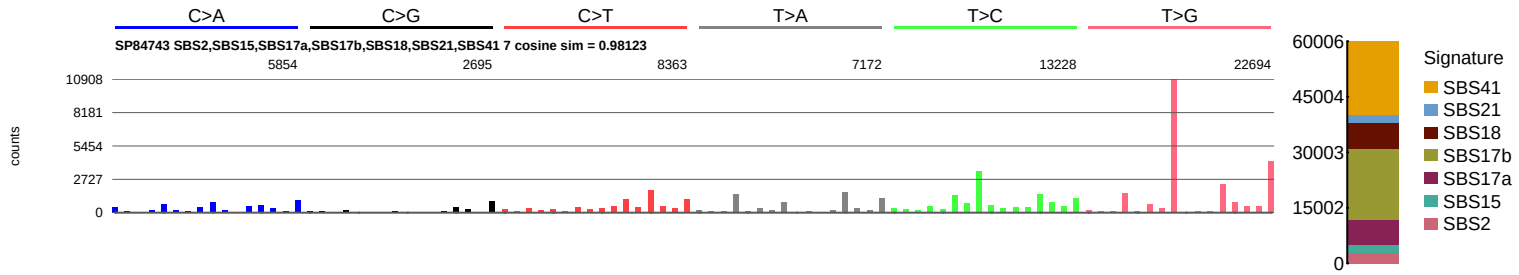

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98144
5943
3347
9052
7107
11986
22567
counts
10912
8184
5456
2728
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

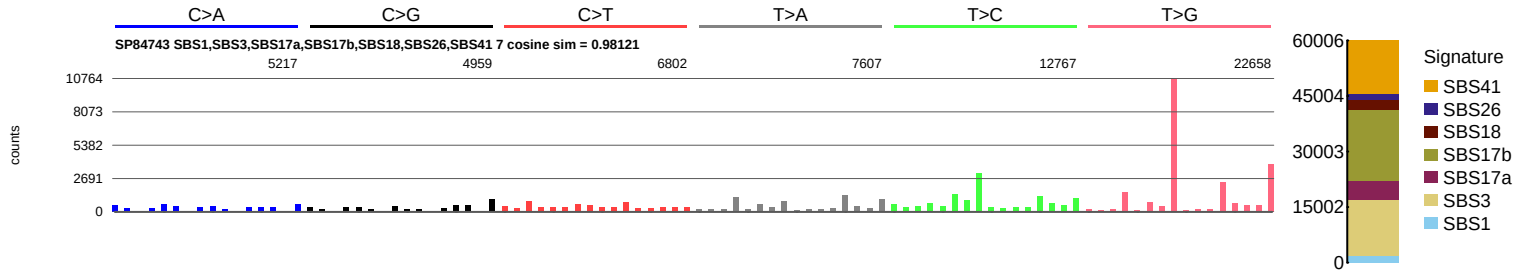

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98128
5192
5041
6908
7509
12747
22606
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

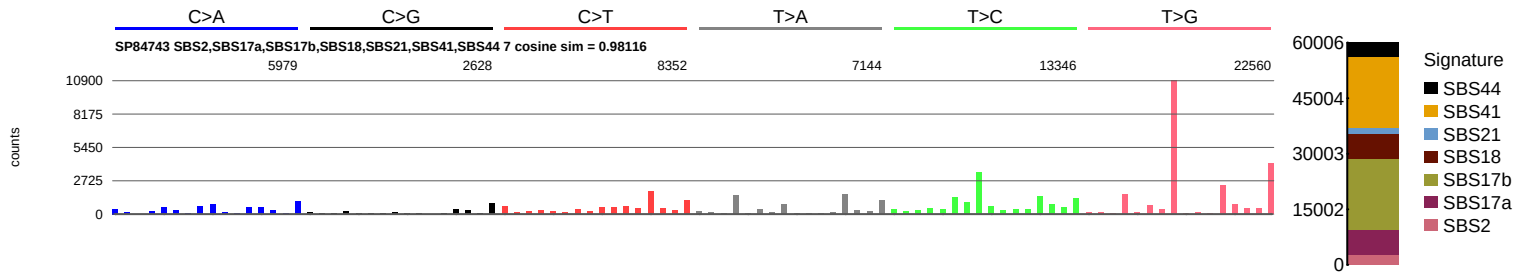

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98127
5892
2832
9112
7487
11764
22920
counts
10916
8187
5458
2729
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98126
5195
4916
6937
7576
12747
22641
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98124
5215
5079
6768
7547
12769
22631
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98123
5854
2695
8363
7172
13228
22694
counts
10908
8181
5454
2727
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98121
5217
4959
6802
7607
12767
22658
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98116
5979
2628
8352
7144
13346
22560
counts
10900
8175
5450
2725
0
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SBS44
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SBS18
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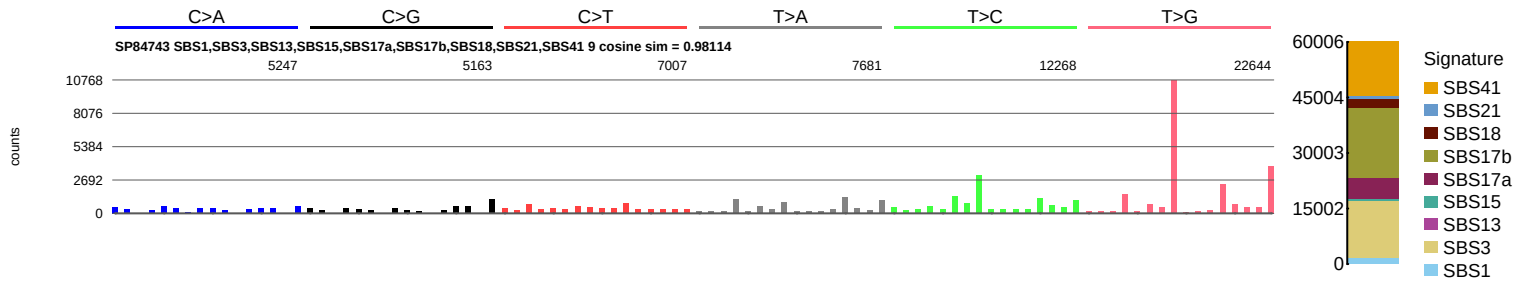
SBS17a
SBS2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SP84743 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98114
5247
5163
7007
7681
12268
22644
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
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SBS18
SBS17b
SBS17a
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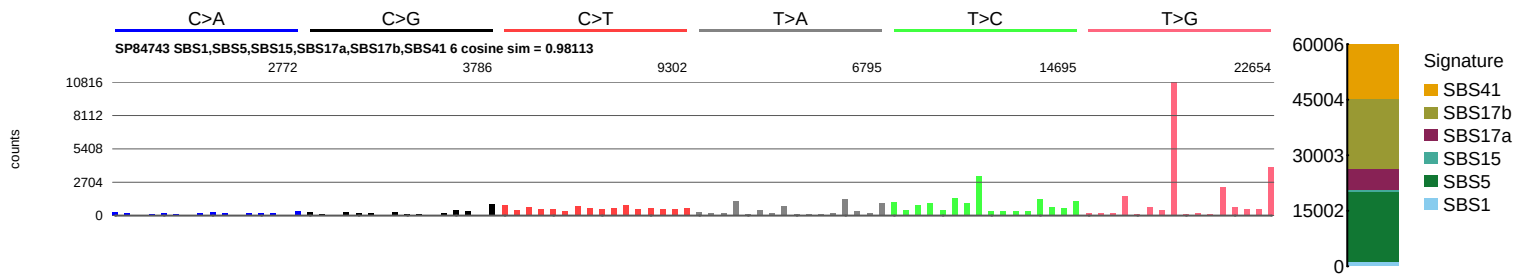
SBS13
SBS3
SBS1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SP84743 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98113
2772
3786
9302
6795
14695
22654
counts
10816
8112
5408
2704
0
60006
45004
30003
15002
0
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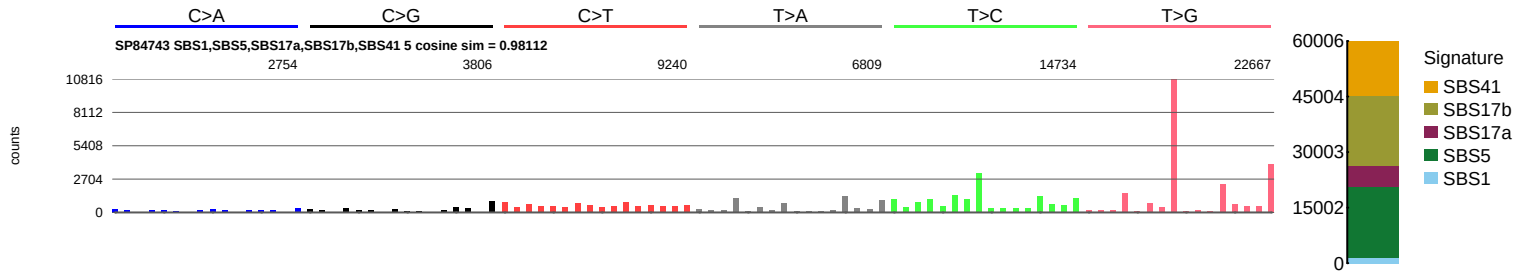
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS1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SP84743 SBS1,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98112
2754
3806
9240
6809
14734
22667
counts
10816
8112
5408
2704
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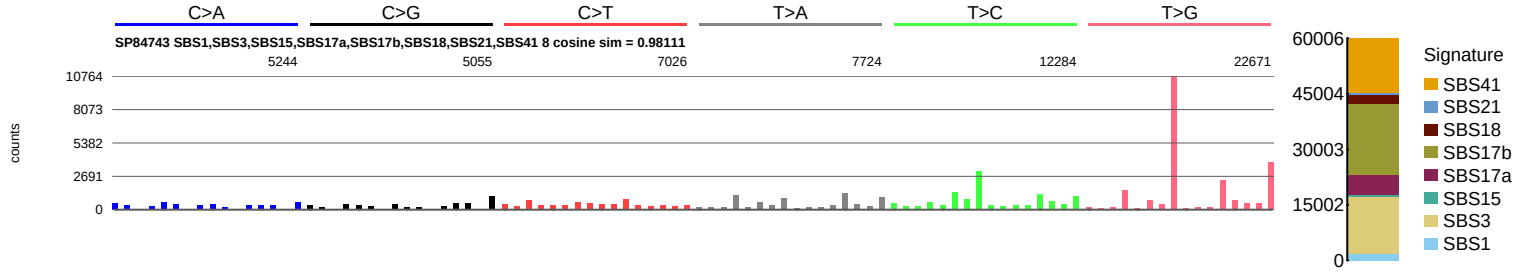
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS1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C>T
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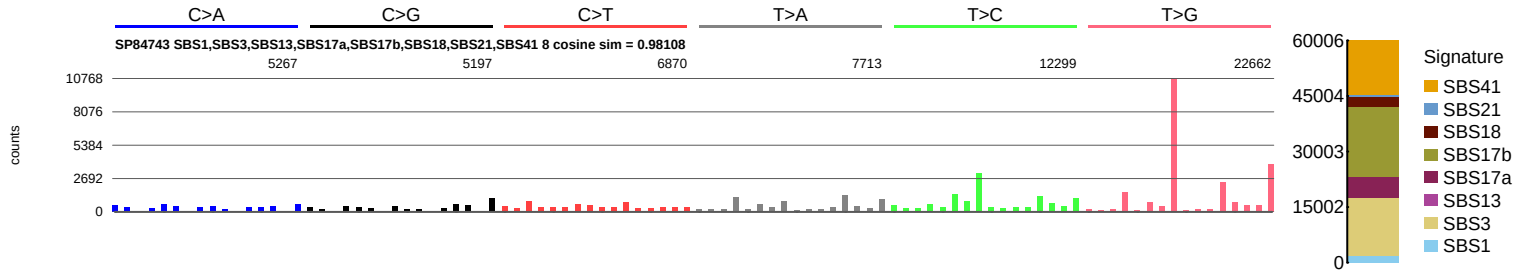
SP84743 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98111
5244
5055
7026
7724
12284
22671
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>T
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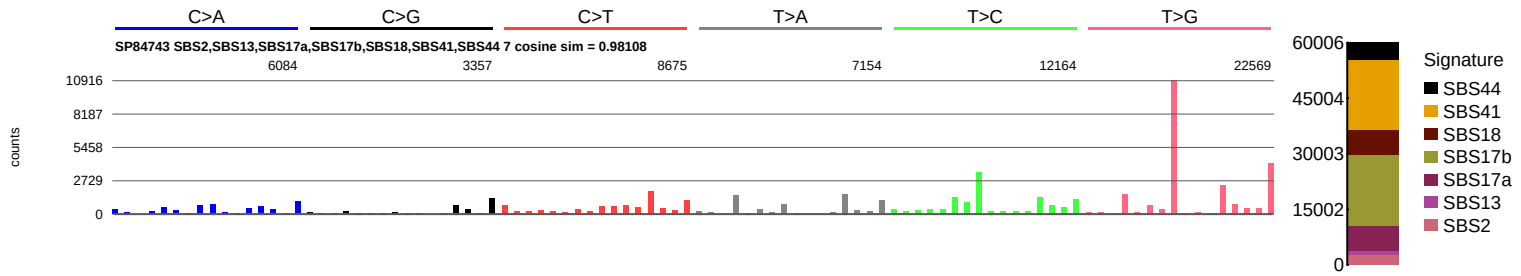
T>G
SP84743 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98108
5267
5197
6870
7713
12299
22662
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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C>G
C>T
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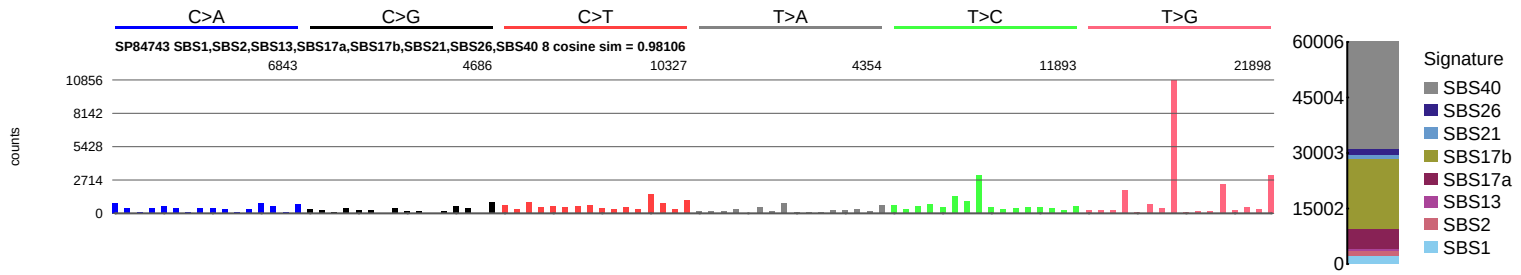
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98108
6084
3357
8675
7154
12164
22569
counts
10916
8187
5458
2729
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2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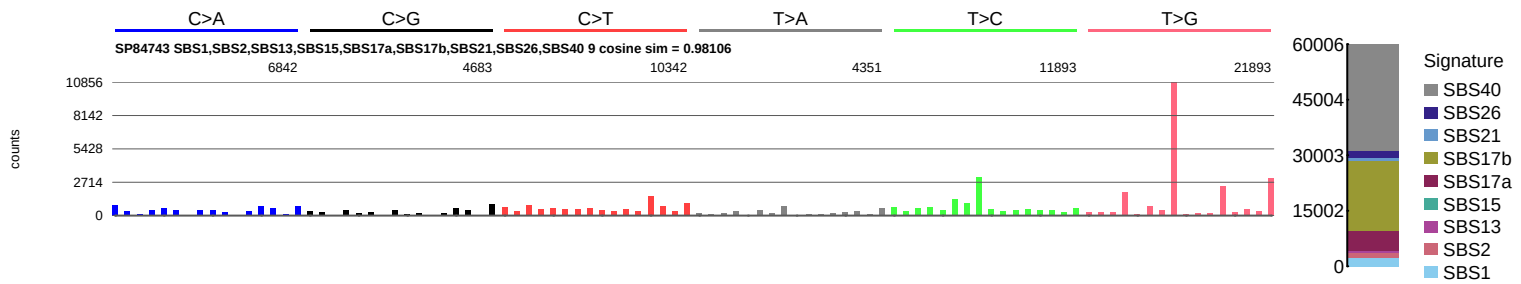
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98106
6843
4686
10327
4354
11893
21898
counts
10856
8142
5428
2714
0
60006
45004
30003
15002
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1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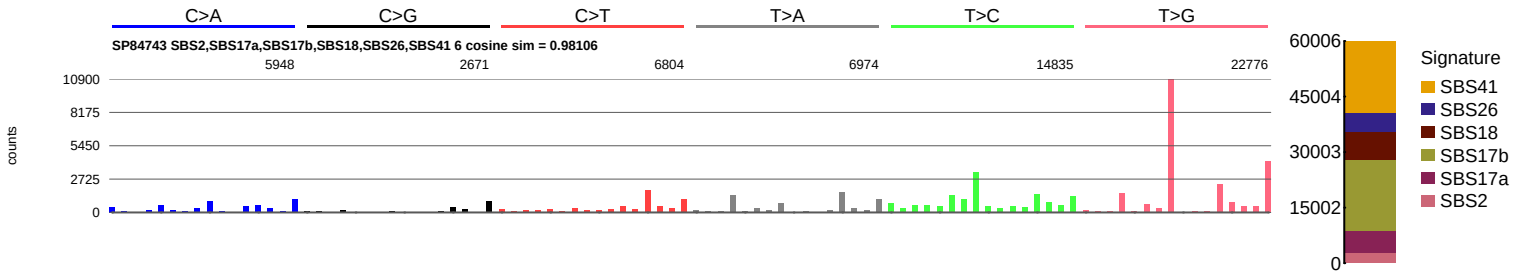
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.98106
6842
4683
10342
4351
11893
21893
counts
10856
8142
5428
2714
0
60006
45004
30003
15002
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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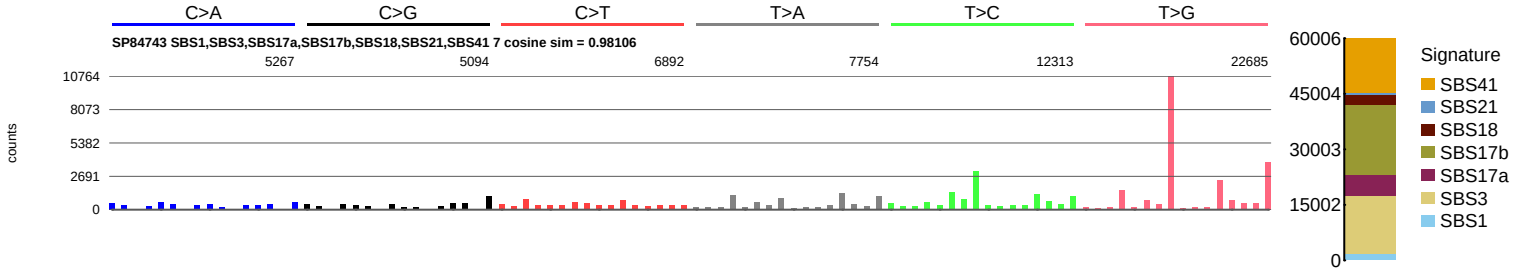
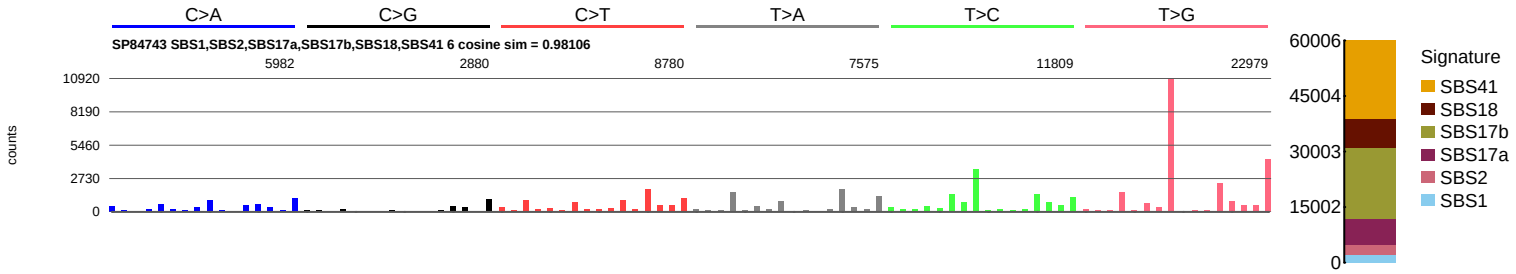
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98106
5948
2671
6804
6974
14835
22776
counts
10900
8175
5450
2725
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98106
5267
5094
6892
7754
12313
22685
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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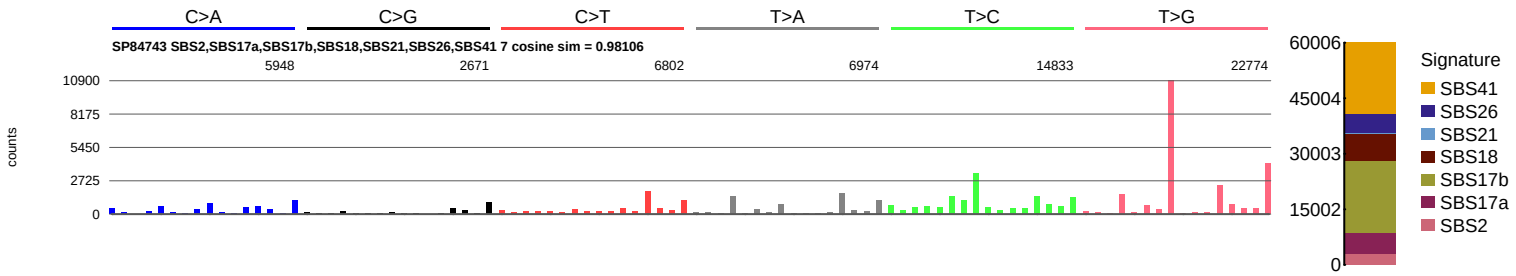
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98106
5982
2880
8780
7575
11809
22979
counts
10920
8190
5460
2730
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1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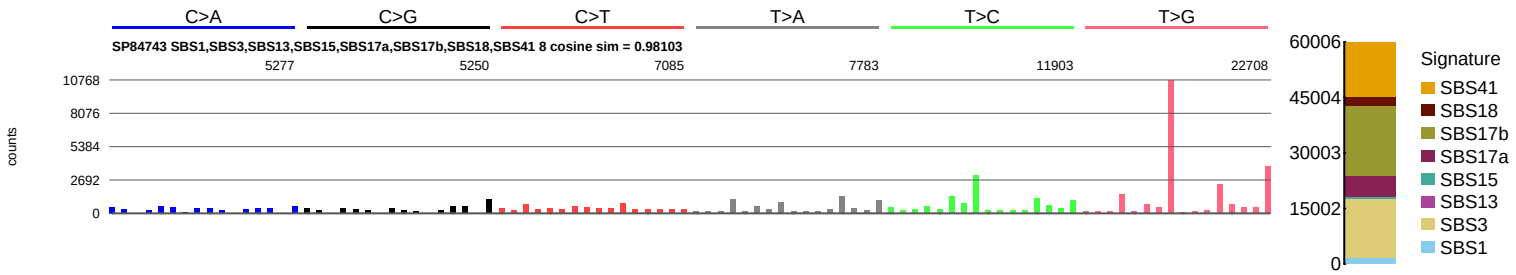
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98106
5948
2671
6802
6974
14833
22774
counts
10900
8175
5450
2725
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98103
5277
5250
7085
7783
11903
22708
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
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SBS41
SBS18
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SBS17a
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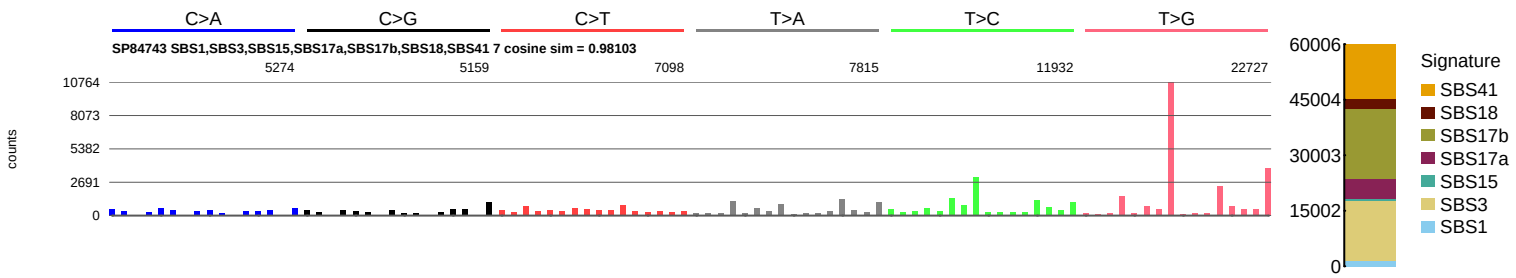
SBS13
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98103
5274
5159
7098
7815
11932
22727
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
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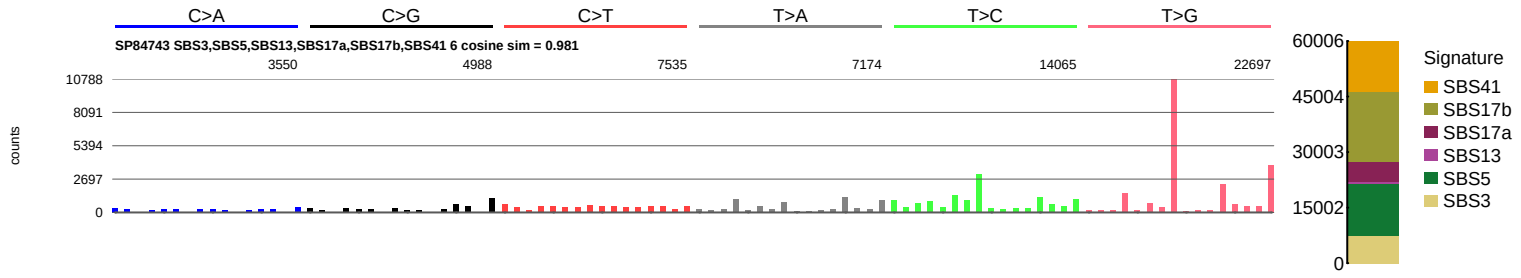
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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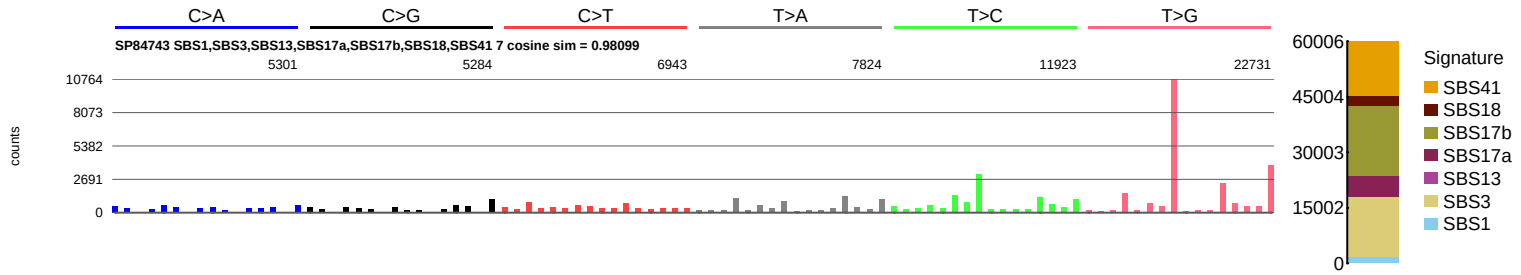
C>T
T>A
T>C
T>G
SP84743 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.981
3550
4988
7535
7174
14065
22697
counts
10788
8091
5394
2697
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98099
5301
5284
6943
7824
11923
22731
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
0
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SBS18
SBS17b
SBS17a
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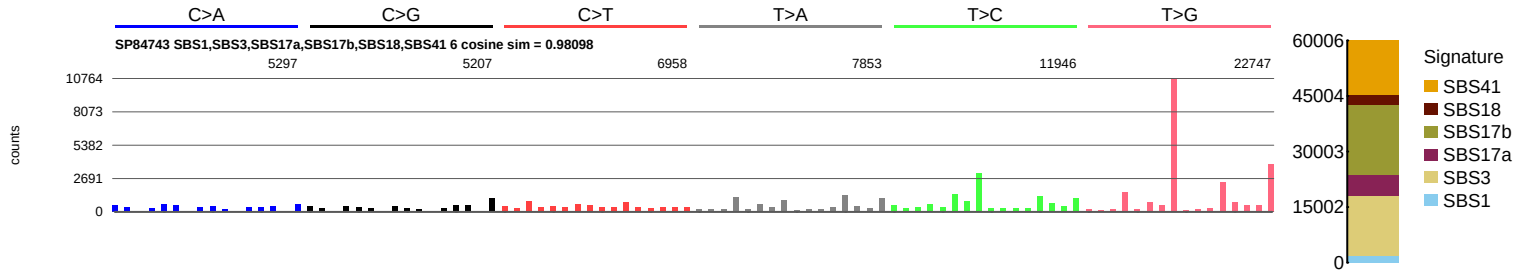
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98098
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5207
6958
7853
11946
22747
counts
10764
8073
5382
2691
0
60006
45004
30003
15002
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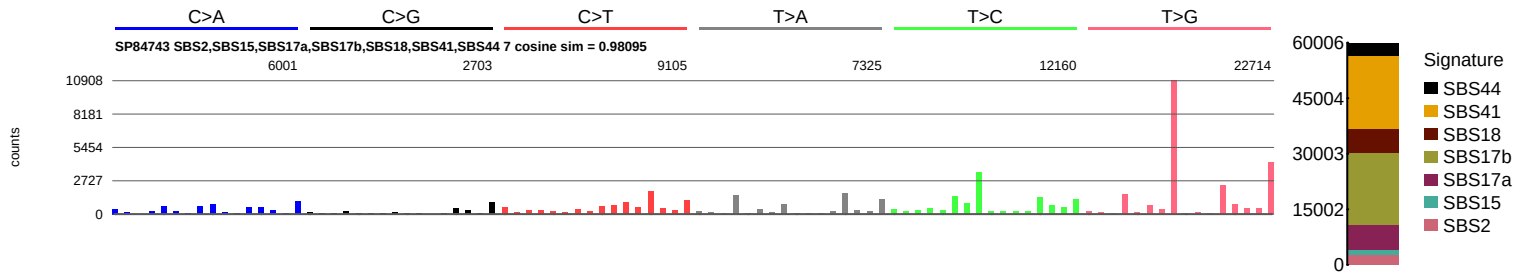
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
C>T
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T>C
T>G
SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98095
6001
2703
9105
7325
12160
22714
counts
10908
8181
5454
2727
0
60006
45004
30003
15002
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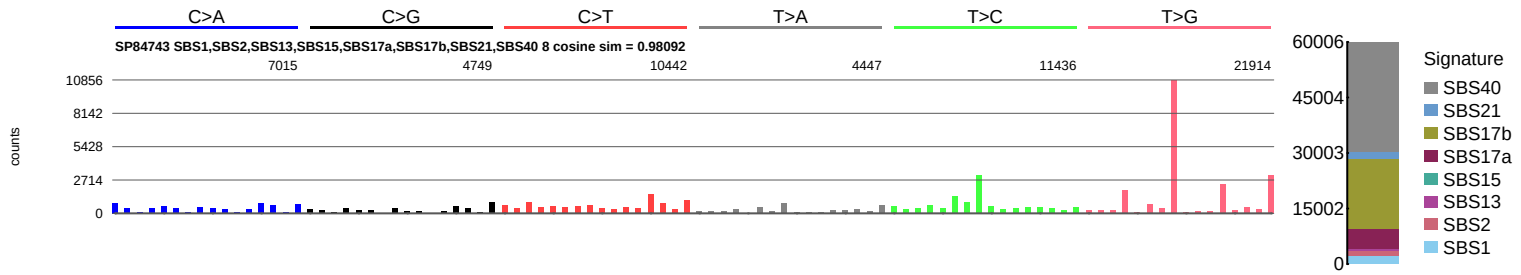
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98092
7015
4749
10442
4447
11436
21914
counts
10856
8142
5428
2714
0
60006
45004
30003
15002
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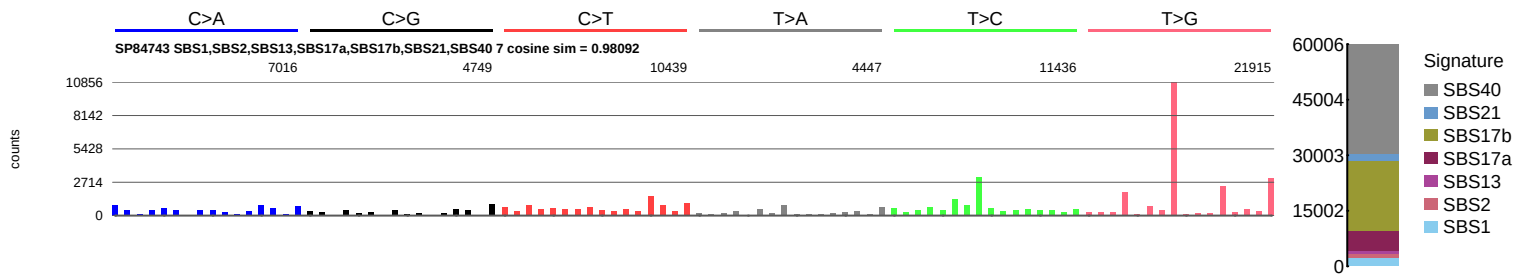
SBS2
SBS1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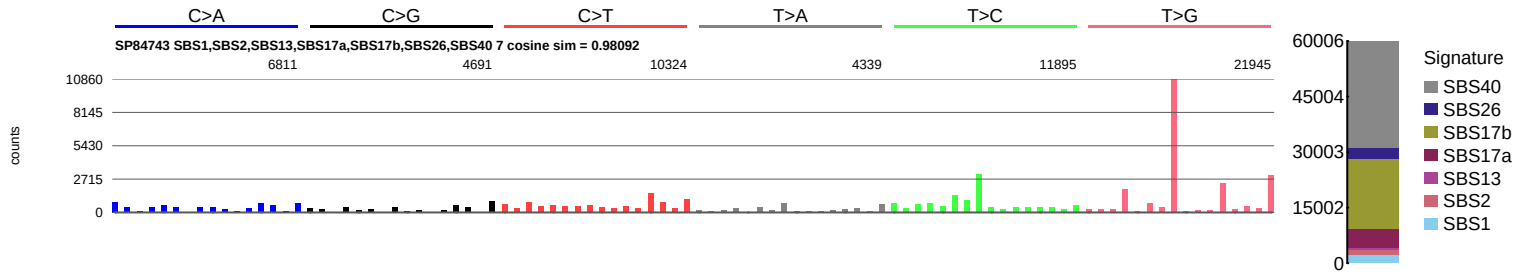
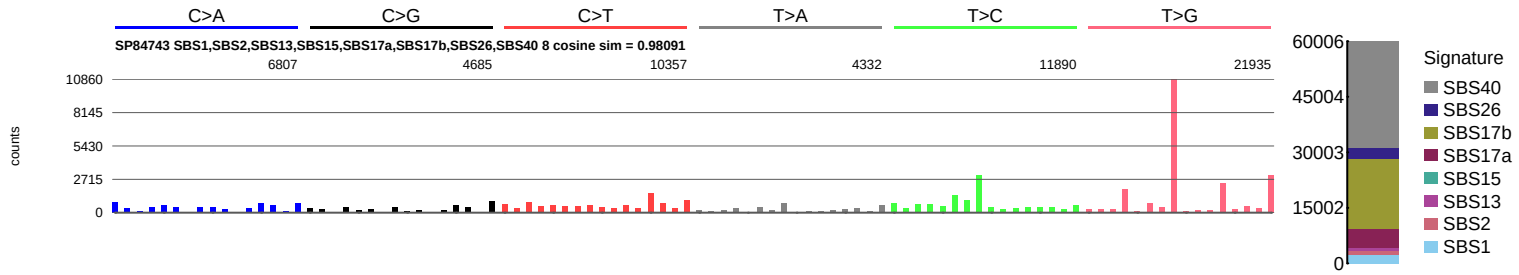
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98092
7016
4749
10439
4447
11436
21915
counts
10856
8142
5428
2714
0
60006
45004
30003
15002
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98092
6811
4691
10324
4339
11895
21945
counts
10860
8145
5430
2715
0
60006
45004
30003
15002
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98091
6807
4685
10357
4332
11890
21935
counts
10860
8145
5430
2715
0
60006
45004
30003
15002
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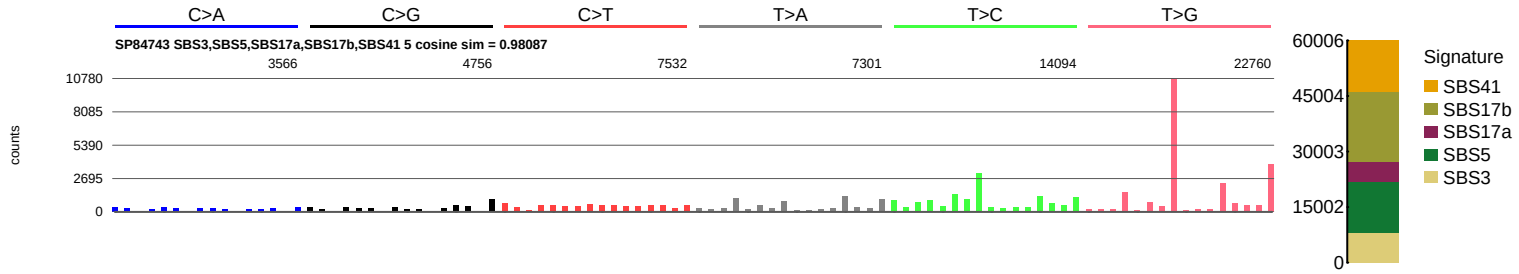
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS3,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98087
3566
4756
7532
7301
14094
22760
counts
10780
8085
5390
2695
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60006
45004
30003
15002
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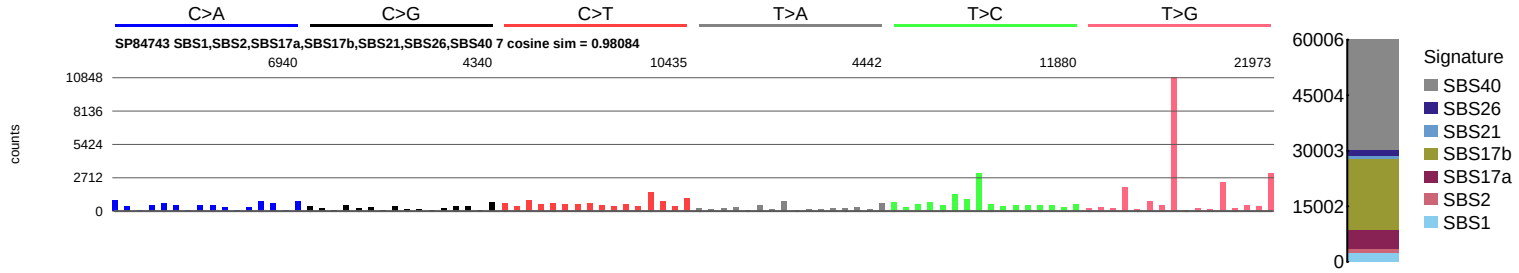
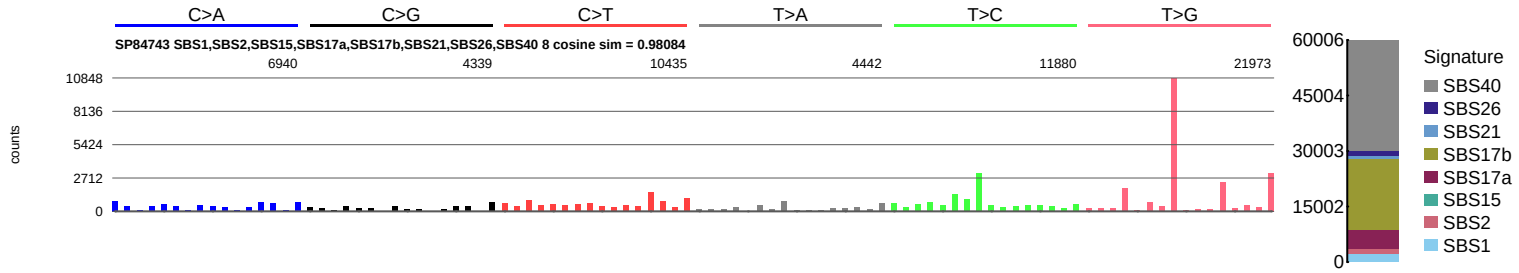
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98084
6940
4340
10435
4442
11880
21973
counts
10848
8136
5424
2712
0
60006
45004
30003
15002
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1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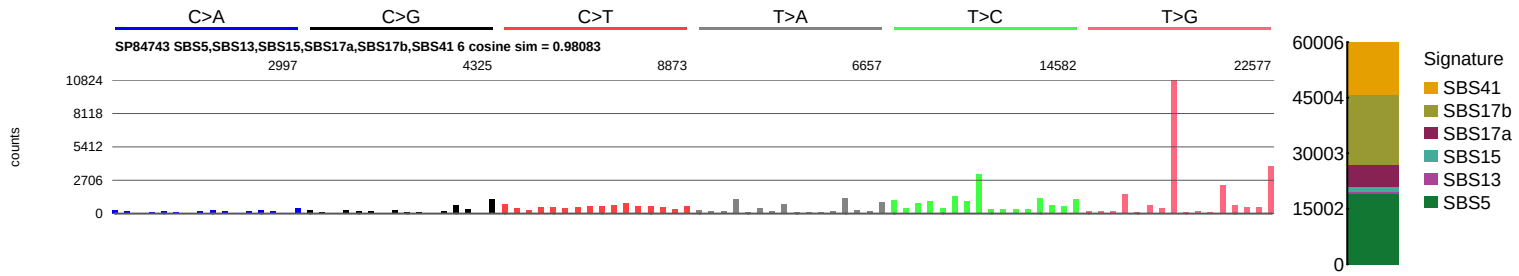
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98084
6940
4339
10435
4442
11880
21973
counts
10848
8136
5424
2712
0
60006
45004
30003
15002
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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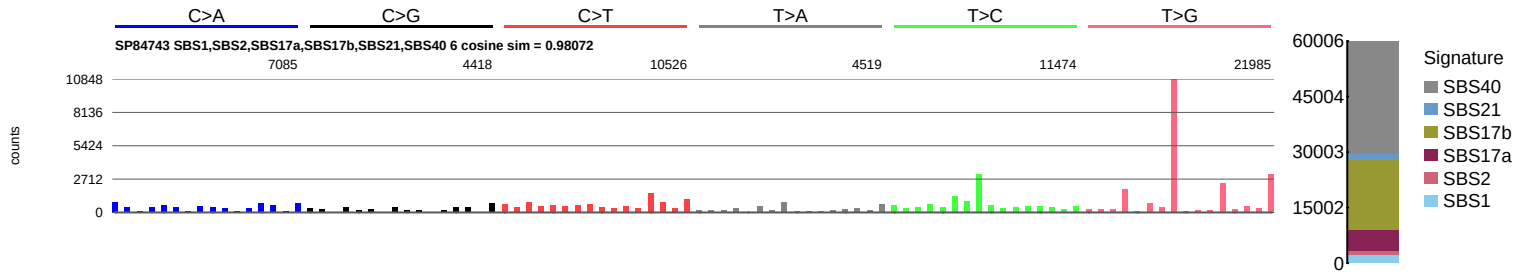
SP84743 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98083
2997
4325
8873
6657
14582
22577
counts
10824
8118
5412
2706
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5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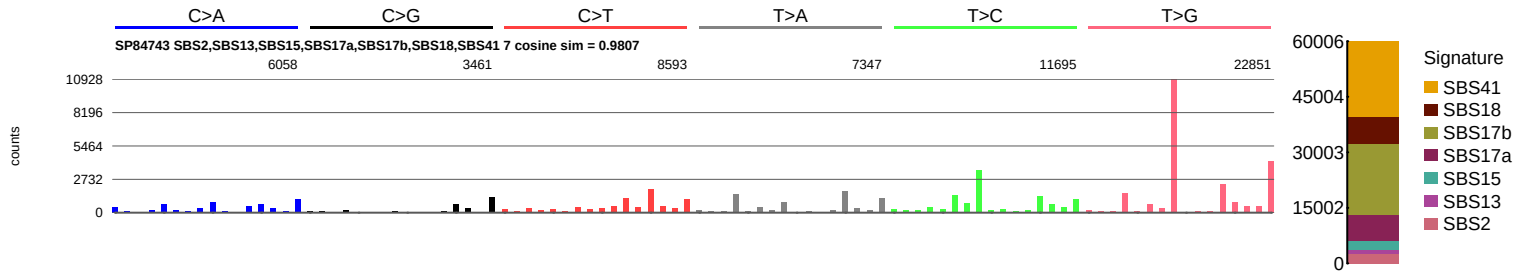
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98072
7085
4418
10526
4519
11474
21985
counts
10848
8136
5424
2712
0
60006
45004
30003
15002
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1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SP84743 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9807
6058
3461
8593
7347
11695
22851
counts
10928
8196
5464
2732
0
60006
45004
30003
15002
0
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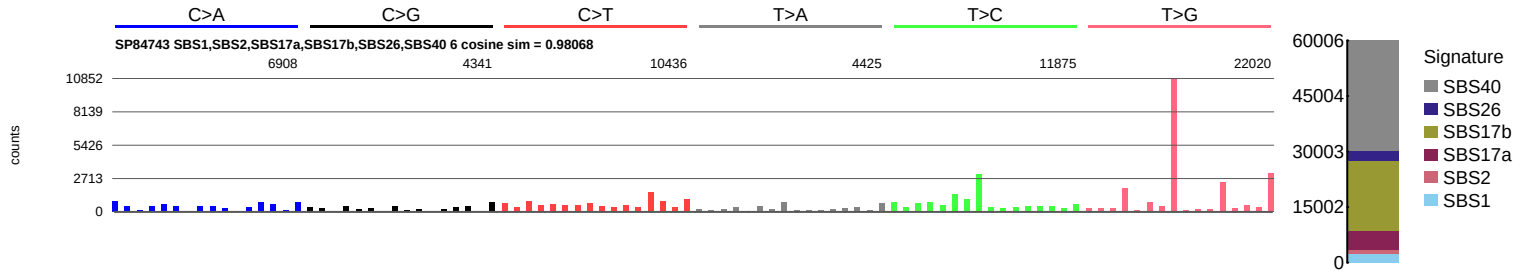
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2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C>T
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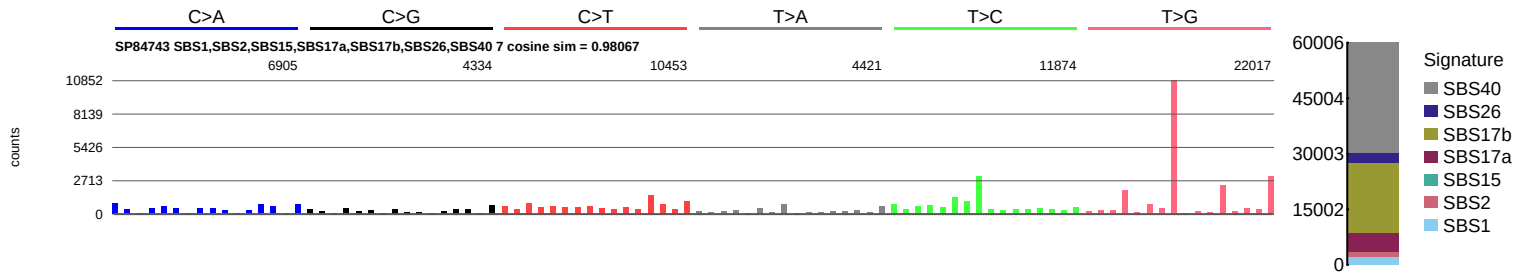
T>C
T>G
SP84743 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98068
6908
4341
10436
4425
11875
22020
counts
10852
8139
5426
2713
0
60006
45004
30003
15002
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1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SP84743 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98067
6905
4334
10453
4421
11874
22017
counts
10852
8139
5426
2713
0
60006
45004
30003
15002
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SBS40
SBS26
SBS17b
SBS17a
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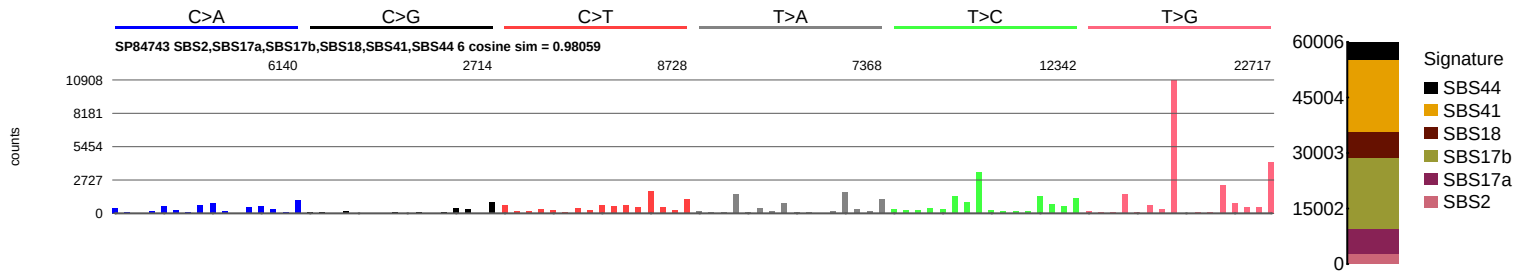
SBS2
SBS1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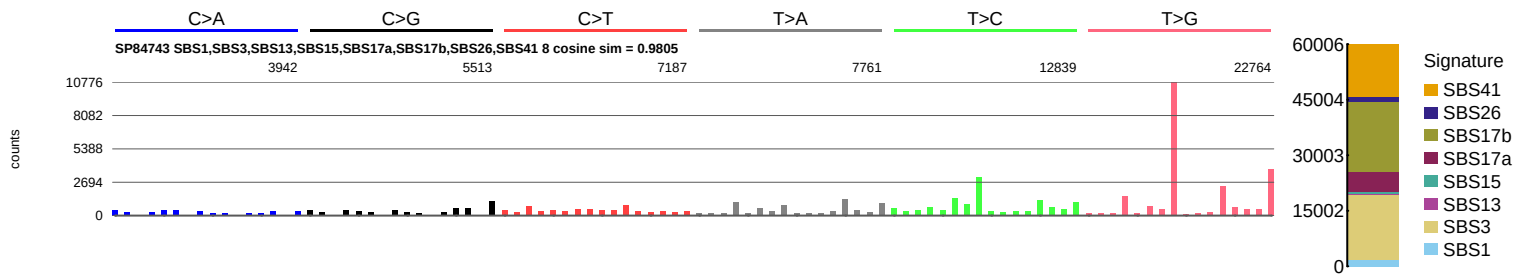
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98059
6140
2714
8728
7368
12342
22717
counts
10908
8181
5454
2727
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.9805
3942
5513
7187
7761
12839
22764
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
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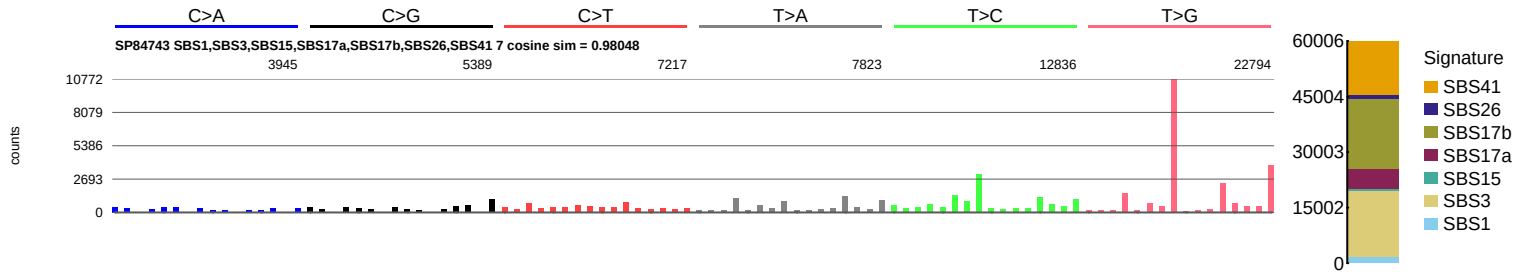
SBS3
SBS1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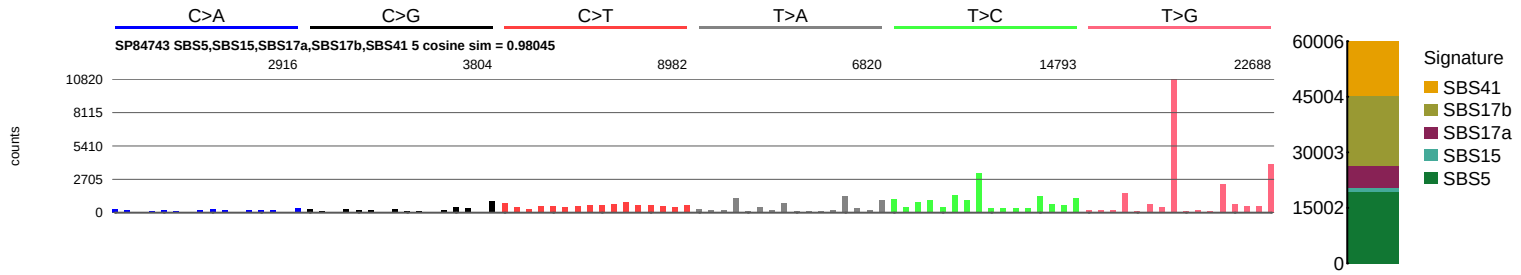
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98048
3945
5389
7217
7823
12836
22794
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1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T>A
T>C
T>G
SP84743 SBS5,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98045
2916
3804
8982
6820
14793
22688
counts
10820
8115
5410
2705
0
60006
45004
30003
15002
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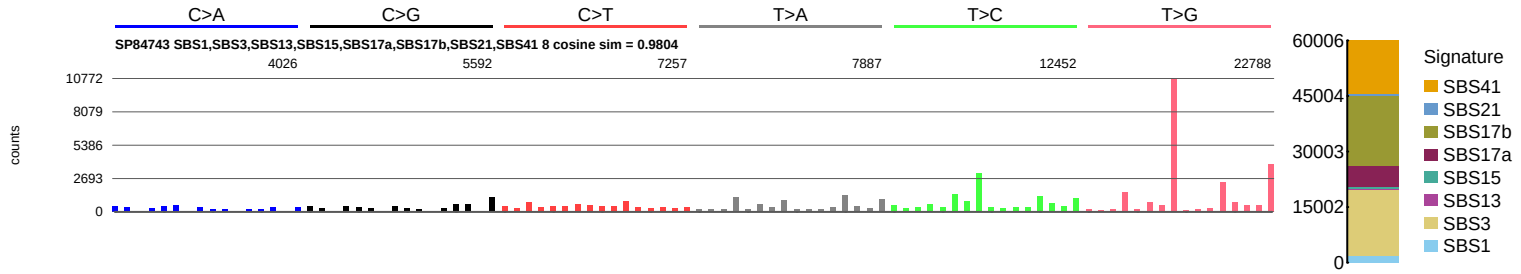
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5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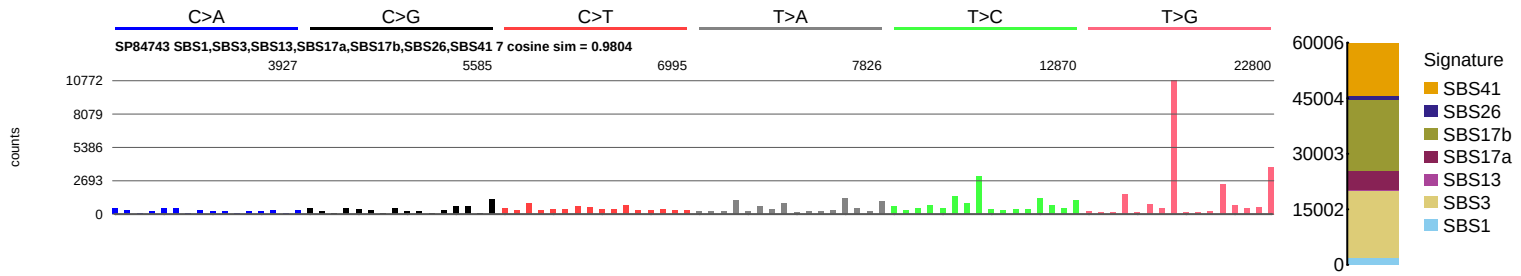
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9804
4026
5592
7257
7887
12452
22788
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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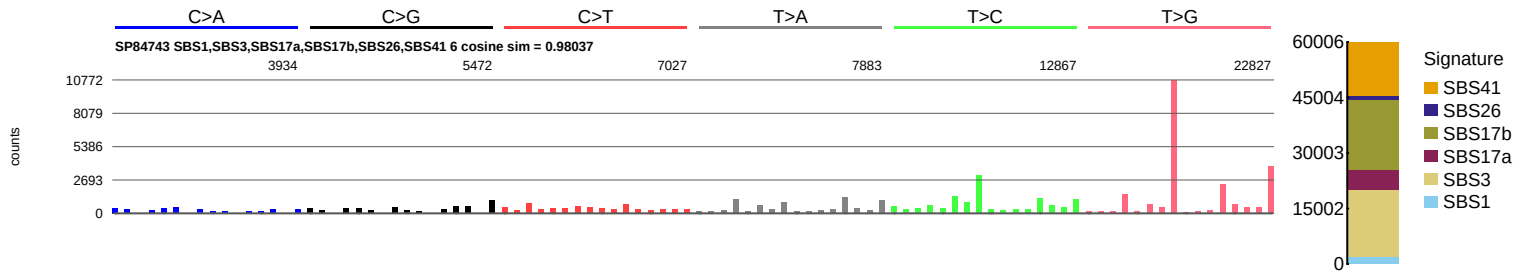
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.9804
3927
5585
6995
7826
12870
22800
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1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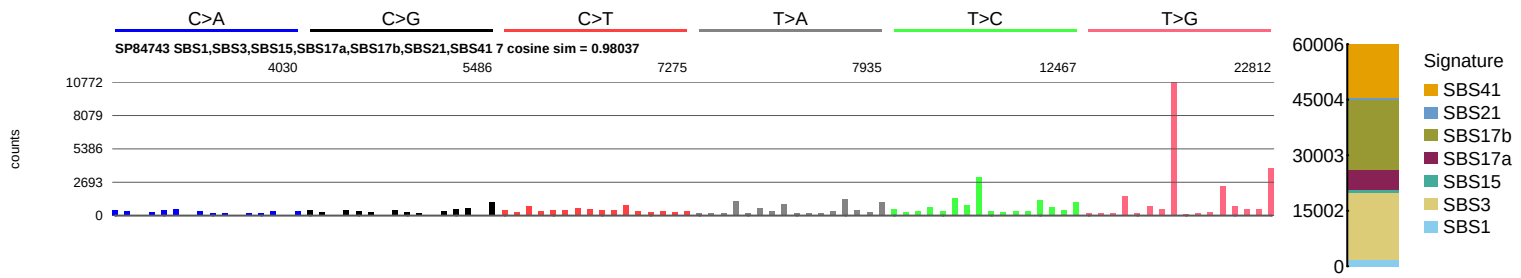
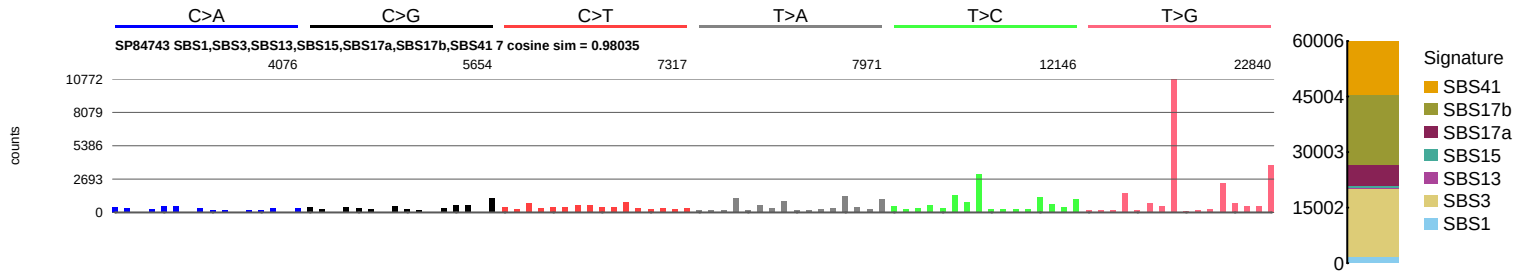
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98037
3934
5472
7027
7883
12867
22827
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98037
4030
5486
7275
7935
12467
22812
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1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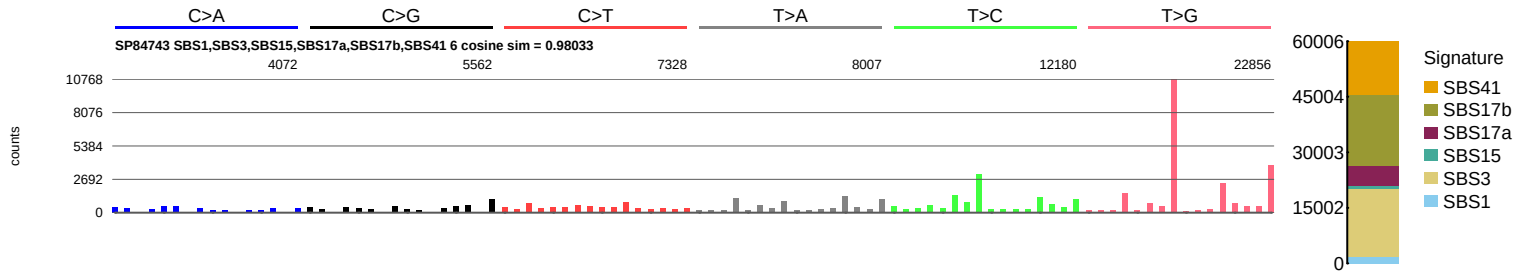
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98035
4076
5654
7317
7971
12146
22840
10772
8079
5386
2693
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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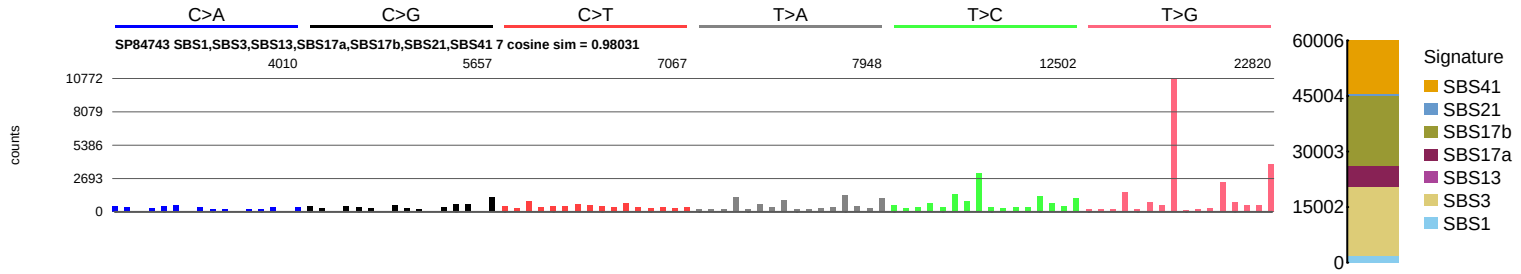
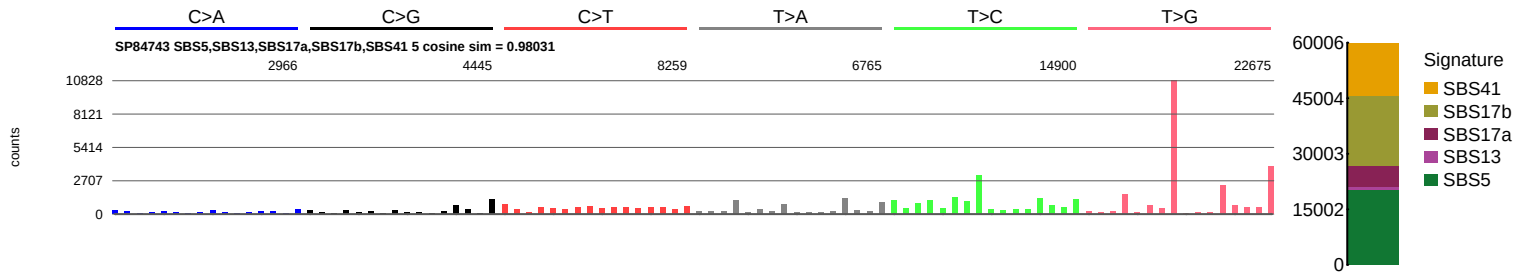
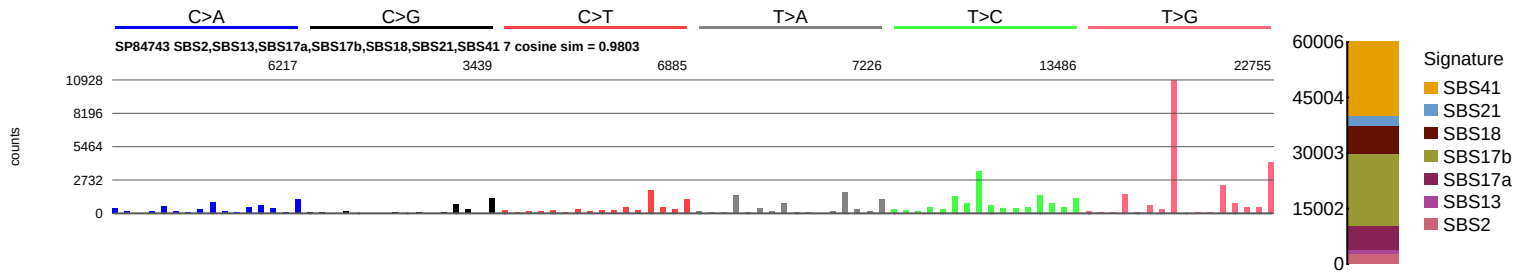
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98033
4072
5562
7328
8007
12180
22856
10768
8076
5384
2692
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98031
4010
5657
7067
7948
12502
22820
10772
8079
5386
2693
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS5,SBS13,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98031
2966
4445
8259
6765
14900
22675
10828
8121
5414
2707
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5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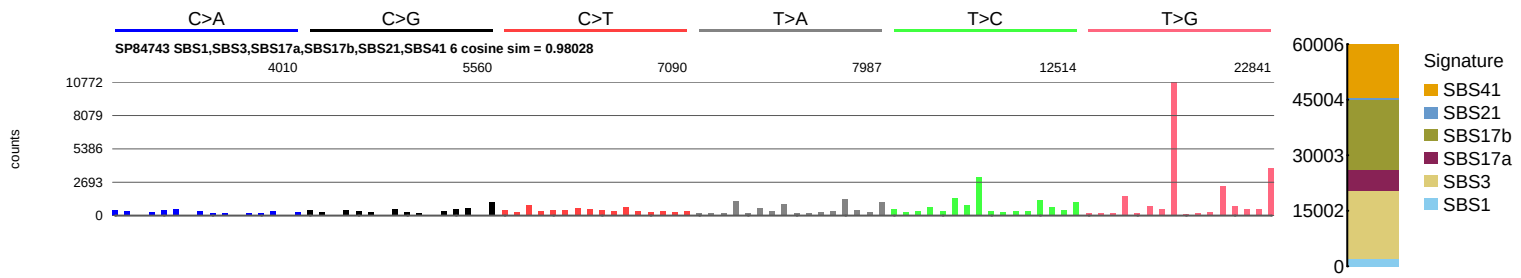
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9803
6217
3439
6885
7226
13486
22755
10928
8196
5464
2732
0
counts
60006
45004
30003
15002
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98028
4010
5560
7090
7987
12514
22841
10772
8079
5386
2693
0
counts
60006
45004
30003
15002
0
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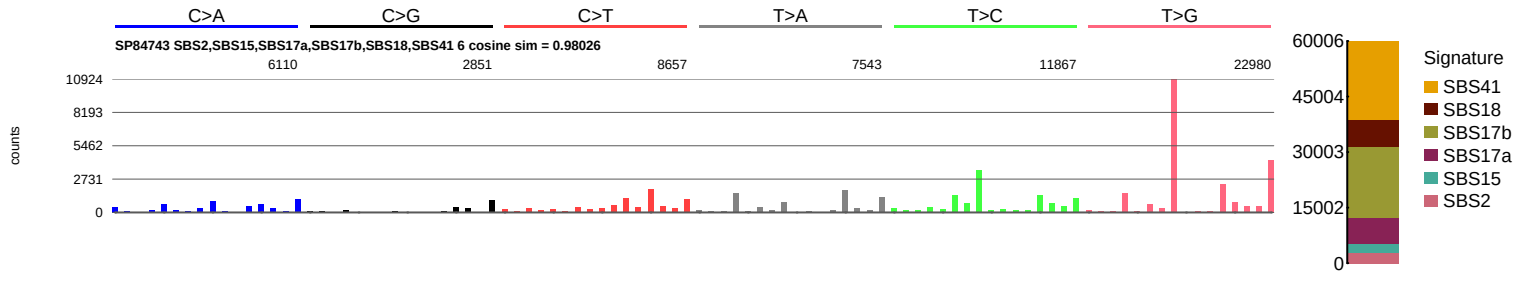
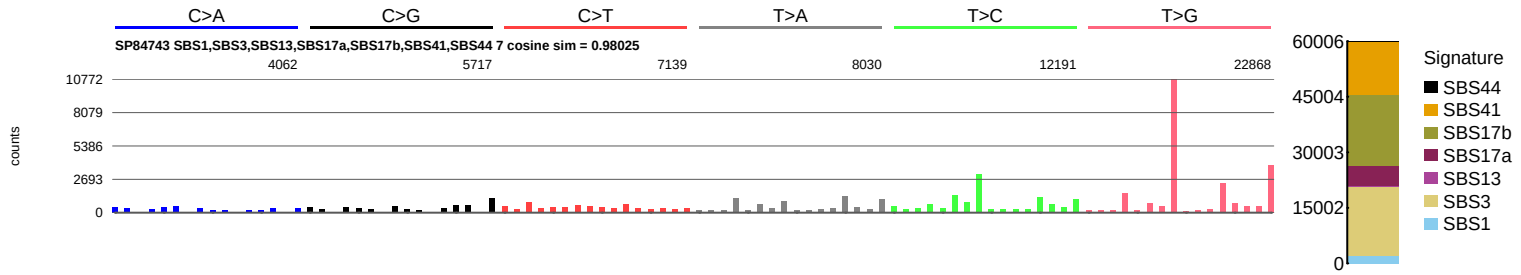
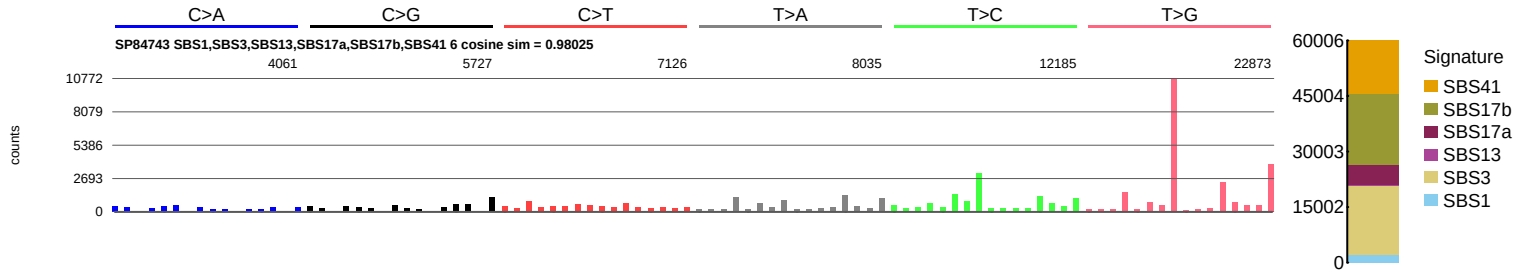
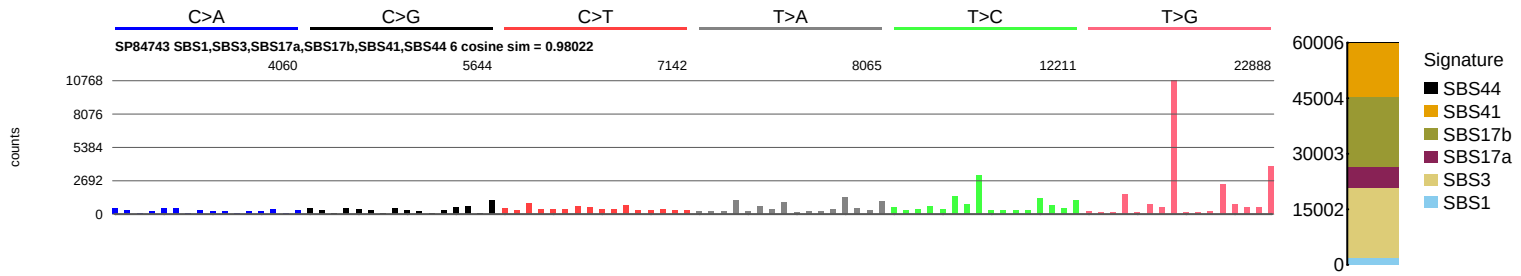
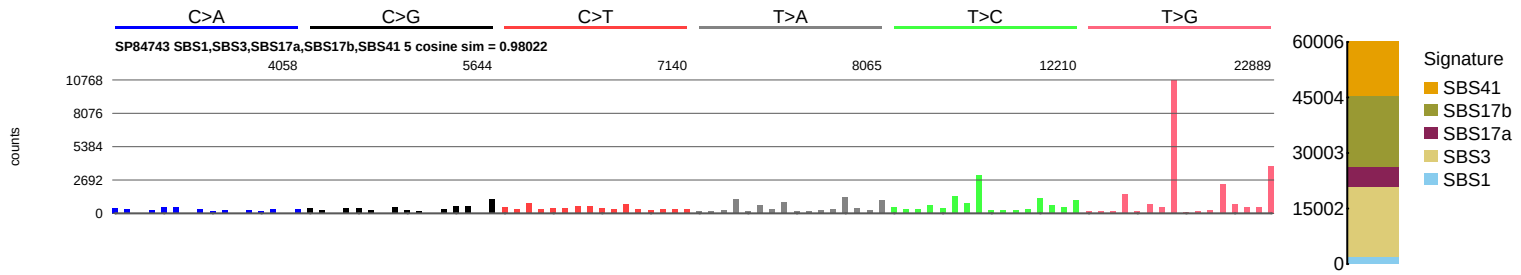
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1

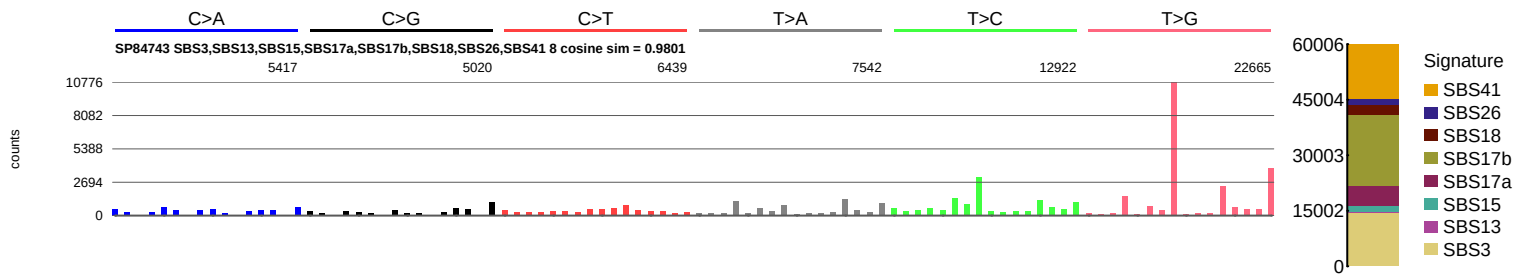

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98026
6110
2851
8657
7543
11867
22980
counts
10924
8193
5462
2731
0
60006
45004
30003
15002
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2

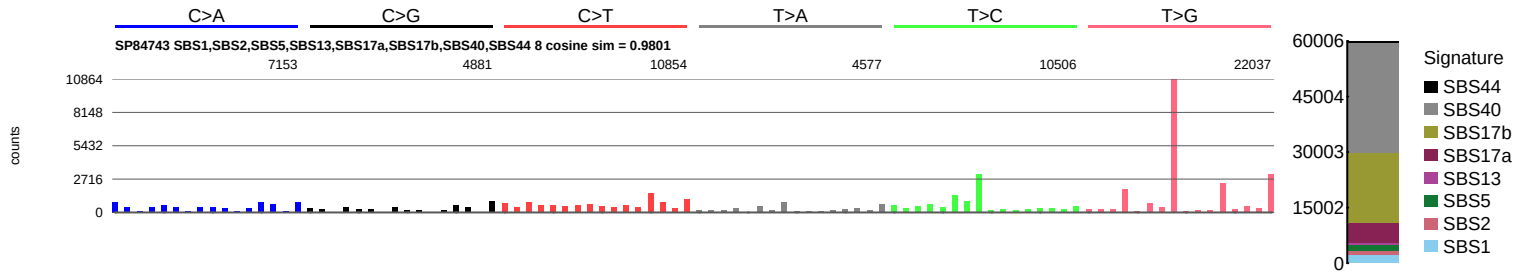

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98025
4062
5717
7139
8030
12191
22868
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS1

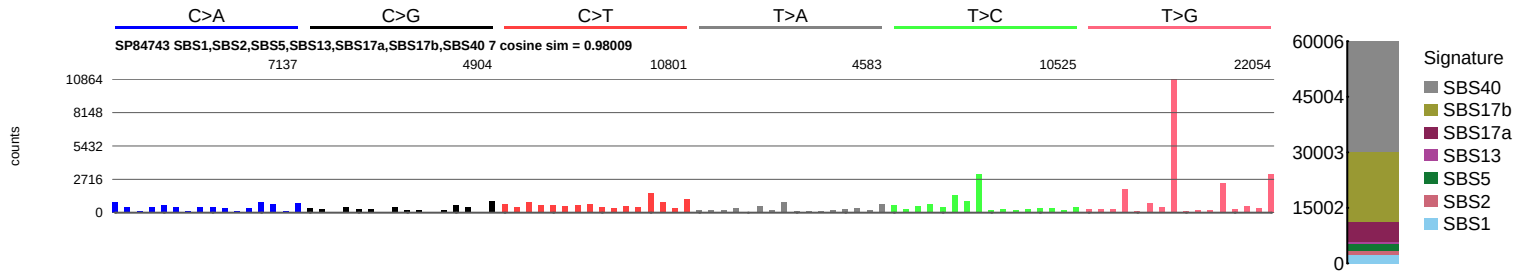

C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98025
4061
5727
7126
8035
12185
22873
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98022
4060
5644
7142
8065
12211
22888
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98022
4058
5644
7140
8065
12210
22889
counts
10768
8076
5384
2692
0
60006
45004
30003
15002
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.9801
5417
5020
6439
7542
12922
22665
counts
10776
8082
5388
2694
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.9801
7153
4881
10854
4577
10506
22037
counts
10864
8148
5432
2716
0
60006
45004
30003
15002
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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SP84743 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98009
7137
4904
10801
4583
10525
22054
counts
10864
8148
5432
2716
0
60006
45004
30003
15002
0
Signature
SBS40
SBS17b
SBS17a
SBS13
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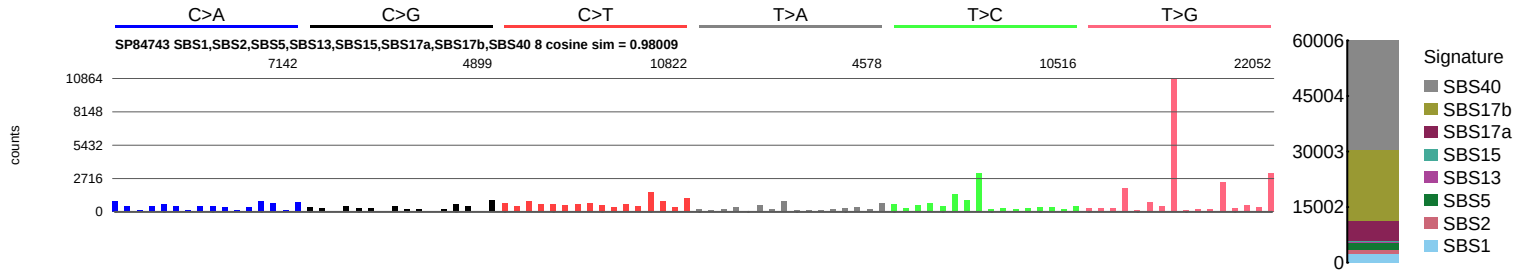
SBS2
SBS1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C>T
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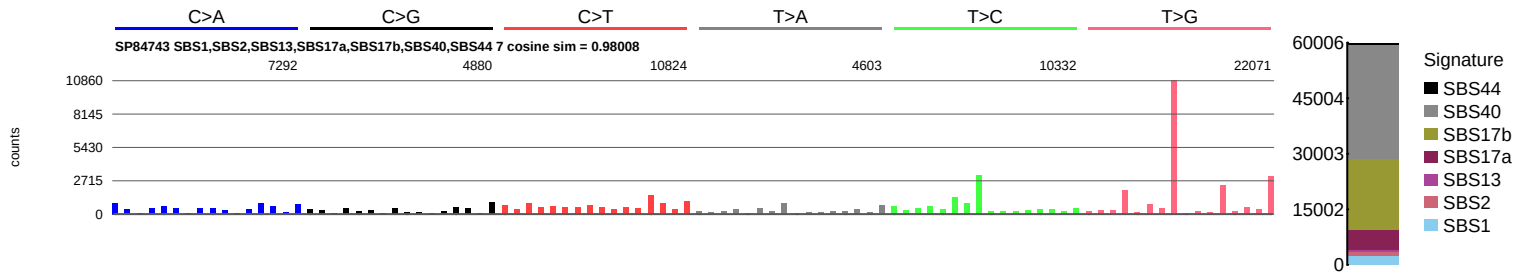
SP84743 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98009
7142
4899
10822
4578
10516
22052
counts
10864
8148
5432
2716
0
60006
45004
30003
15002
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98008
7292
4880
10824
4603
10332
22071
counts
10860
8145
5430
2715
0
60006
45004
30003
15002
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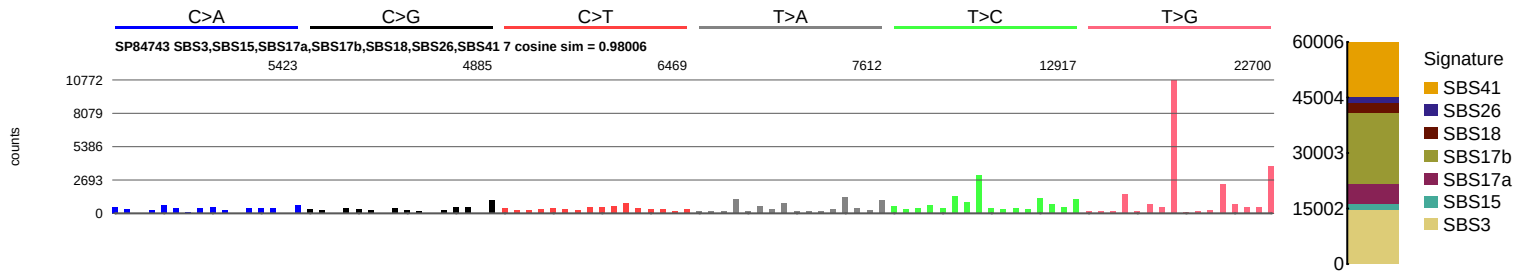
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1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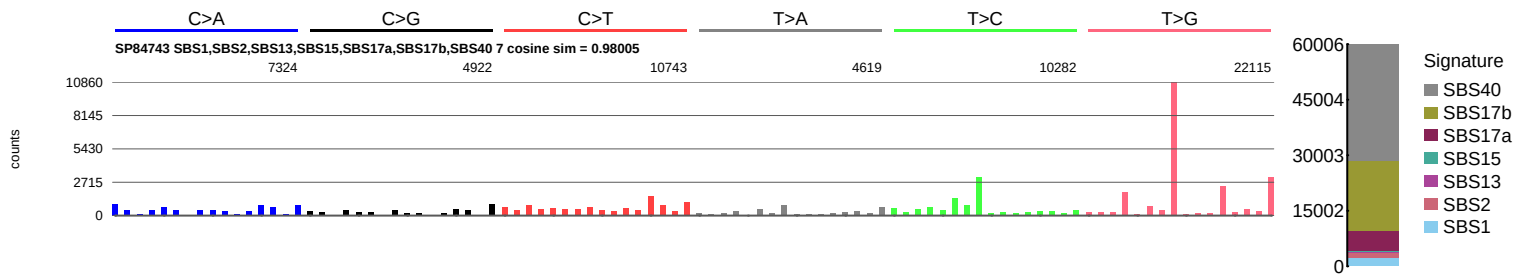
SP84743 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98006
5423
4885
6469
7612
12917
22700
counts
10772
8079
5386
2693
0
60006
45004
30003
15002
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3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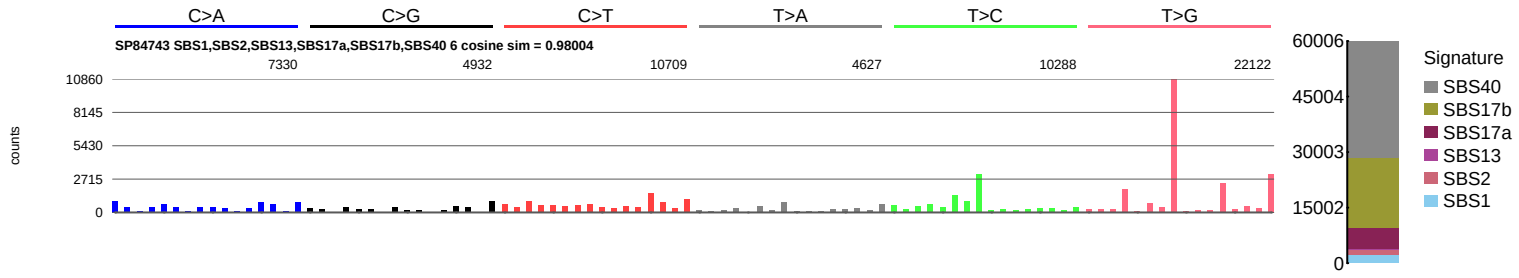
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98005
7324
4922
10743
4619
10282
22115
counts
10860
8145
5430
2715
0
60006
45004
30003
15002
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98004
7330
4932
10709
4627
10288
22122
counts
10860
8145
5430
2715
0
60006
45004
30003
15002
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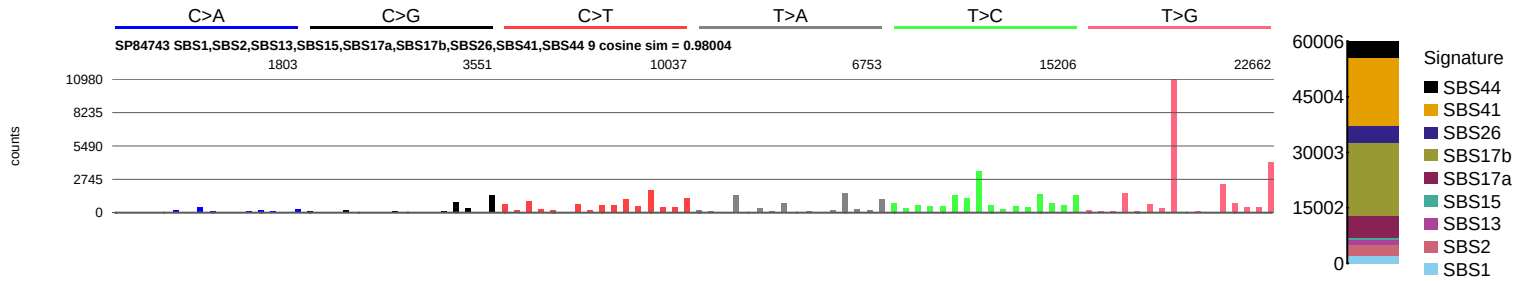
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1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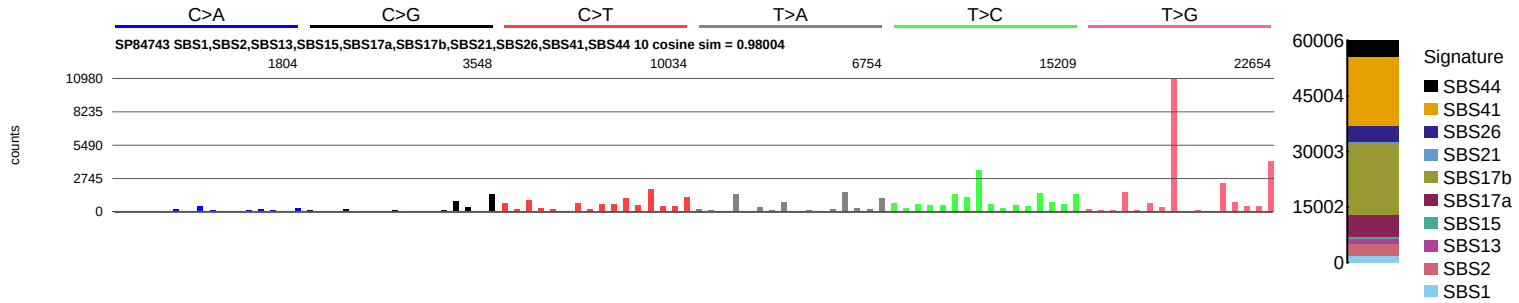
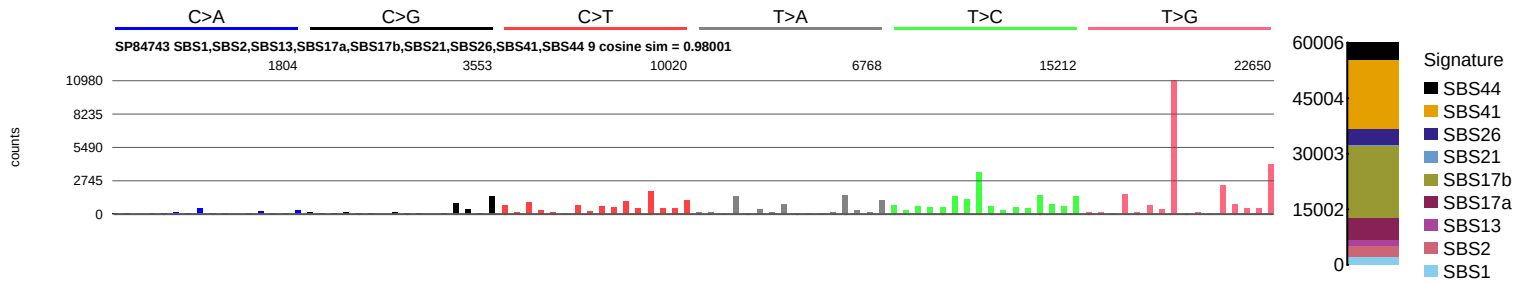
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98004
1803
3551
10037
6753
15206
22662
counts
10980
8235
5490
2745
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98004
1804
3548
10034
6754
15209
22654
counts
10980
8235
5490
2745
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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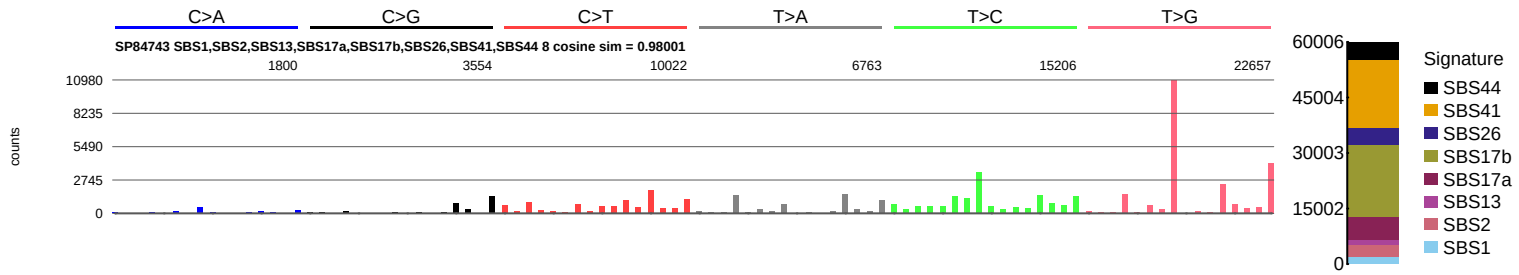
C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98001
1804
3553
10020
6768
15212
22650
counts
10980
8235
5490
2745
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84743 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98001
1800
3554
10022
6763
15206
22657
counts
10980
8235
5490
2745
0
60006
45004
30003
15002
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1